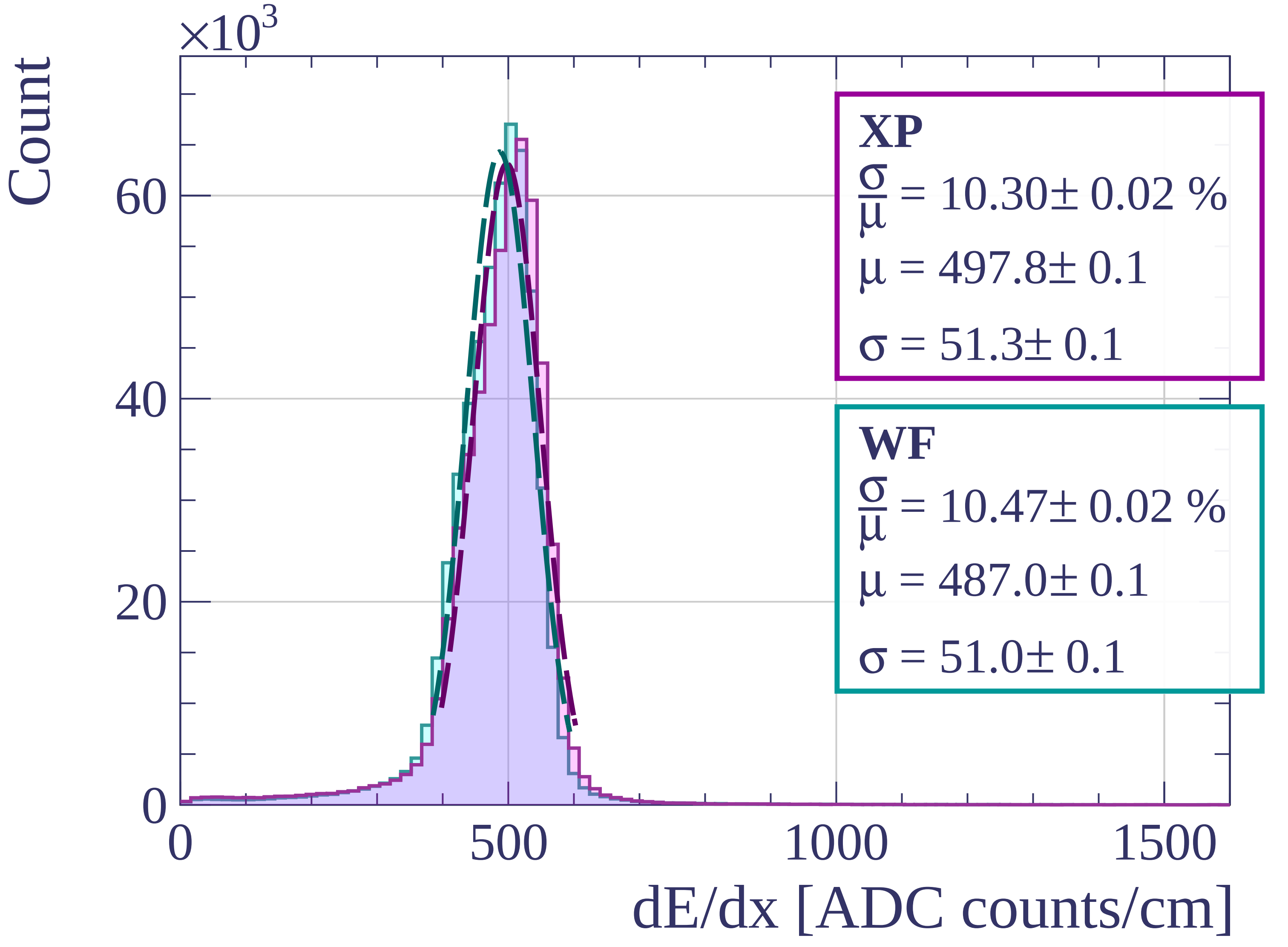
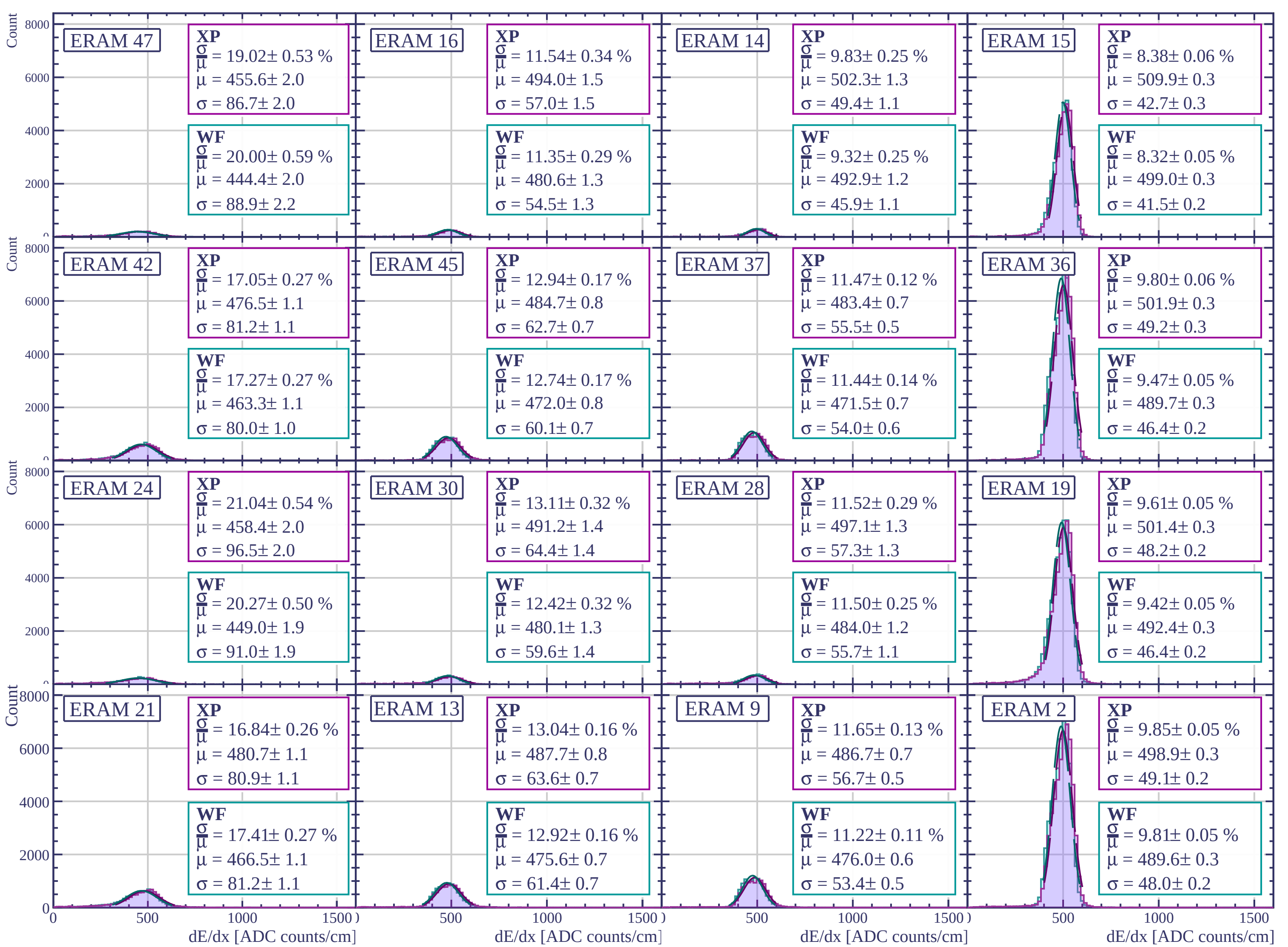
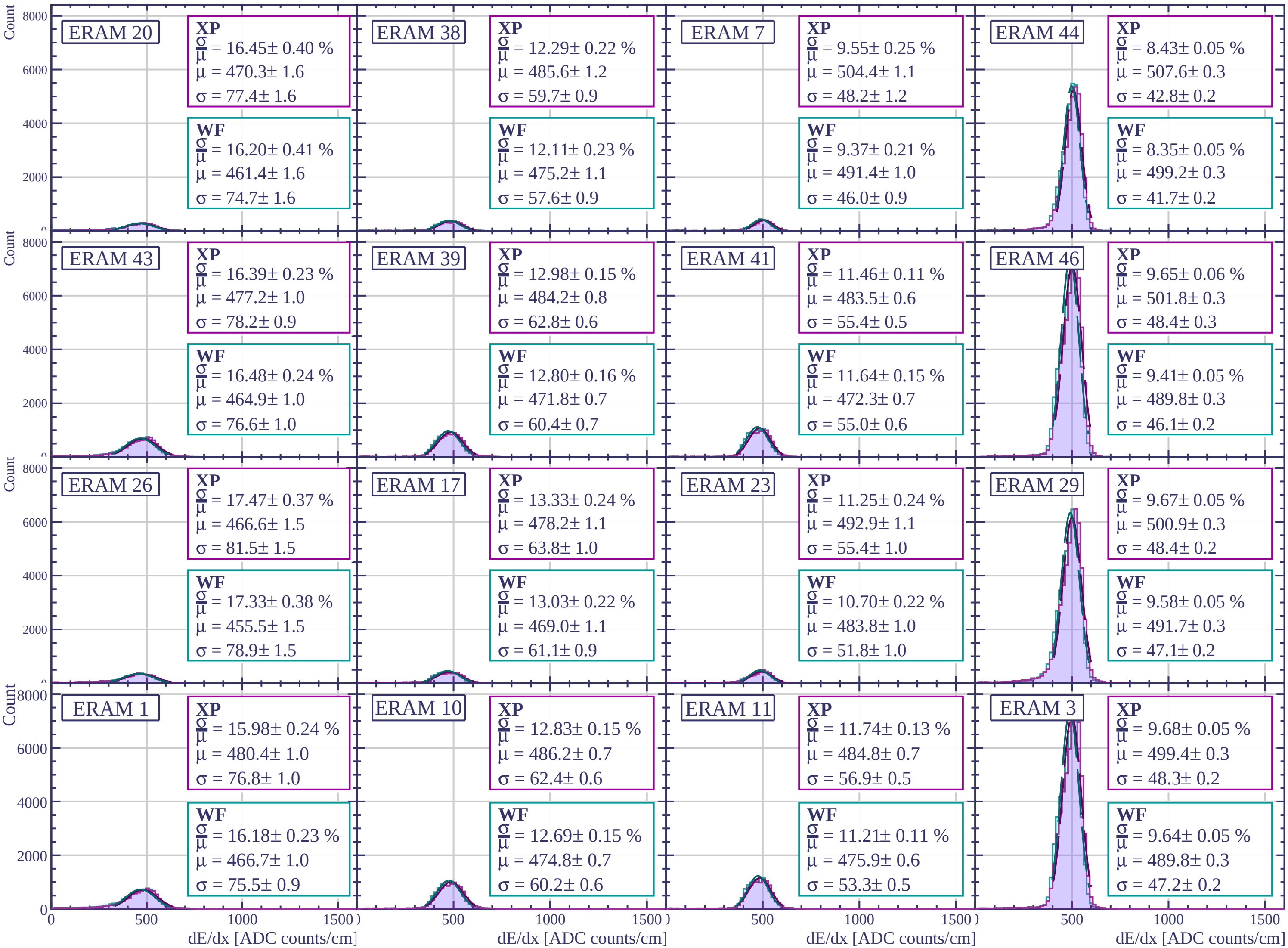


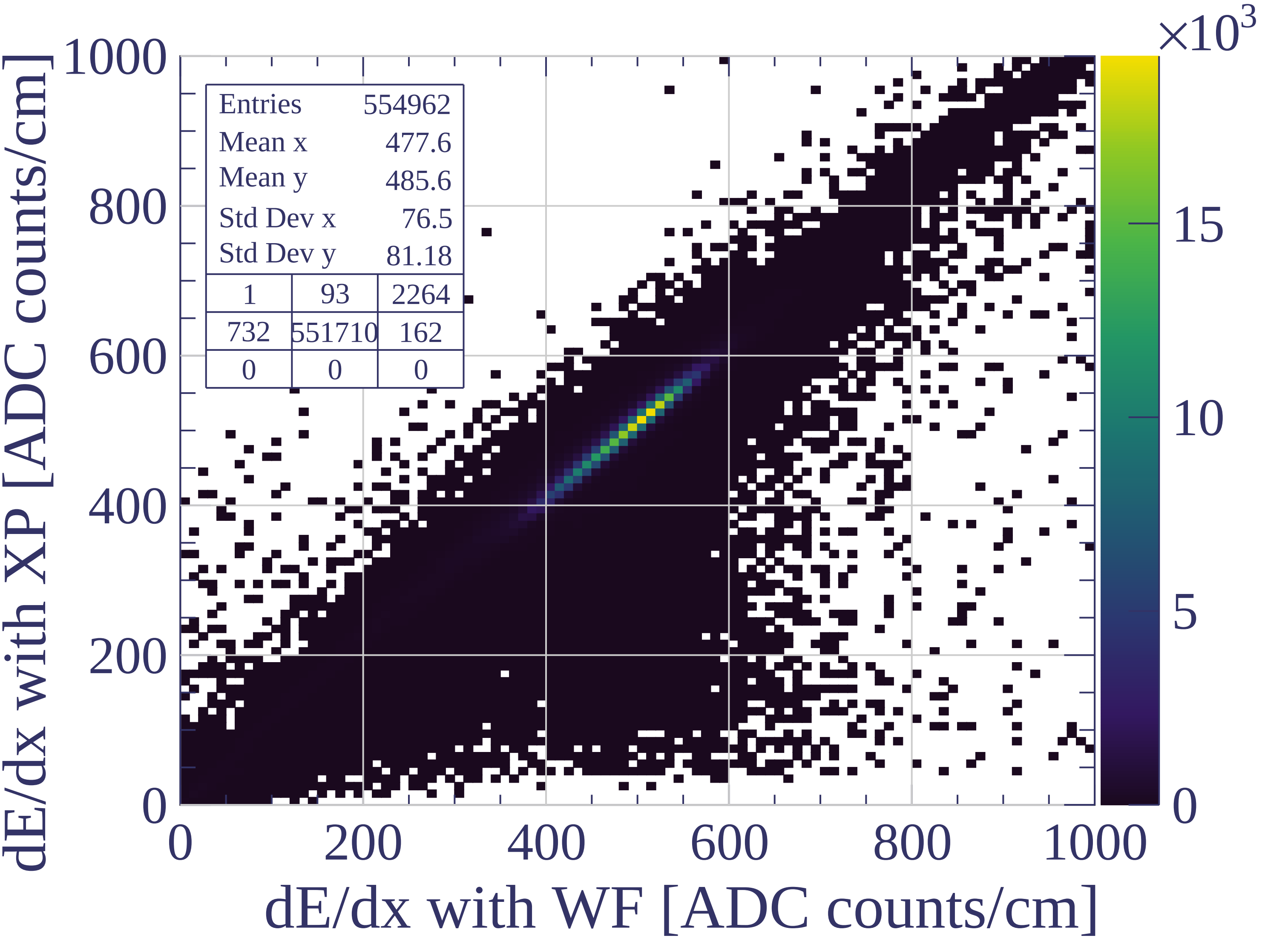
×10^3
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 10.30± 0.02 %
μ = 497.8± 0.1
σ = 51.3± 0.1
WF
σ/μ = 10.47± 0.02 %
μ = 487.0± 0.1
σ = 51.0± 0.1



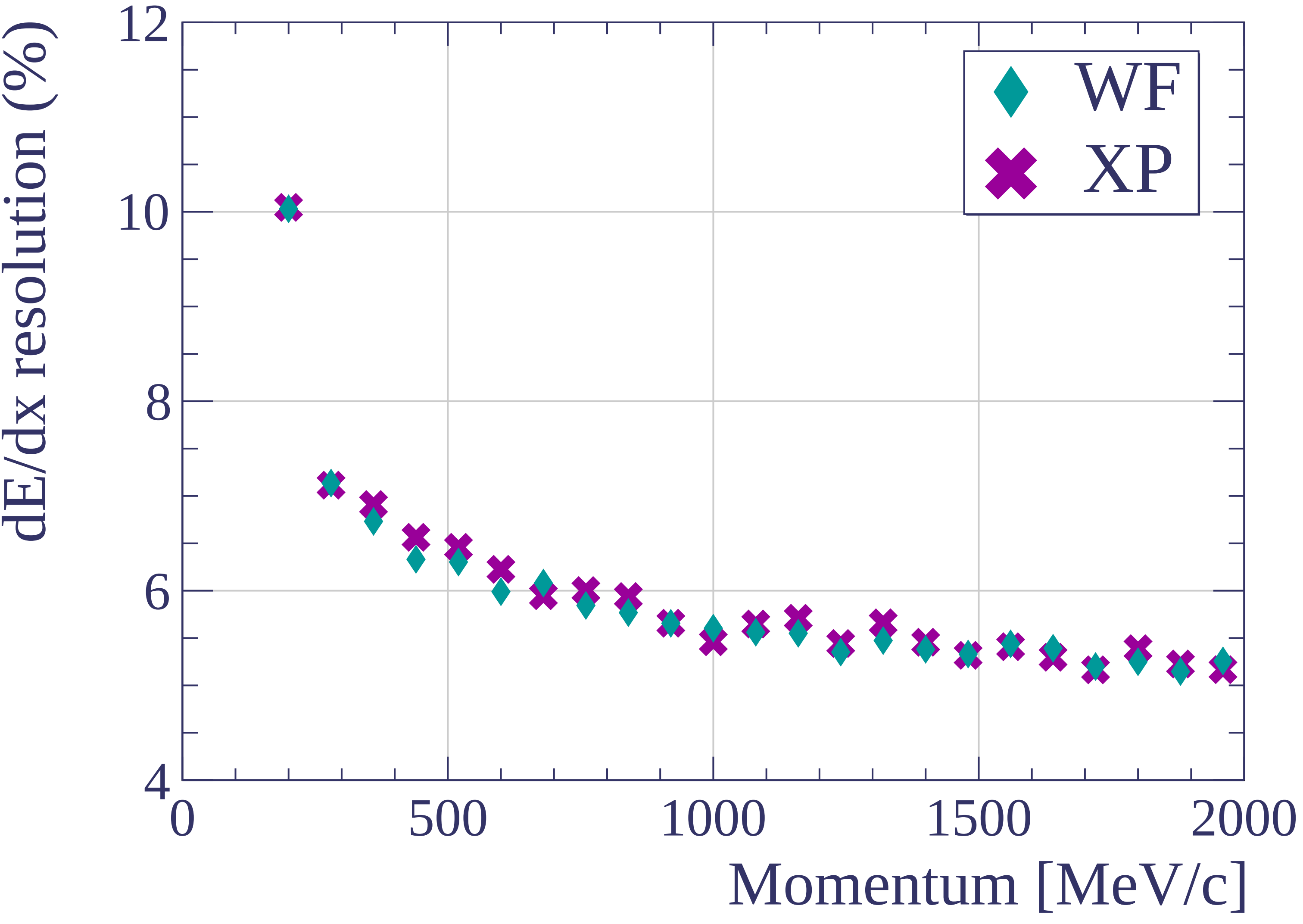
ERAM 47
XP
σ/μ = 19.02± 0.53 %
μ = 455.6± 2.0
σ = 86.7± 2.0
WF
σ/μ = 20.00± 0.59 %
μ = 444.4± 2.0
σ = 88.9± 2.2
ERAM 16
XP
σ/μ = 11.54± 0.34 %
μ = 494.0± 1.5
σ = 57.0± 1.5
WF
σ/μ = 11.35± 0.29 %
μ = 480.6± 1.3
σ = 54.5± 1.3
ERAM 14
XP
σ/μ = 9.83± 0.25 %
μ = 502.3± 1.3
σ = 49.4± 1.1
WF
σ/μ = 9.32± 0.25 %
μ = 492.9± 1.2
σ = 45.9± 1.1
ERAM 15
XP
σ/μ = 8.38± 0.06 %
μ = 509.9± 0.3
σ = 42.7± 0.3
WF
σ/μ = 8.32± 0.05 %
μ = 499.0± 0.3
σ = 41.5± 0.2
ERAM 42
XP
σ/μ = 17.05± 0.27 %
μ = 476.5± 1.1
σ = 81.2± 1.1
WF
σ/μ = 17.27± 0.27 %
μ = 463.3± 1.1
σ = 80.0± 1.0
ERAM 45
XP
σ/μ = 12.94± 0.17 %
μ = 484.7± 0.8
σ = 62.7± 0.7
WF
σ/μ = 12.74± 0.17 %
μ = 472.0± 0.8
σ = 60.1± 0.7
ERAM 37
XP
σ/μ = 11.47± 0.12 %
μ = 483.4± 0.7
σ = 55.5± 0.5
WF
σ/μ = 11.44± 0.14 %
μ = 471.5± 0.7
σ = 54.0± 0.6
ERAM 36
XP
σ/μ = 9.80± 0.06 %
μ = 501.9± 0.3
σ = 49.2± 0.3
WF
σ/μ = 9.47± 0.05 %
μ = 489.7± 0.3
σ = 46.4± 0.2
ERAM 24
XP
σ/μ = 21.04± 0.54 %
μ = 458.4± 2.0
σ = 96.5± 2.0
WF
σ/μ = 20.27± 0.50 %
μ = 449.0± 1.9
σ = 91.0± 1.9
ERAM 30
XP
σ/μ = 13.11± 0.32 %
μ = 491.2± 1.4
σ = 64.4± 1.4
WF
σ/μ = 12.42± 0.32 %
μ = 480.1± 1.3
σ = 59.6± 1.4
ERAM 28
XP
σ/μ = 11.52± 0.29 %
μ = 497.1± 1.3
σ = 57.3± 1.3
WF
σ/μ = 11.50± 0.25 %
μ = 484.0± 1.2
σ = 55.7± 1.1
ERAM 19
XP
σ/μ = 9.61± 0.05 %
μ = 501.4± 0.3
σ = 48.2± 0.2
WF
σ/μ = 9.42± 0.05 %
μ = 492.4± 0.3
σ = 46.4± 0.2
ERAM 21
XP
σ/μ = 16.84± 0.26 %
μ = 480.7± 1.1
σ = 80.9± 1.1
WF
σ/μ = 17.41± 0.27 %
μ = 466.5± 1.1
σ = 81.2± 1.1
ERAM 13
XP
σ/μ = 13.04± 0.16 %
μ = 487.7± 0.8
σ = 63.6± 0.7
WF
σ/μ = 12.92± 0.16 %
μ = 475.6± 0.7
σ = 61.4± 0.7
ERAM 9
XP
σ/μ = 11.65± 0.13 %
μ = 486.7± 0.7
σ = 56.7± 0.5
WF
σ/μ = 11.22± 0.11 %
μ = 476.0± 0.6
σ = 53.4± 0.5
ERAM 2
XP
σ/μ = 9.85± 0.05 %
μ = 498.9± 0.3
σ = 49.1± 0.2
WF
σ/μ = 9.81± 0.05 %
μ = 489.6± 0.3
σ = 48.0± 0.2
Count
8000
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]

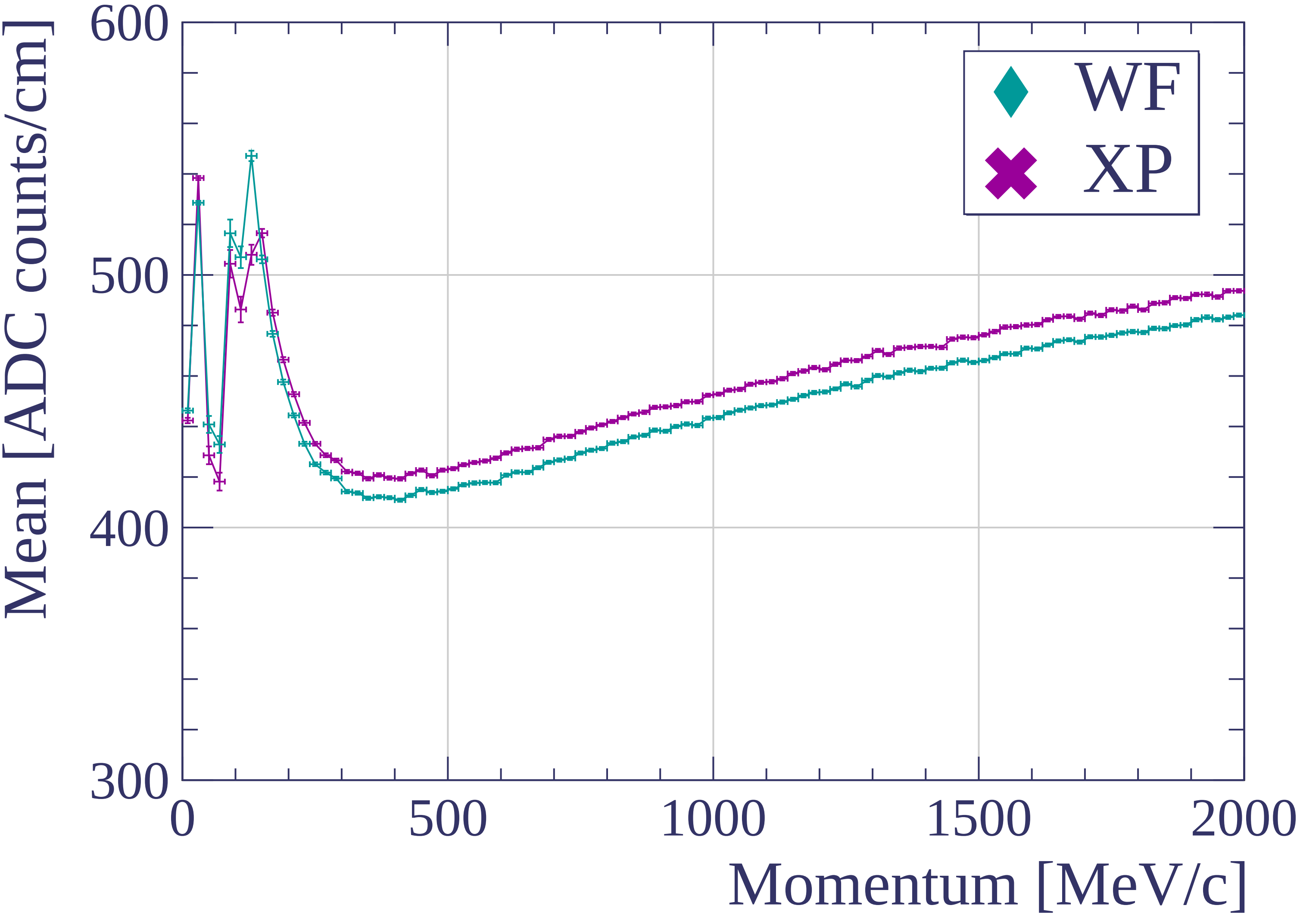


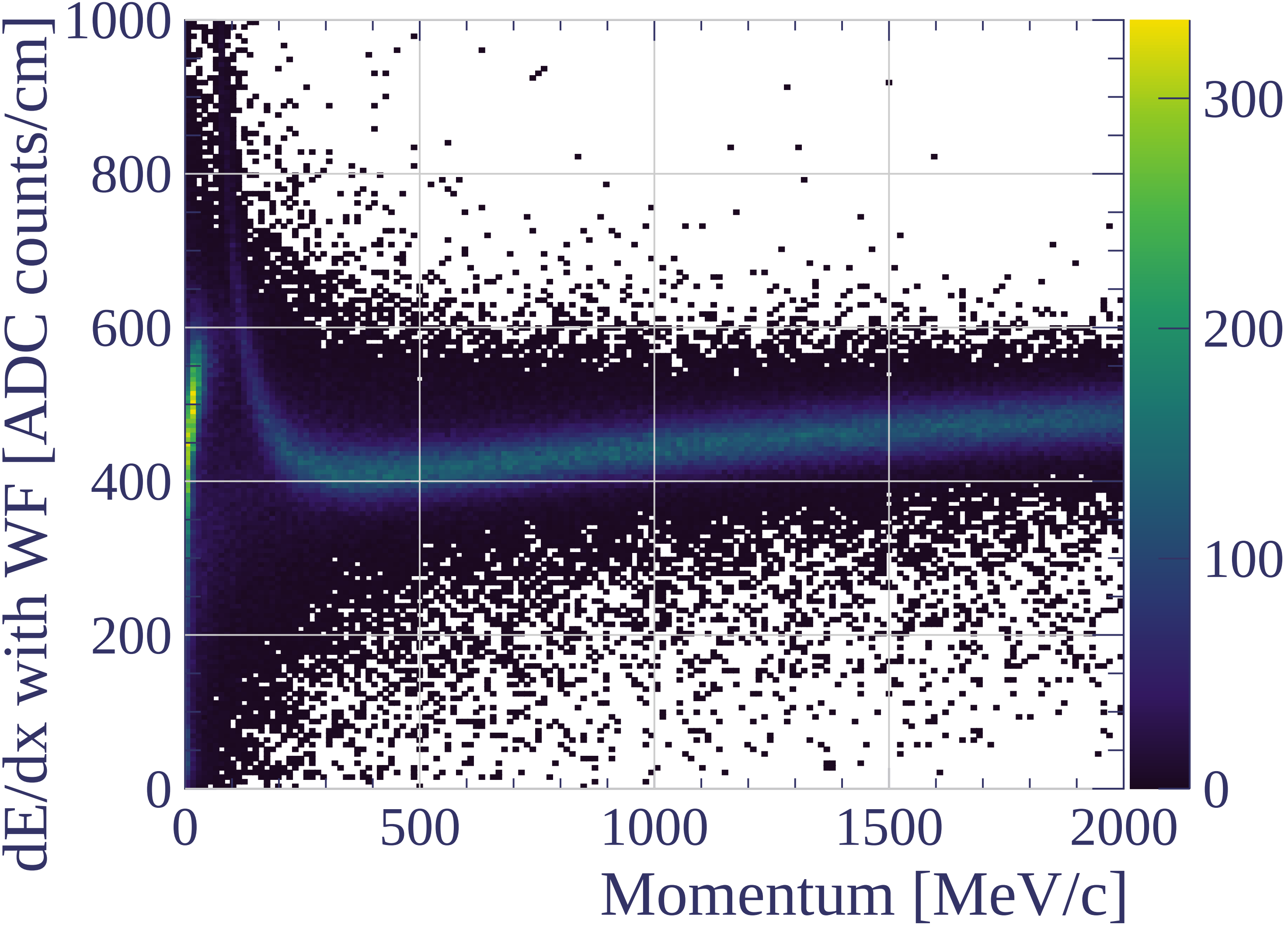
ERAM 20
XP
σ/μ = 16.45± 0.40 %
μ = 470.3± 1.6
σ = 77.4± 1.6
WF
σ/μ = 16.20± 0.41 %
μ = 461.4± 1.6
σ = 74.7± 1.6
ERAM 38
XP
σ/μ = 12.29± 0.22 %
μ = 485.6± 1.2
σ = 59.7± 0.9
WF
σ/μ = 12.11± 0.23 %
μ = 475.2± 1.1
σ = 57.6± 0.9
ERAM 7
XP
σ/μ = 9.55± 0.25 %
μ = 504.4± 1.1
σ = 48.2± 1.2
WF
σ/μ = 9.37± 0.21 %
μ = 491.4± 1.0
σ = 46.0± 0.9
ERAM 44
XP
σ/μ = 8.43± 0.05 %
μ = 507.6± 0.3
σ = 42.8± 0.2
WF
σ/μ = 8.35± 0.05 %
μ = 499.2± 0.3
σ = 41.7± 0.2
ERAM 43
XP
σ/μ = 16.39± 0.23 %
μ = 477.2± 1.0
σ = 78.2± 0.9
WF
σ/μ = 16.48± 0.24 %
μ = 464.9± 1.0
σ = 76.6± 1.0
ERAM 39
XP
σ/μ = 12.98± 0.15 %
μ = 484.2± 0.8
σ = 62.8± 0.6
WF
σ/μ = 12.80± 0.16 %
μ = 471.8± 0.7
σ = 60.4± 0.7
ERAM 41
XP
σ/μ = 11.46± 0.11 %
μ = 483.5± 0.6
σ = 55.4± 0.5
WF
σ/μ = 11.64± 0.15 %
μ = 472.3± 0.7
σ = 55.0± 0.6
ERAM 46
XP
σ/μ = 9.65± 0.06 %
μ = 501.8± 0.3
σ = 48.4± 0.3
WF
σ/μ = 9.41± 0.05 %
μ = 489.8± 0.3
σ = 46.1± 0.2
ERAM 26
XP
σ/μ = 17.47± 0.37 %
μ = 466.6± 1.5
σ = 81.5± 1.5
WF
σ/μ = 17.33± 0.38 %
μ = 455.5± 1.5
σ = 78.9± 1.5
ERAM 17
XP
σ/μ = 13.33± 0.24 %
μ = 478.2± 1.1
σ = 63.8± 1.0
WF
σ/μ = 13.03± 0.22 %
μ = 469.0± 1.1
σ = 61.1± 0.9
ERAM 23
XP
σ/μ = 11.25± 0.24 %
μ = 492.9± 1.1
σ = 55.4± 1.0
WF
σ/μ = 10.70± 0.22 %
μ = 483.8± 1.0
σ = 51.8± 1.0
ERAM 29
XP
σ/μ = 9.67± 0.05 %
μ = 500.9± 0.3
σ = 48.4± 0.2
WF
σ/μ = 9.58± 0.05 %
μ = 491.7± 0.3
σ = 47.1± 0.2
ERAM 1
XP
σ/μ = 15.98± 0.24 %
μ = 480.4± 1.0
σ = 76.8± 1.0
WF
σ/μ = 16.18± 0.23 %
μ = 466.7± 1.0
σ = 75.5± 0.9
ERAM 10
XP
σ/μ = 12.83± 0.15 %
μ = 486.2± 0.7
σ = 62.4± 0.6
WF
σ/μ = 12.69± 0.15 %
μ = 474.8± 0.7
σ = 60.2± 0.6
ERAM 11
XP
σ/μ = 11.74± 0.13 %
μ = 484.8± 0.7
σ = 56.9± 0.5
WF
σ/μ = 11.21± 0.11 %
μ = 475.9± 0.6
σ = 53.3± 0.5
ERAM 3
XP
σ/μ = 9.68± 0.05 %
μ = 499.4± 0.3
σ = 48.3± 0.2
WF
σ/μ = 9.64± 0.05 %
μ = 489.8± 0.3
σ = 47.2± 0.2
Count
0
2000
4000
6000
8000
0
500
1000
1500
dE/dx [ADC counts/cm]



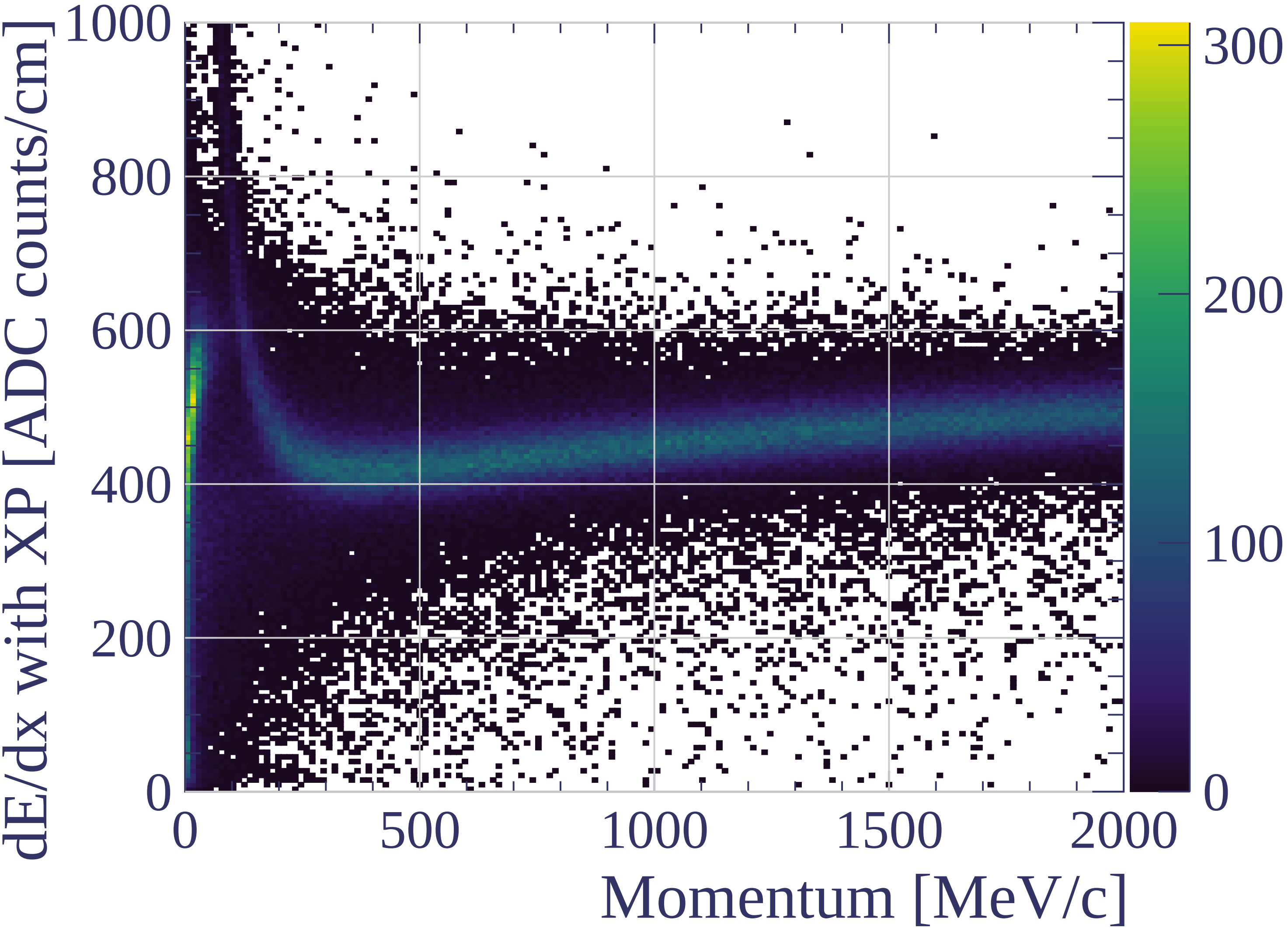
Entries 554962
Mean x 477.6
Mean y 485.6
Std Dev x 76.5
Std Dev y 81.18
1 | 93 | 2264
732 | 551710 | 162
0 | 0 | 0
dE/dx with XP [ADC counts/cm]
dE/dx with WF [ADC counts/cm]
0
200
400
600
800
1000
$\times 10^3$
15
10
5
0


dE/dx resolution (%)
Momentum [MeV/c]
WF
XP
12
10
8
6
4
0
500
1000
1500
2000

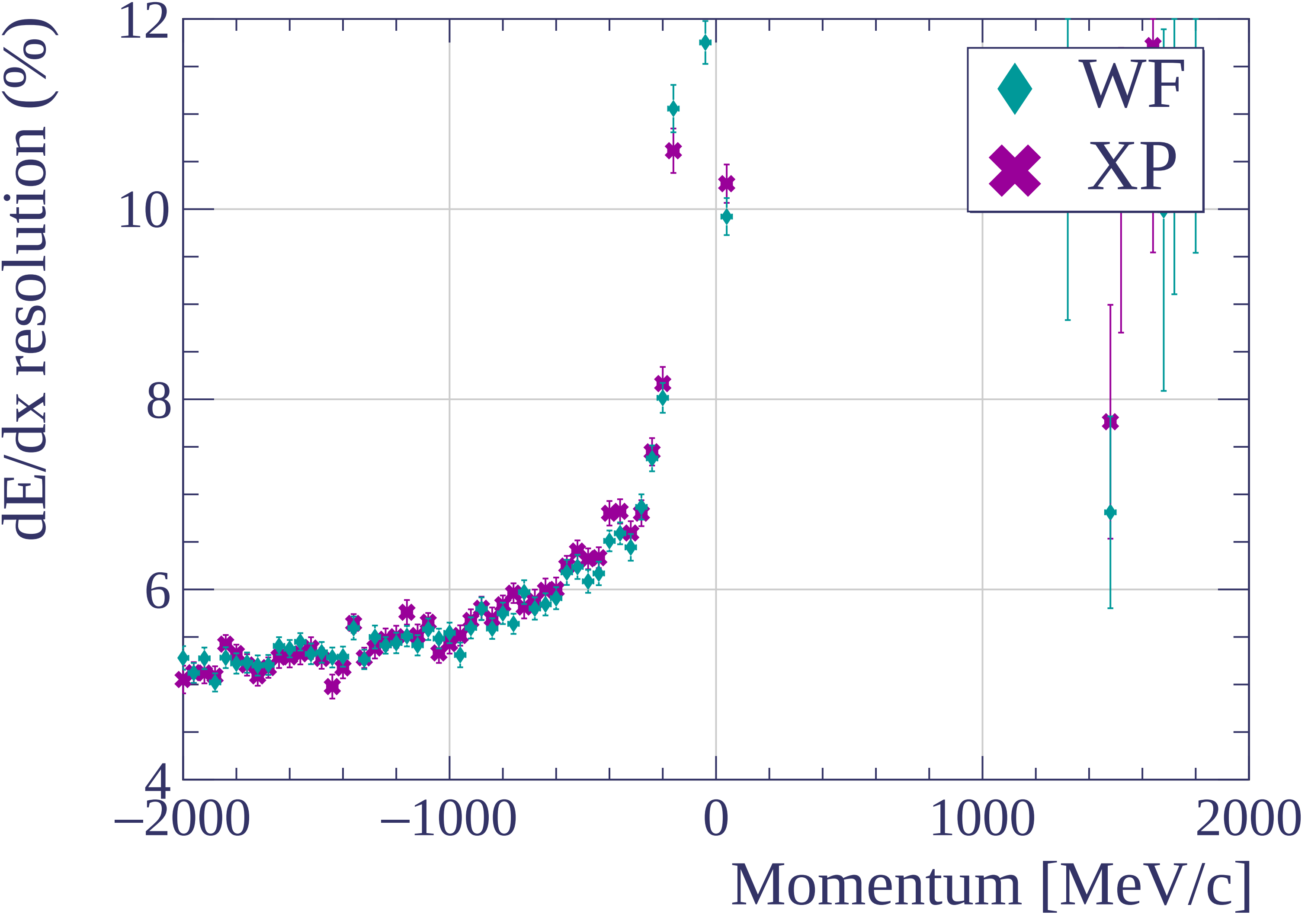

Mean [ADC counts/cm]
Momentum [MeV/c]
WF
XP
600
500
400
300
0
500
1000
1500
2000



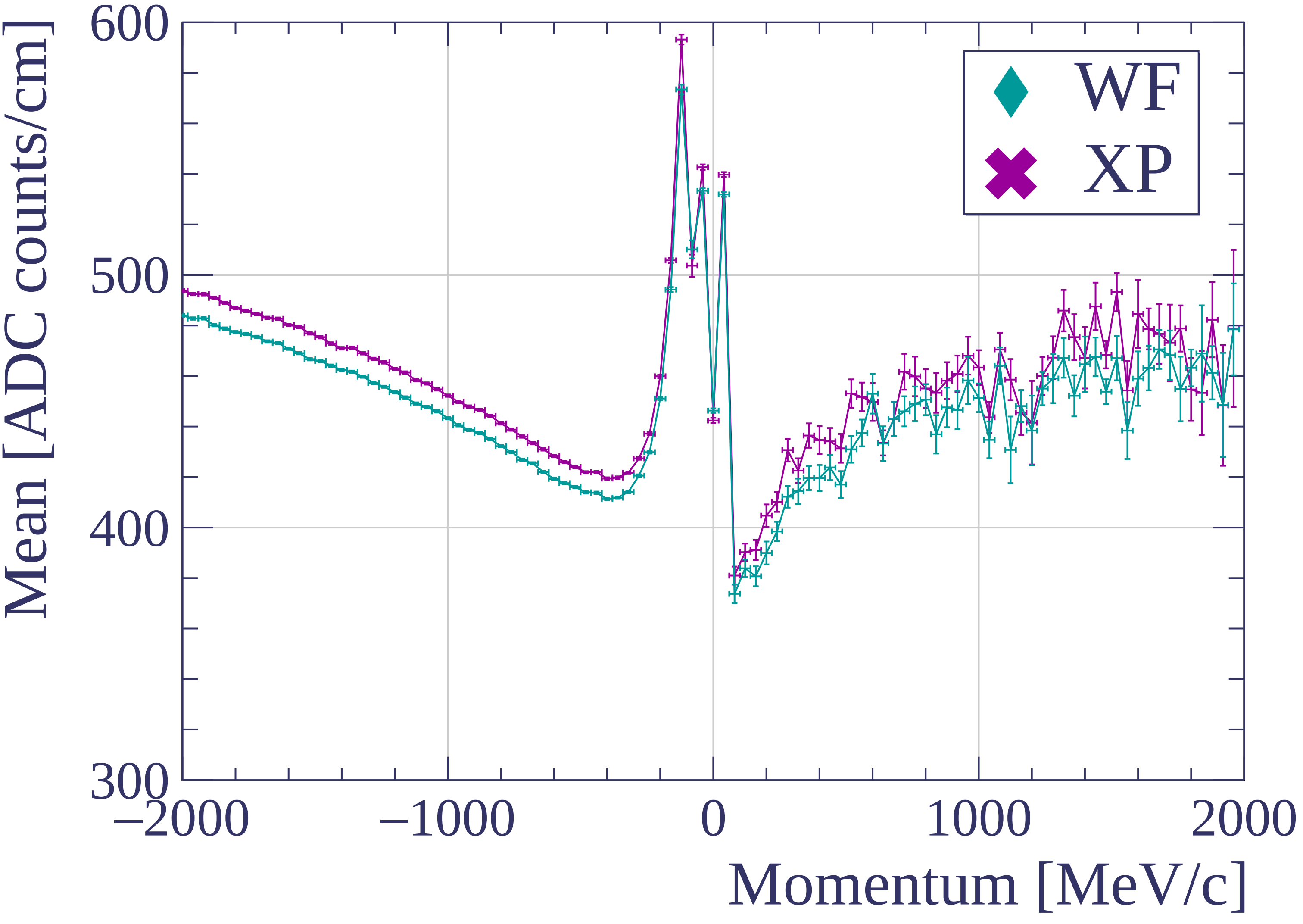
dE/dx with WF [ADC counts/cm]
Momentum [MeV/c]
0
200
400
600
800
1000
500
1000
1500
2000
100
200
300



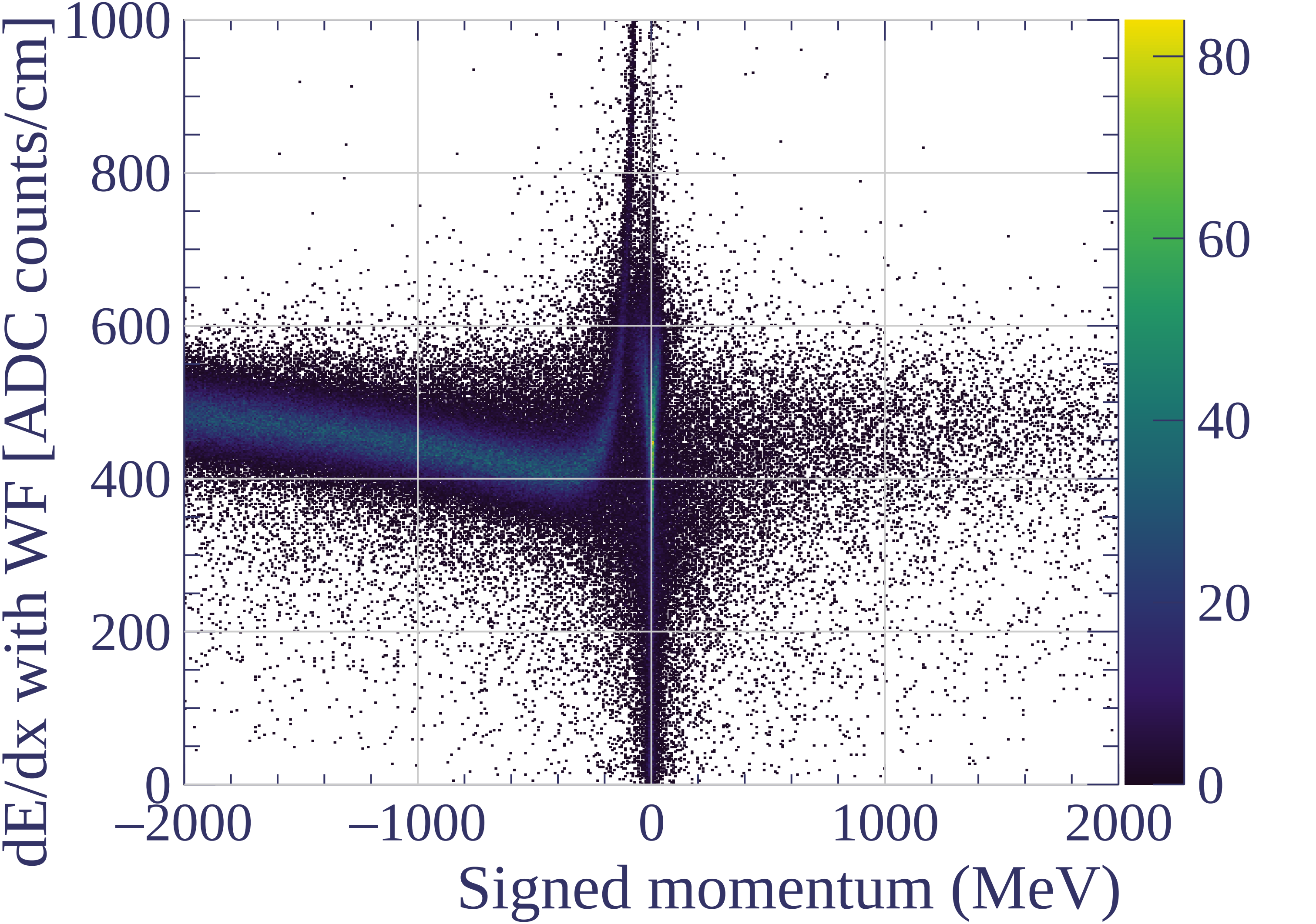
dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
0
500
1000
1500
2000
Momentum [MeV/c]
300
200
100
0



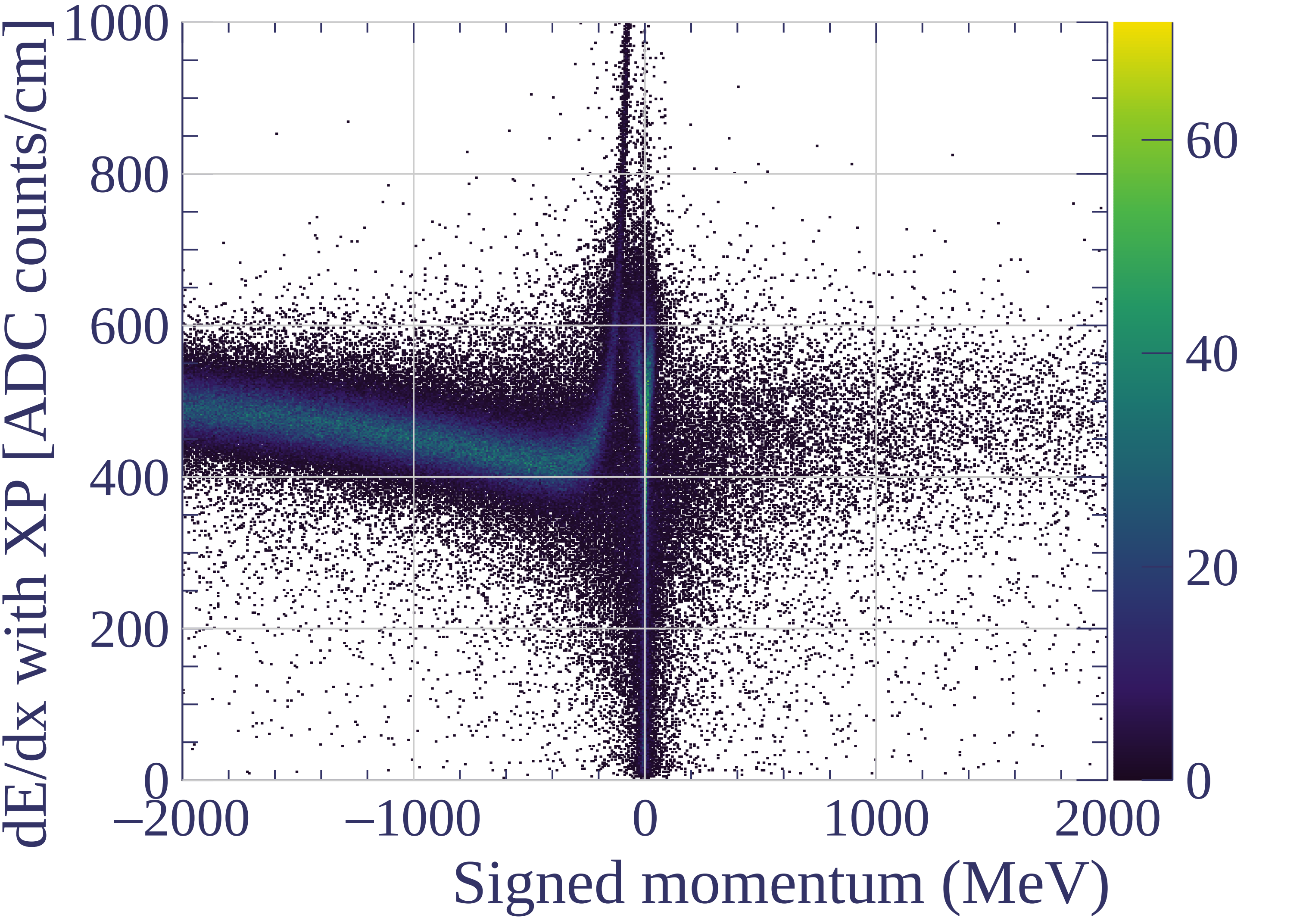
dE/dx resolution (%)
12
10
8
6
4
−2000
−1000
0
1000
2000
Momentum [MeV/c]
WF
XP



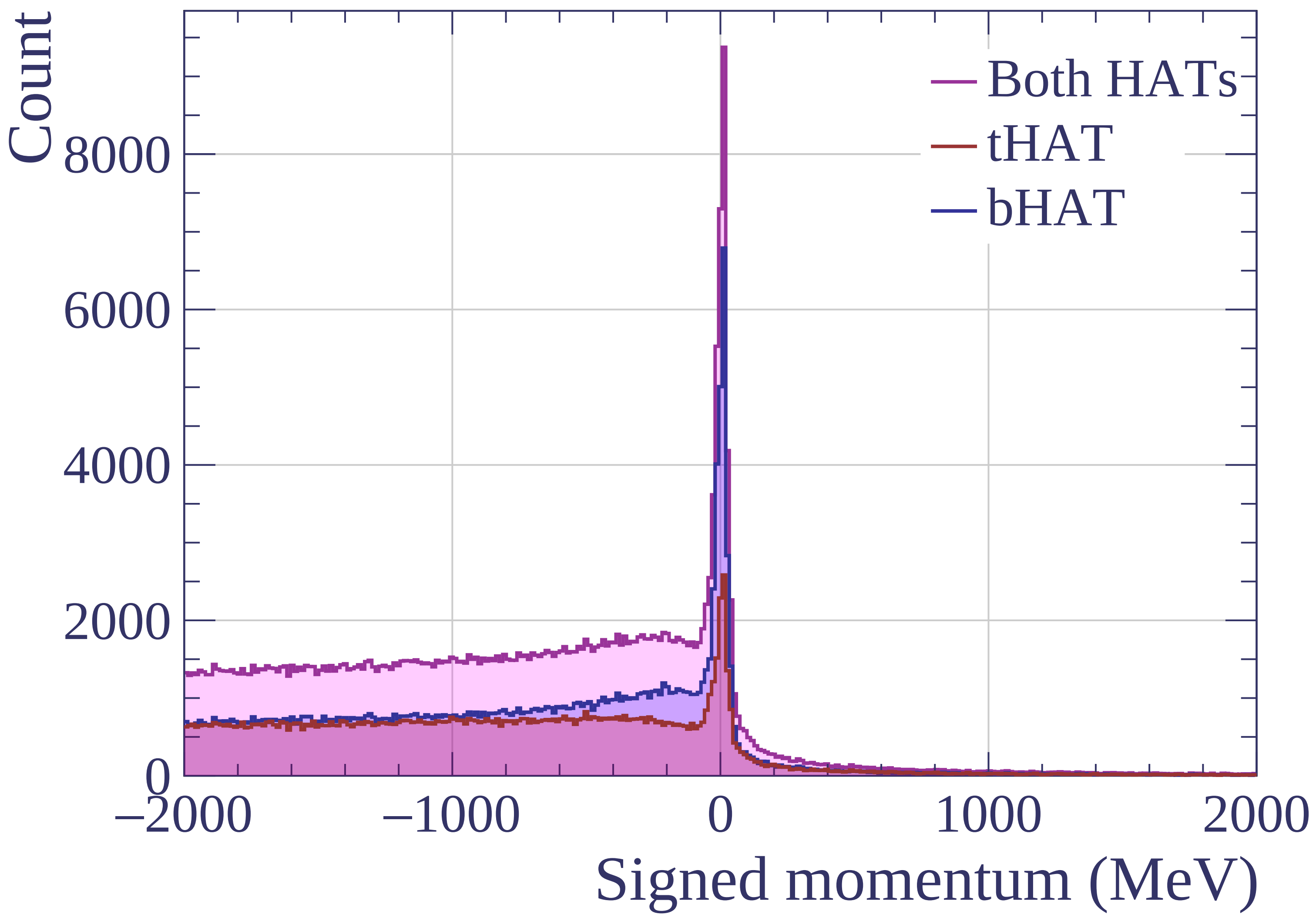
WF
XP
Mean [ADC counts/cm]
600
500
400
300
−2000
−1000
0
1000
2000
Momentum [MeV/c]



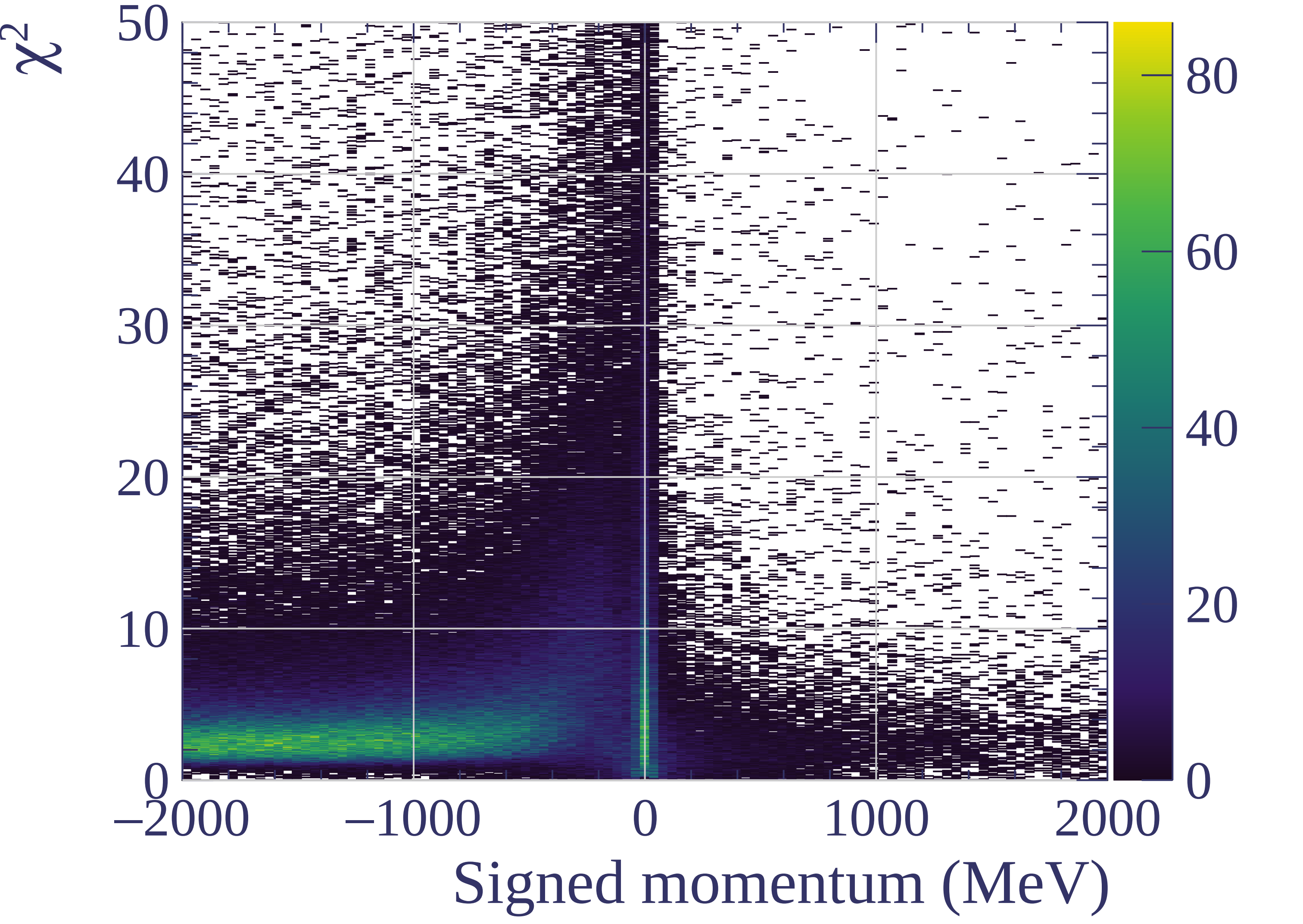
dE/dx with WF [ADC counts/cm]
Signed momentum (MeV)
1000
800
600
400
200
0
−2000
−1000
0
1000
2000
80
60
40
20
0



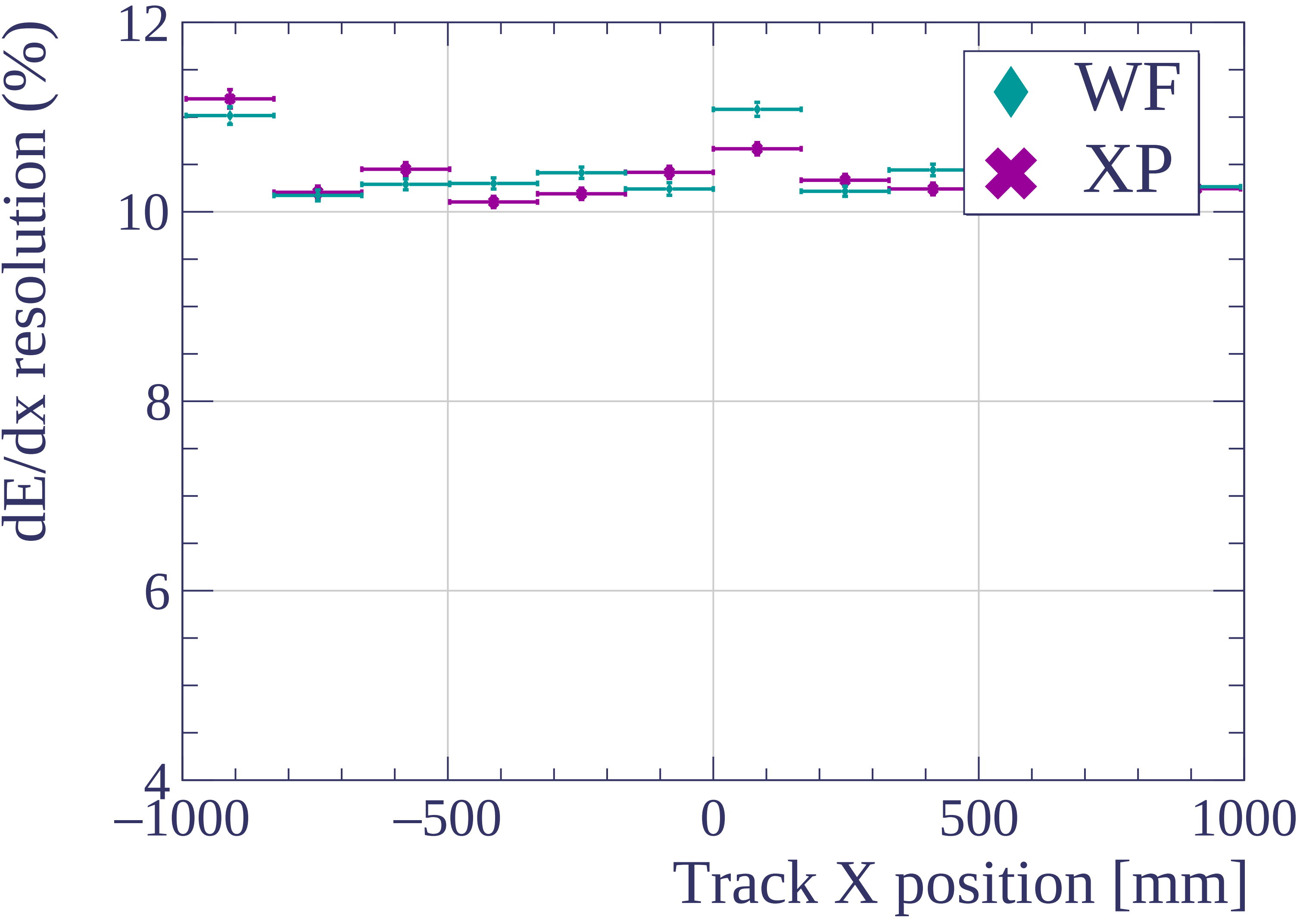
dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
−2000
−1000
0
1000
2000
Signed momentum (MeV)
60
40
20
0



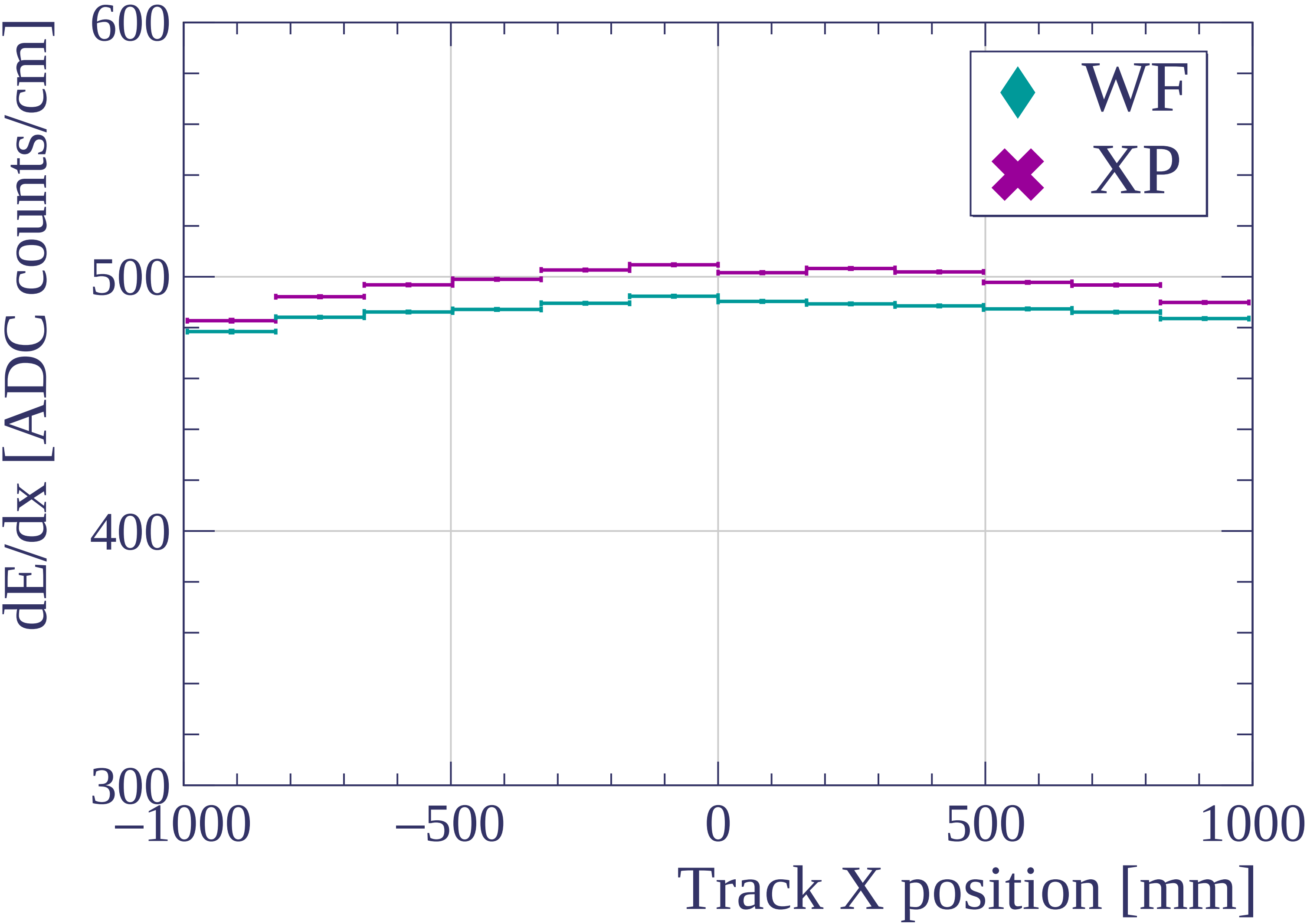
Count
8000
6000
4000
2000
0
−2000
−1000
0
1000
2000
Signed momentum (MeV)
Both HATs
tHAT
bHAT



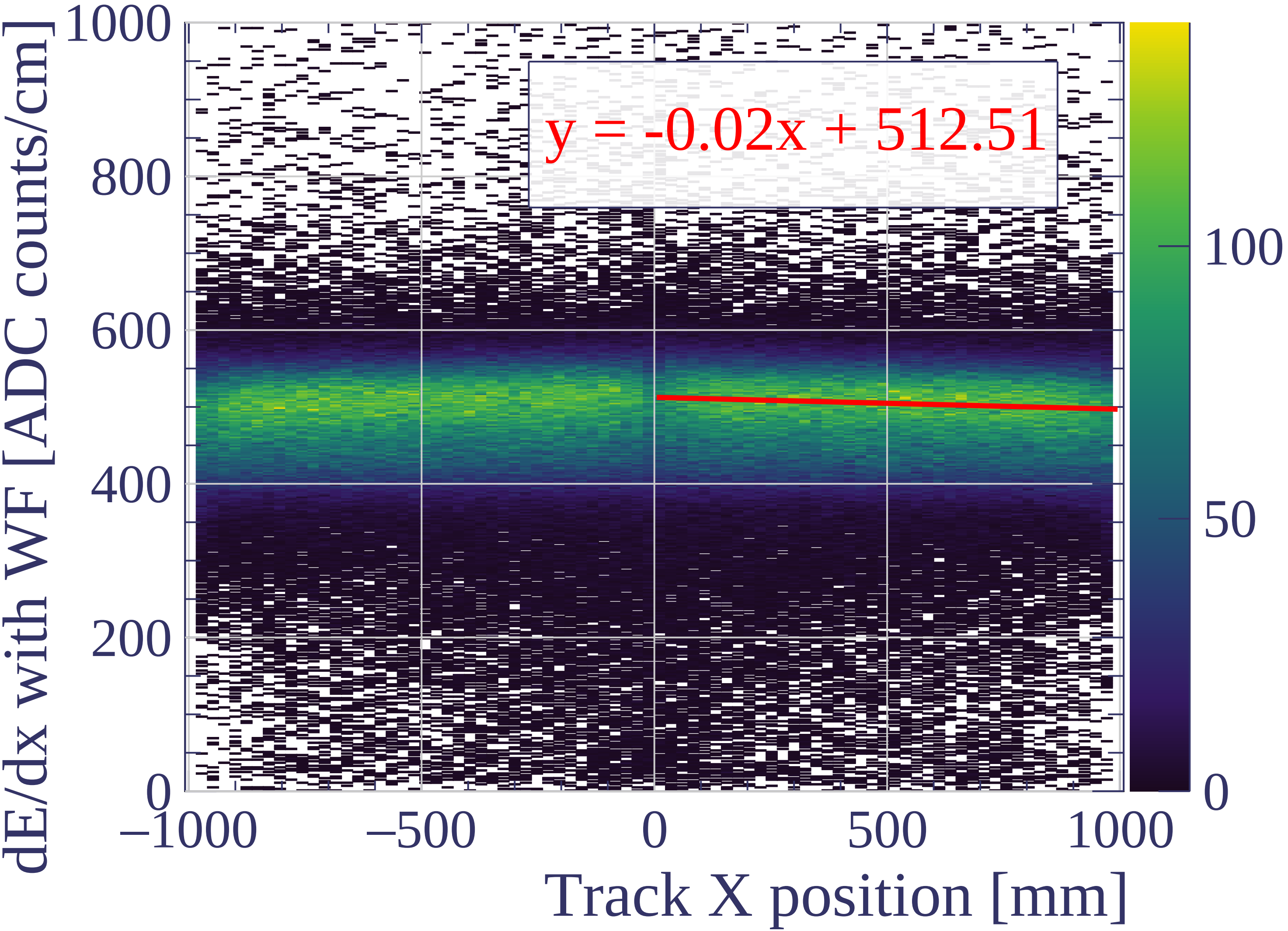
$\chi^2$
50
40
30
20
10
0
−2000
−1000
0
1000
2000
Signed momentum (MeV)
80
60
40
20
0



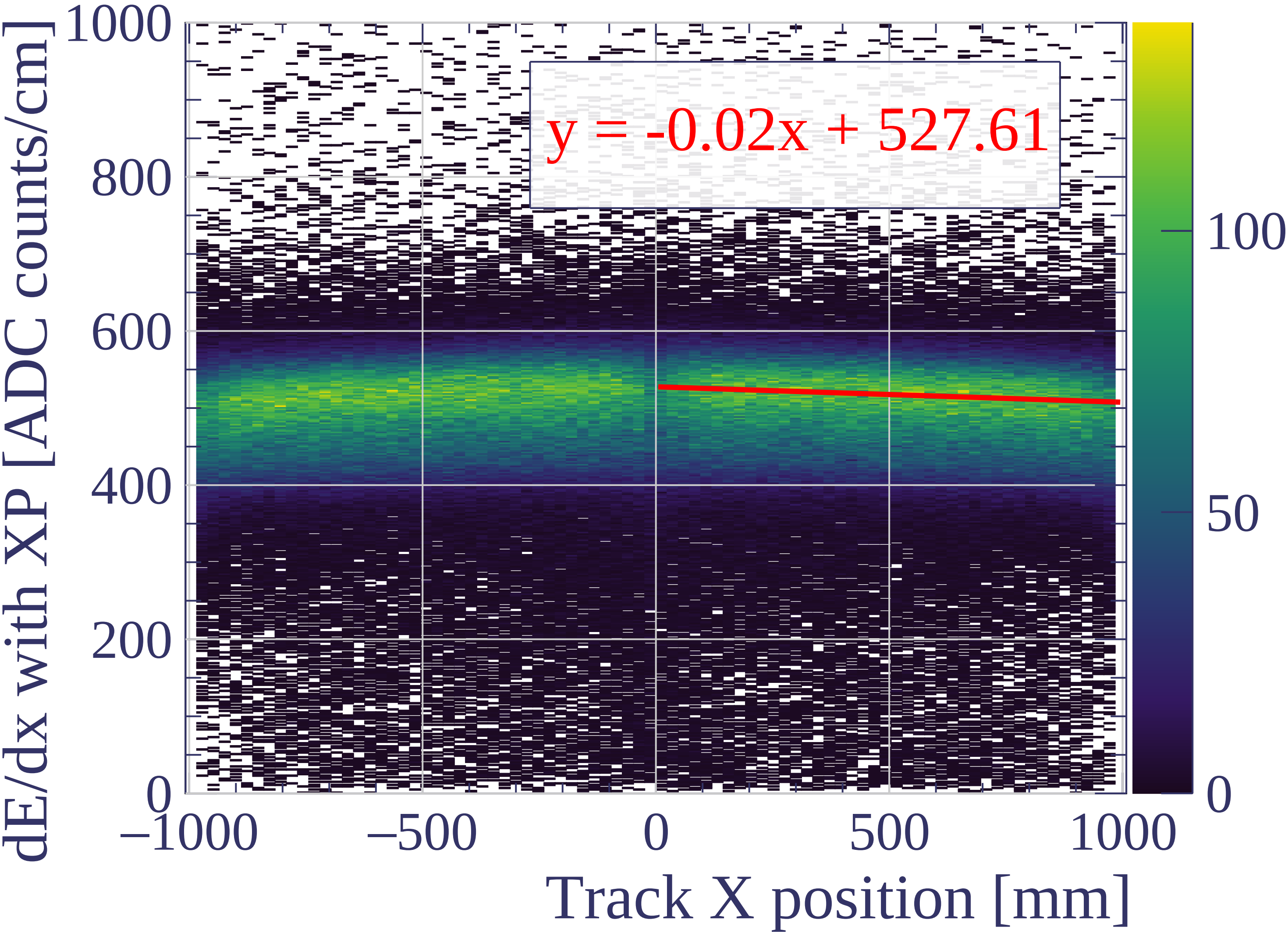
dE/dx resolution (%)
12
10
8
6
4
−1000
−500
0
500
1000
Track X position [mm]
WF
XP



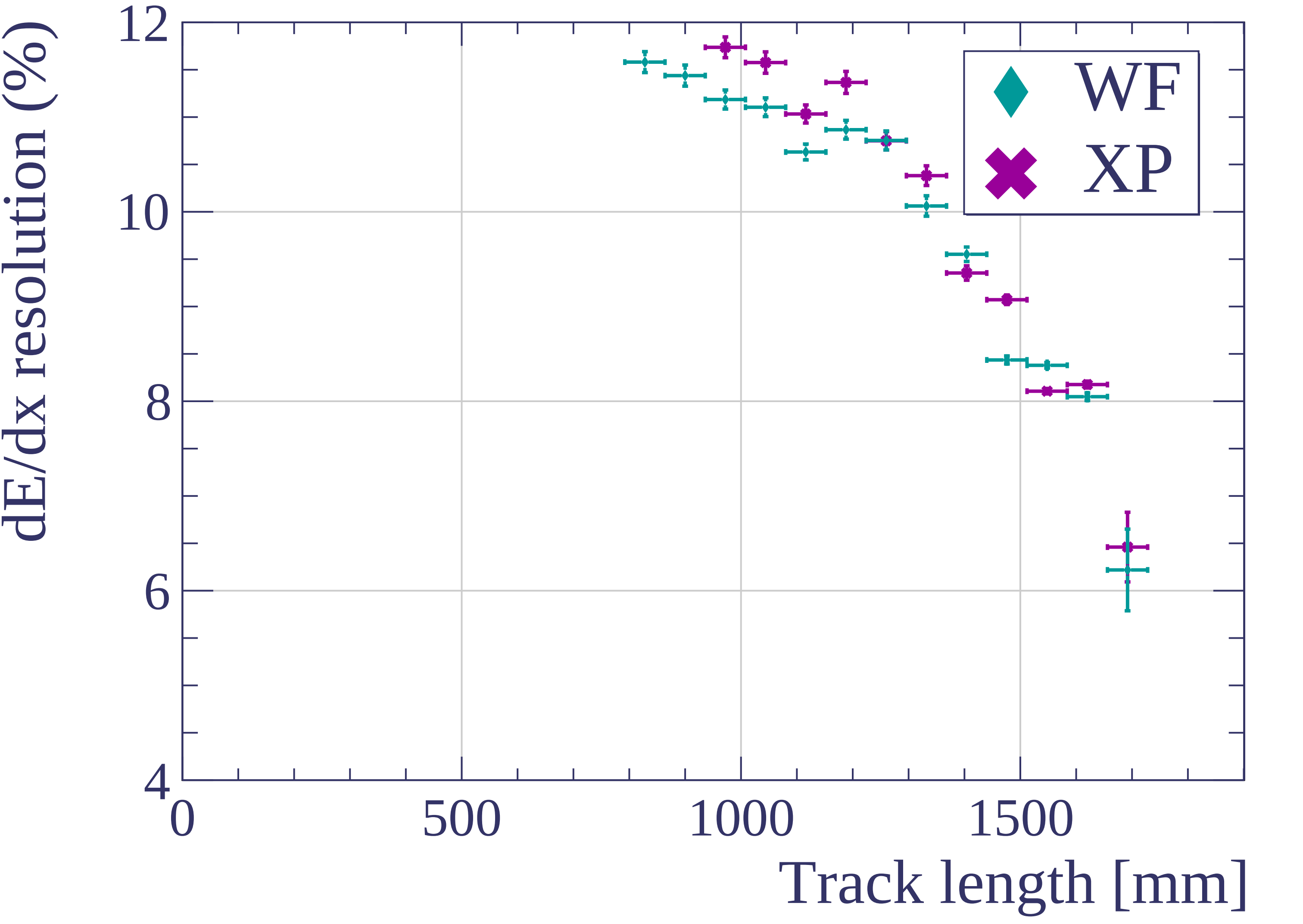
dE/dx [ADC counts/cm]
600
500
400
300
−1000
−500
0
500
1000
Track X position [mm]
WF
XP

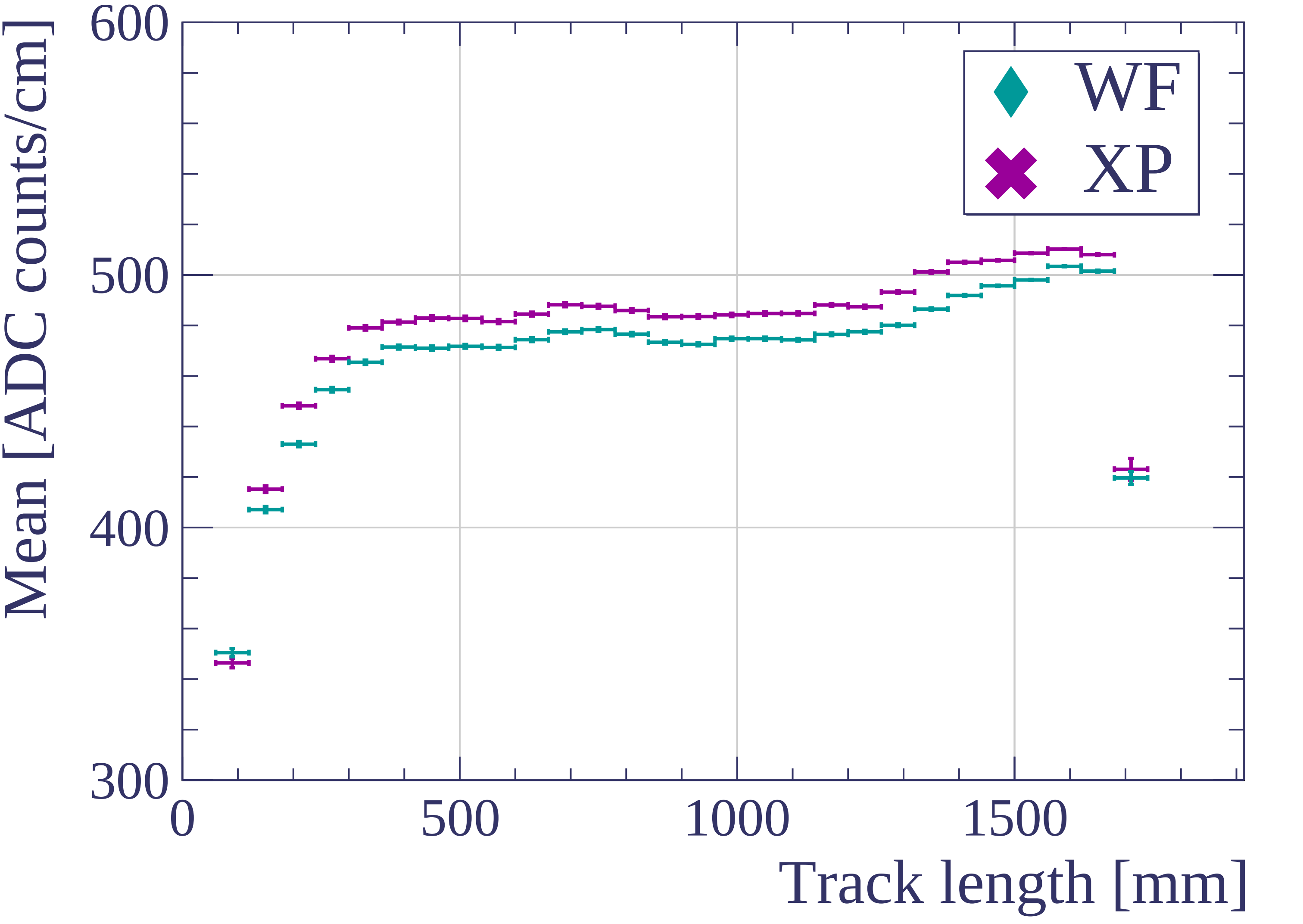


y = -0.02x + 512.51
dE/dx with WF [ADC counts/cm]
Track X position [mm]
1000
800
600
400
200
0
−1000
−500
0
500
1000
100
50
0

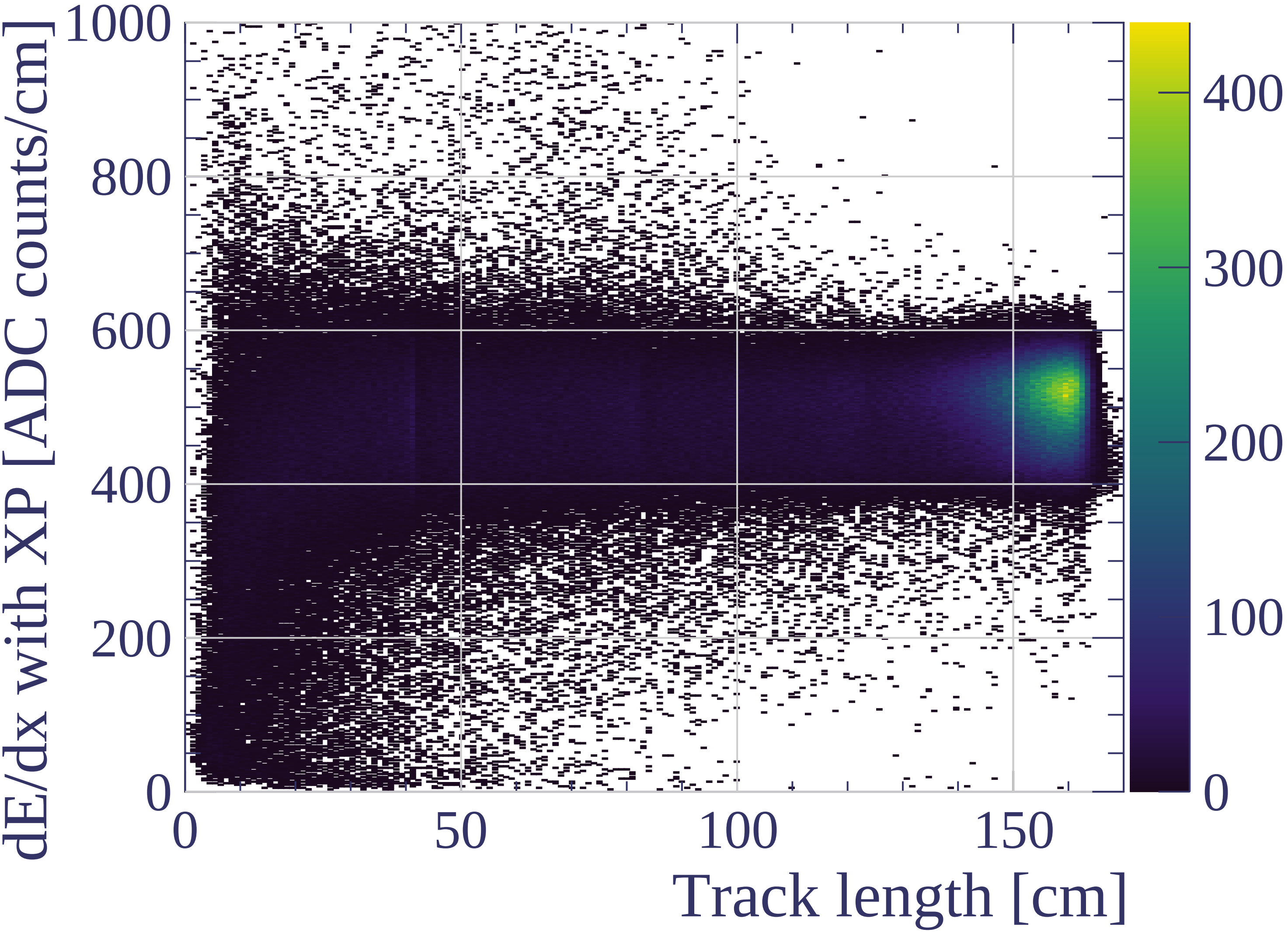


y = -0.02x + 527.61
dE/dx with XP [ADC counts/cm]
Track X position [mm]
0
200
400
600
800
1000
−1000
−500
0
500
1000
0
50
100

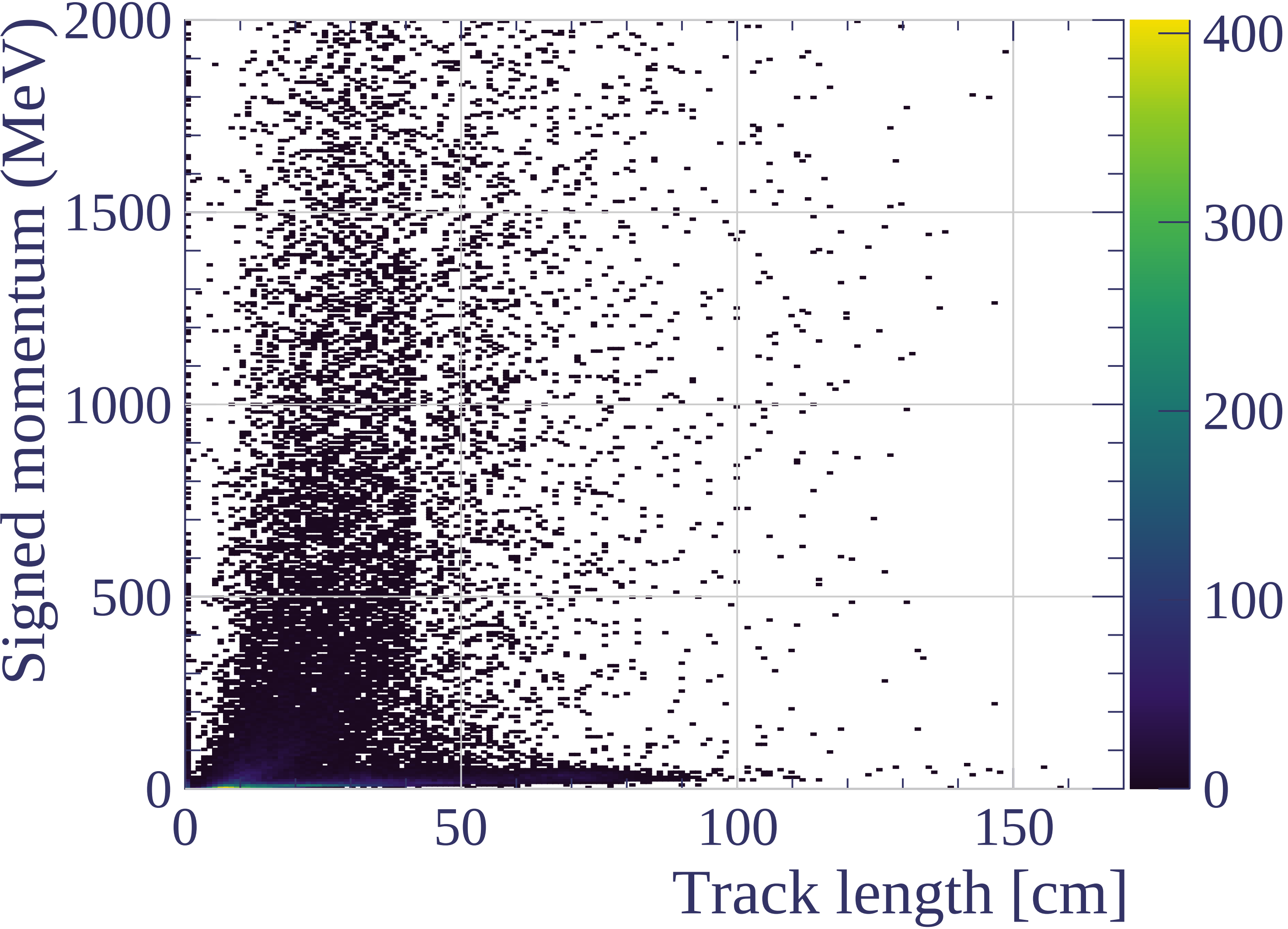

dE/dx resolution (%)
12
10
8
6
4
0
500
1000
1500
Track length [mm]
WF
XP



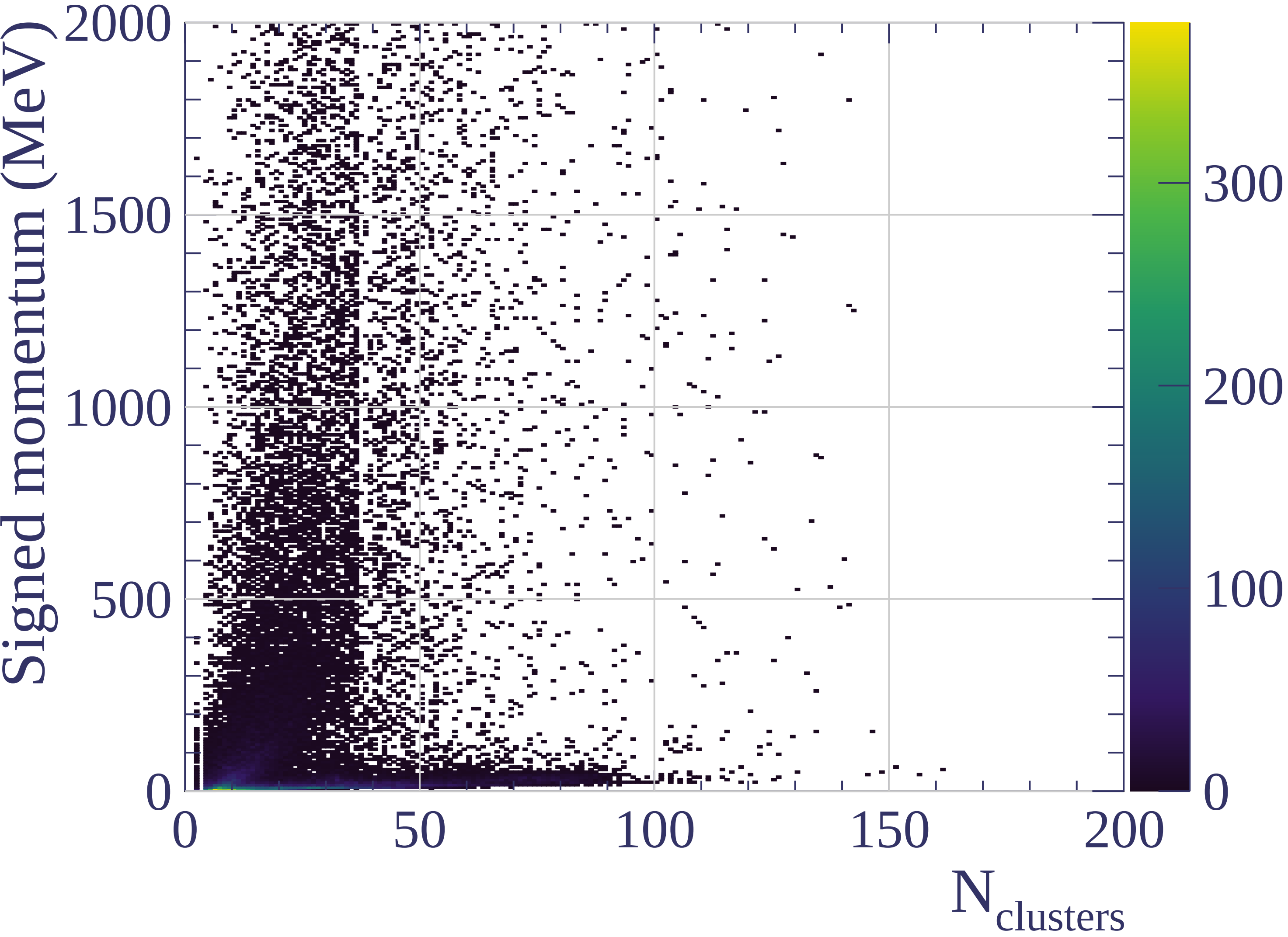
WF
XP
Mean [ADC counts/cm]
600
500
400
300
0
500
1000
1500
Track length [mm]



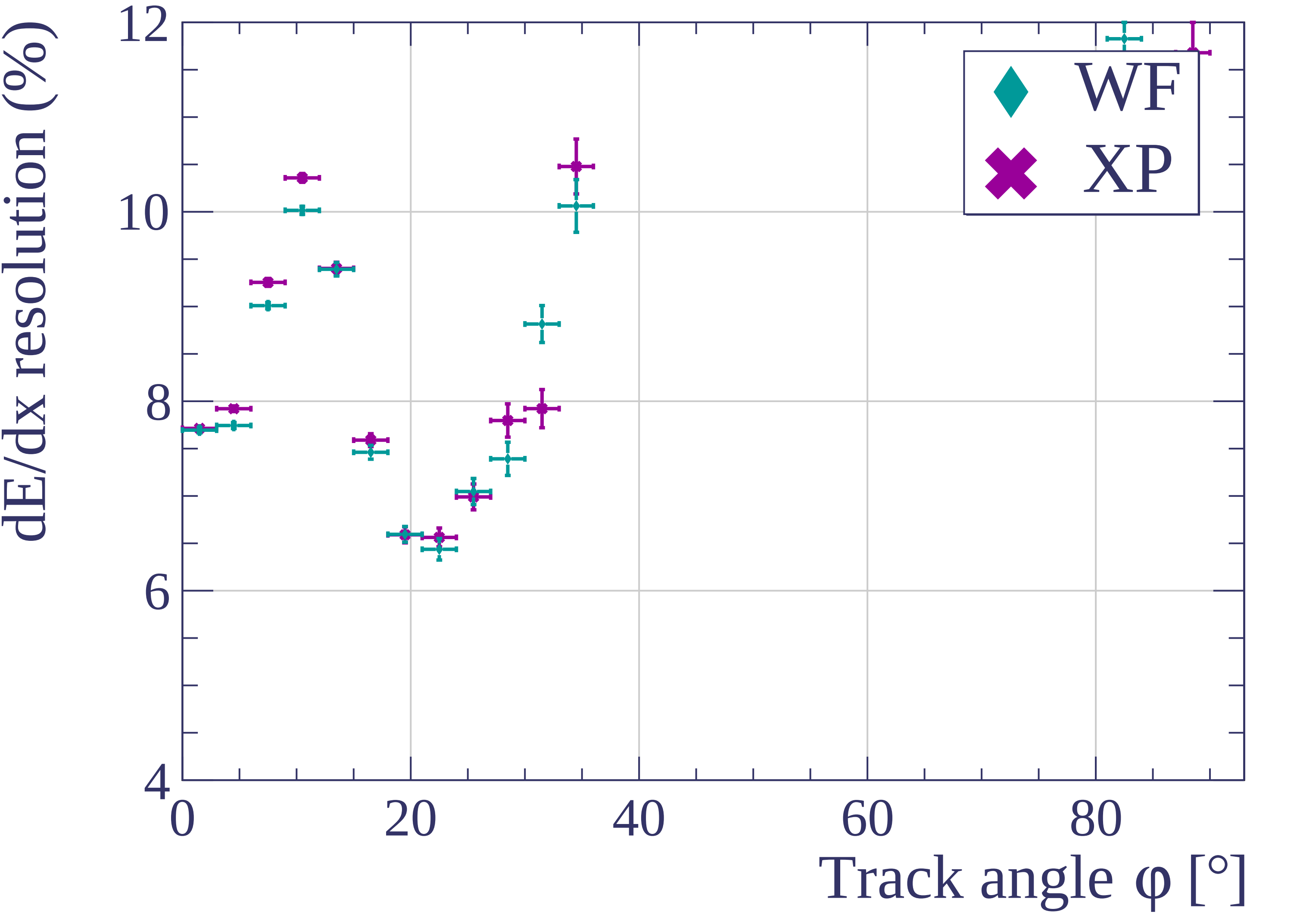
dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
0
50
100
150
Track length [cm]
400
300
200
100
0



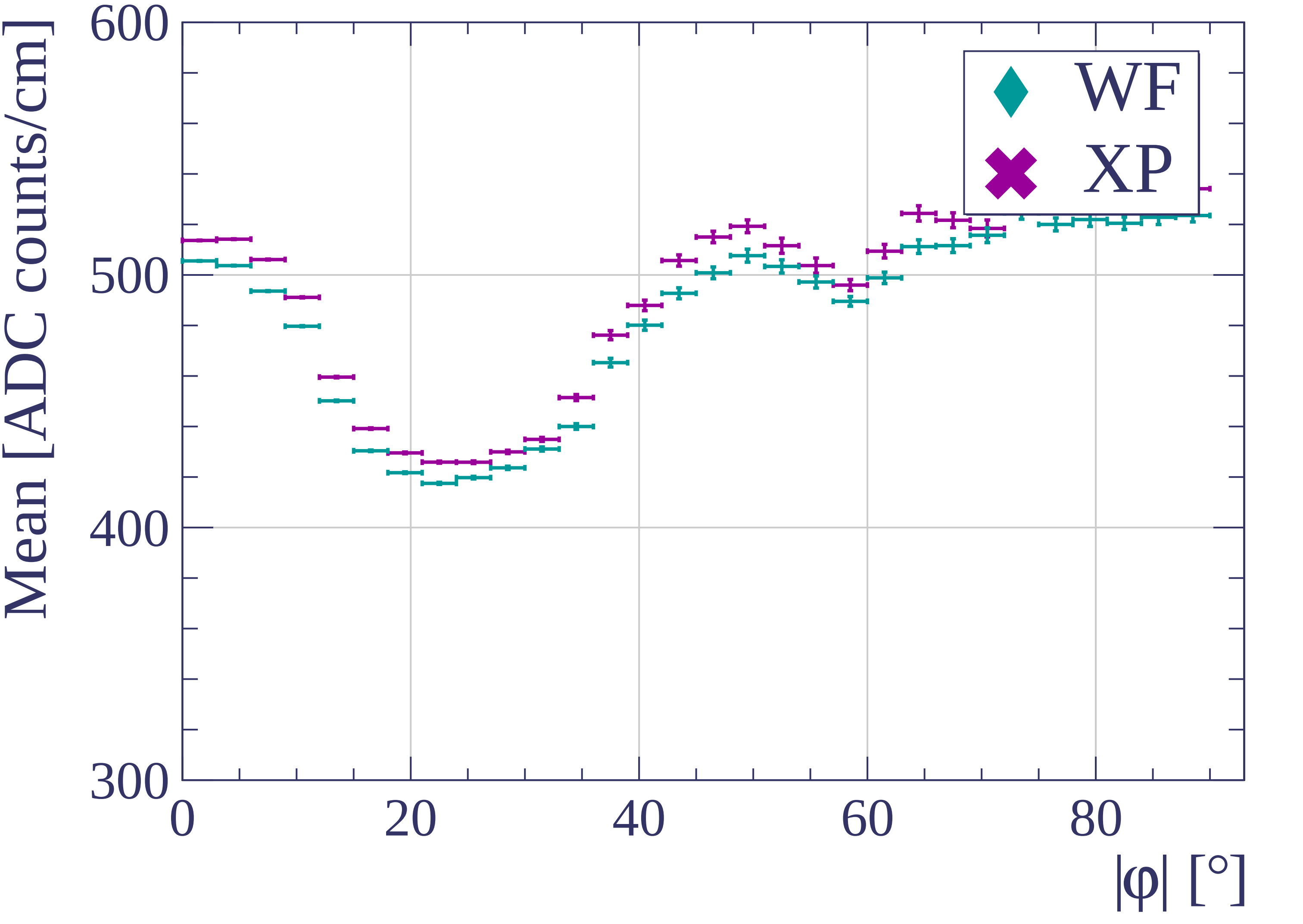
Signed momentum (MeV)
2000
1500
1000
500
0
0
50
100
150
Track length [cm]
400
300
200
100
0



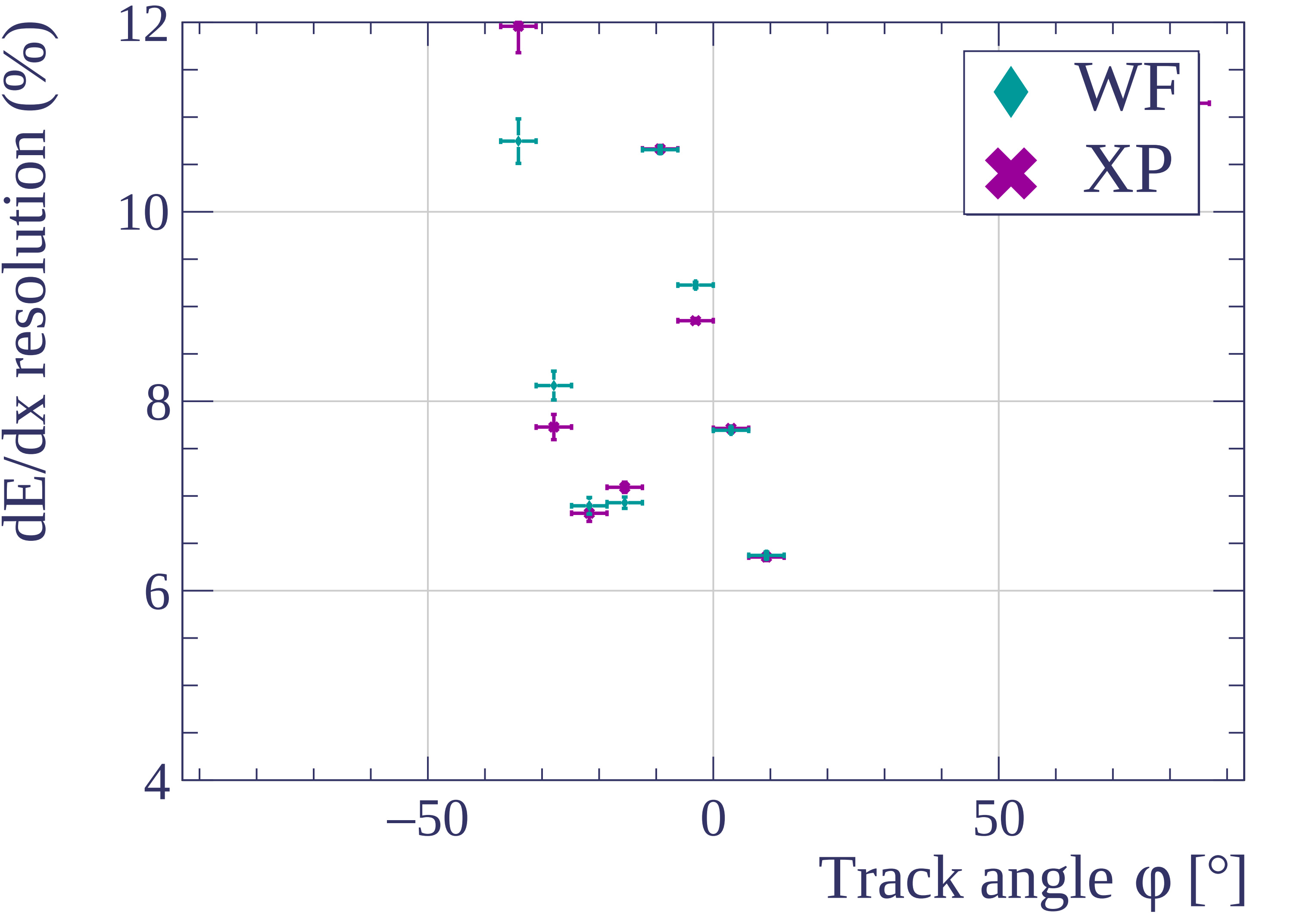
Signed momentum (MeV)
2000
1500
1000
500
0
0
50
100
150
200
$N_{clusters}$
300
200
100
0



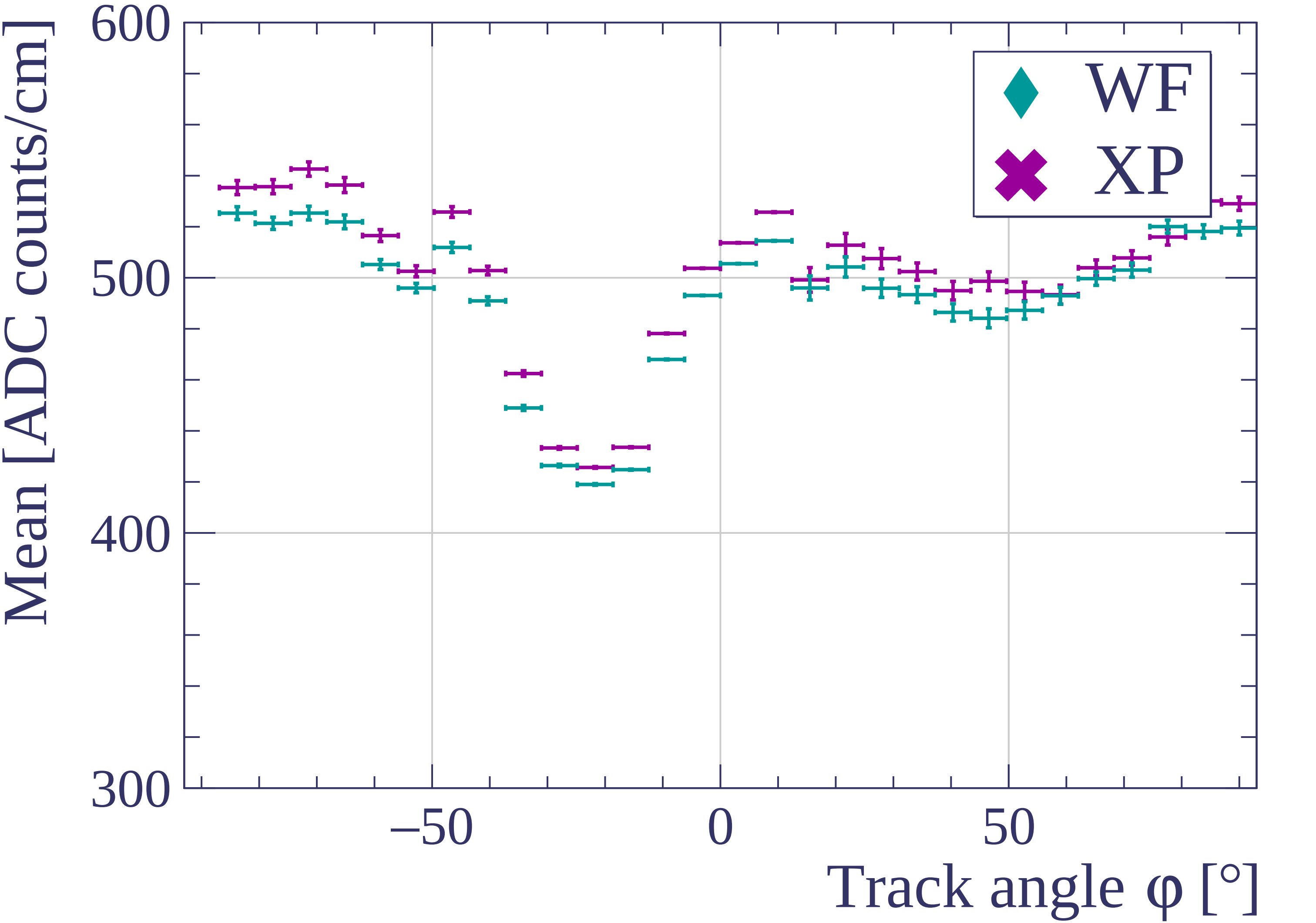
dE/dx resolution (%)
12
10
8
6
4
0
20
40
60
80
Track angle φ [°]
WF
XP



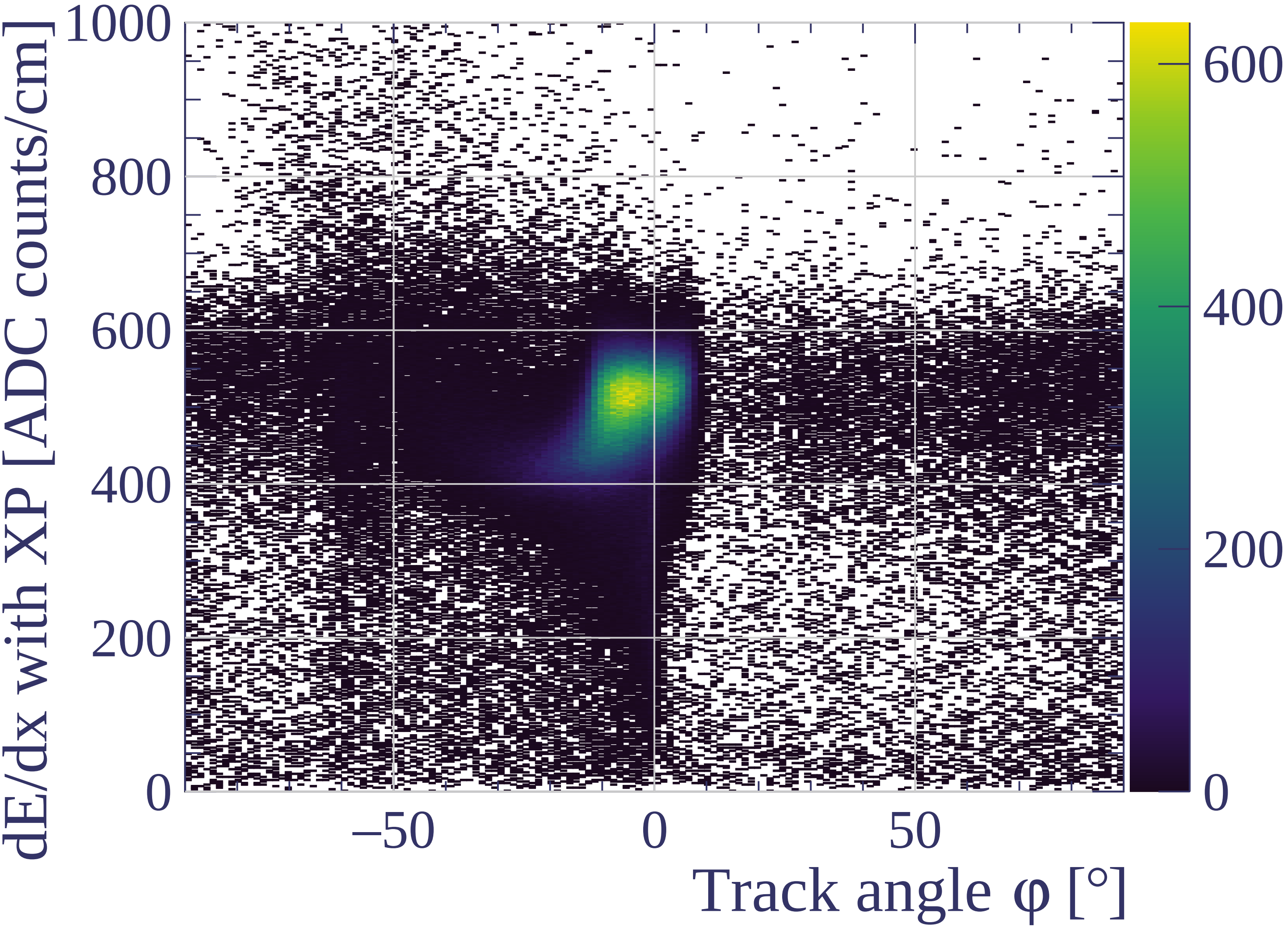
Mean [ADC counts/cm]
600
500
400
300
0
20
40
60
80
|φ| [°]
WF
XP



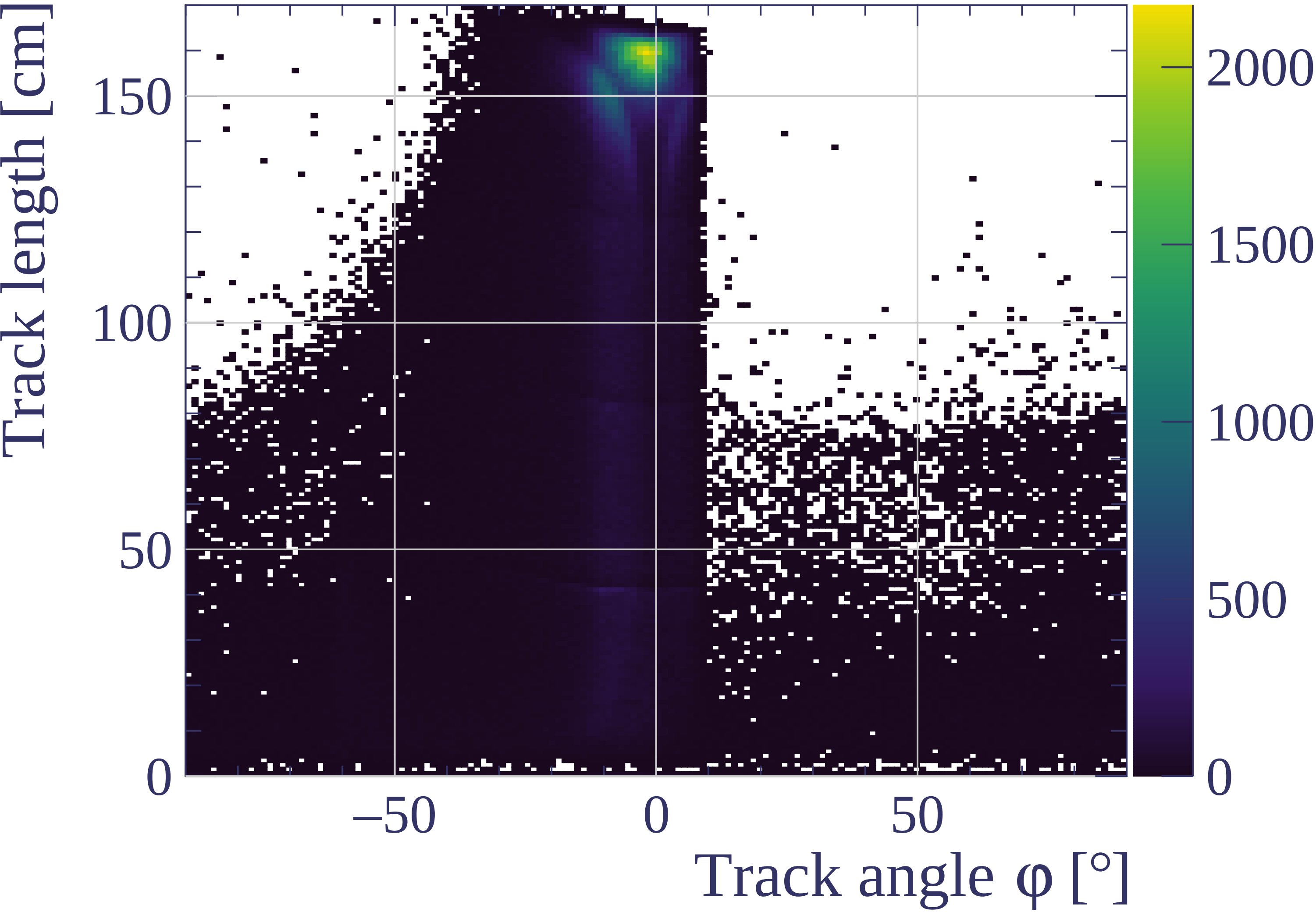
dE/dx resolution (%)
12
10
8
6
4
−50
0
50
Track angle φ [°]
WF
XP



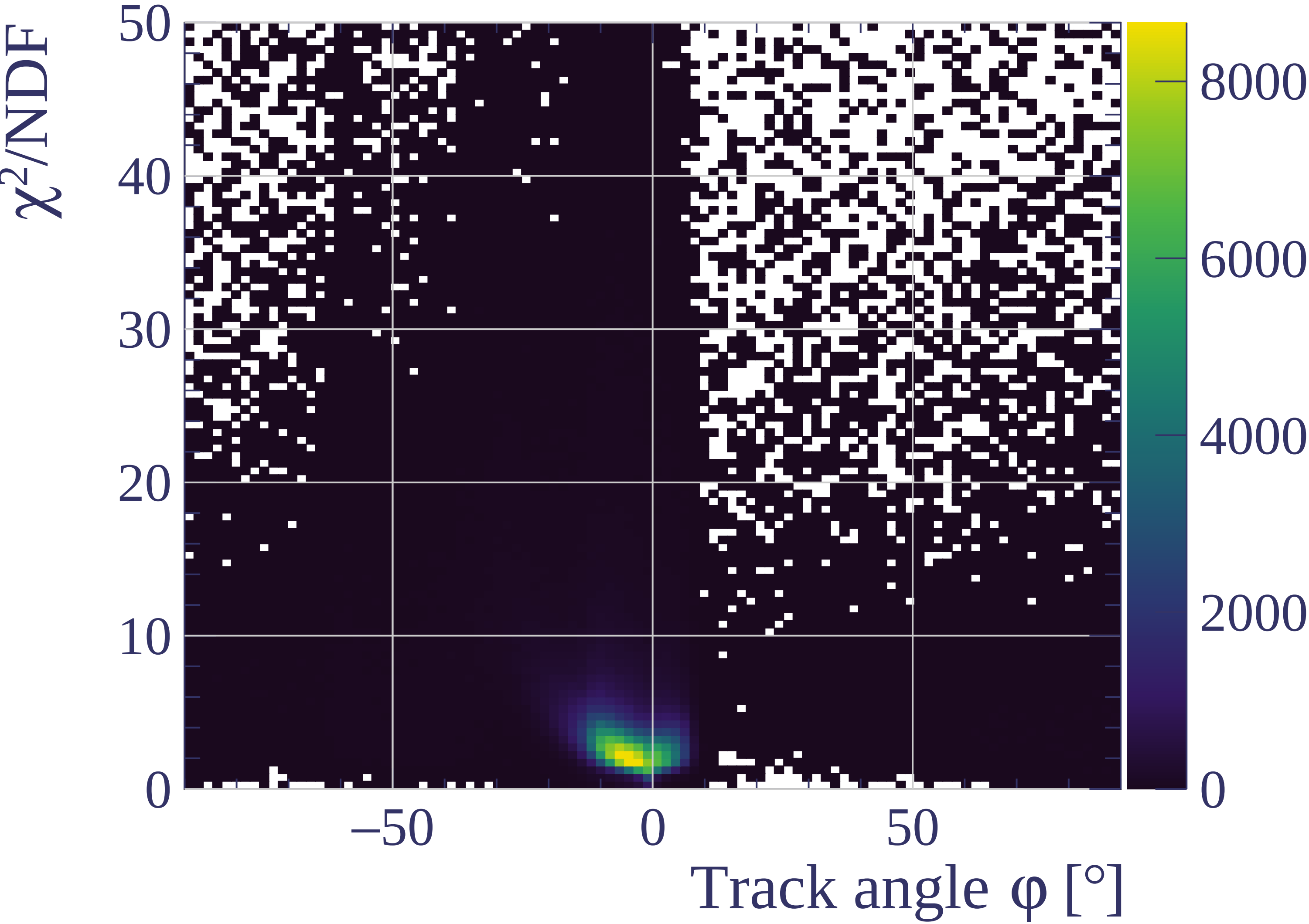
Mean [ADC counts/cm]
600
500
400
300
−50
0
50
Track angle φ [°]
WF
XP



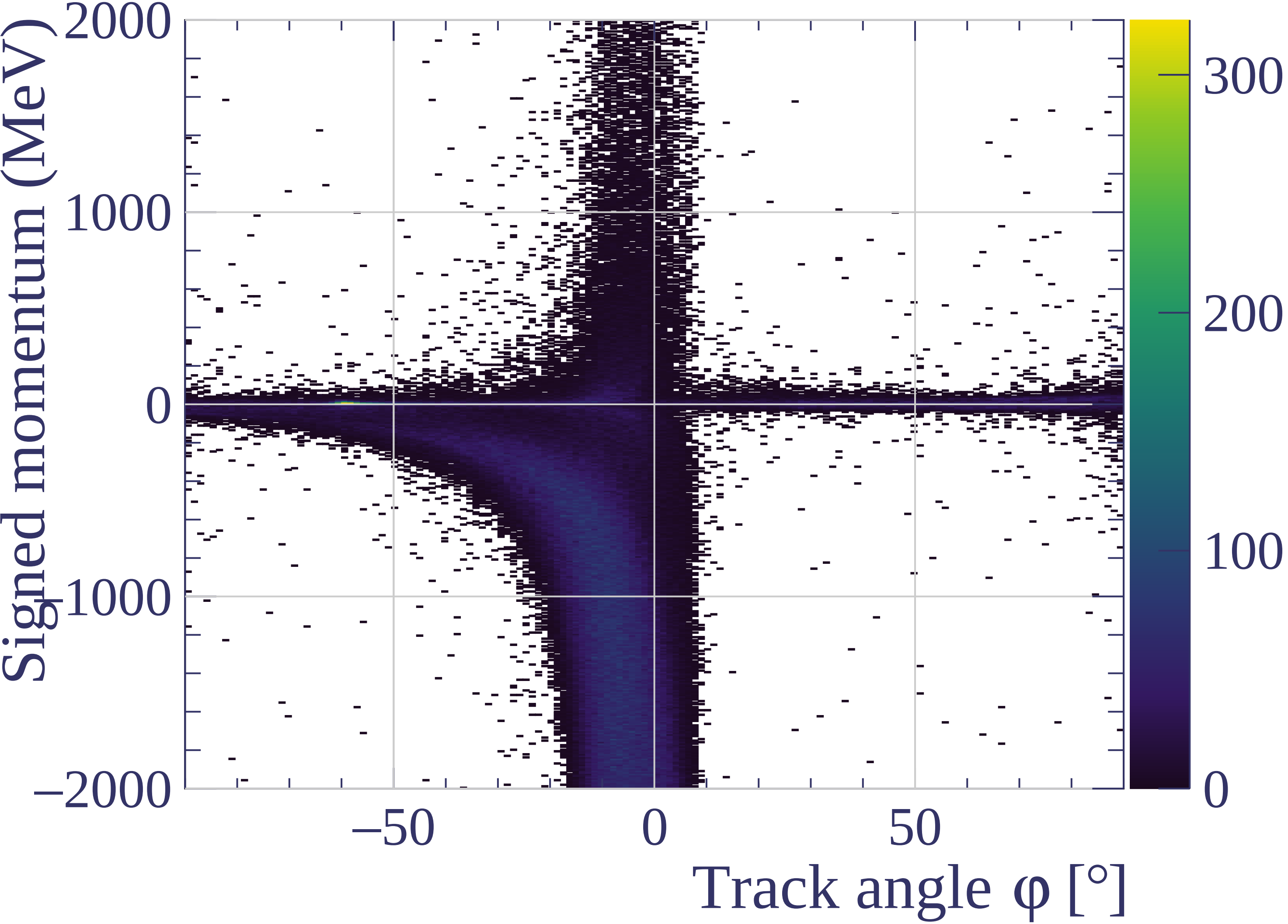
dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
−50
0
50
Track angle φ [°]
600
400
200
0



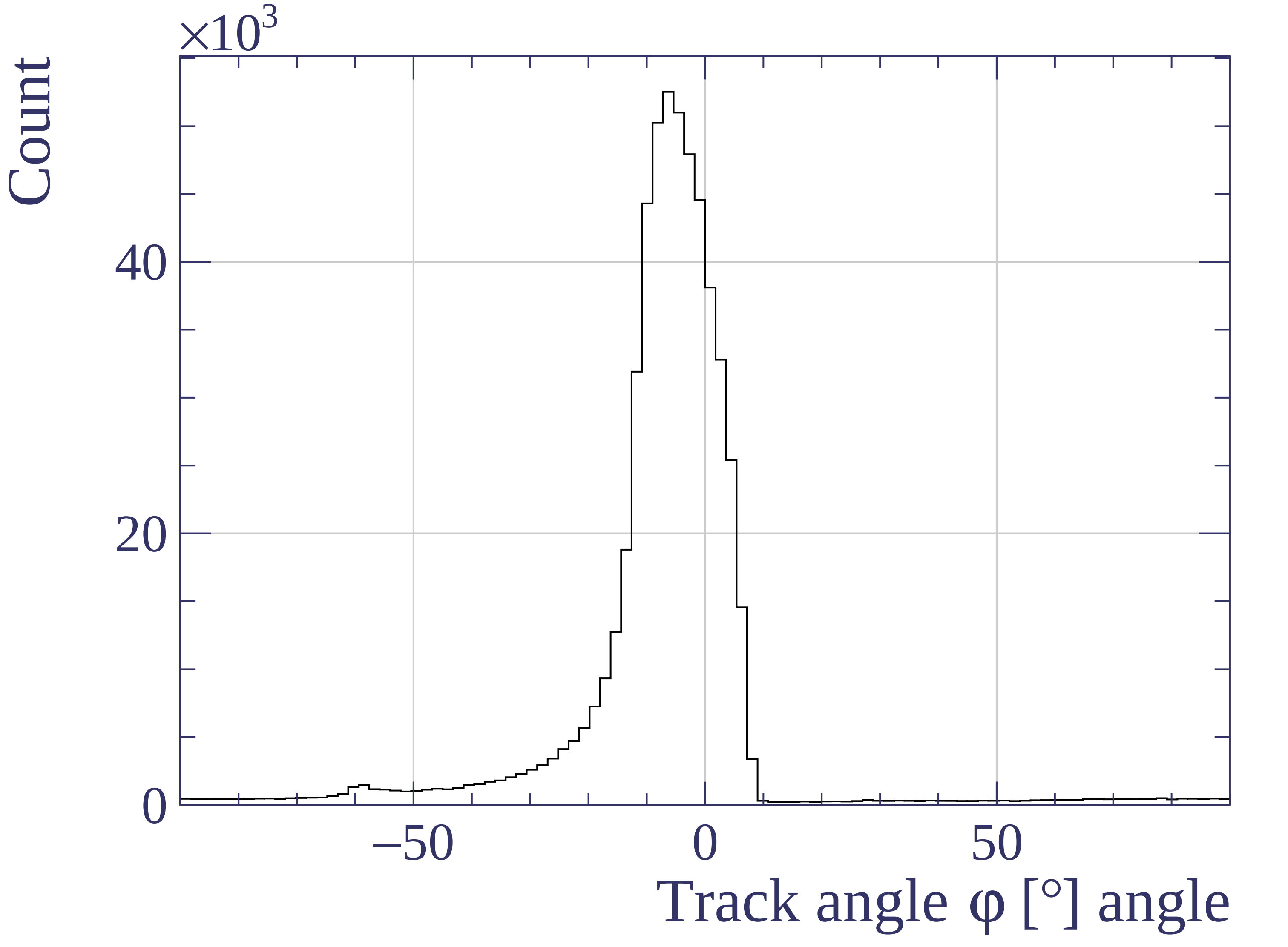
Track length [cm]
150
100
50
0
−50
0
50
Track angle φ [°]
2000
1500
1000
500
0



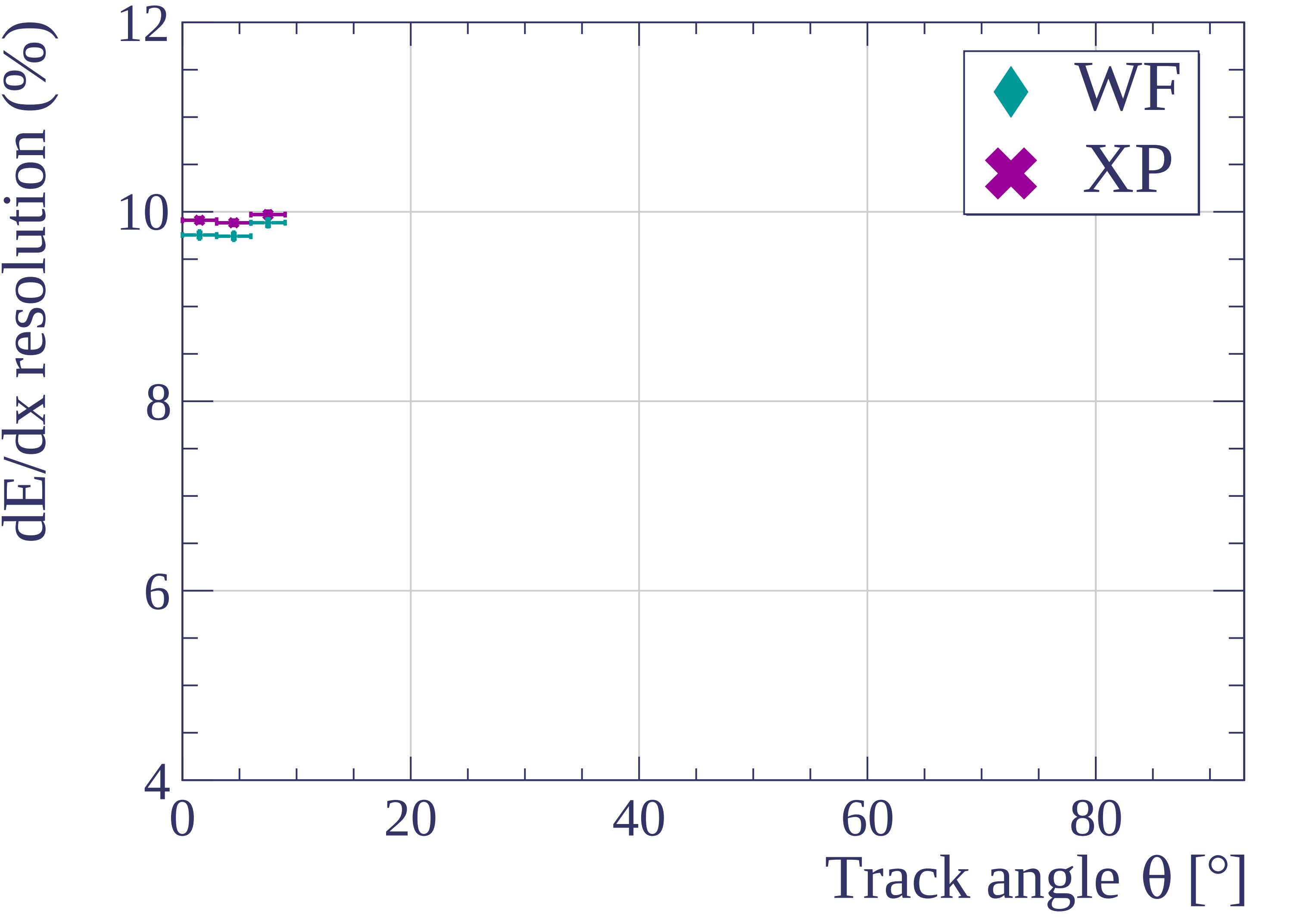
$\chi^2$/NDF
50
40
30
20
10
0
−50
0
50
Track angle φ [°]
8000
6000
4000
2000
0



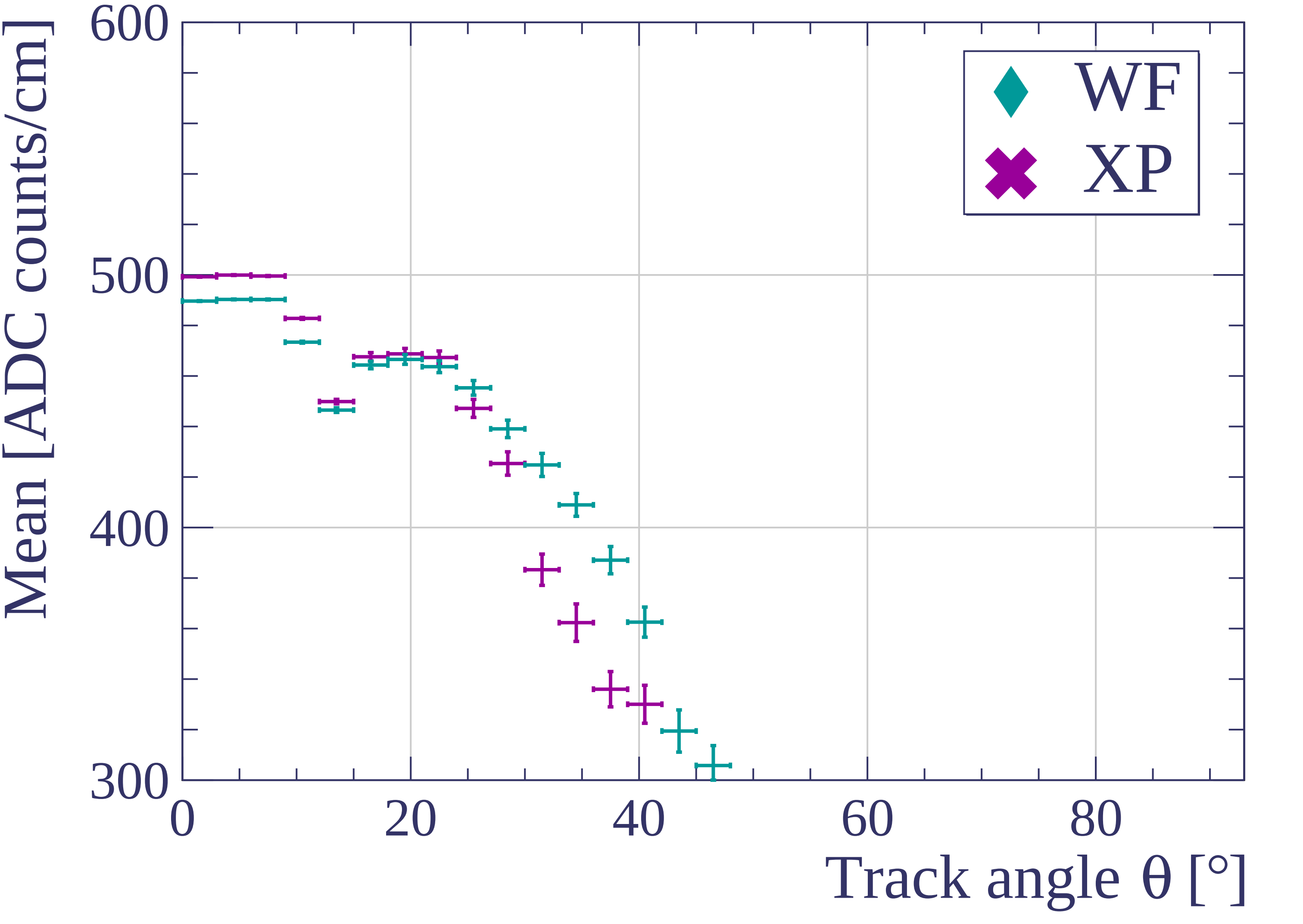
Signed momentum (MeV)
2000
1000
0
−1000
−2000
−50
0
50
Track angle φ [°]
300
200
100
0



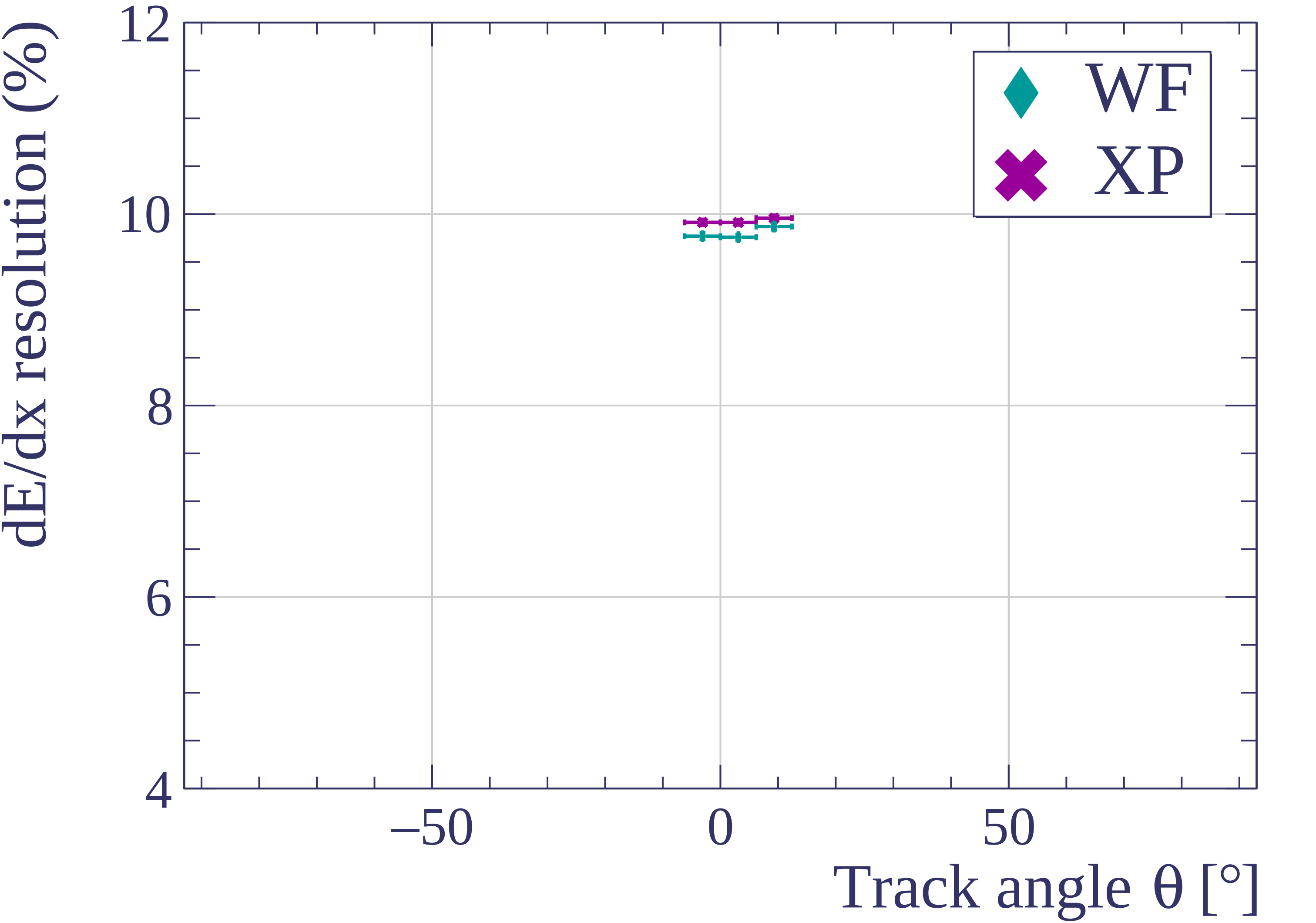
×10^3
Count
40
20
0
−50
0
50
Track angle φ [°] angle

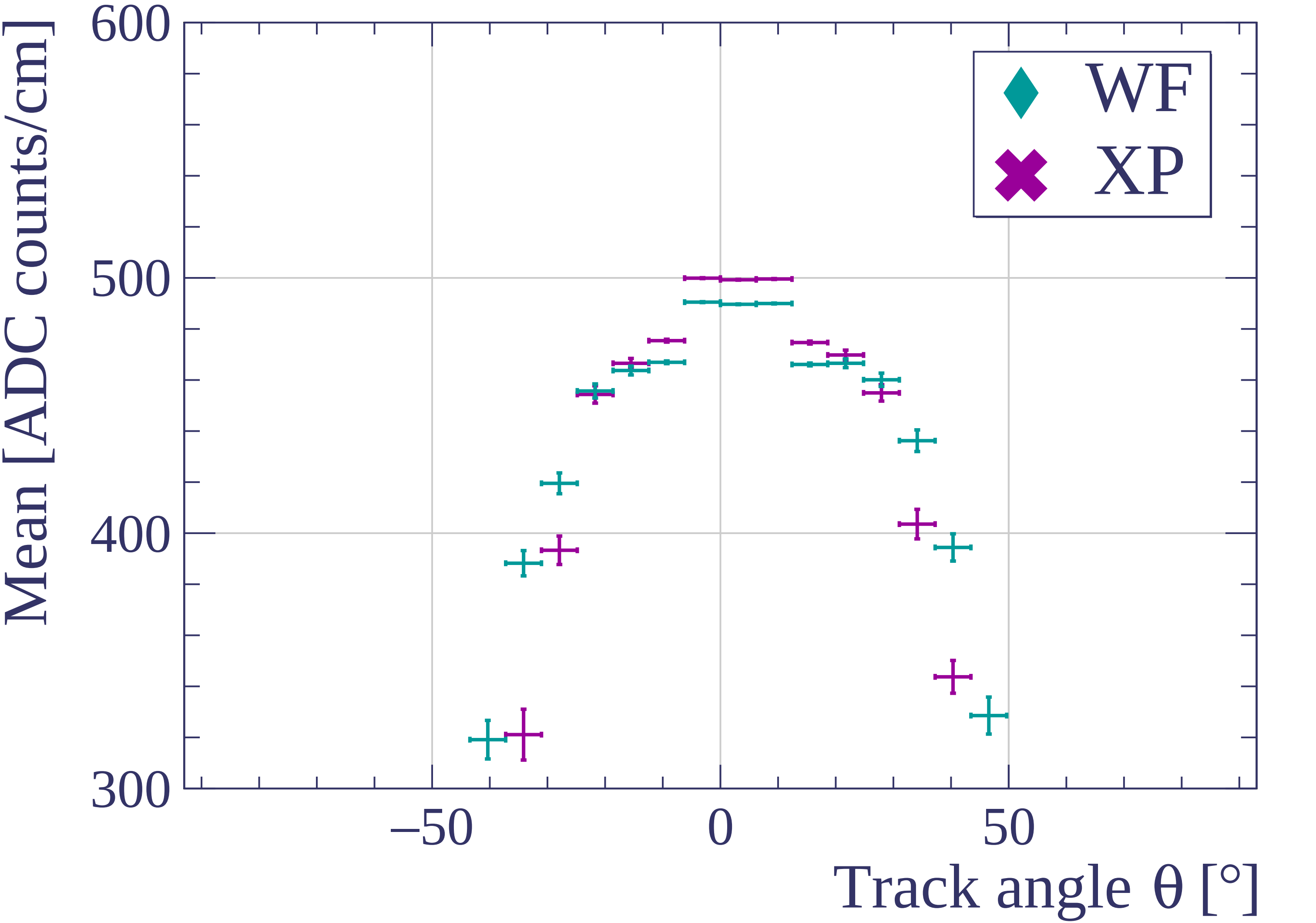


dE/dx resolution (%)
12
10
8
6
4
0
20
40
60
80
Track angle θ [°]
WF
XP

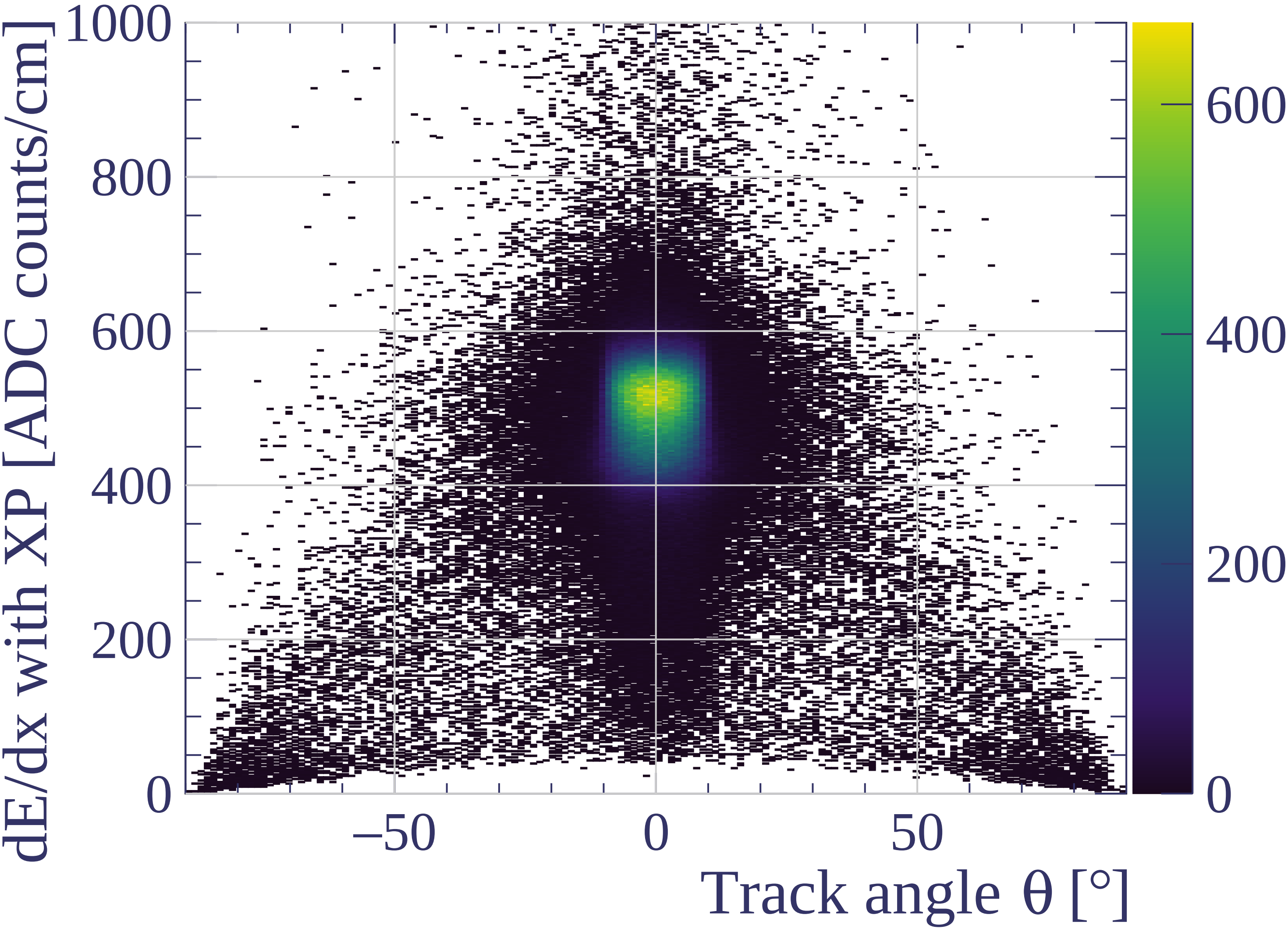


Mean [ADC counts/cm]
600
500
400
300
0
20
40
60
80
Track angle θ [°]
WF
XP

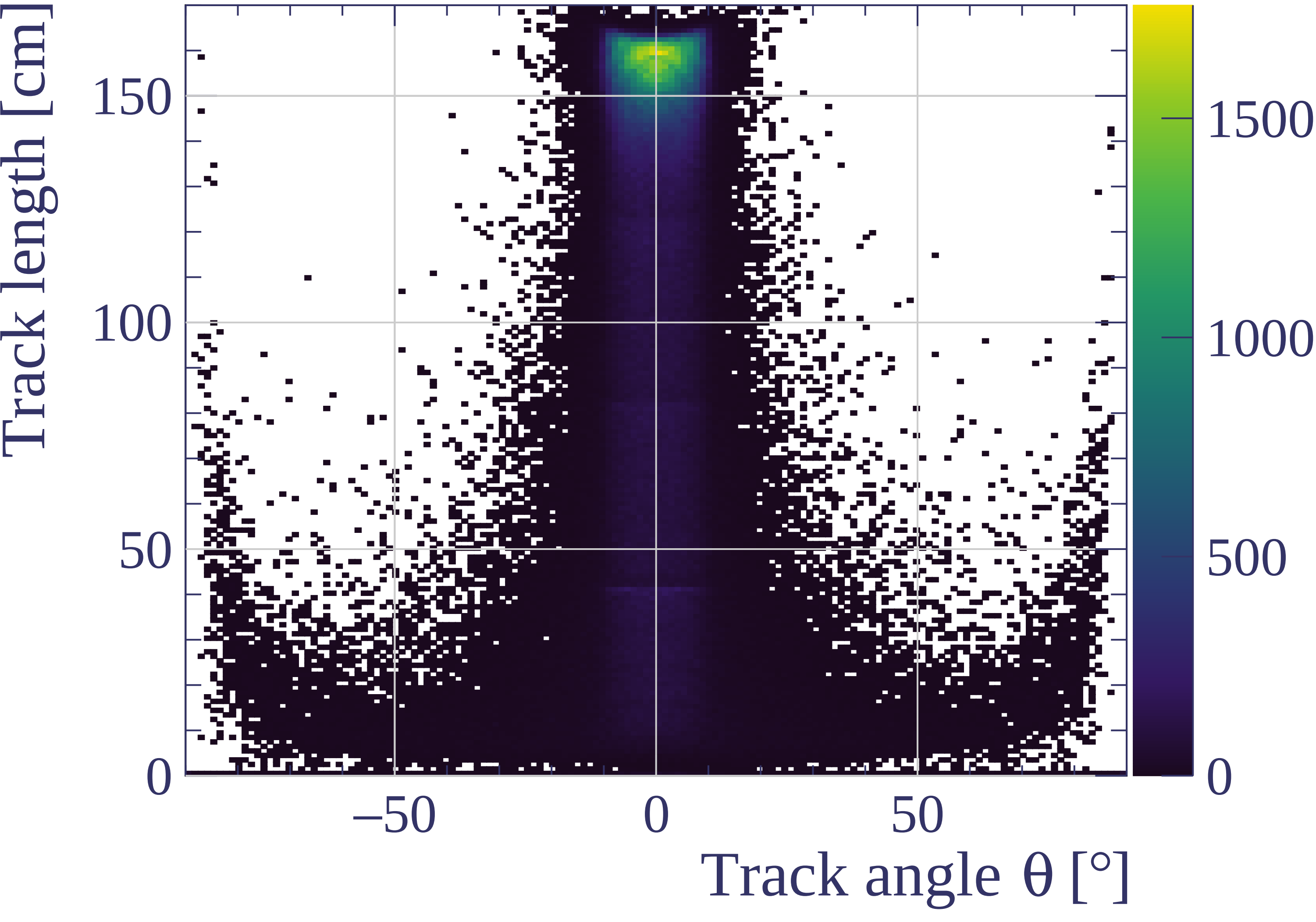

dE/dx resolution (%)
12
10
8
6
4
−50
0
50
Track angle θ [°]
WF
XP



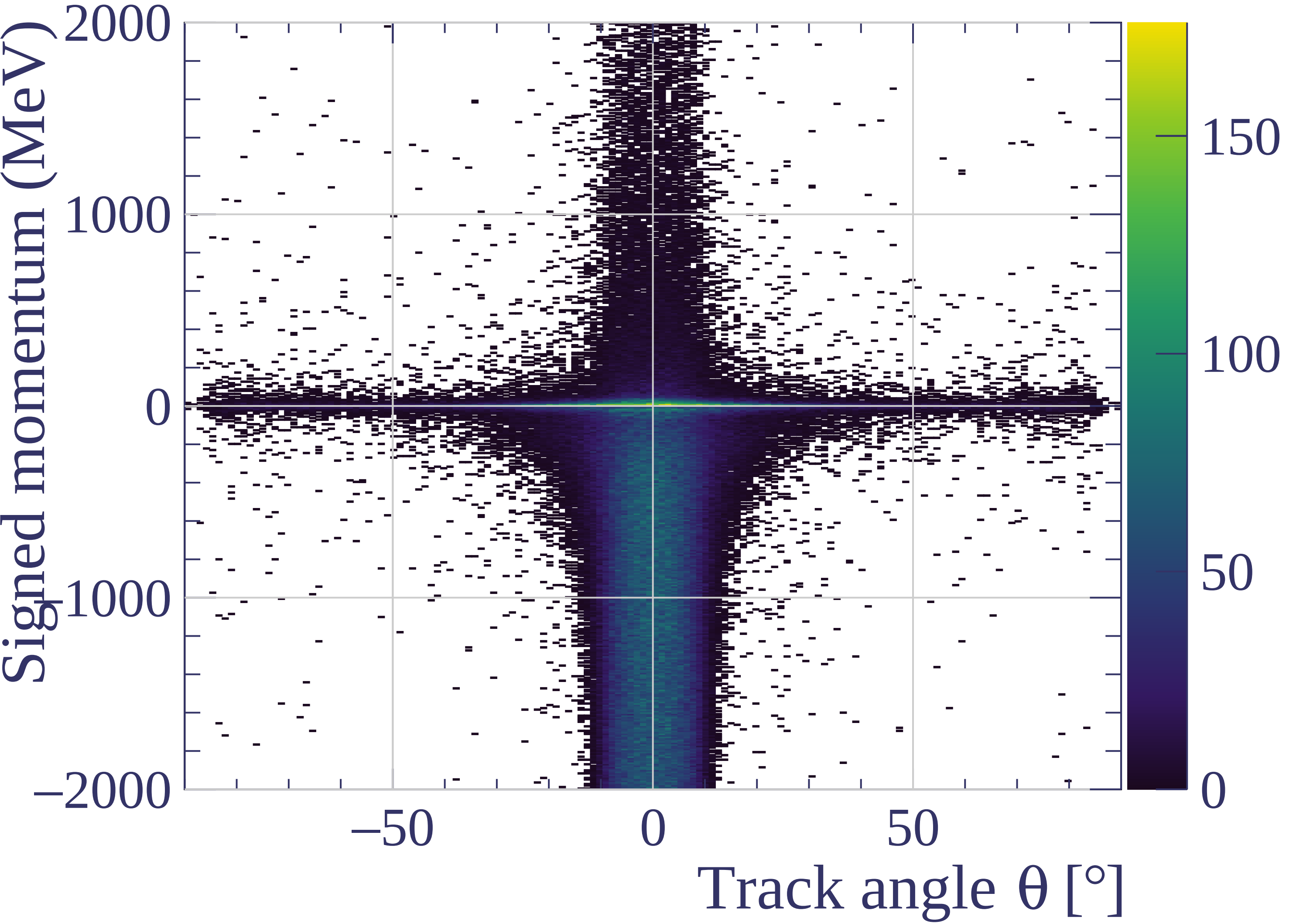
Mean [ADC counts/cm]
600
500
400
300
−50
0
50
Track angle θ [°]
WF
XP



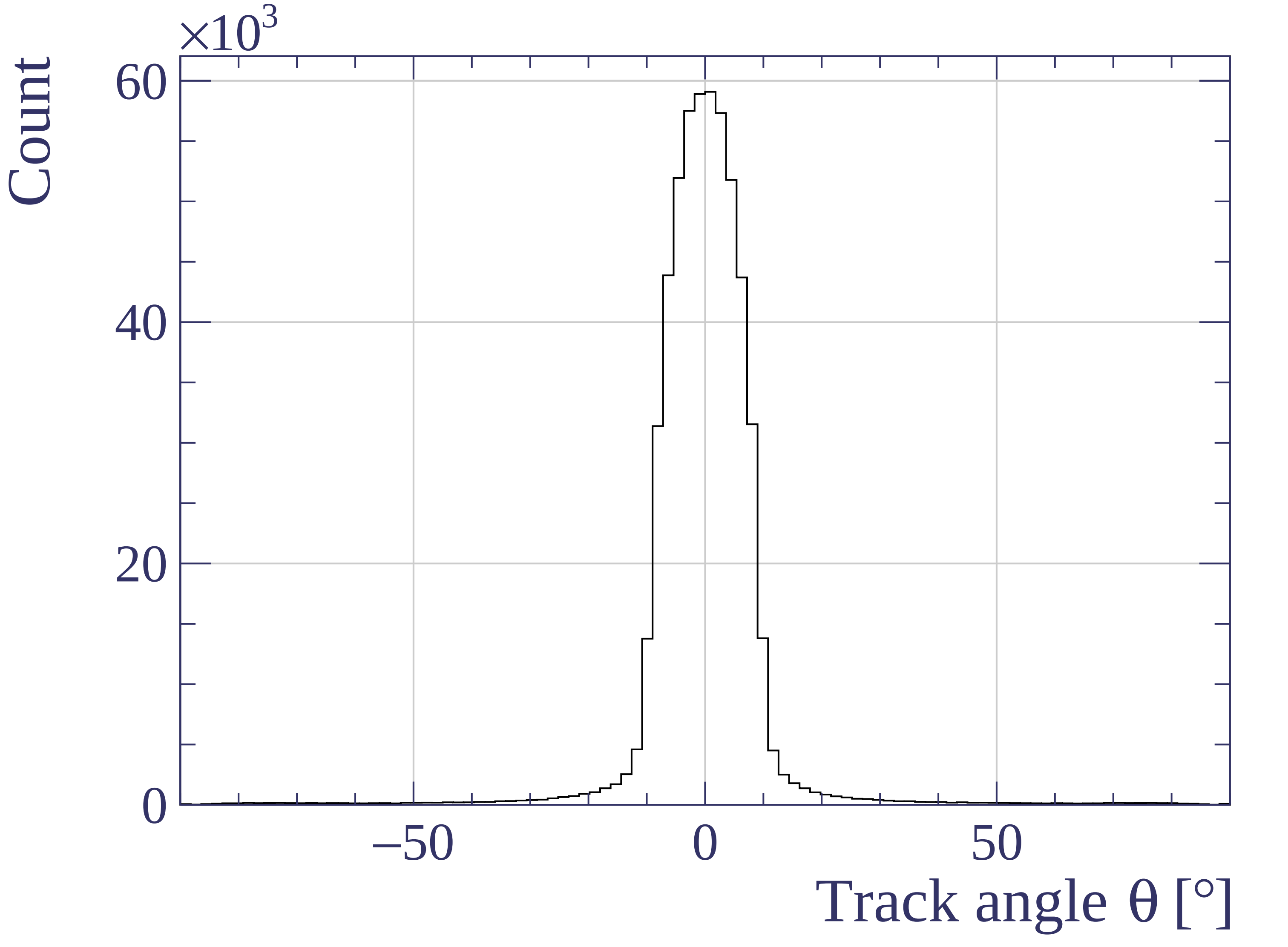
dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
−50
0
50
Track angle θ [°]
600
400
200
0



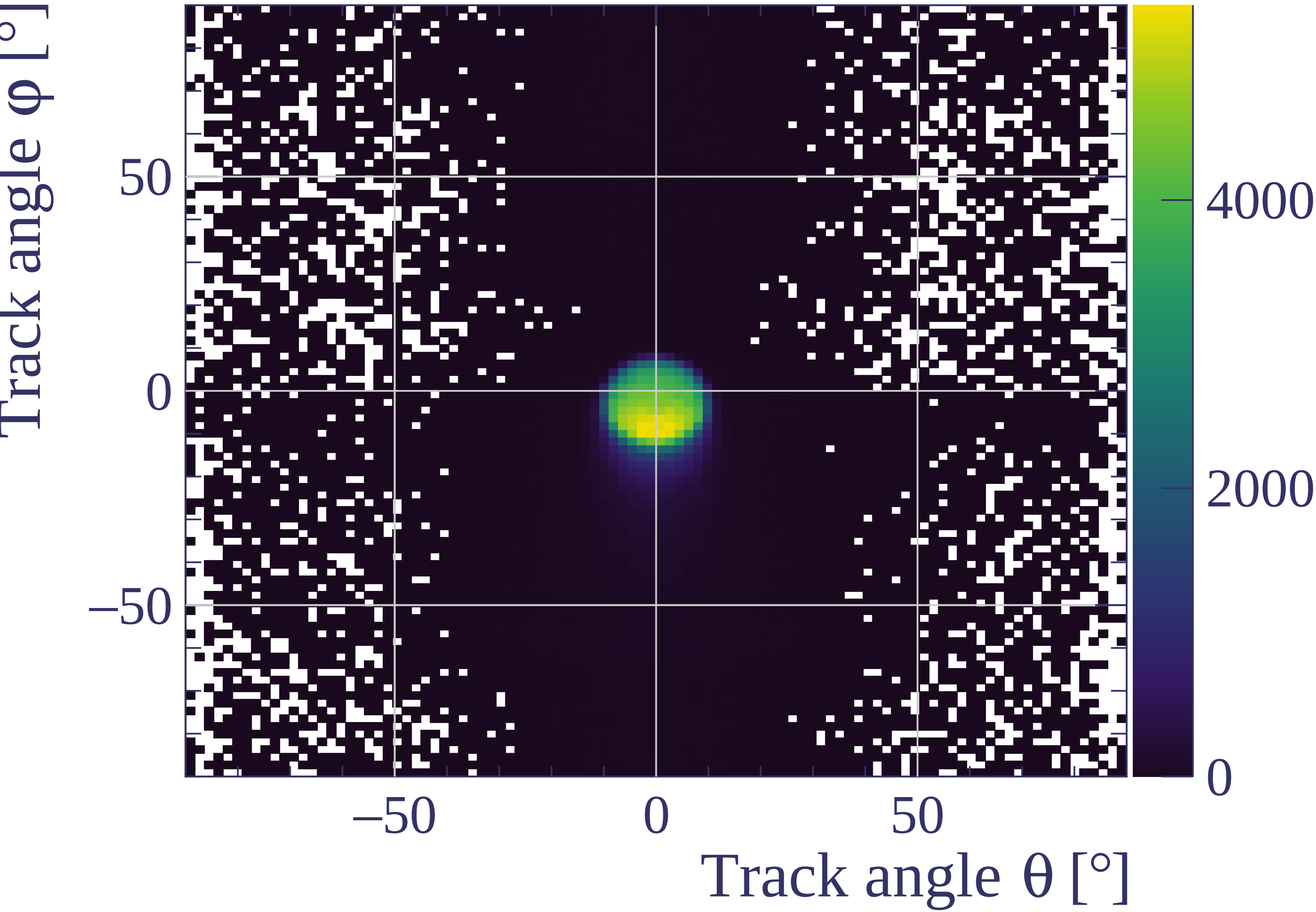
Track length [cm]
150
100
50
0
−50
0
50
Track angle θ [°]
1500
1000
500
0



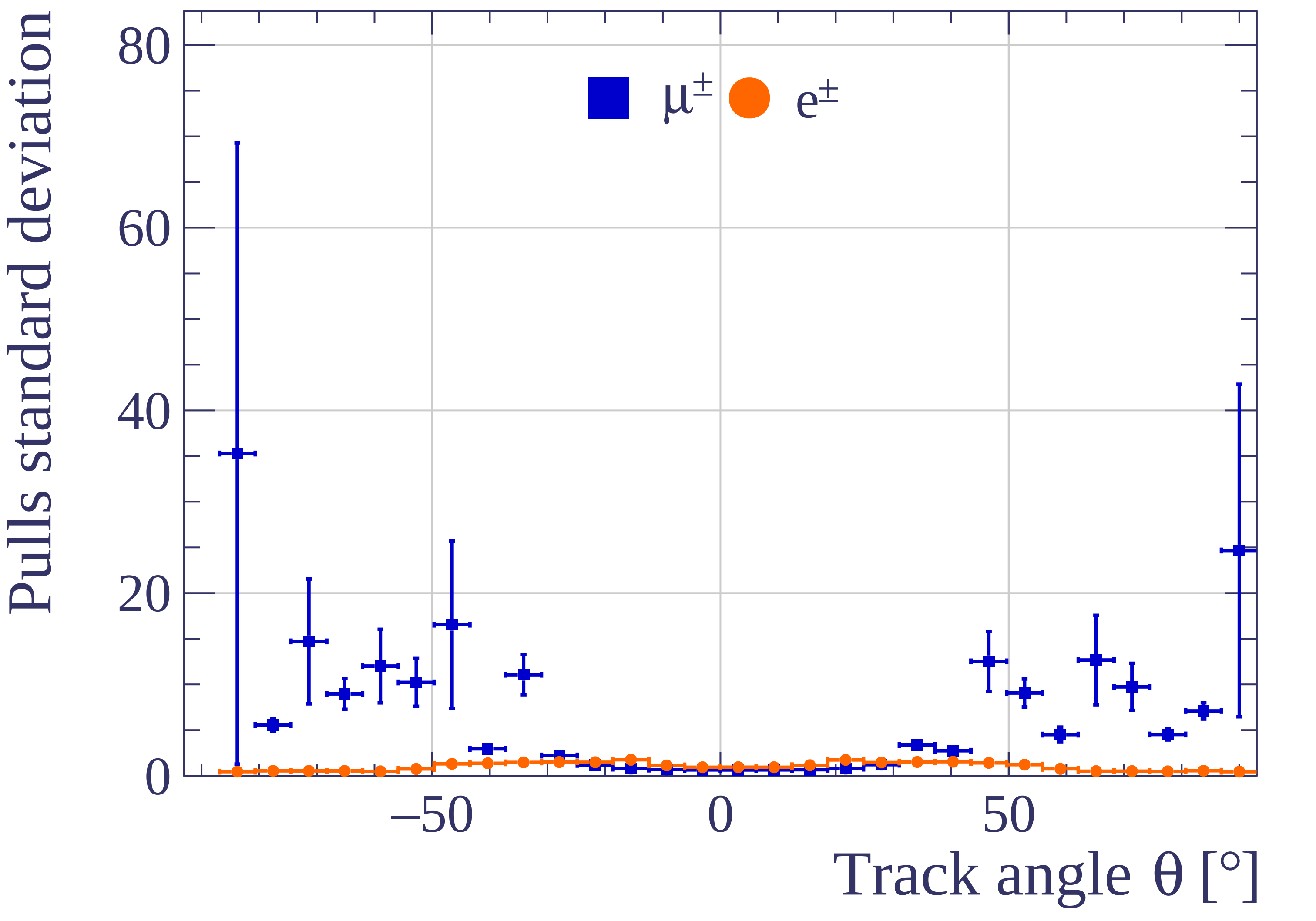
Signed momentum (MeV)
2000
1000
0
−1000
−2000
−50
0
50
Track angle θ [°]
150
100
50
0



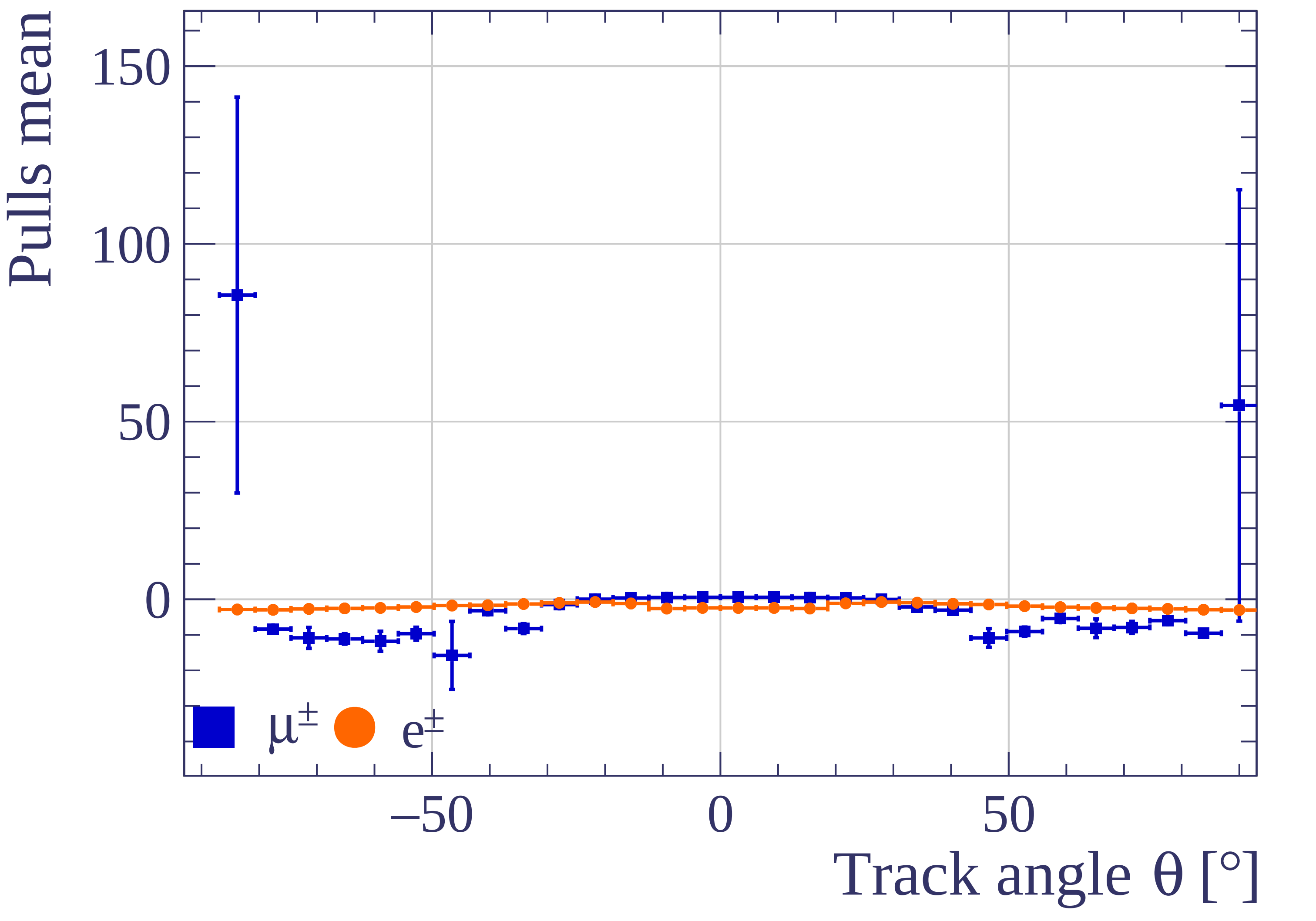
$\times 10^3$
Count
60
40
20
0
−50
0
50
Track angle θ [°]

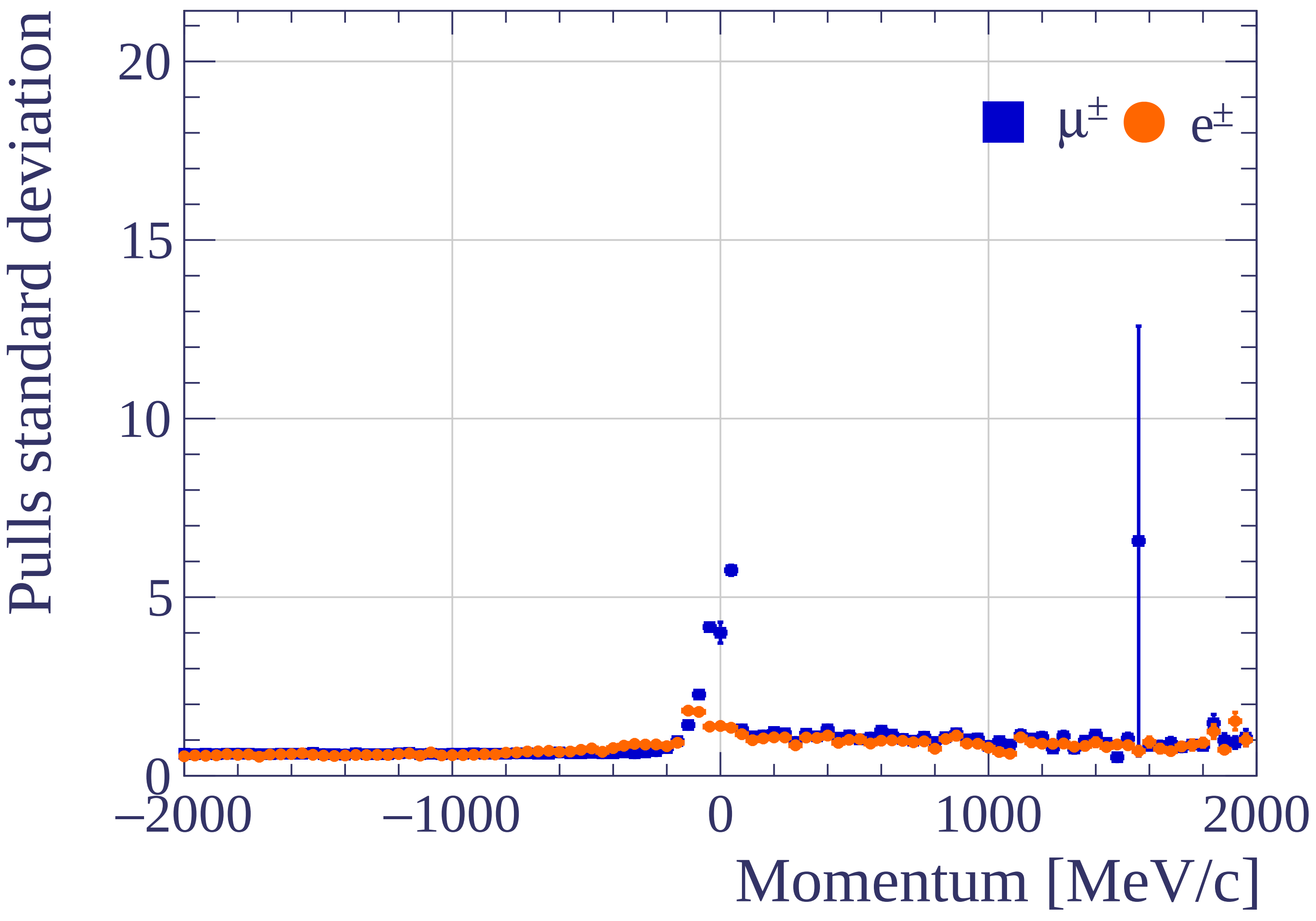


Track angle φ [°]
50
0
−50
−50
0
50
Track angle θ [°]
4000
2000
0

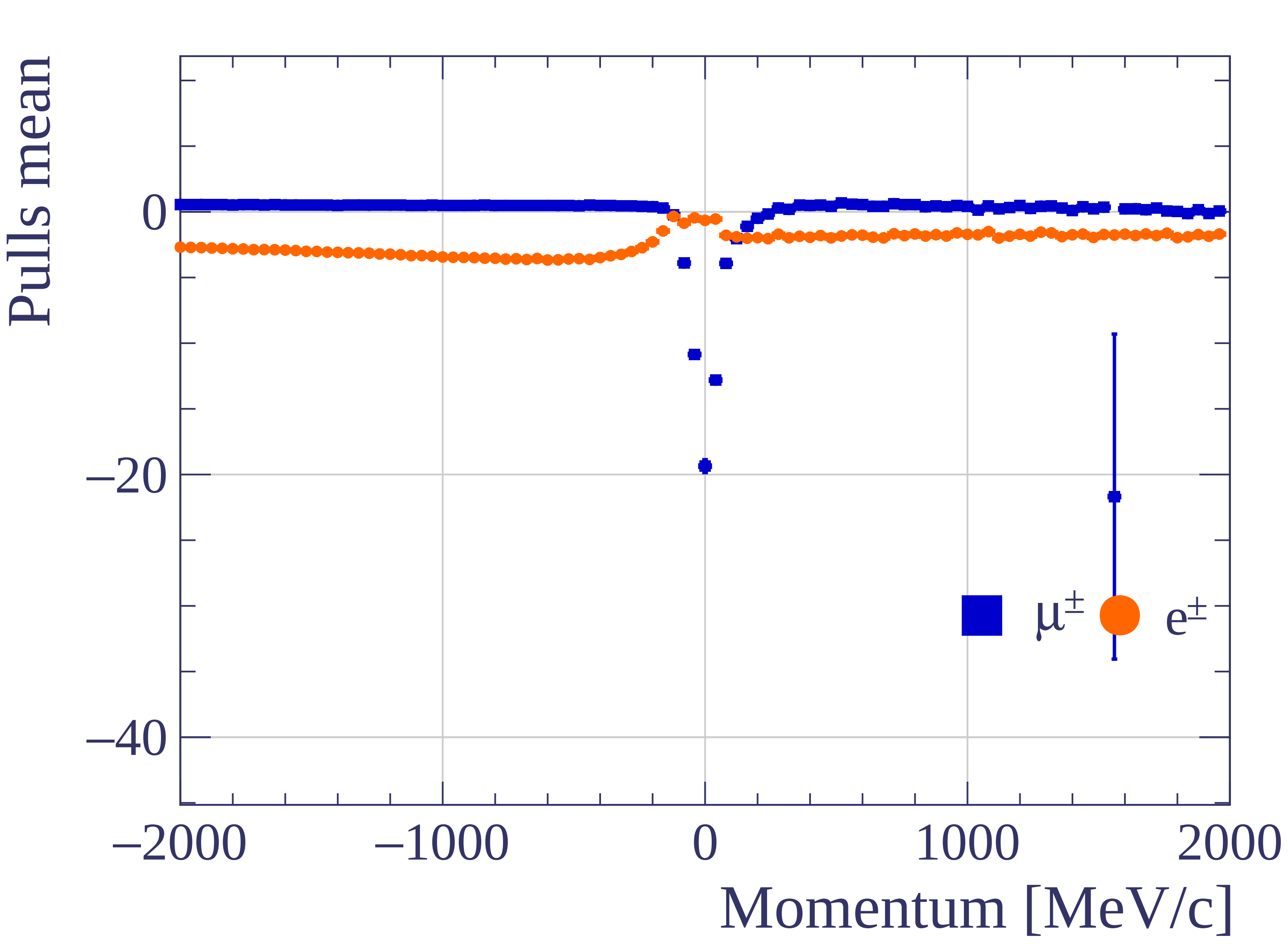


μ±
e±
Pulls standard deviation
0
20
40
60
80
−50
0
50
Track angle θ [°]

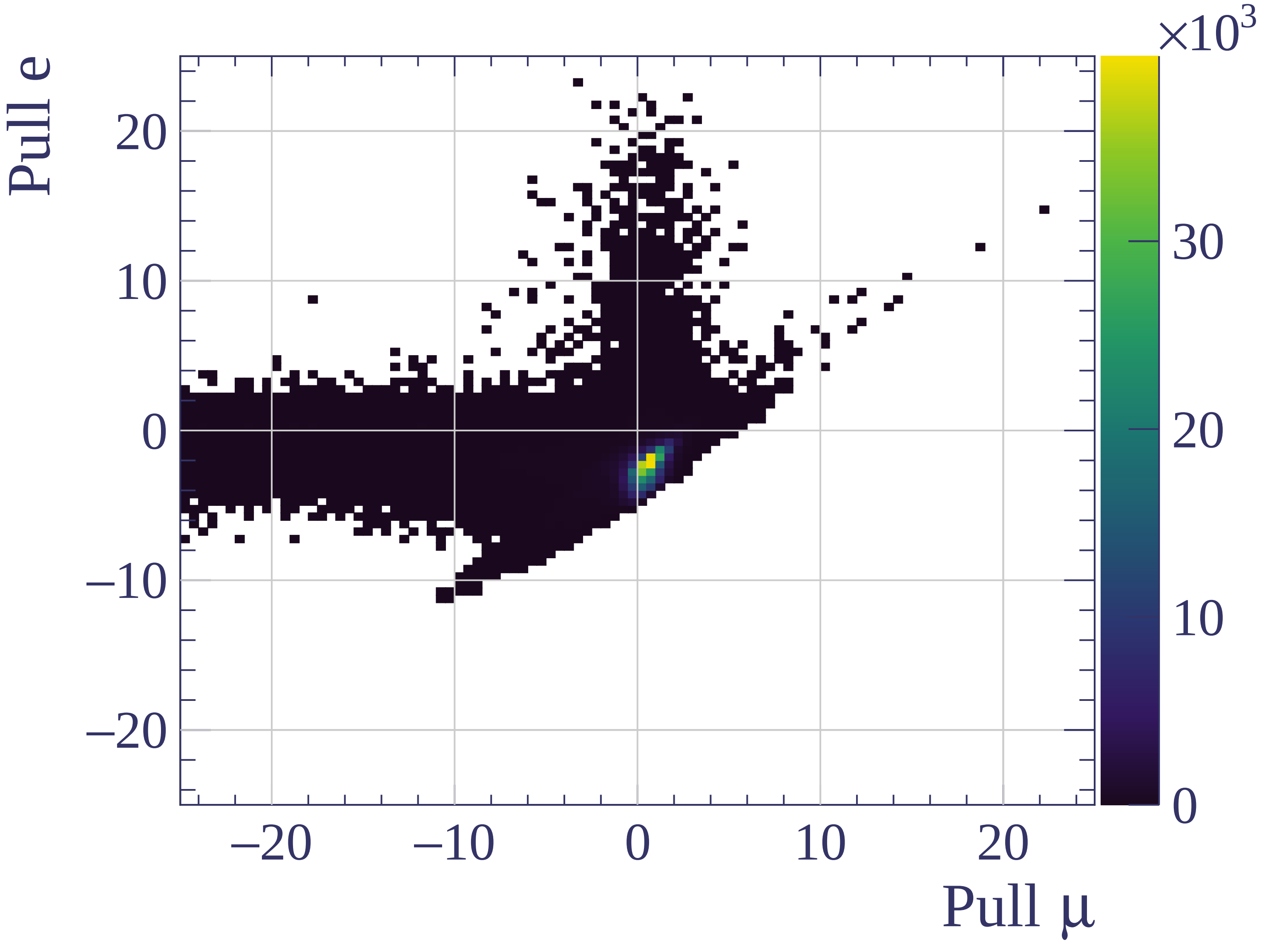

Pulls mean
150
100
50
0
−50
0
50
Track angle θ [°]
μ±
e±



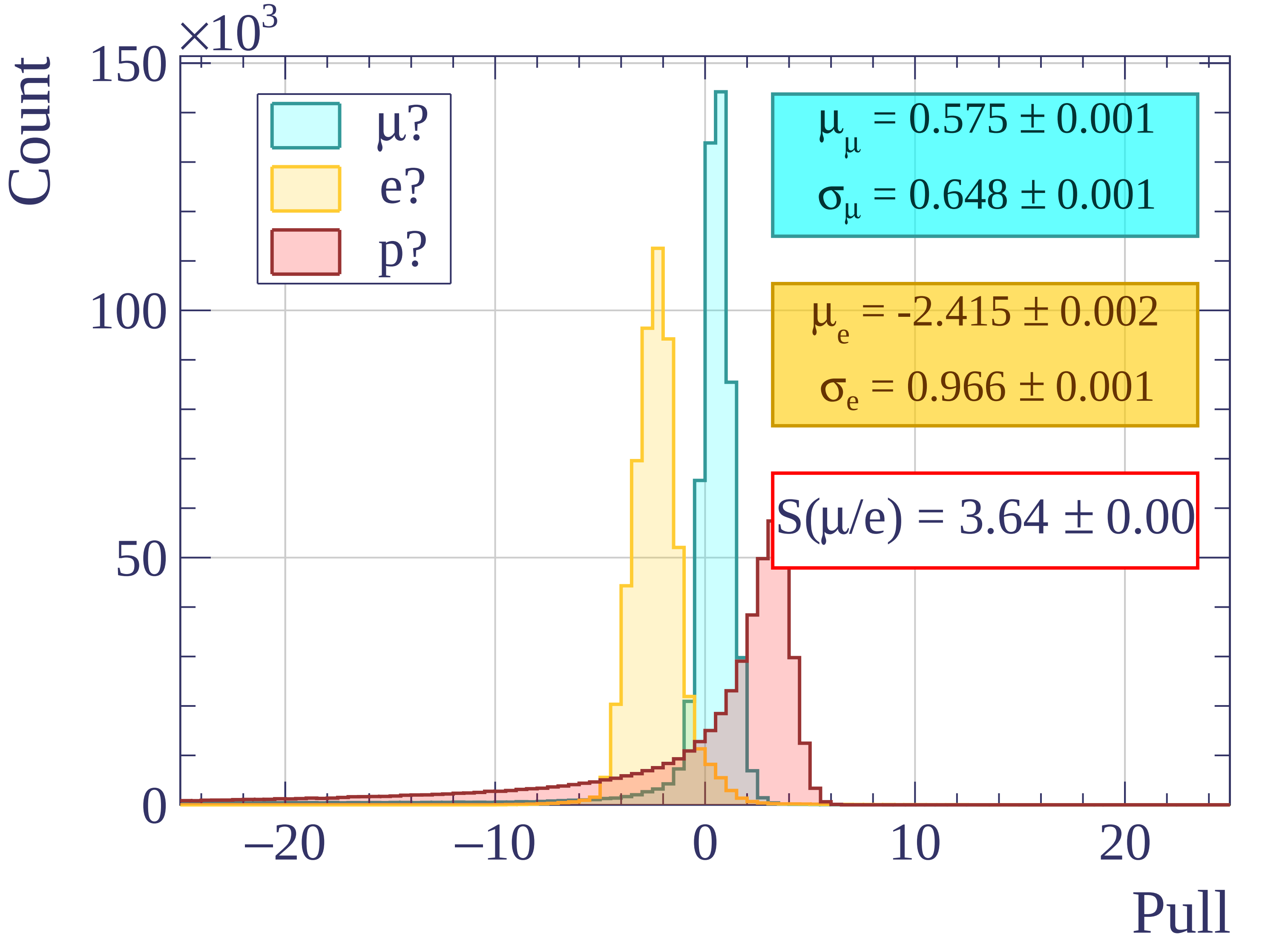
Pulls standard deviation
Momentum [MeV/c]
μ±
e±
0
5
10
15
20
−2000
−1000
0
1000
2000



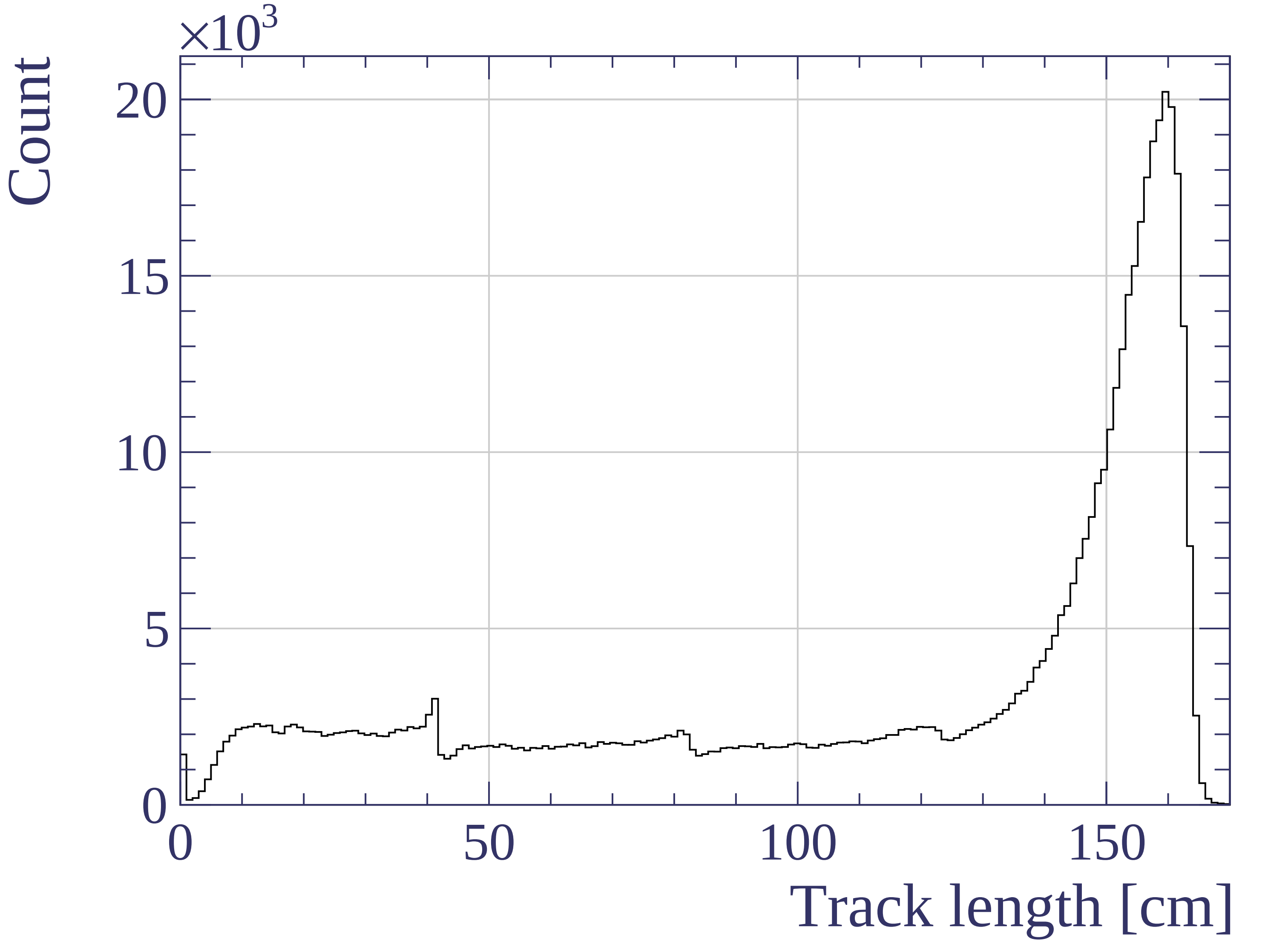
Pulls mean
0
−20
−40
−2000
−1000
0
1000
2000
Momentum [MeV/c]
$\mu^{\pm}$
$e^{\pm}$



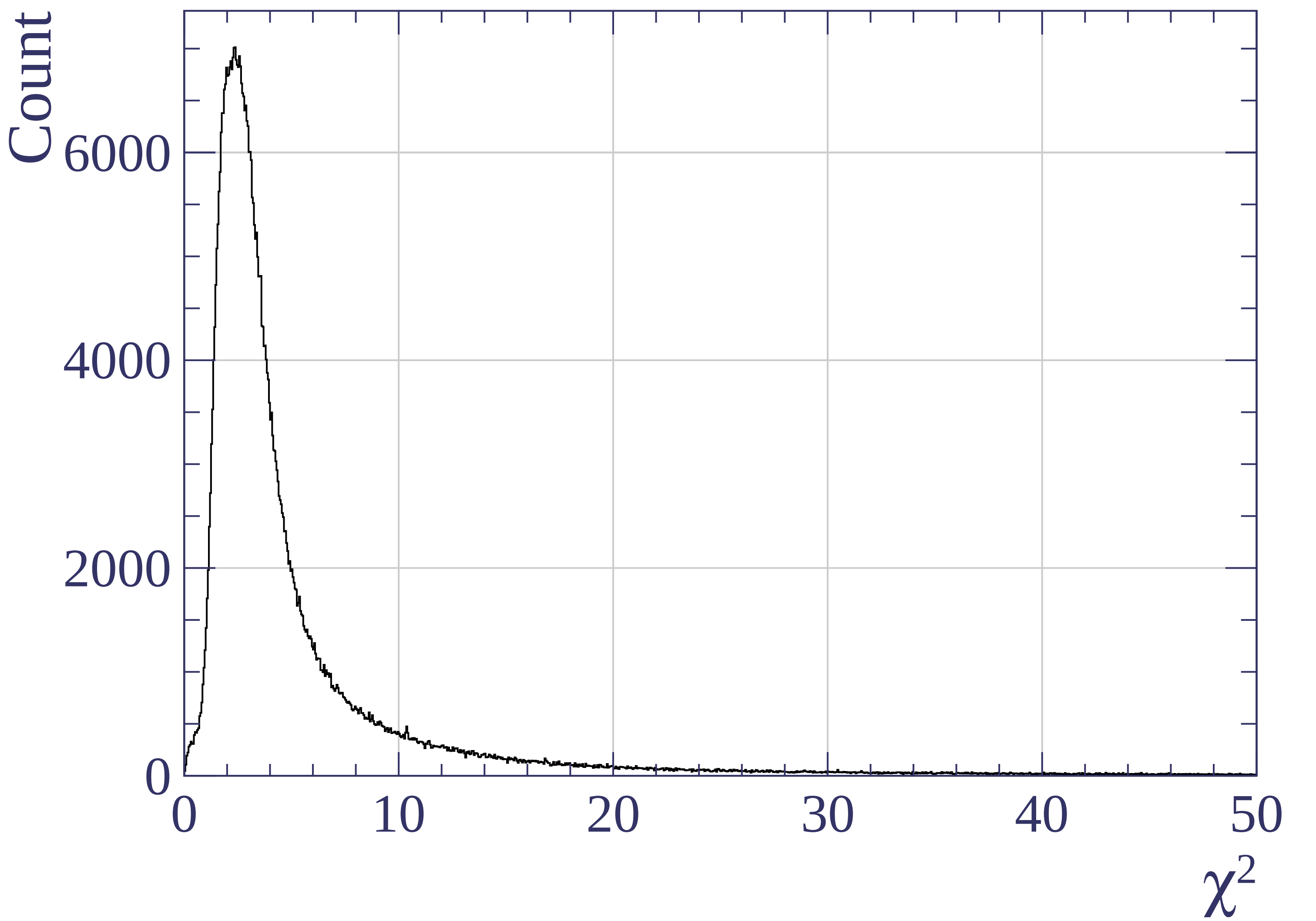
Pull e
20
10
0
−10
−20
−20
−10
0
10
20
Pull μ
×10^3
30
20
10
0



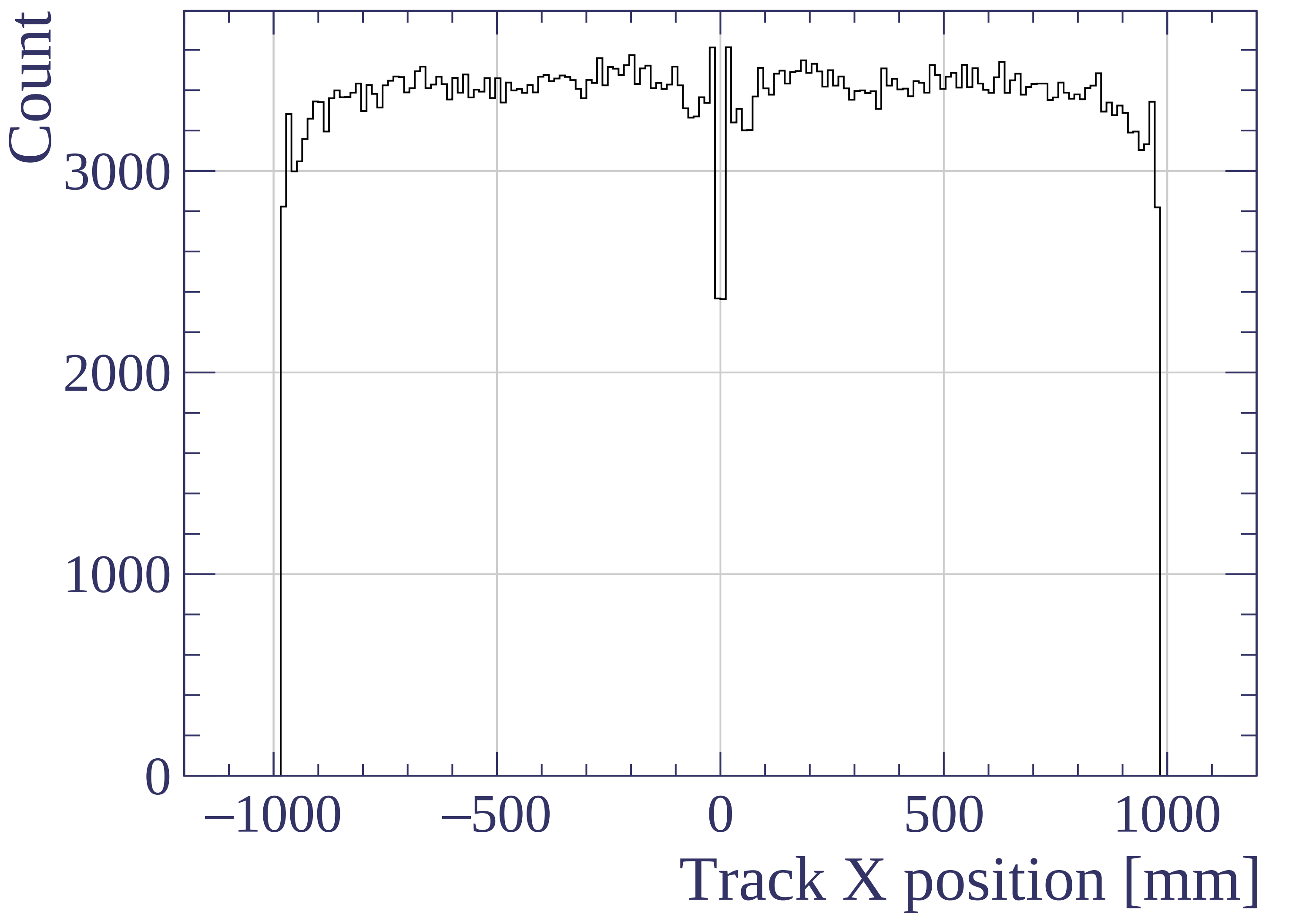
×10^3
150
100
50
0
Count
μ?
e?
p?
μ_μ = 0.575 ± 0.001
σ_μ = 0.648 ± 0.001
μ_e = -2.415 ± 0.002
σ_e = 0.966 ± 0.001
S(μ/e) = 3.64 ± 0.00
−20
−10
0
10
20
Pull

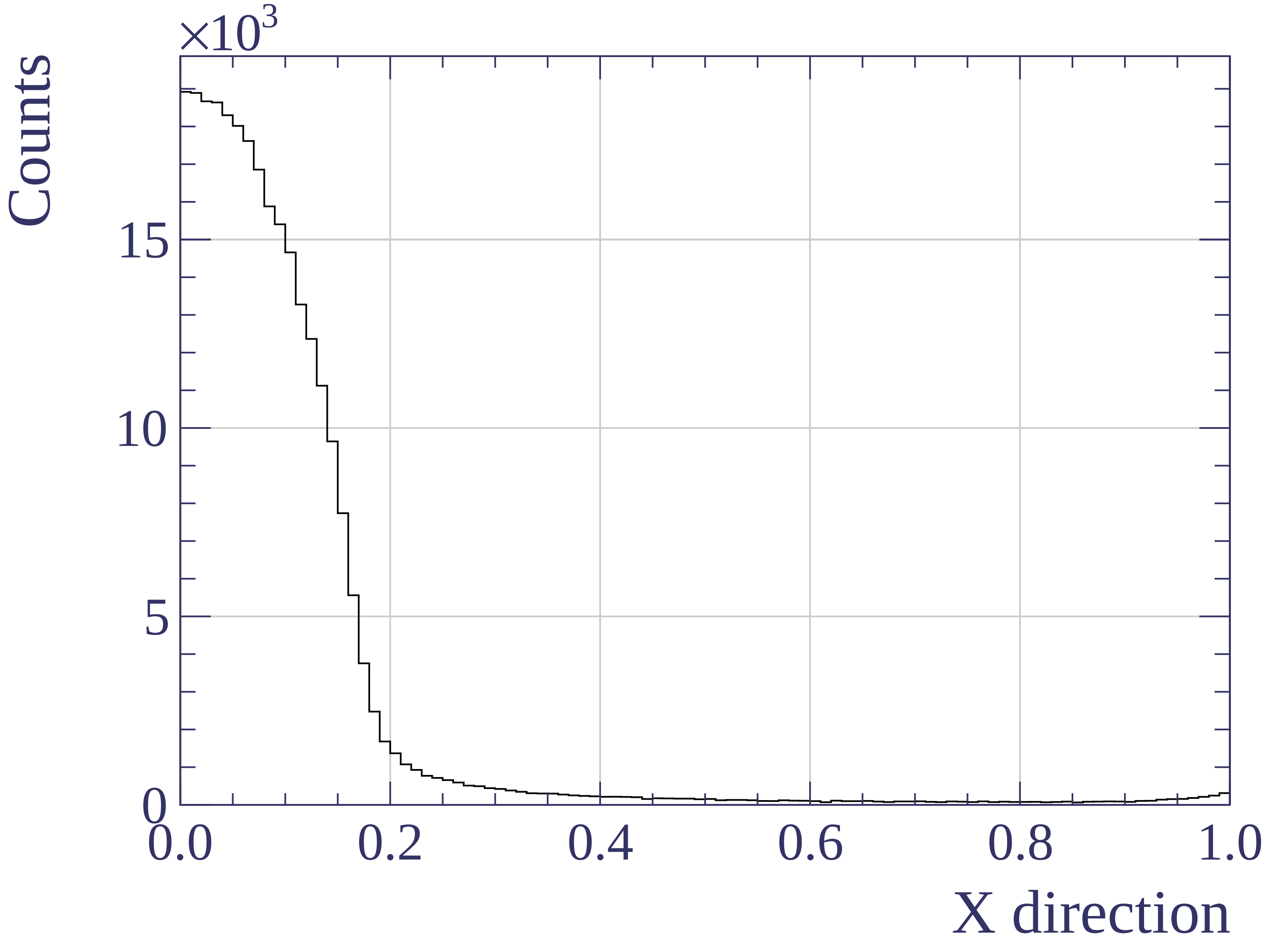


$\times 10^3$
Count
20
15
10
5
0
0
50
100
150
Track length [cm]

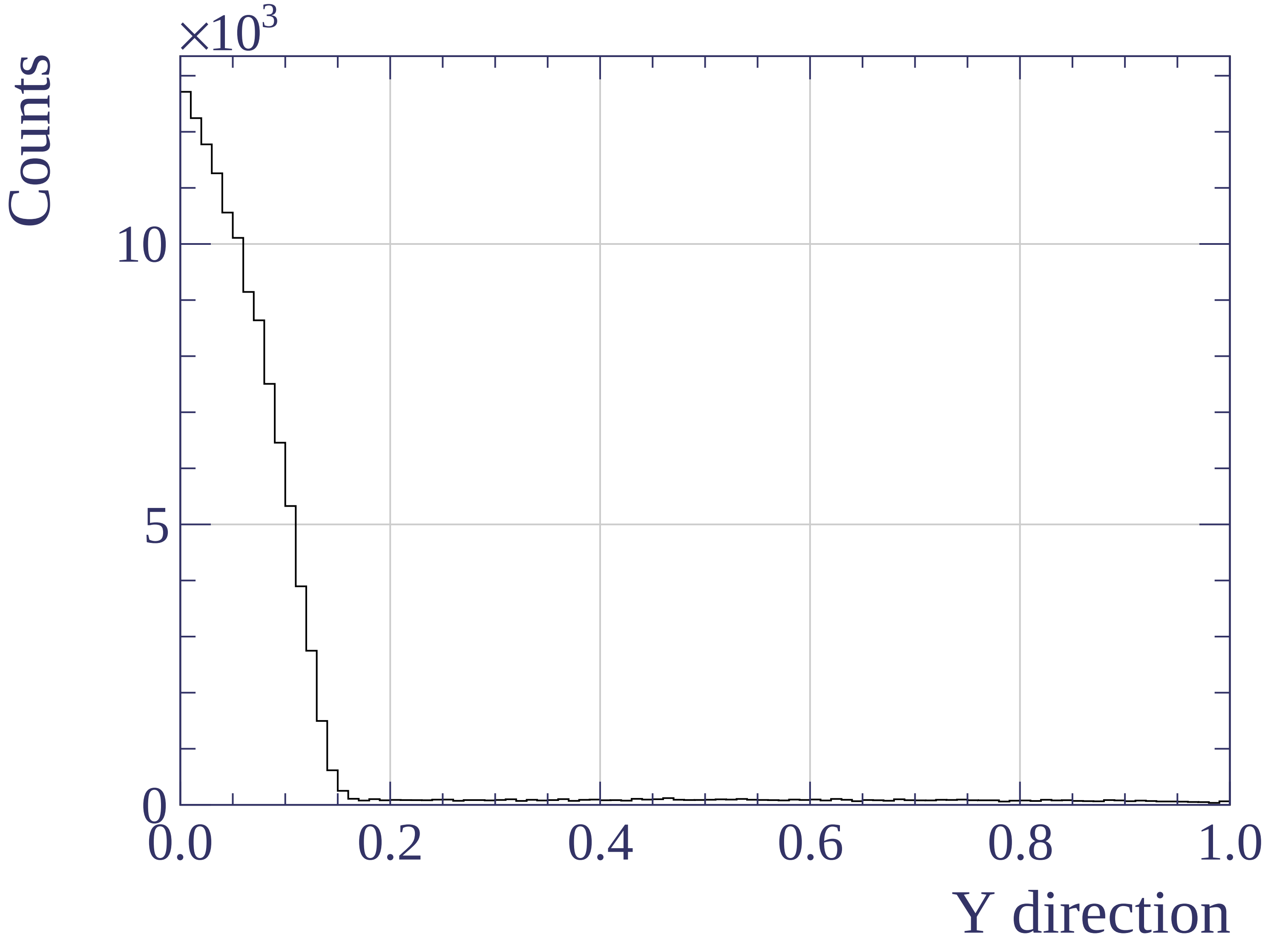


Count
6000
4000
2000
0
0
10
20
30
40
50
$\chi^2$

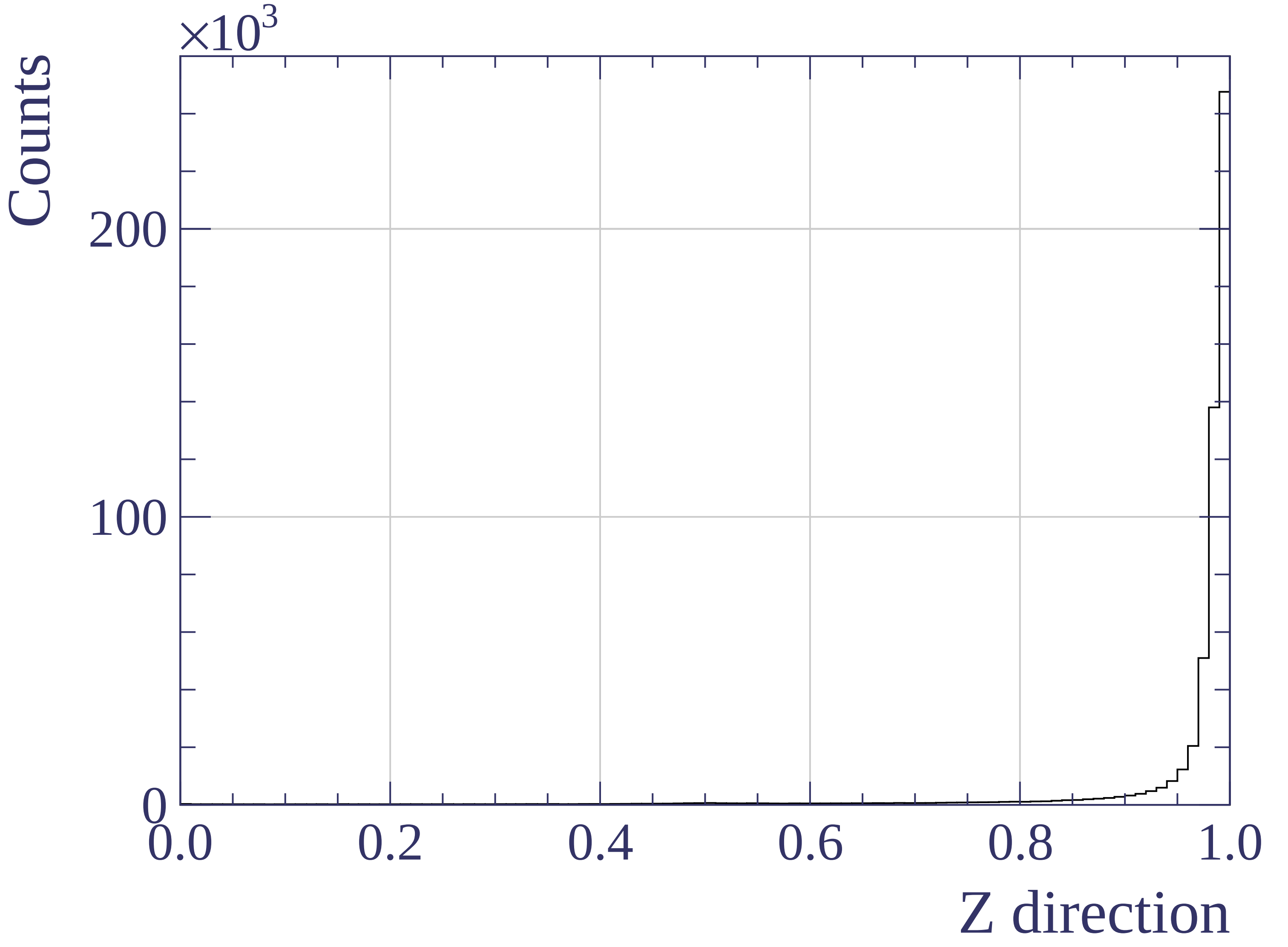

Count
0
1000
2000
3000
−1000
−500
0
500
1000
Track X position [mm]



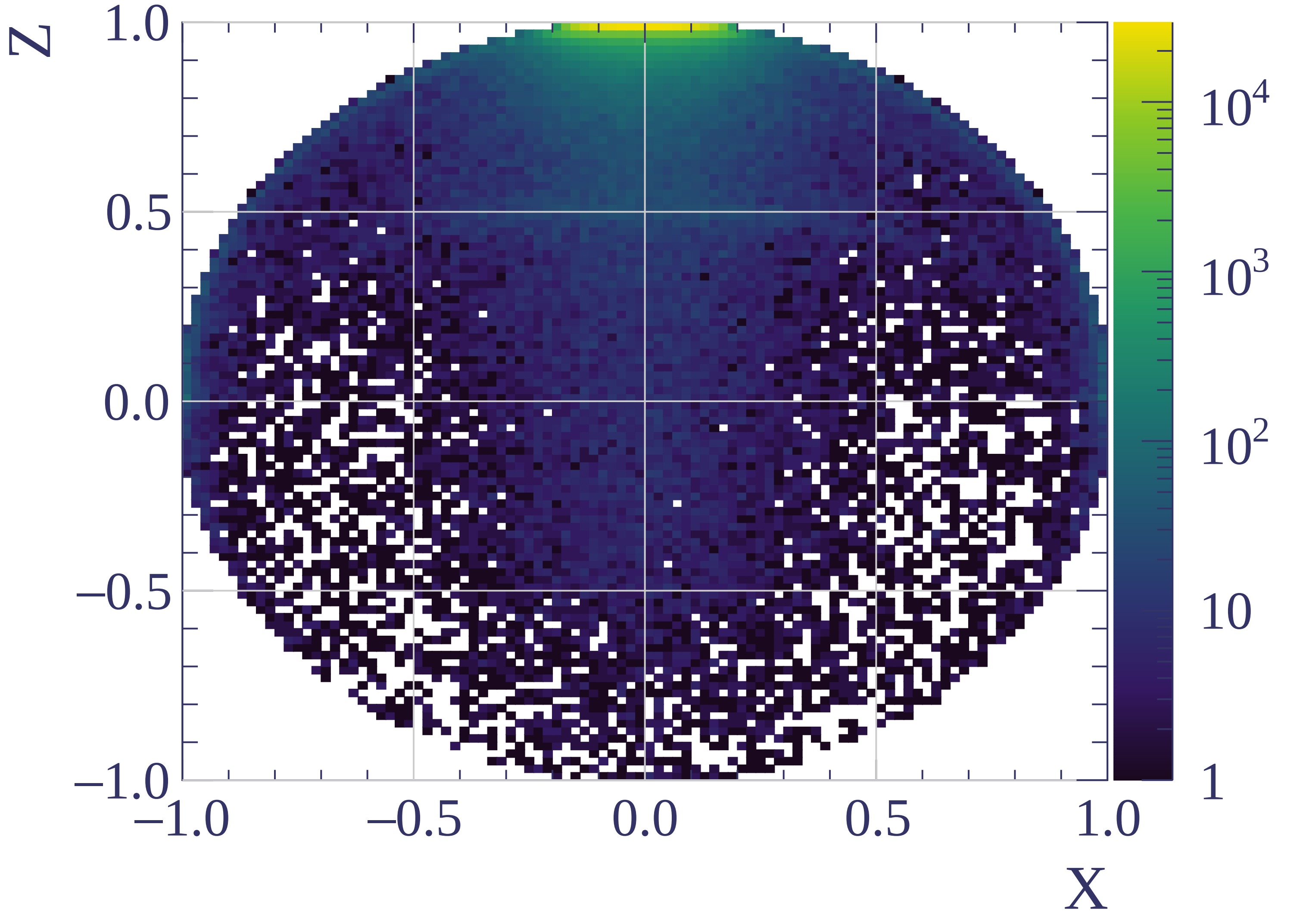
×10^3
Counts
15
10
5
0
0.0
0.2
0.4
0.6
0.8
1.0
X direction

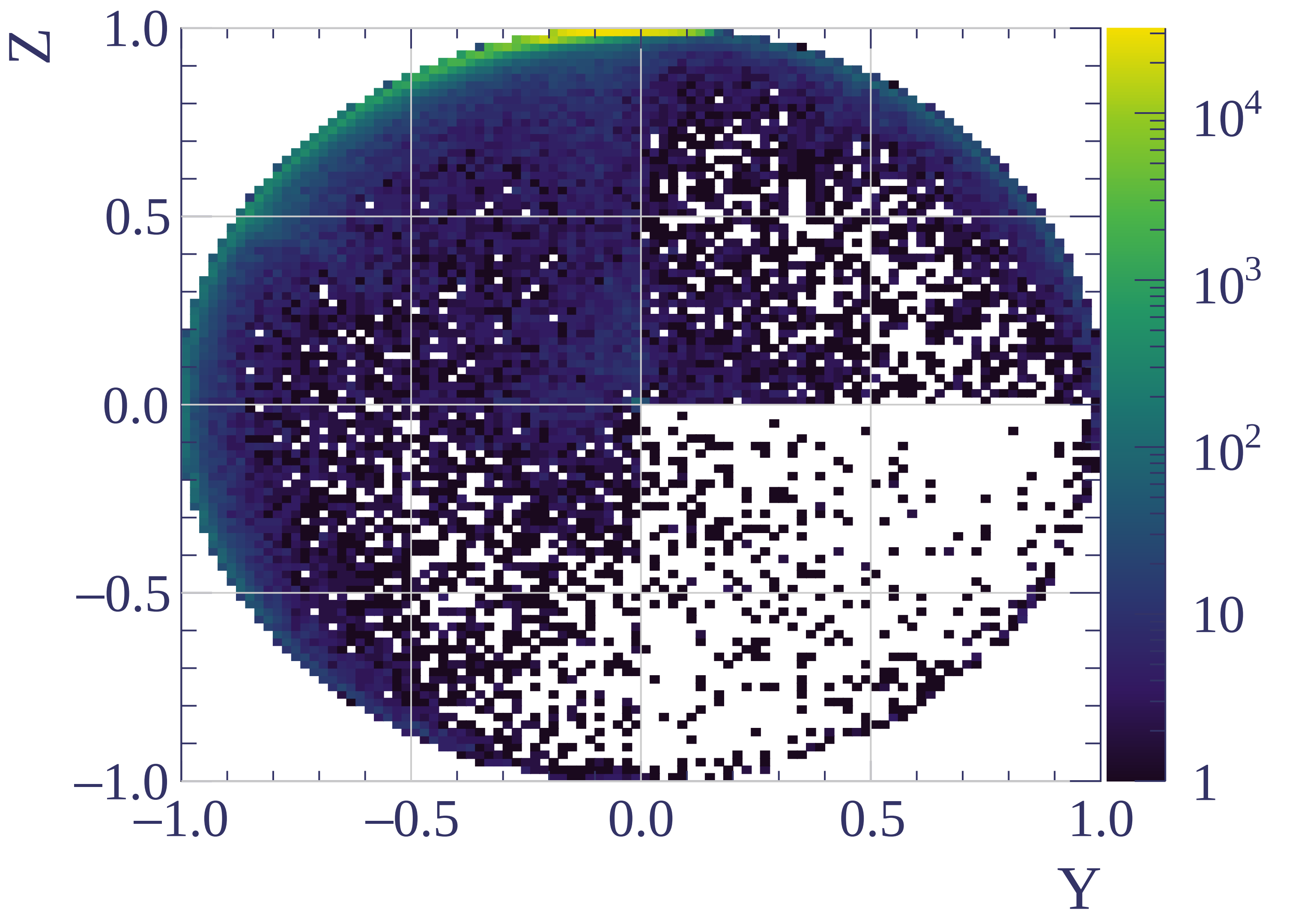


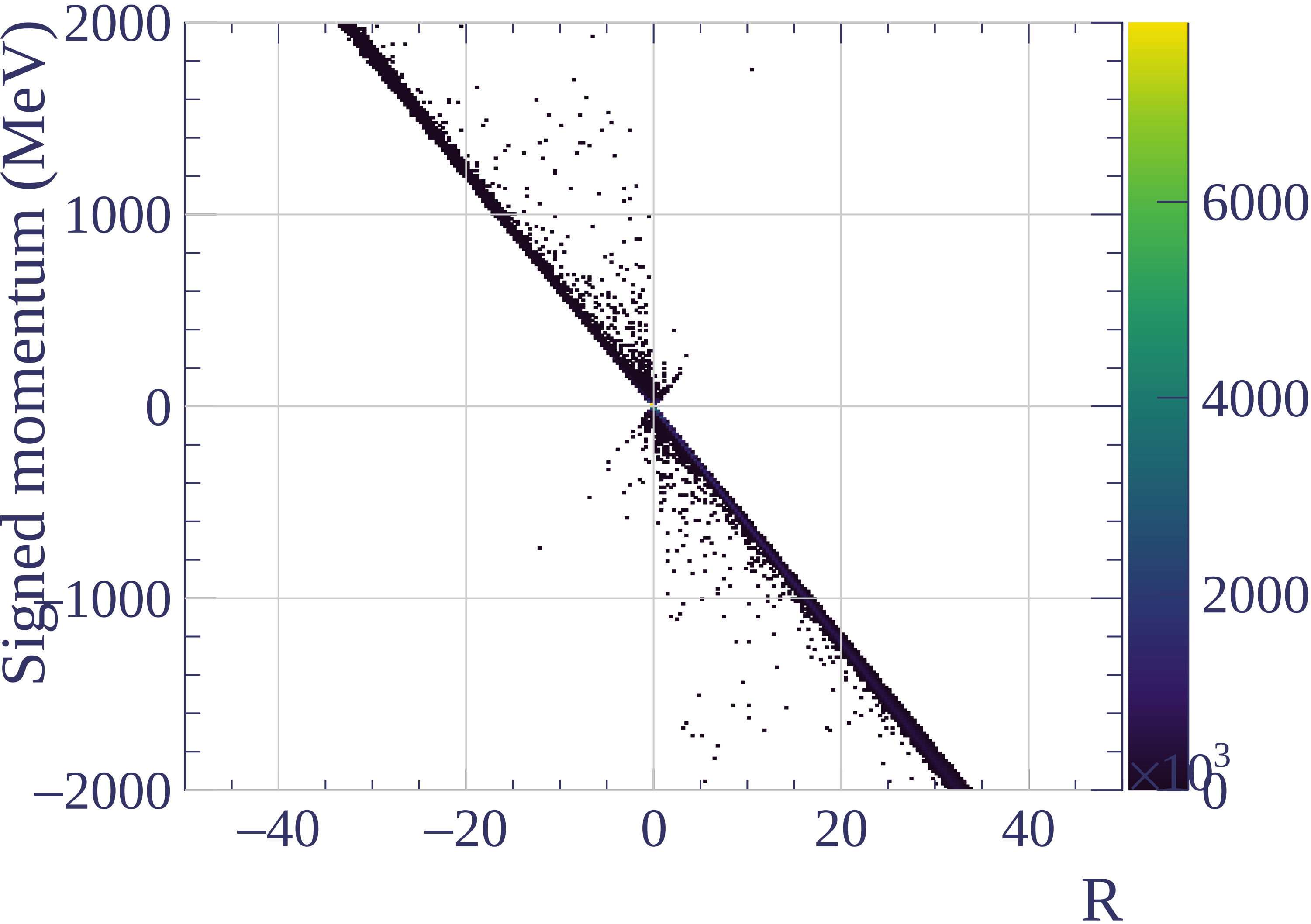
×10^3
Counts
10
5
0
0.0
0.2
0.4
0.6
0.8
1.0
Y direction

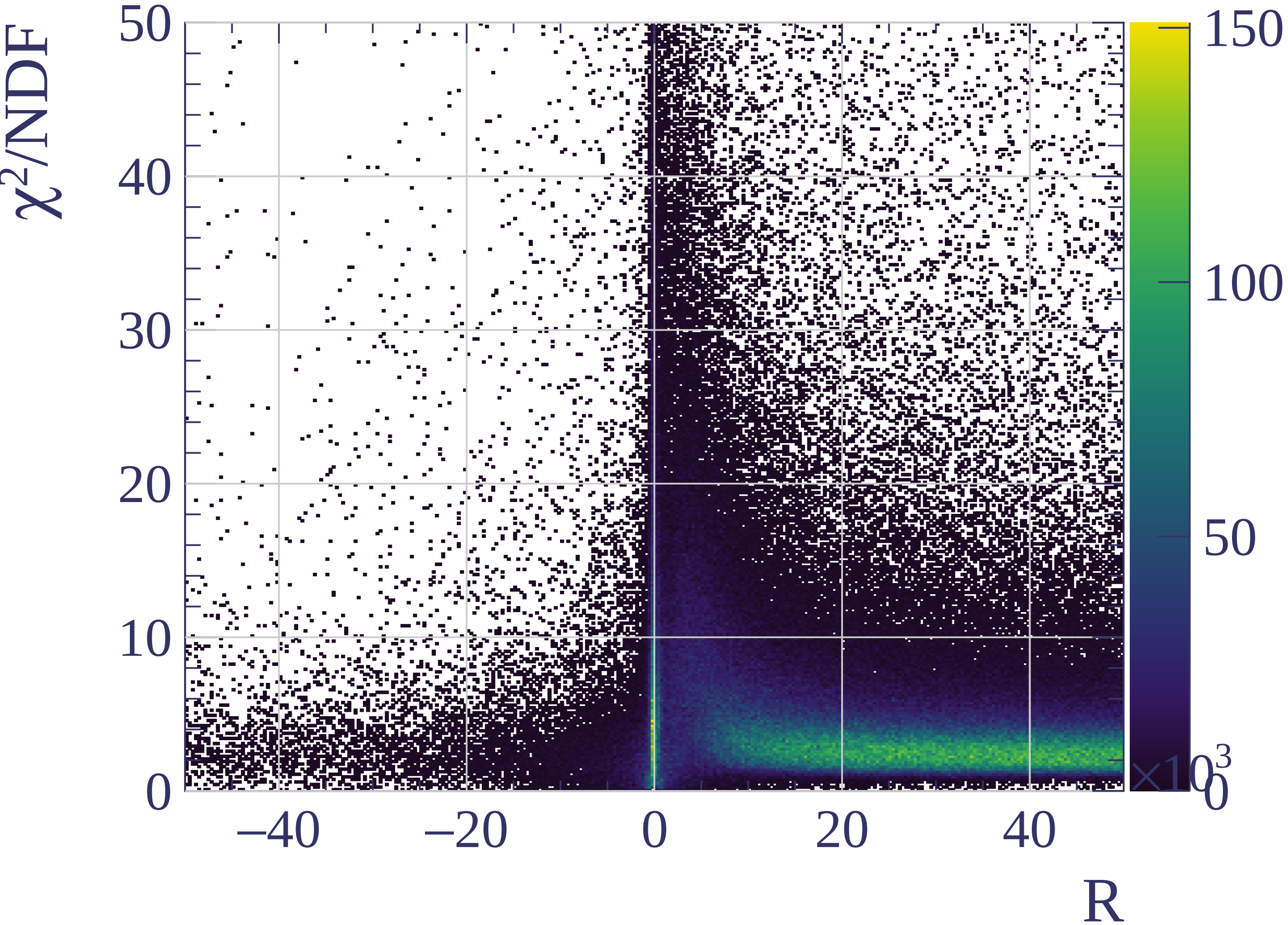


×10^3
Counts
200
100
0
0.0
0.2
0.4
0.6
0.8
1.0
Z direction

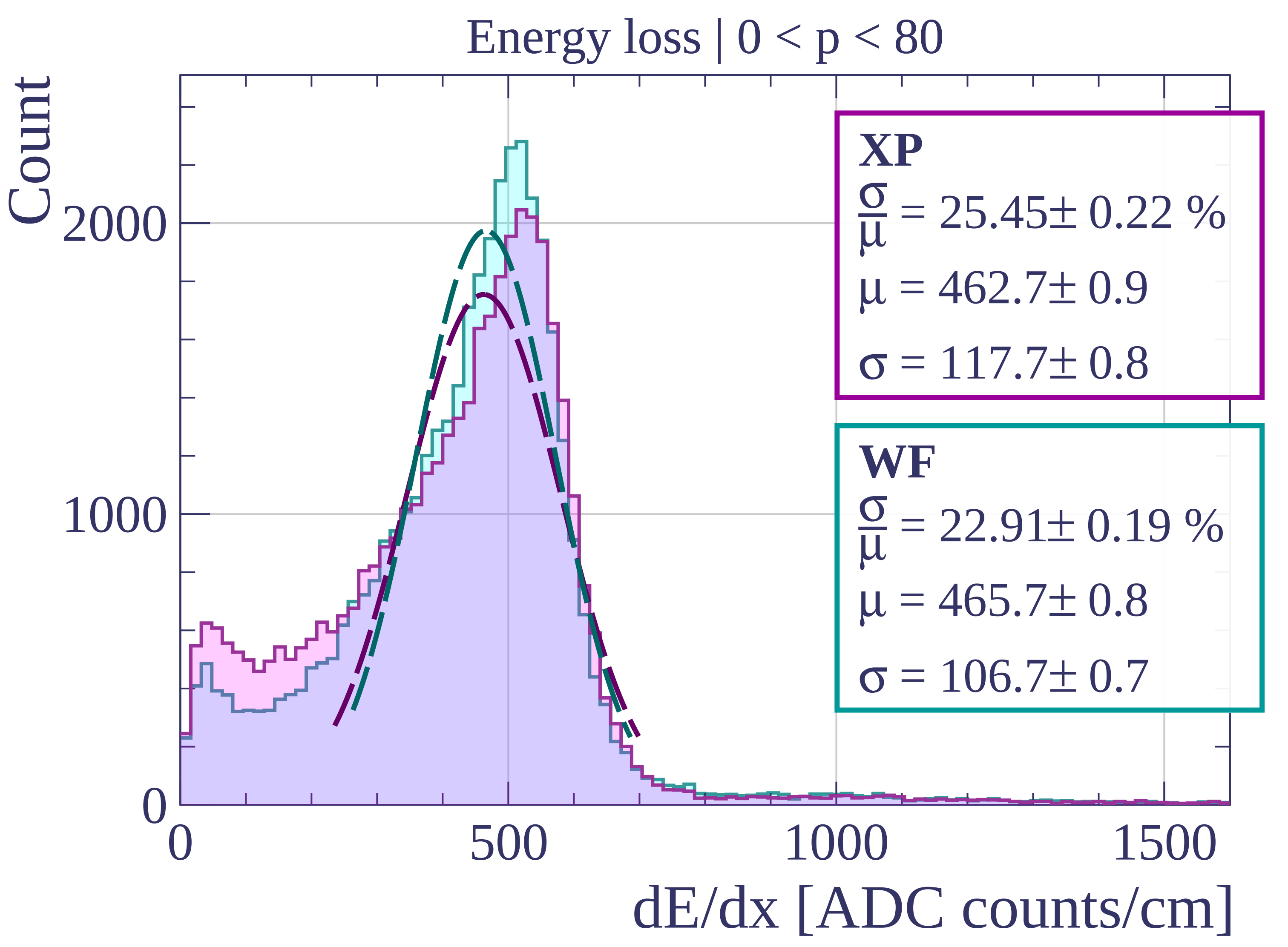

Z
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
X
$10^4$
$10^3$
$10^2$
10
1


Z
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
Y
$10^4$
$10^3$
$10^2$
10
1

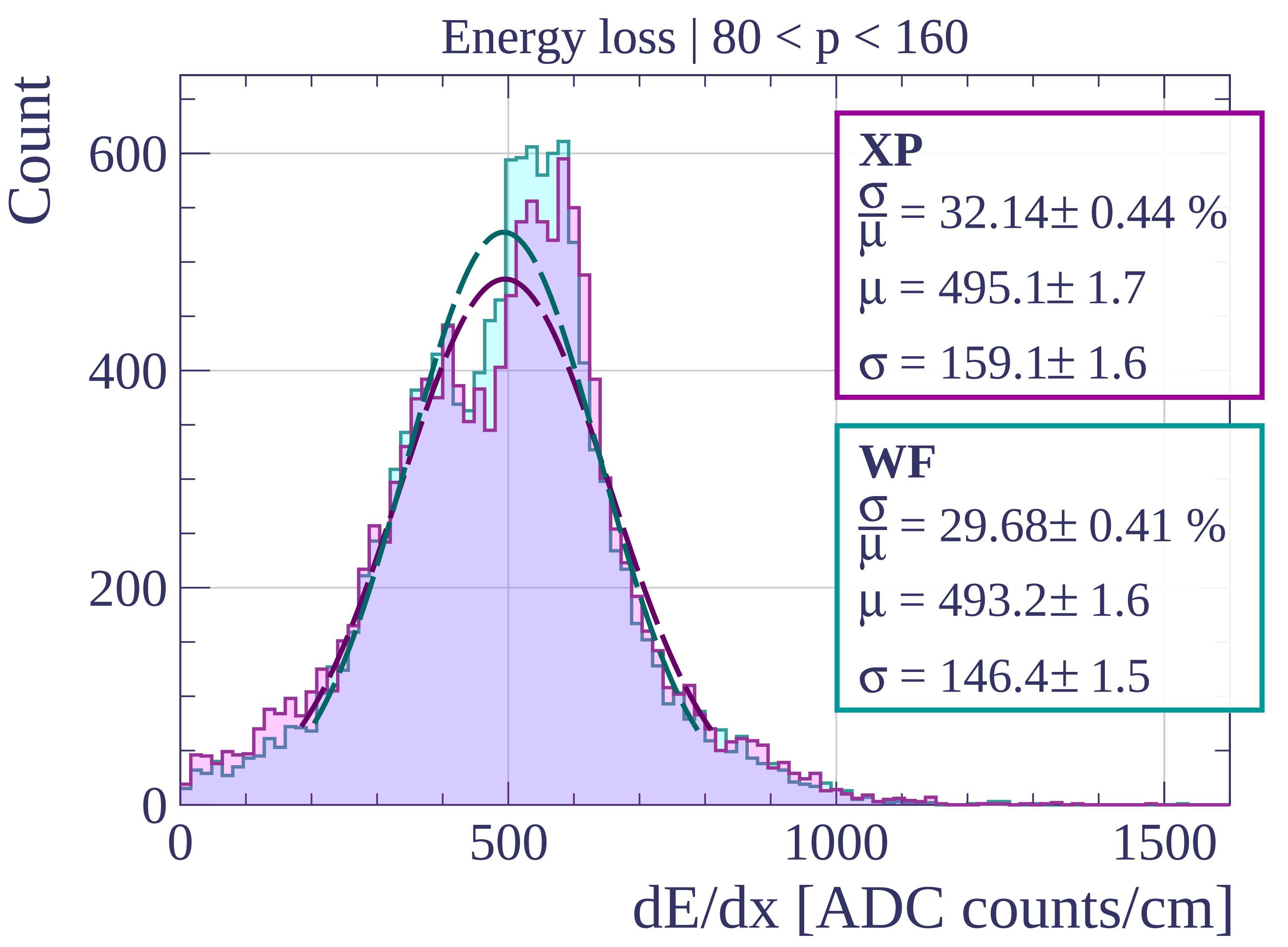

Signed momentum (MeV)
2000
1000
0
−1000
−2000
−40
−20
0
20
40
R
6000
4000
2000
0
×10^3

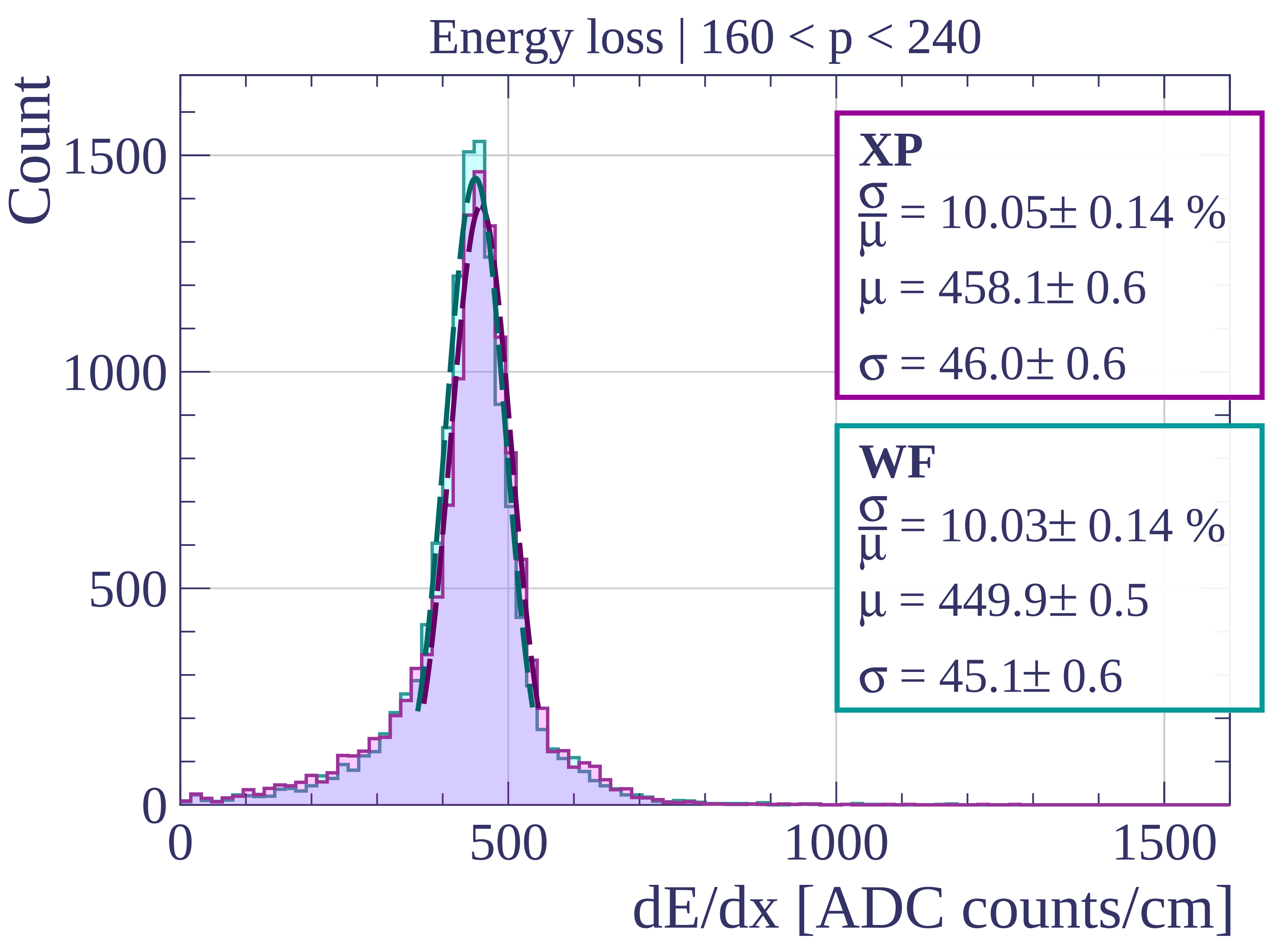


$\chi^2$/NDF
50
40
30
20
10
0
−40
−20
0
20
40
R
150
100
50
0
$\times 10^3$

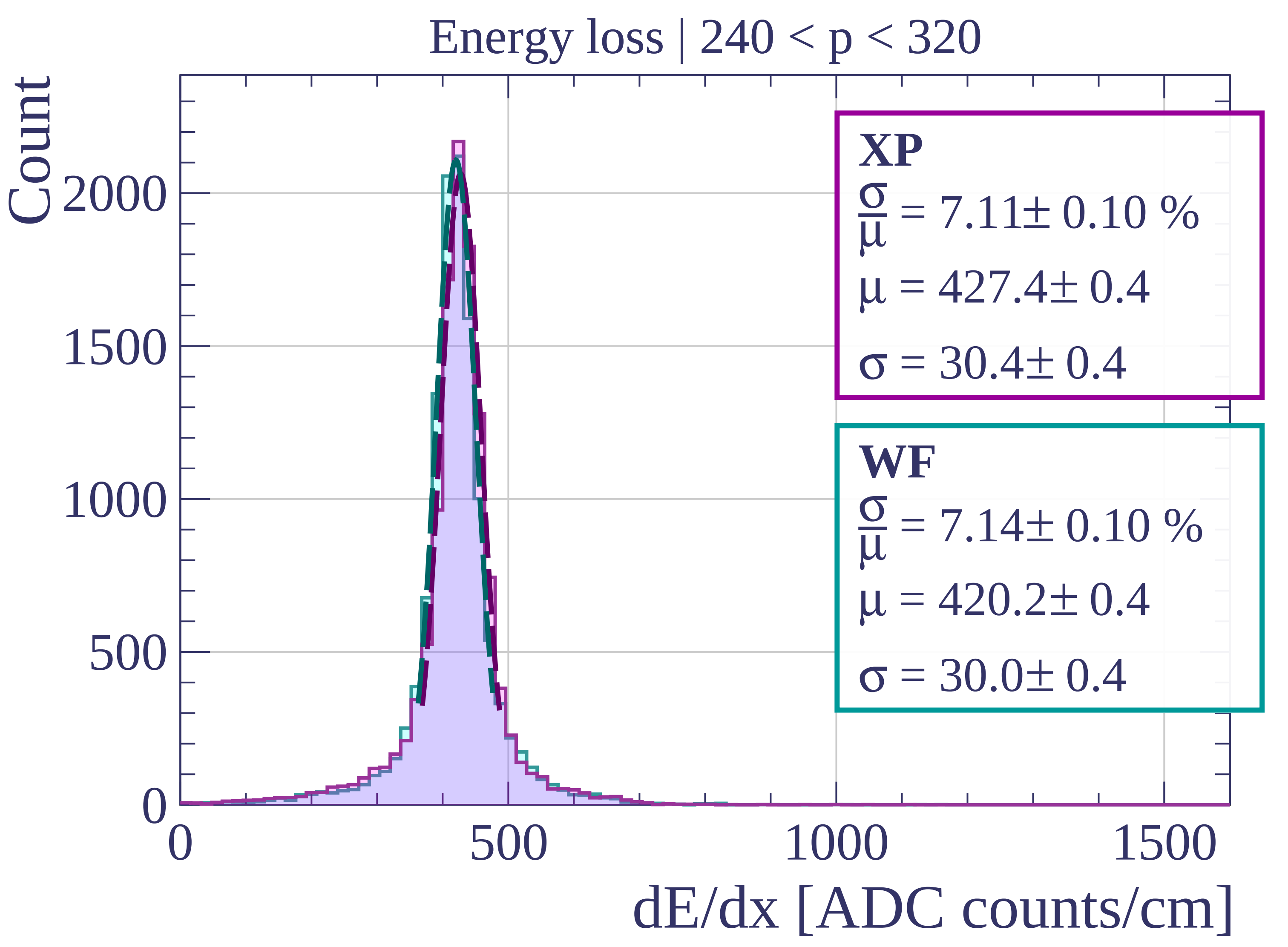


Energy loss | 0 < p < 80
Count
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 25.45± 0.22 %
μ = 462.7± 0.9
σ = 117.7± 0.8
WF
σ/μ = 22.91± 0.19 %
μ = 465.7± 0.8
σ = 106.7± 0.7

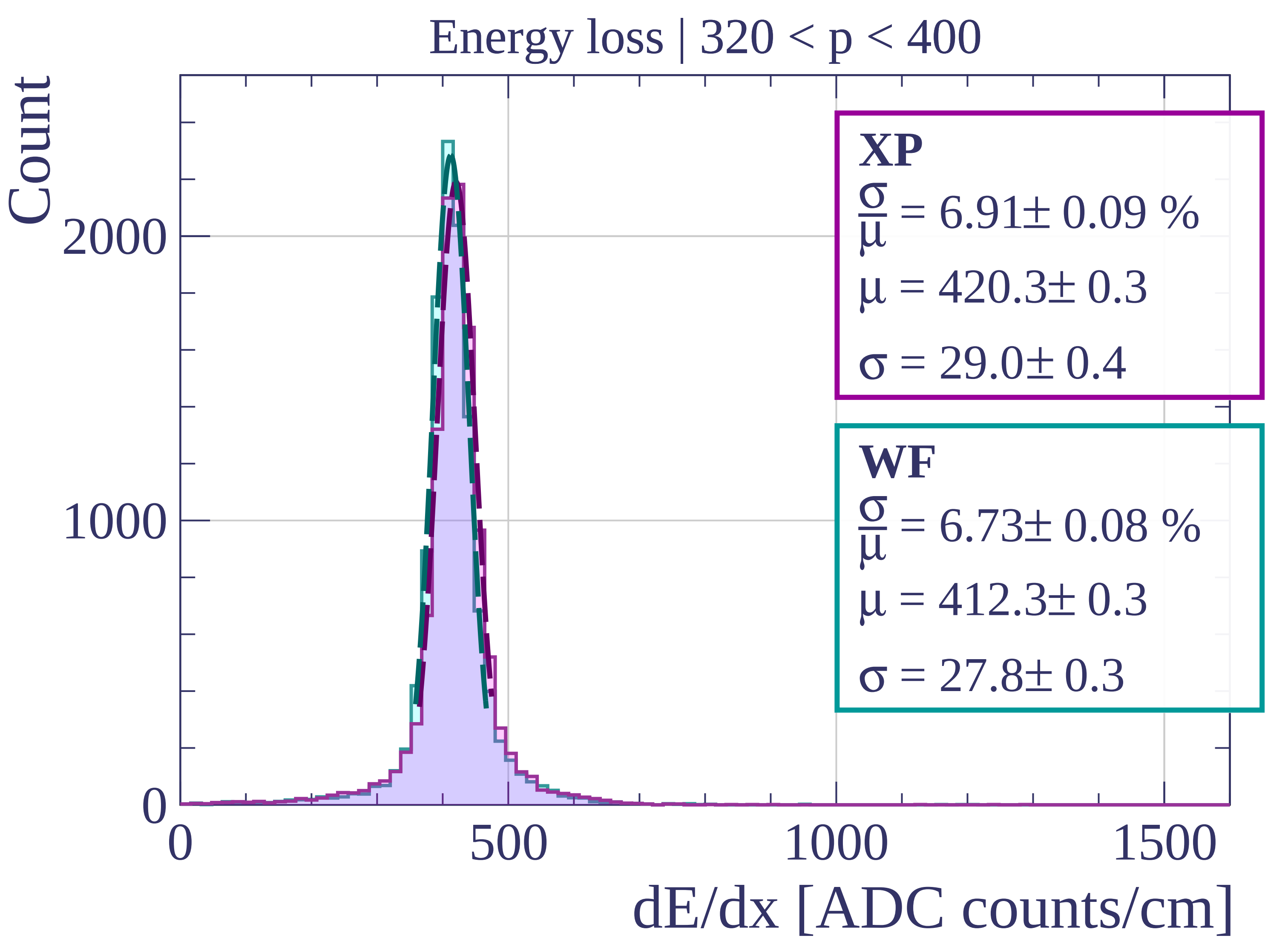

Energy loss | 80 < p < 160
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 32.14± 0.44 %
μ = 495.1± 1.7
σ = 159.1± 1.6
WF
σ/μ = 29.68± 0.41 %
μ = 493.2± 1.6
σ = 146.4± 1.5



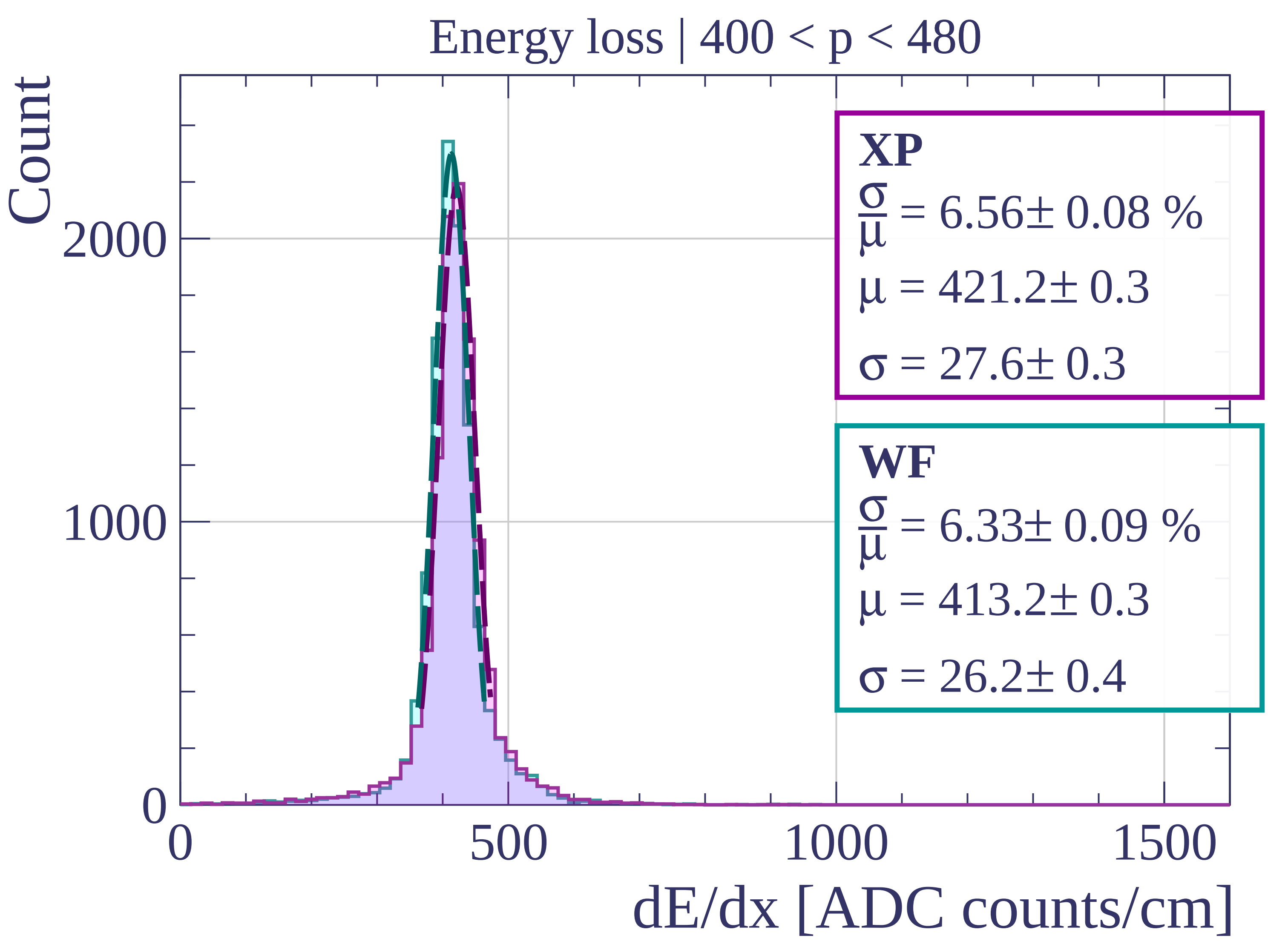
Energy loss | 160 < p < 240
Count
dE/dx [ADC counts/cm]
XP
σ/μ = 10.05± 0.14 %
μ = 458.1± 0.6
σ = 46.0± 0.6
WF
σ/μ = 10.03± 0.14 %
μ = 449.9± 0.5
σ = 45.1± 0.6



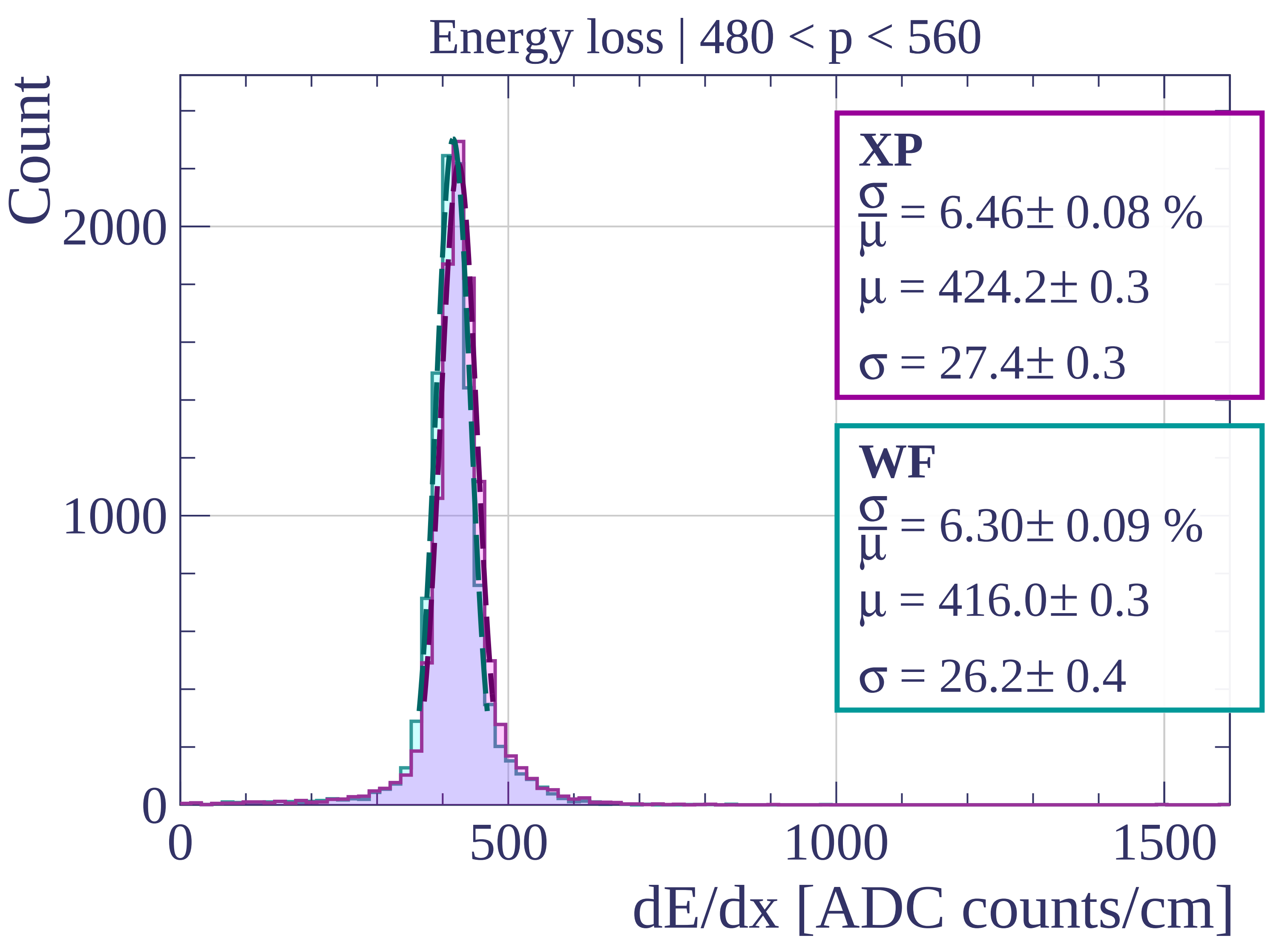
Energy loss | 240 < p < 320
Count
2000
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 7.11± 0.10 %
μ = 427.4± 0.4
σ = 30.4± 0.4
WF
$\frac{\sigma}{\mu}$ = 7.14± 0.10 %
μ = 420.2± 0.4
σ = 30.0± 0.4



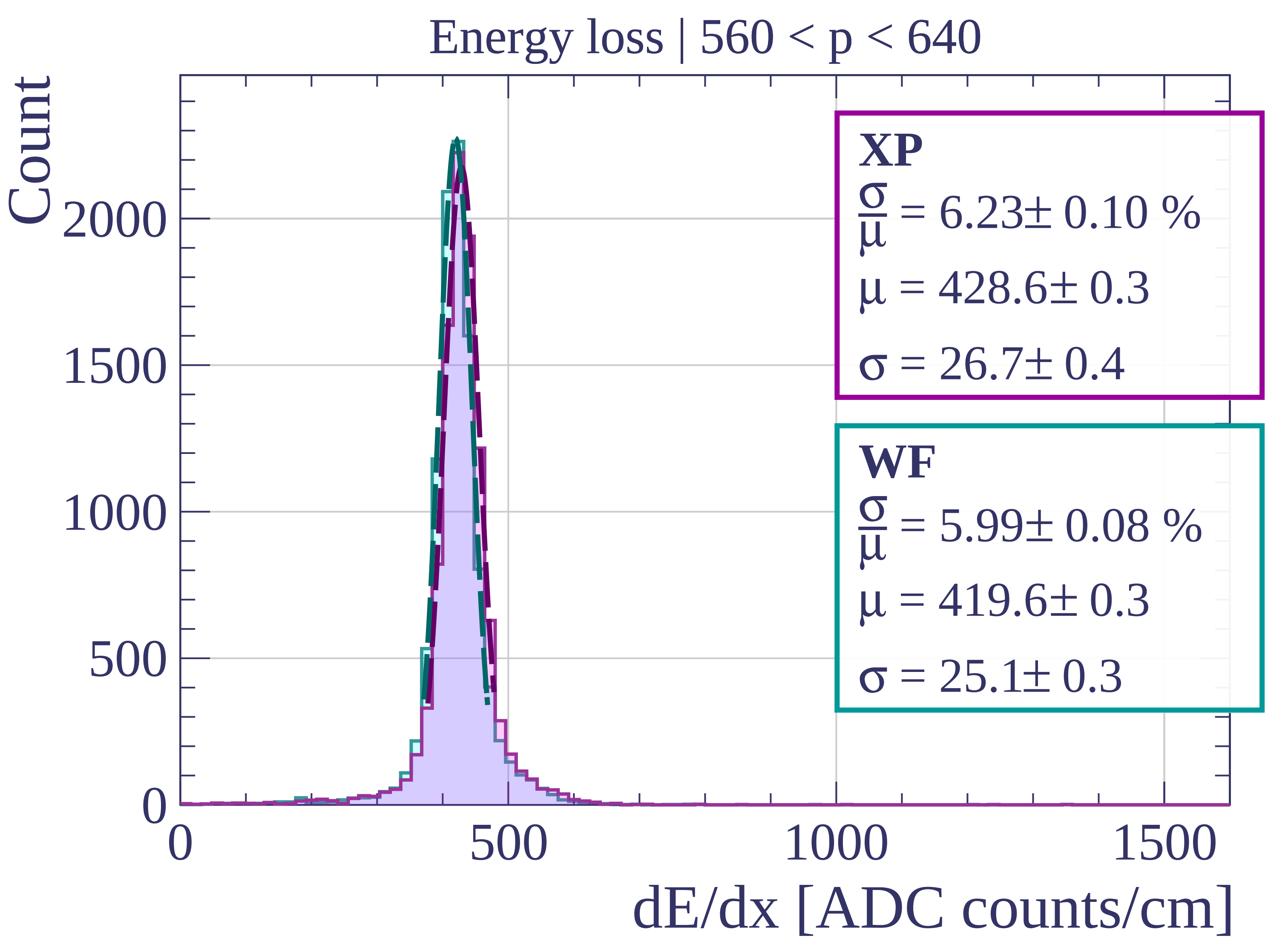
Energy loss | 320 < p < 400
Count
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 6.91± 0.09 %
μ = 420.3± 0.3
σ = 29.0± 0.4
WF
σ/μ = 6.73± 0.08 %
μ = 412.3± 0.3
σ = 27.8± 0.3

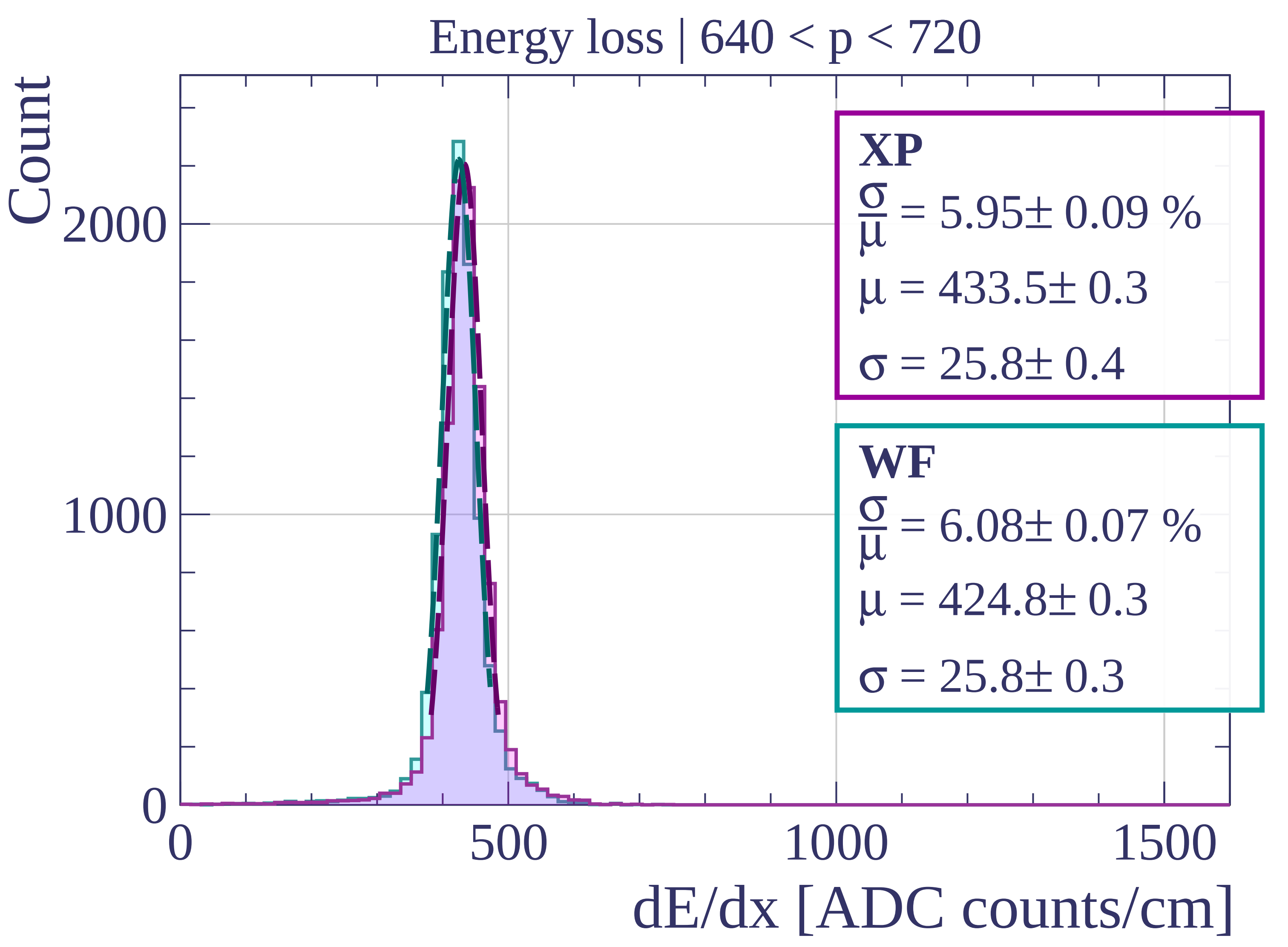


Energy loss | 400 < p < 480
Count
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 6.56± 0.08 %
μ = 421.2± 0.3
σ = 27.6± 0.3
WF
σ/μ = 6.33± 0.09 %
μ = 413.2± 0.3
σ = 26.2± 0.4

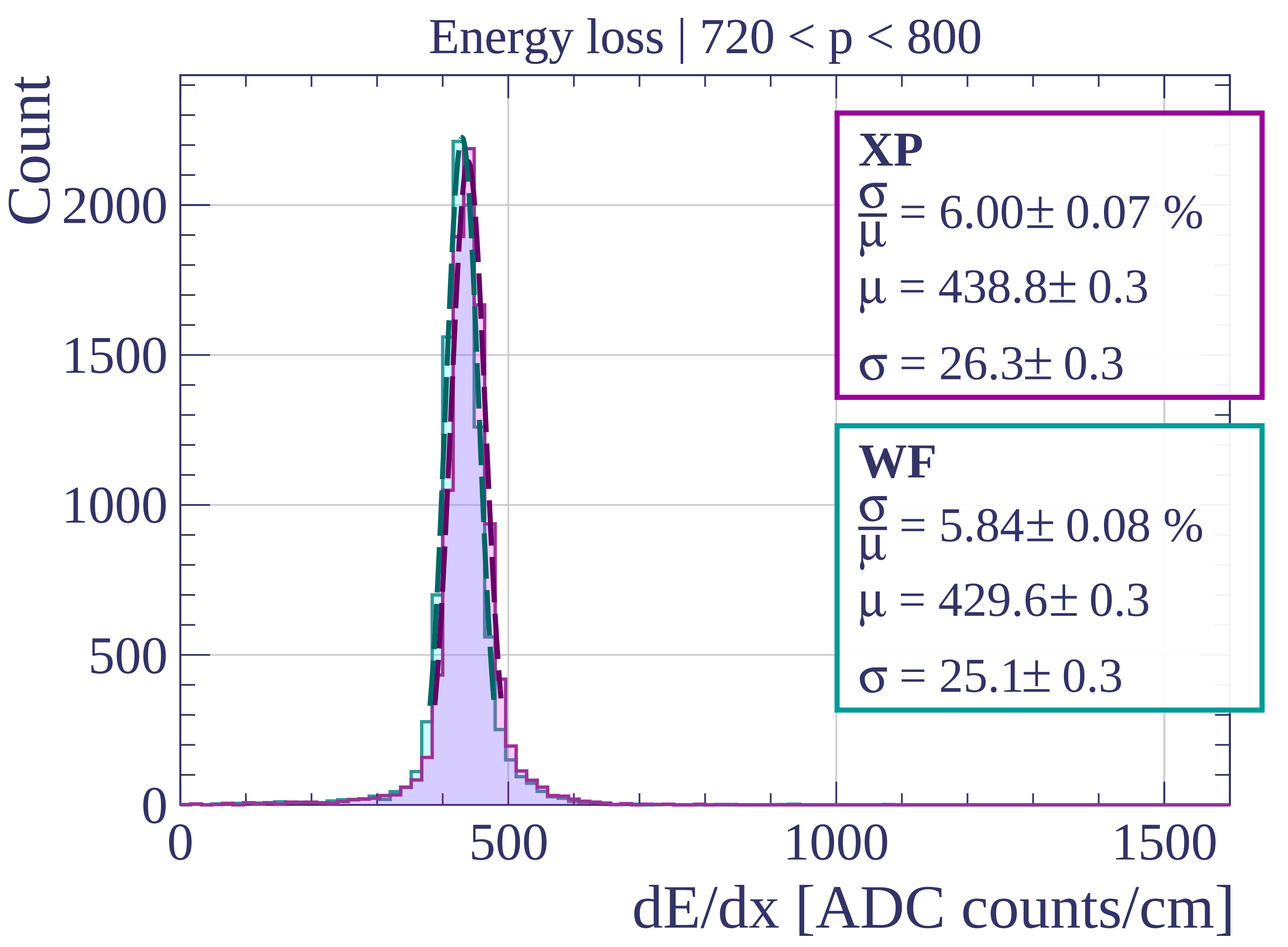


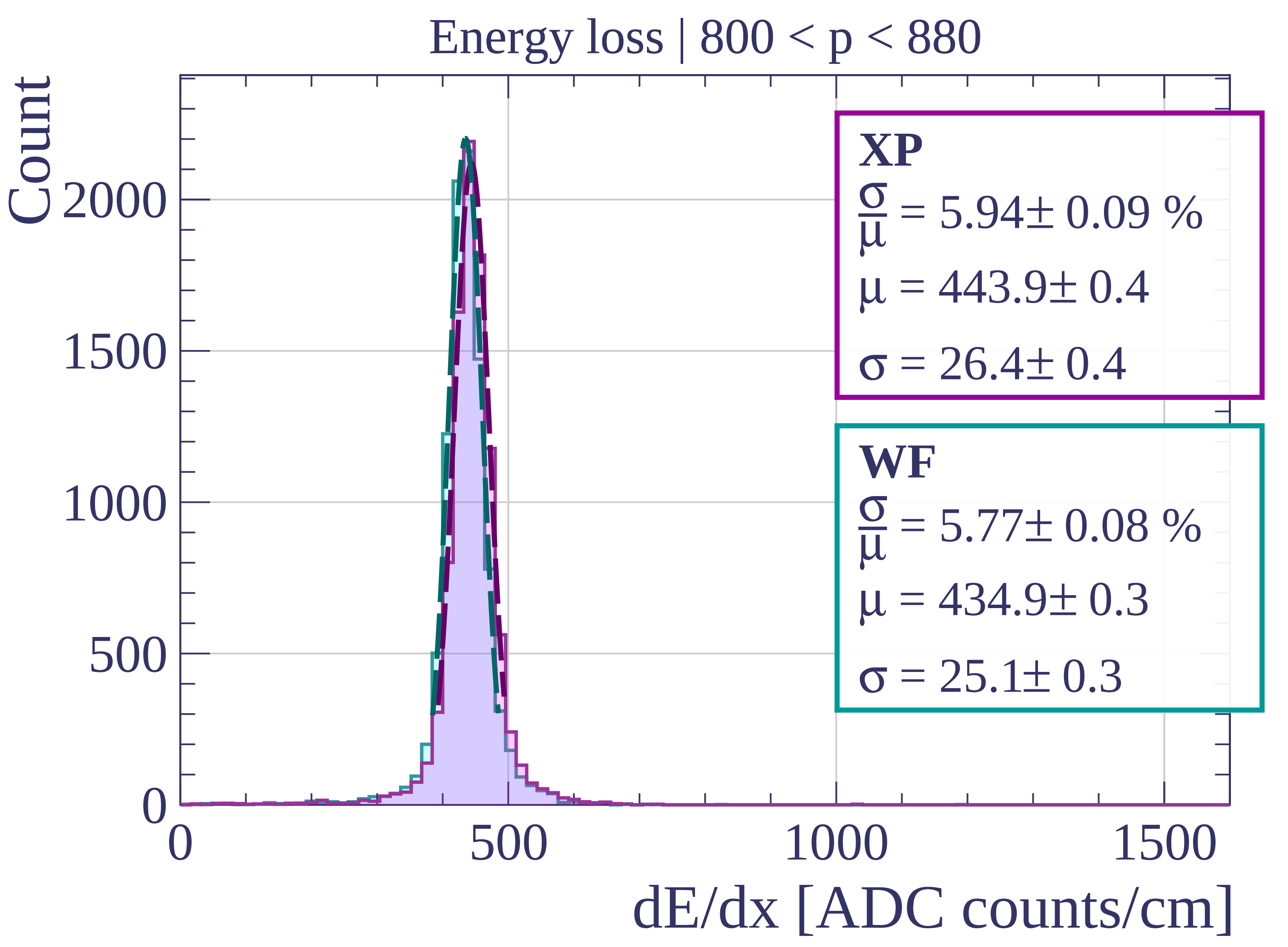
Energy loss | 480 < p < 560
Count
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 6.46± 0.08 %
μ = 424.2± 0.3
σ = 27.4± 0.3
WF
$\frac{\sigma}{\mu}$ = 6.30± 0.09 %
μ = 416.0± 0.3
σ = 26.2± 0.4


Energy loss | 560 < p < 640
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
2000
XP
$\frac{\sigma}{\mu}$ = 6.23± 0.10 %
μ = 428.6± 0.3
σ = 26.7± 0.4
WF
$\frac{\sigma}{\mu}$ = 5.99± 0.08 %
μ = 419.6± 0.3
σ = 25.1± 0.3

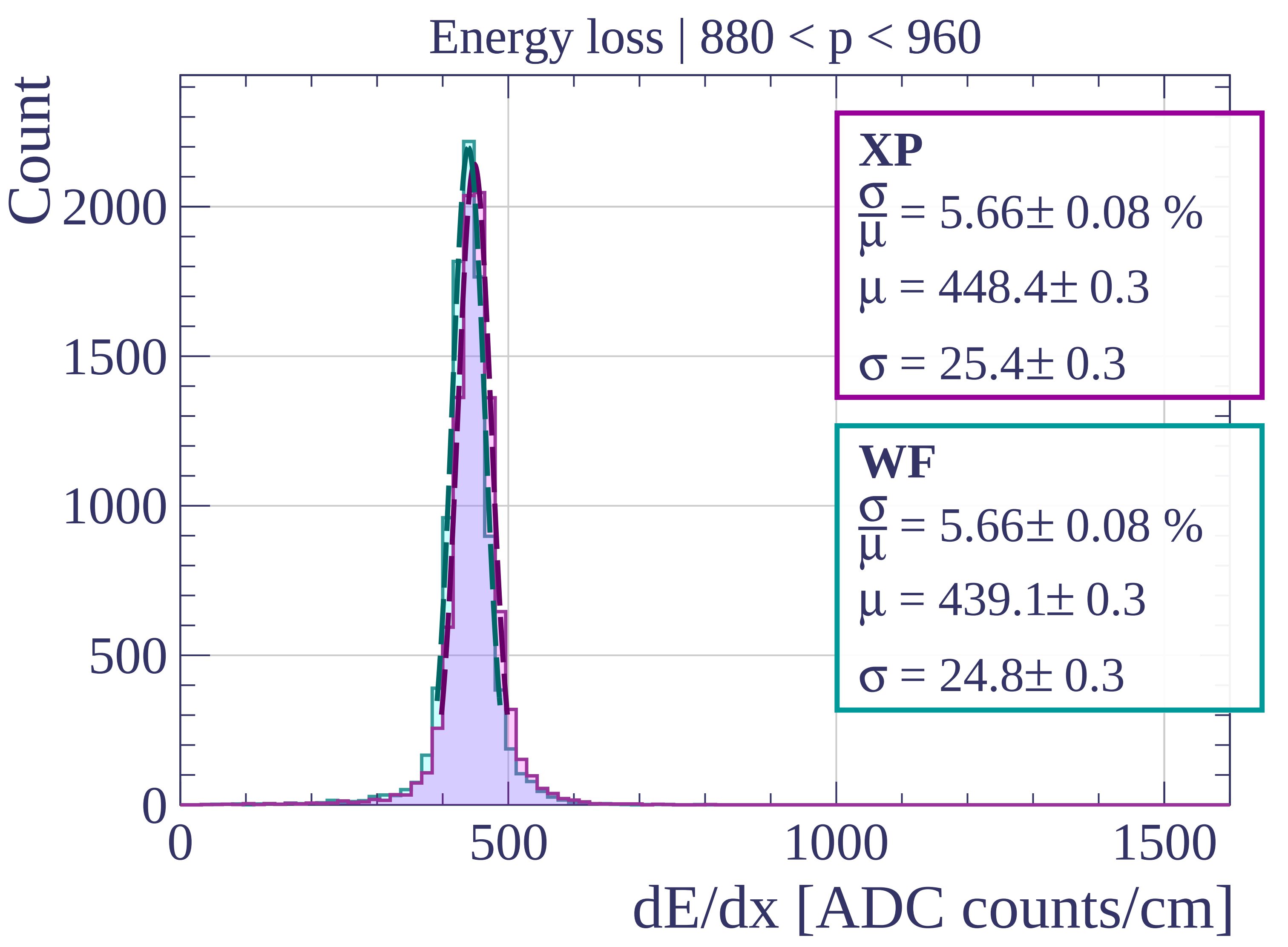


Energy loss | 640 < p < 720
Count
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 5.95± 0.09 %
μ = 433.5± 0.3
σ = 25.8± 0.4
WF
$\frac{\sigma}{\mu}$ = 6.08± 0.07 %
μ = 424.8± 0.3
σ = 25.8± 0.3

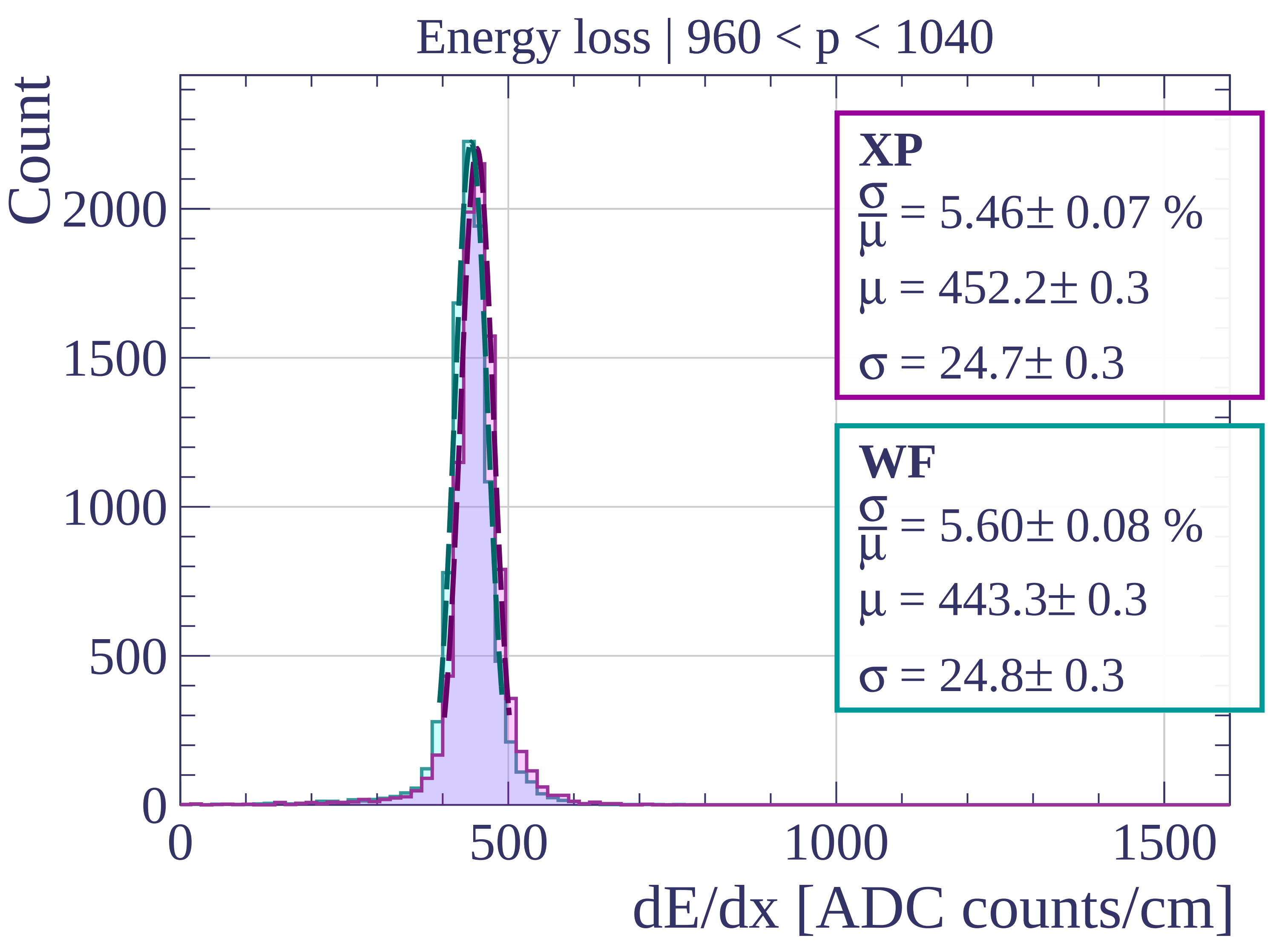

Energy loss | 720 < p < 800
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
2000
0
500
1000
1500
XP
$\frac{\sigma}{\mu}$ = 6.00± 0.07 %
μ = 438.8± 0.3
σ = 26.3± 0.3
WF
$\frac{\sigma}{\mu}$ = 5.84± 0.08 %
μ = 429.6± 0.3
σ = 25.1± 0.3

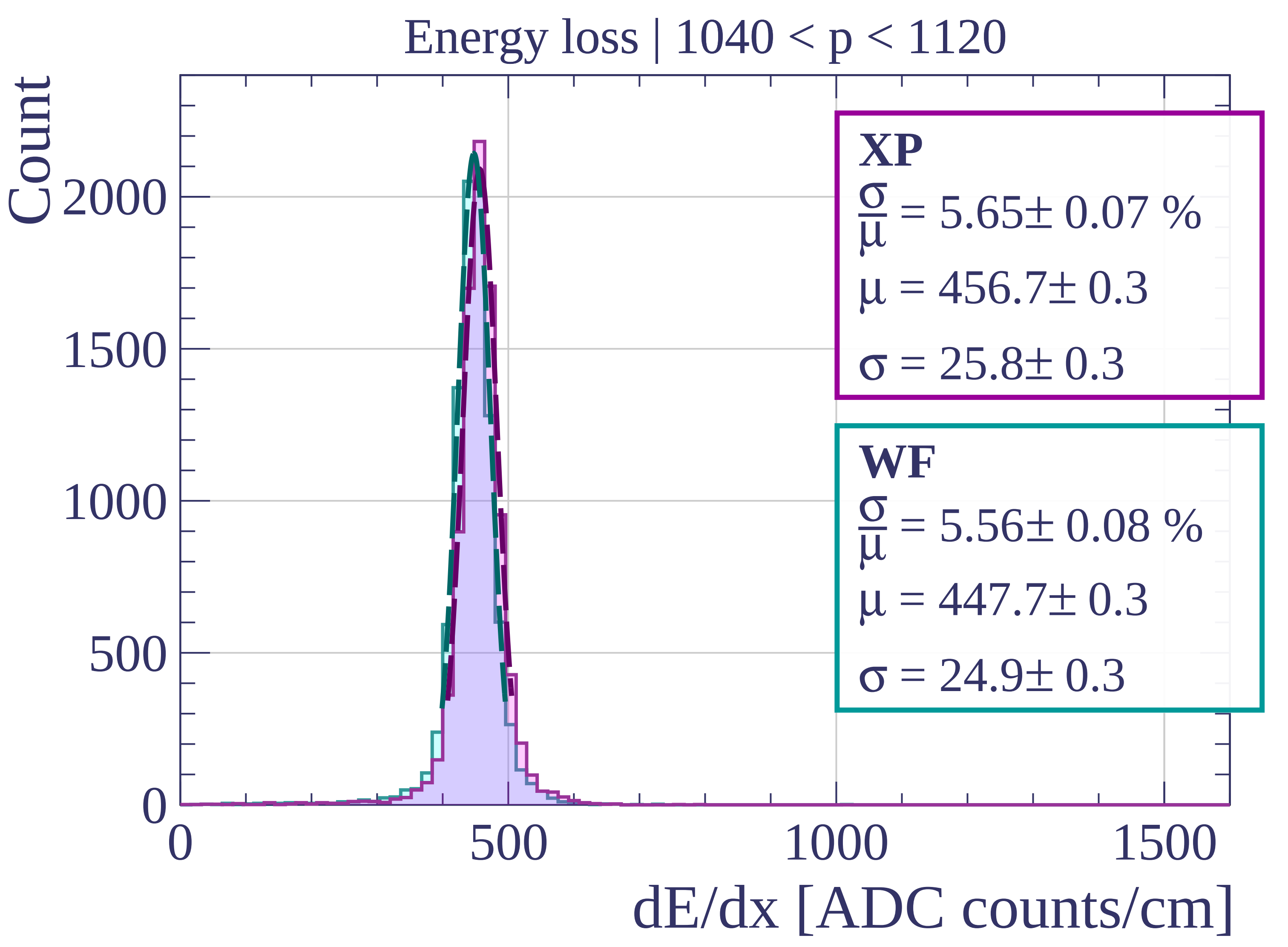


Energy loss | 800 < p < 880
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
2000
XP
σ/μ = 5.94± 0.09 %
μ = 443.9± 0.4
σ = 26.4± 0.4
WF
σ/μ = 5.77± 0.08 %
μ = 434.9± 0.3
σ = 25.1± 0.3

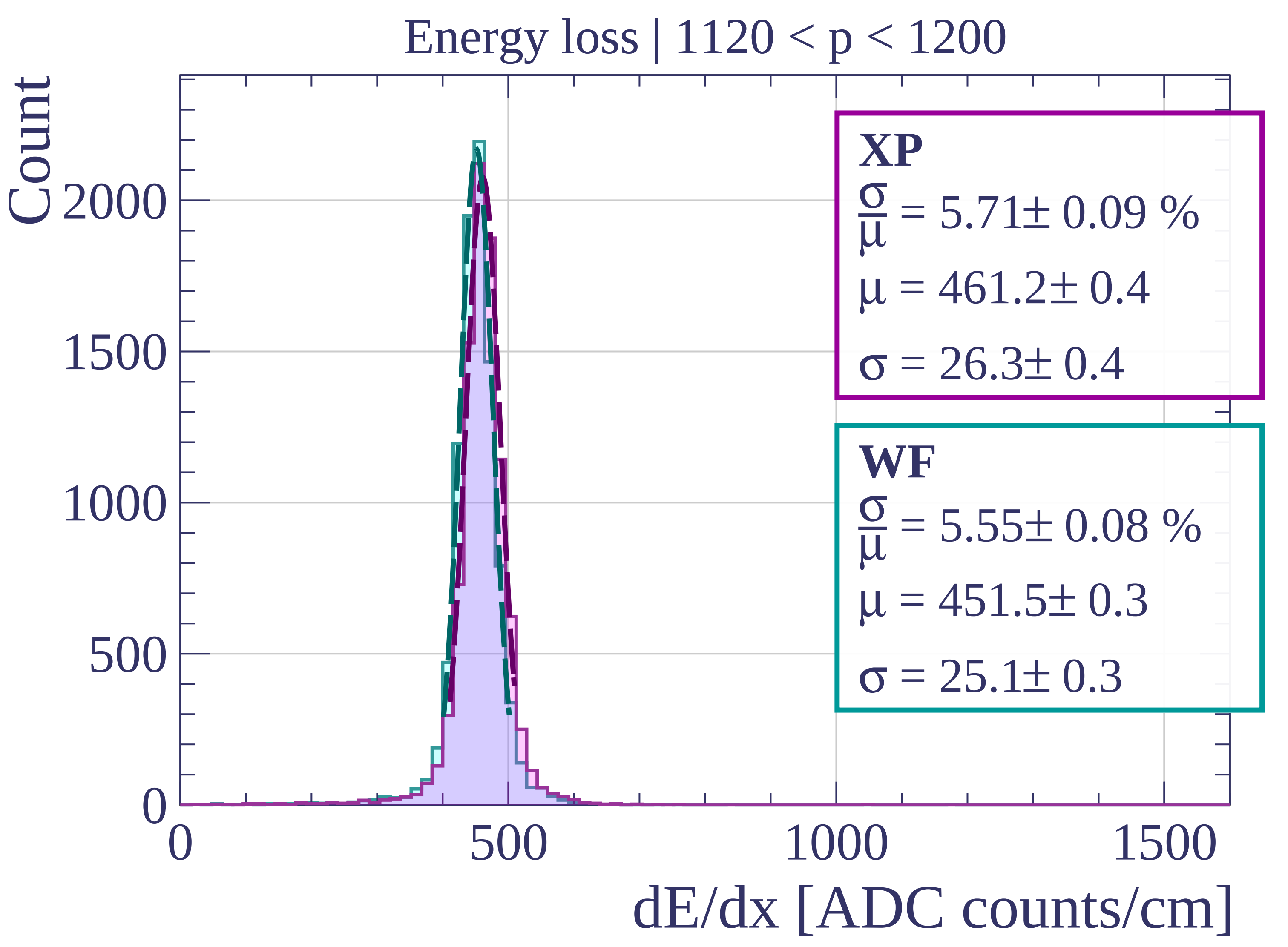


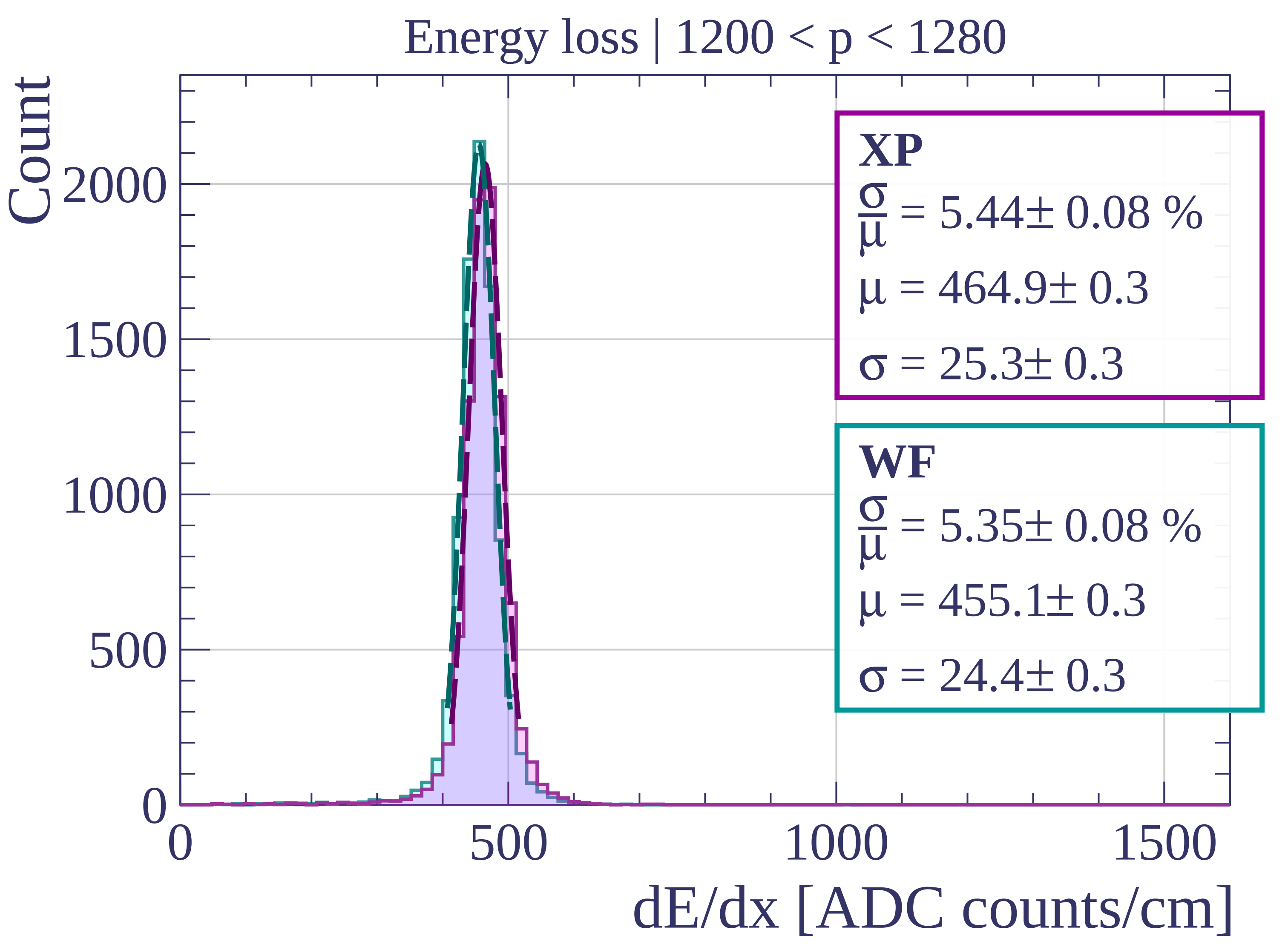
Energy loss | 880 < p < 960
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
2000
XP
σ/μ = 5.66± 0.08 %
μ = 448.4± 0.3
σ = 25.4± 0.3
WF
σ/μ = 5.66± 0.08 %
μ = 439.1± 0.3
σ = 24.8± 0.3


Energy loss | 960 < p < 1040
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
2000
XP
σ/μ = 5.46± 0.07 %
μ = 452.2± 0.3
σ = 24.7± 0.3
WF
σ/μ = 5.60± 0.08 %
μ = 443.3± 0.3
σ = 24.8± 0.3

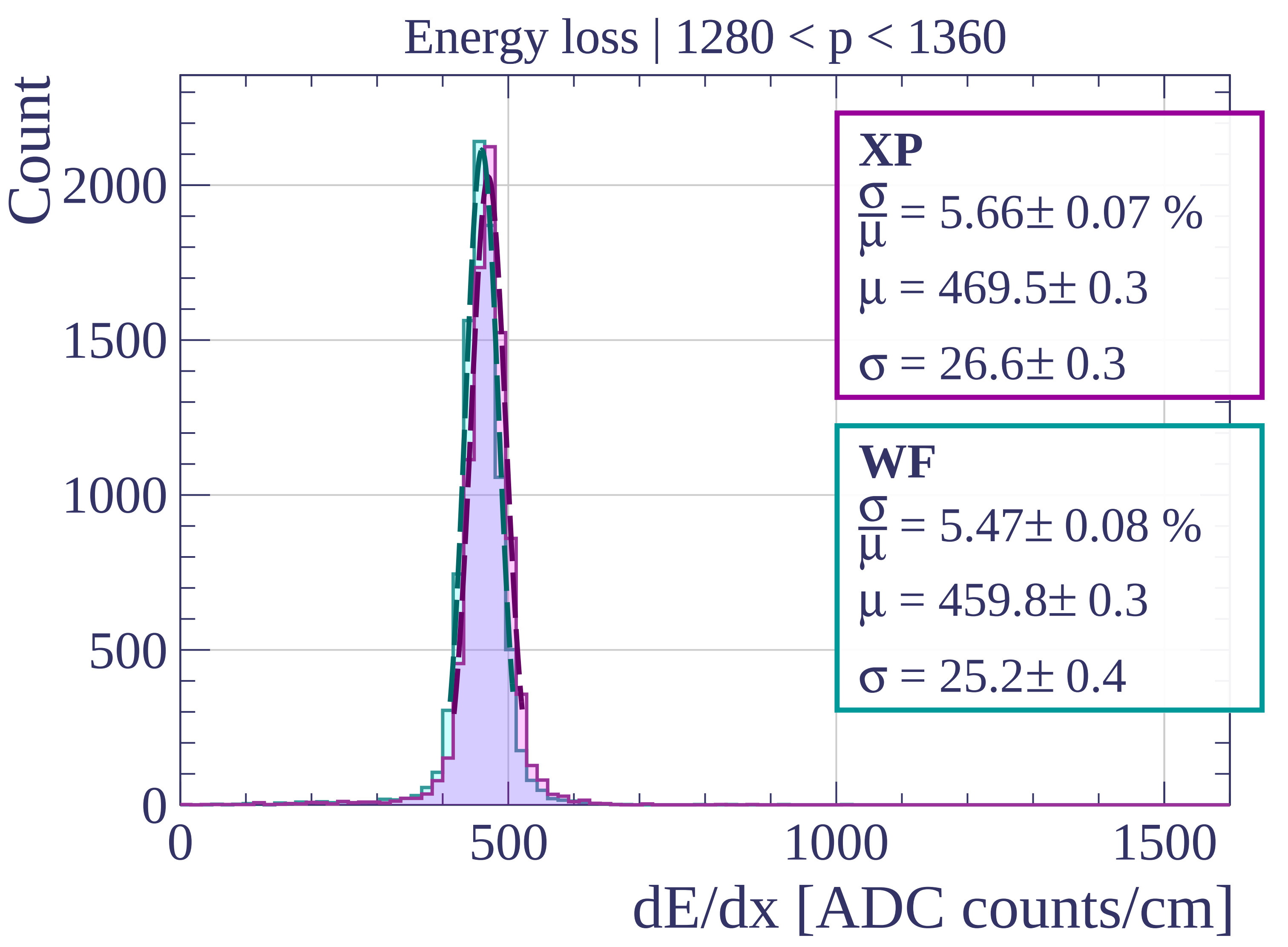


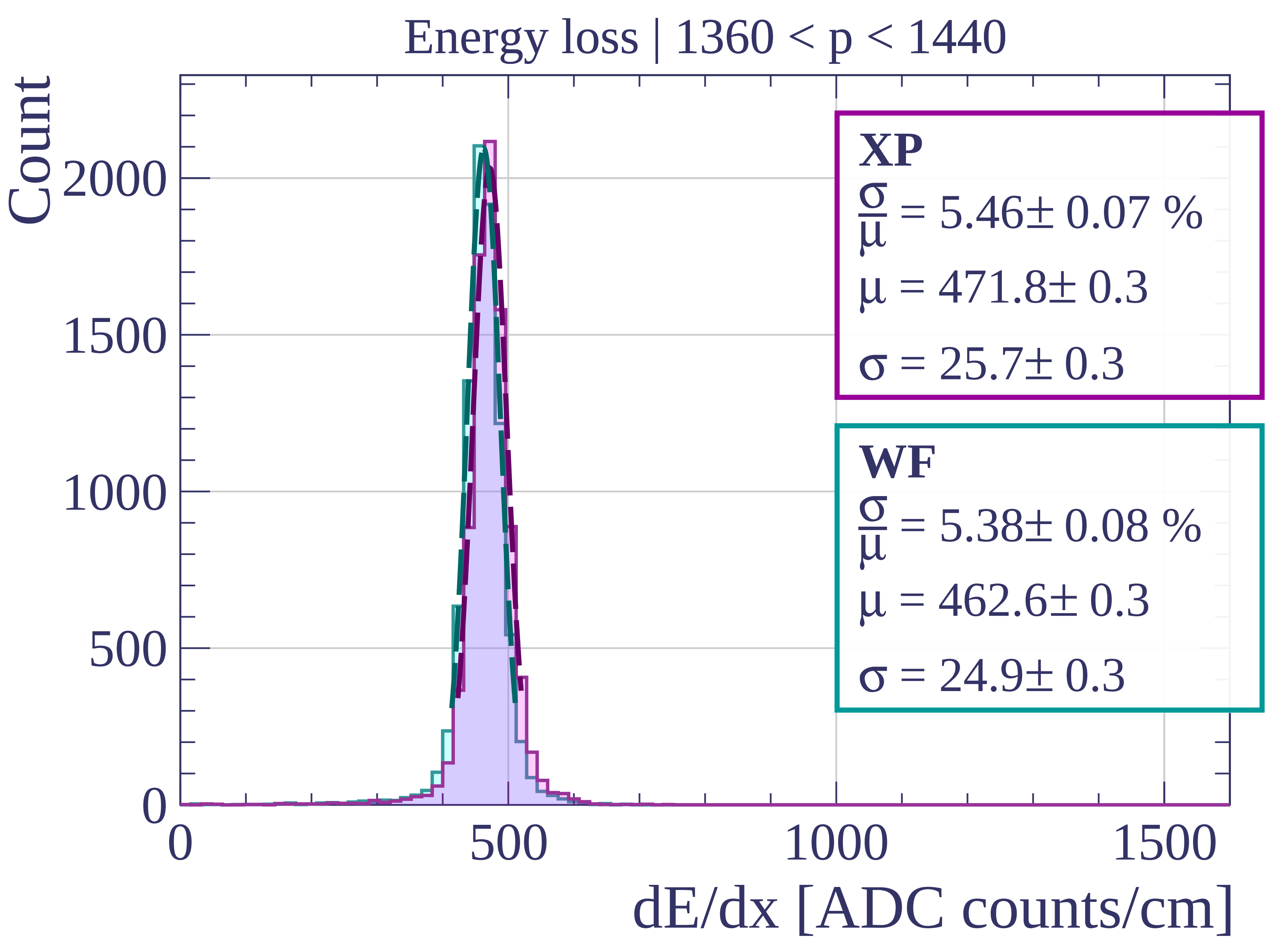
Energy loss | 1040 < p < 1120
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
2000
XP
$\frac{\sigma}{\mu}$ = 5.65± 0.07 %
μ = 456.7± 0.3
σ = 25.8± 0.3
WF
$\frac{\sigma}{\mu}$ = 5.56± 0.08 %
μ = 447.7± 0.3
σ = 24.9± 0.3


Energy loss | 1120 < p < 1200
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
2000
0
500
1000
1500
XP
σ/μ = 5.71± 0.09 %
μ = 461.2± 0.4
σ = 26.3± 0.4
WF
σ/μ = 5.55± 0.08 %
μ = 451.5± 0.3
σ = 25.1± 0.3

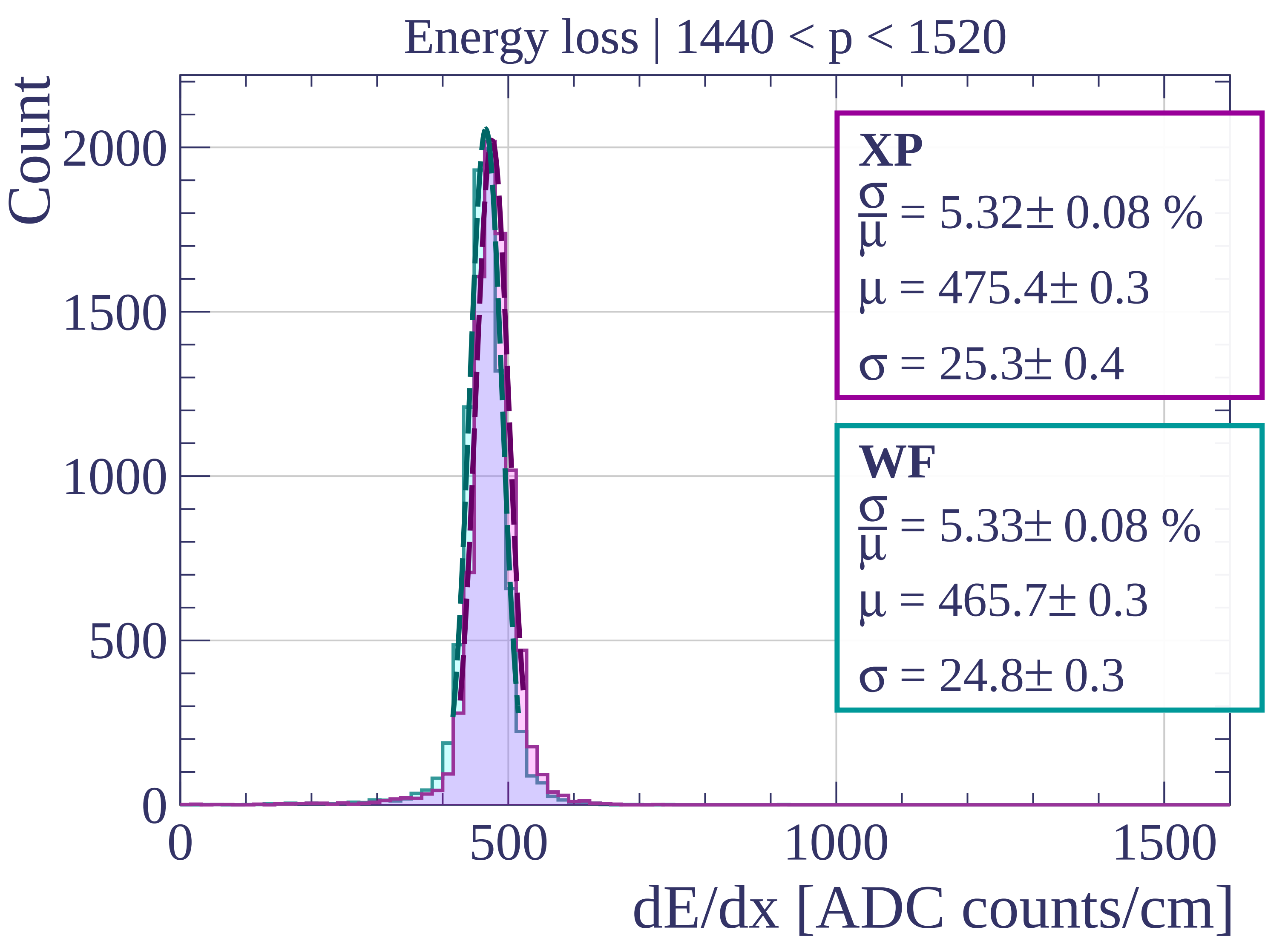


Energy loss | 1200 < p < 1280
Count
2000
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 5.44± 0.08 %
μ = 464.9± 0.3
σ = 25.3± 0.3
WF
σ/μ = 5.35± 0.08 %
μ = 455.1± 0.3
σ = 24.4± 0.3

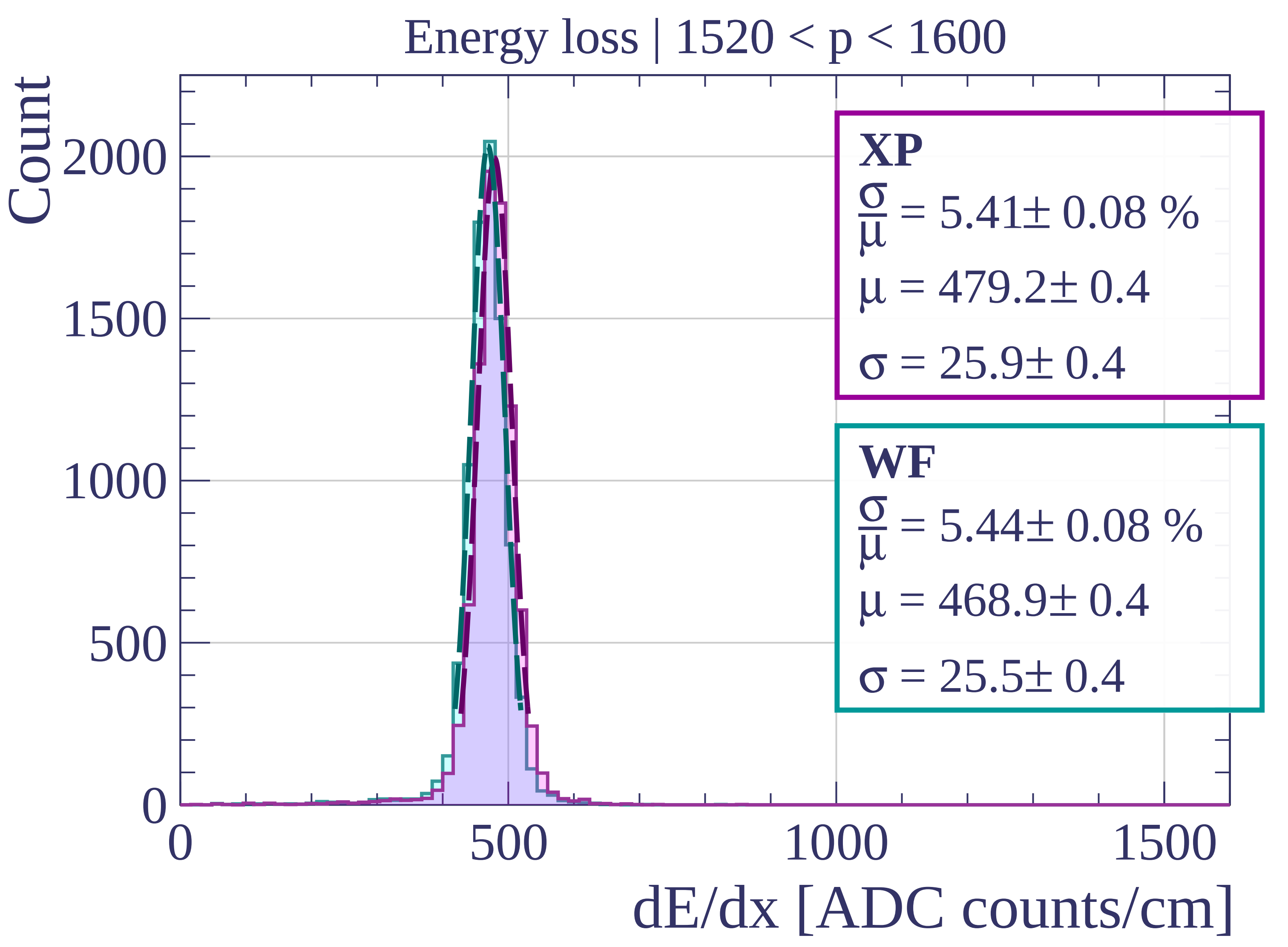

Energy loss | 1280 < p < 1360
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
2000
XP
σ/μ = 5.66± 0.07 %
μ = 469.5± 0.3
σ = 26.6± 0.3
WF
σ/μ = 5.47± 0.08 %
μ = 459.8± 0.3
σ = 25.2± 0.4



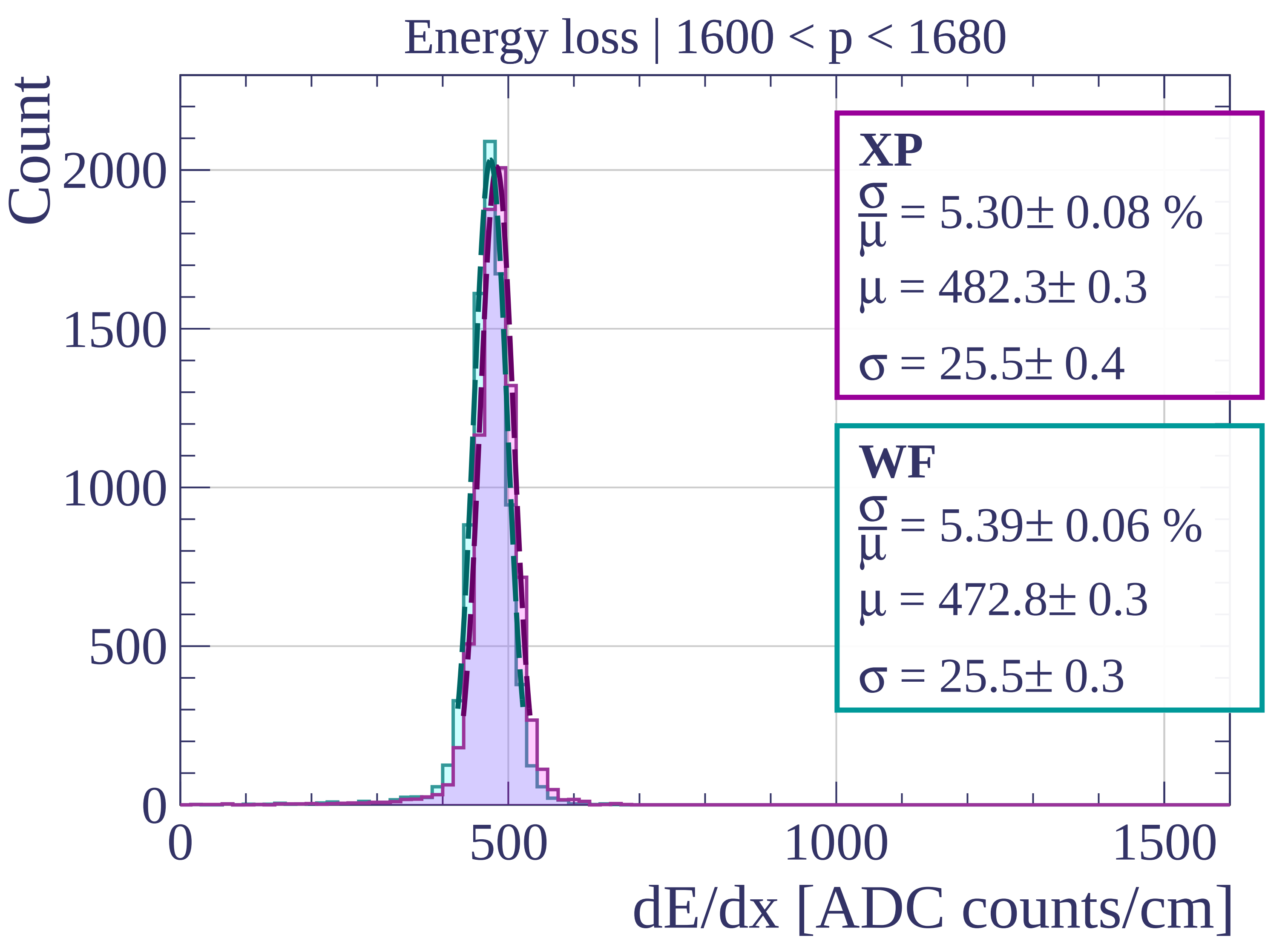
Energy loss | 1360 < p < 1440
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
2000
XP
$\frac{\sigma}{\mu}$ = 5.46± 0.07 %
μ = 471.8± 0.3
σ = 25.7± 0.3
WF
$\frac{\sigma}{\mu}$ = 5.38± 0.08 %
μ = 462.6± 0.3
σ = 24.9± 0.3

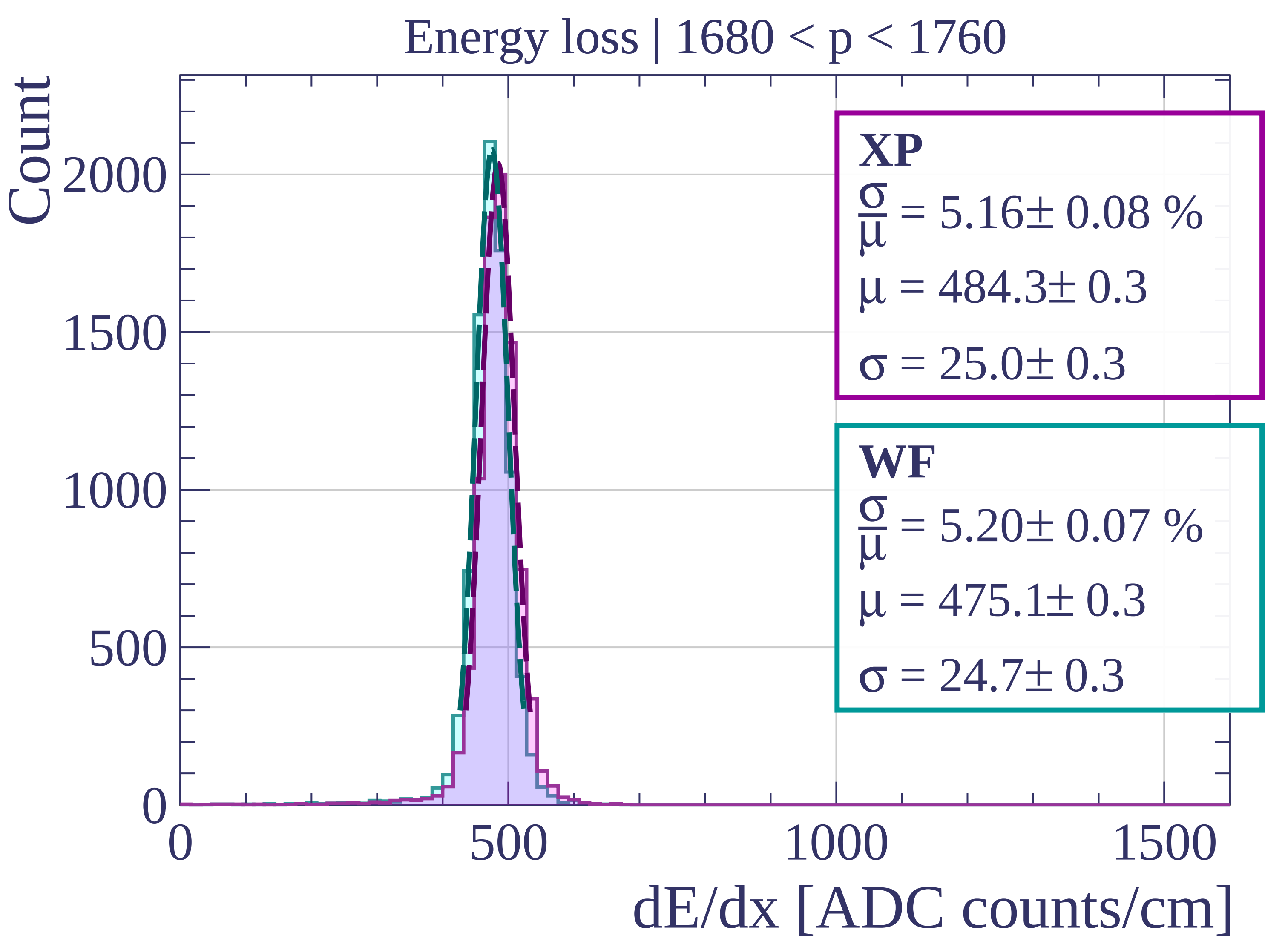


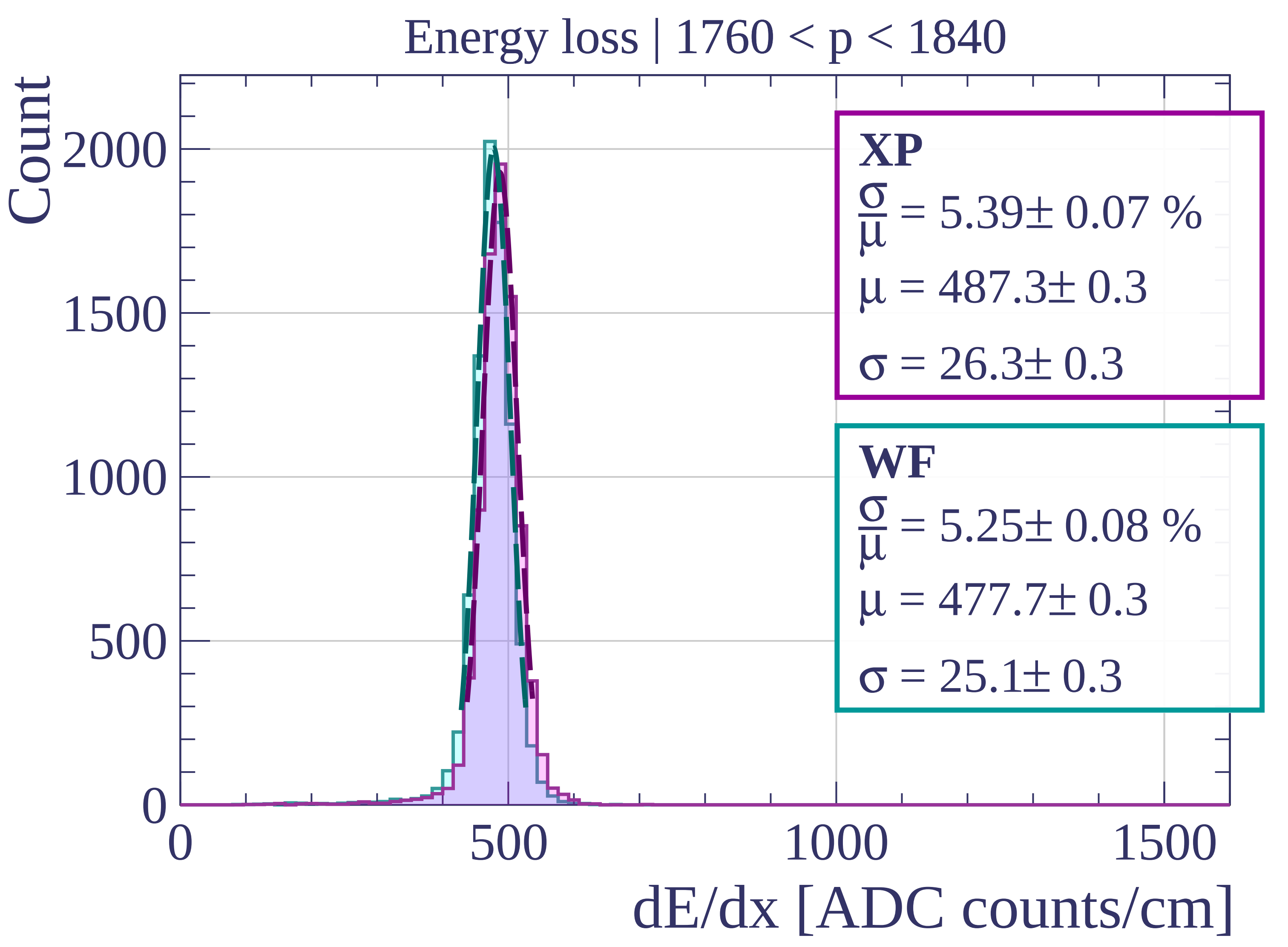
Energy loss | 1440 < p < 1520
Count
0
500
1000
1500
2000
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 5.32± 0.08 %
μ = 475.4± 0.3
σ = 25.3± 0.4
WF
$\frac{\sigma}{\mu}$ = 5.33± 0.08 %
μ = 465.7± 0.3
σ = 24.8± 0.3

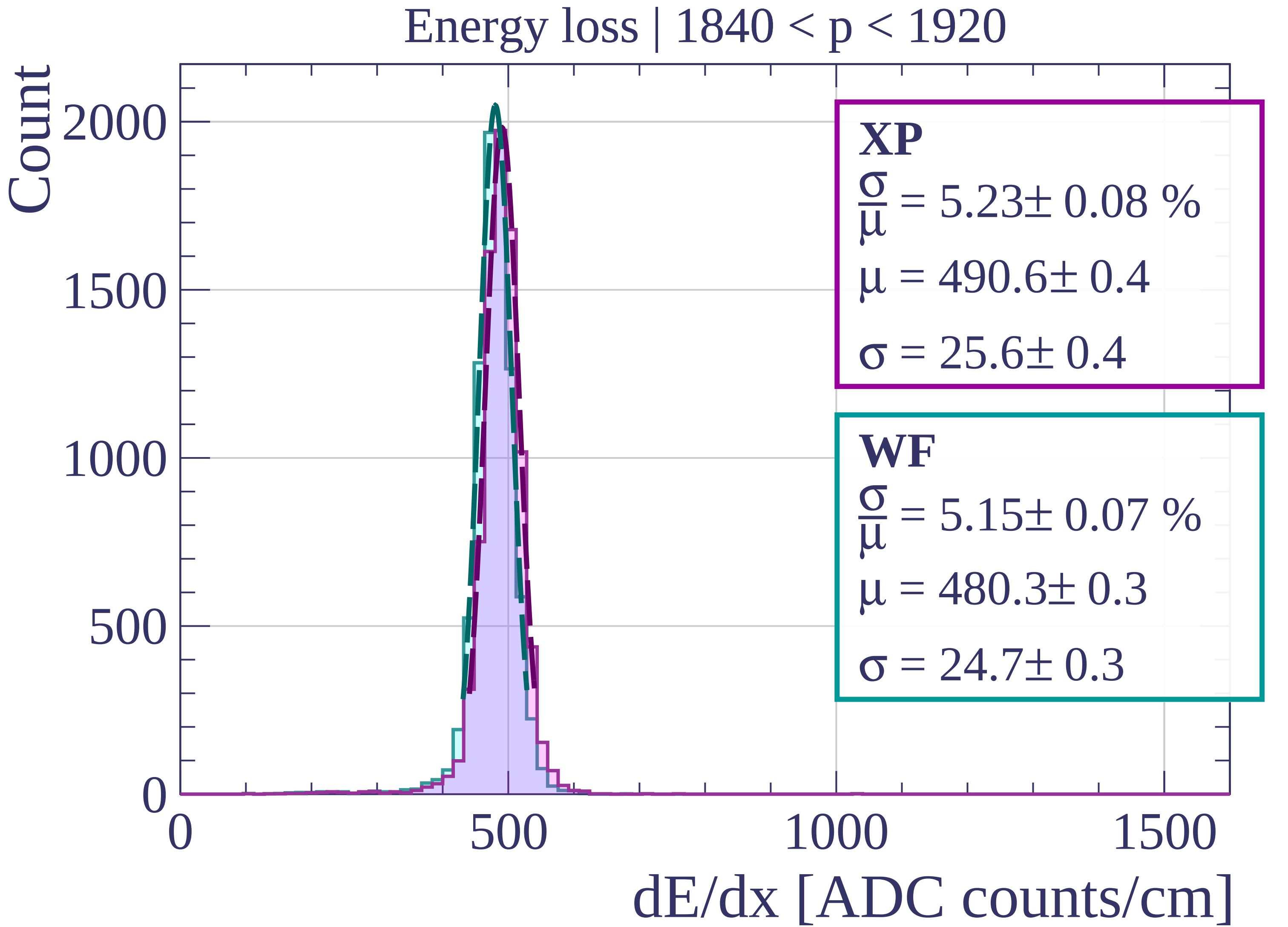


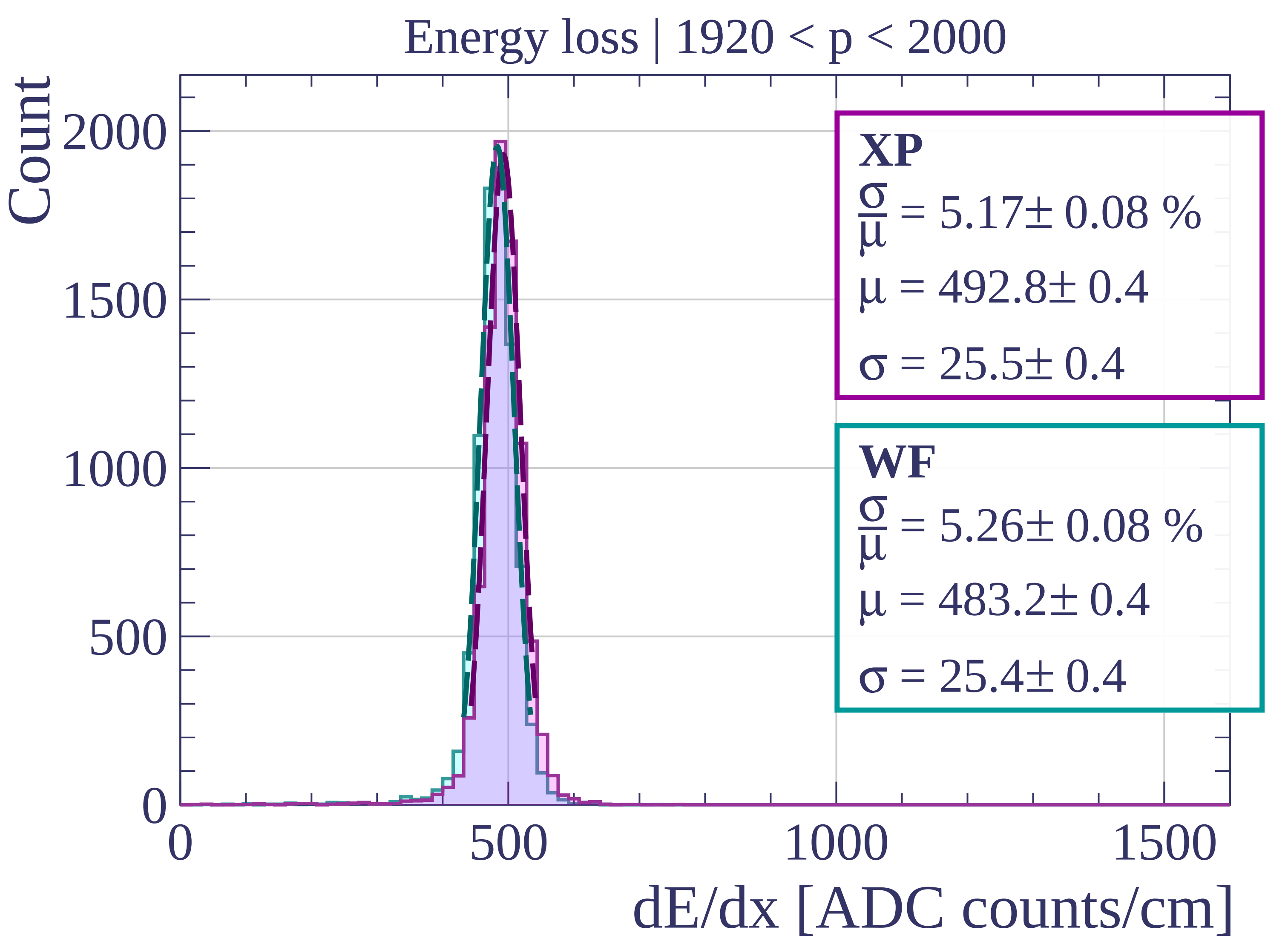
Energy loss | 1520 < p < 1600
Count
2000
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 5.41 ± 0.08 %
μ = 479.2 ± 0.4
σ = 25.9 ± 0.4
WF
$\frac{\sigma}{\mu}$ = 5.44 ± 0.08 %
μ = 468.9 ± 0.4
σ = 25.5 ± 0.4


Energy loss | 1600 < p < 1680
Count
0
500
1000
1500
2000
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 5.30± 0.08 %
μ = 482.3± 0.3
σ = 25.5± 0.4
WF
$\frac{\sigma}{\mu}$ = 5.39± 0.06 %
μ = 472.8± 0.3
σ = 25.5± 0.3

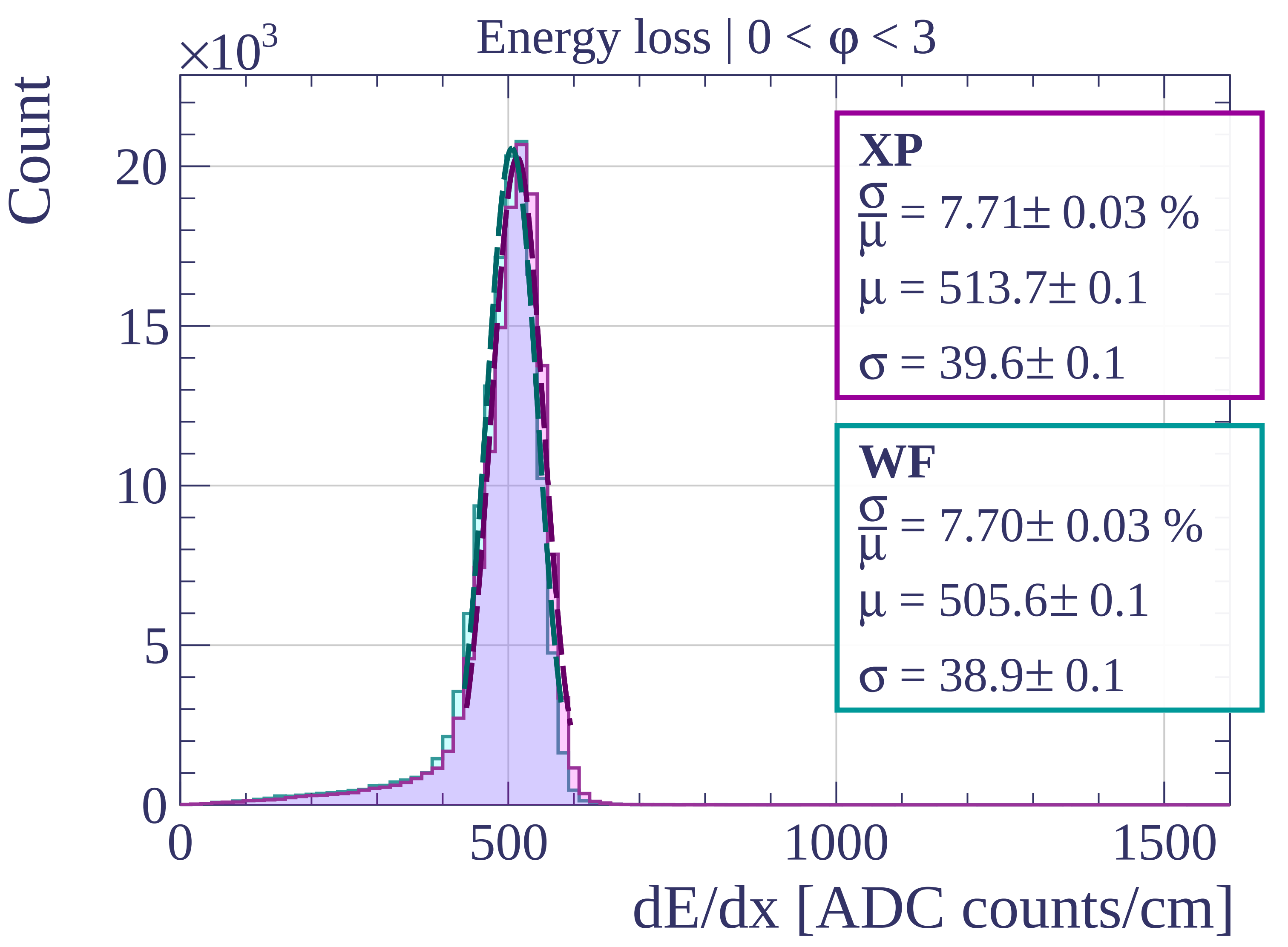

Energy loss | 1680 < p < 1760
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
2000
XP
σ/μ = 5.16± 0.08 %
μ = 484.3± 0.3
σ = 25.0± 0.3
WF
σ/μ = 5.20± 0.07 %
μ = 475.1± 0.3
σ = 24.7± 0.3


Energy loss | 1760 < p < 1840
Count
2000
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 5.39± 0.07 %
μ = 487.3± 0.3
σ = 26.3± 0.3
WF
σ/μ = 5.25± 0.08 %
μ = 477.7± 0.3
σ = 25.1± 0.3

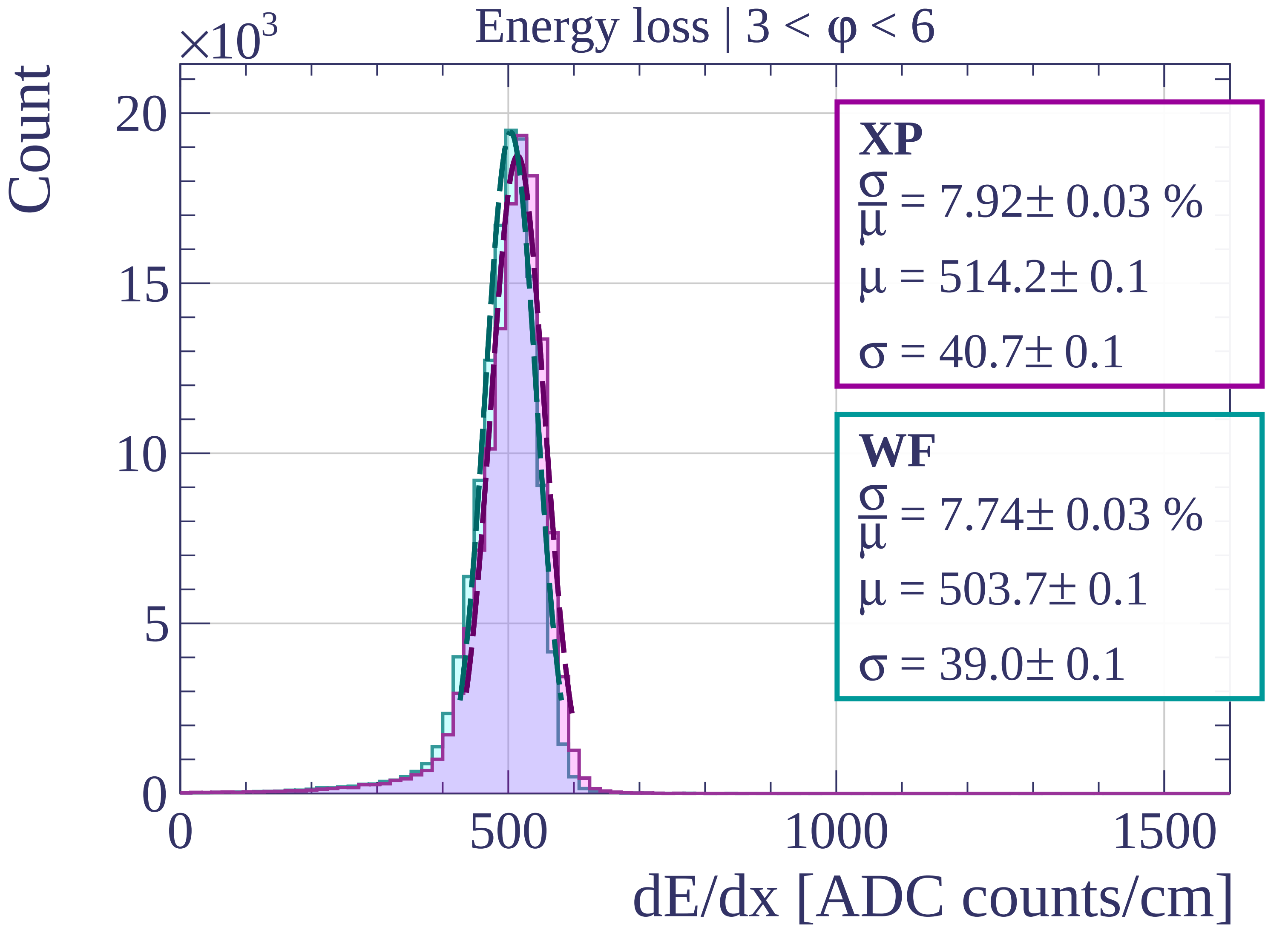

Energy loss | 1840 < p < 1920
Count
2000
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 5.23± 0.08 %
μ = 490.6± 0.4
σ = 25.6± 0.4
WF
$\frac{\sigma}{\mu}$ = 5.15± 0.07 %
μ = 480.3± 0.3
σ = 24.7± 0.3



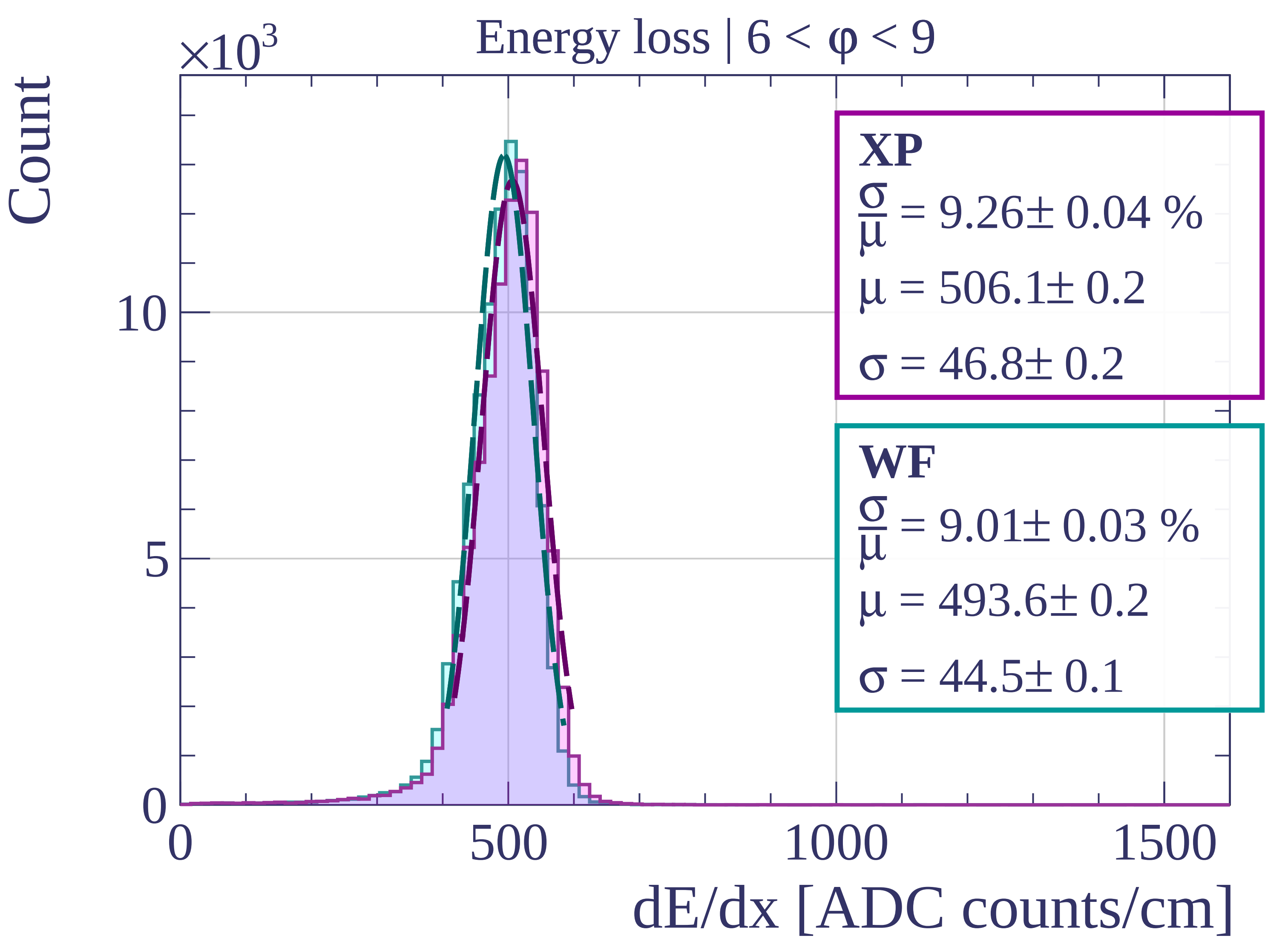
Energy loss | 1920 < p < 2000
Count
2000
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 5.17± 0.08 %
μ = 492.8± 0.4
σ = 25.5± 0.4
WF
σ/μ = 5.26± 0.08 %
μ = 483.2± 0.4
σ = 25.4± 0.4



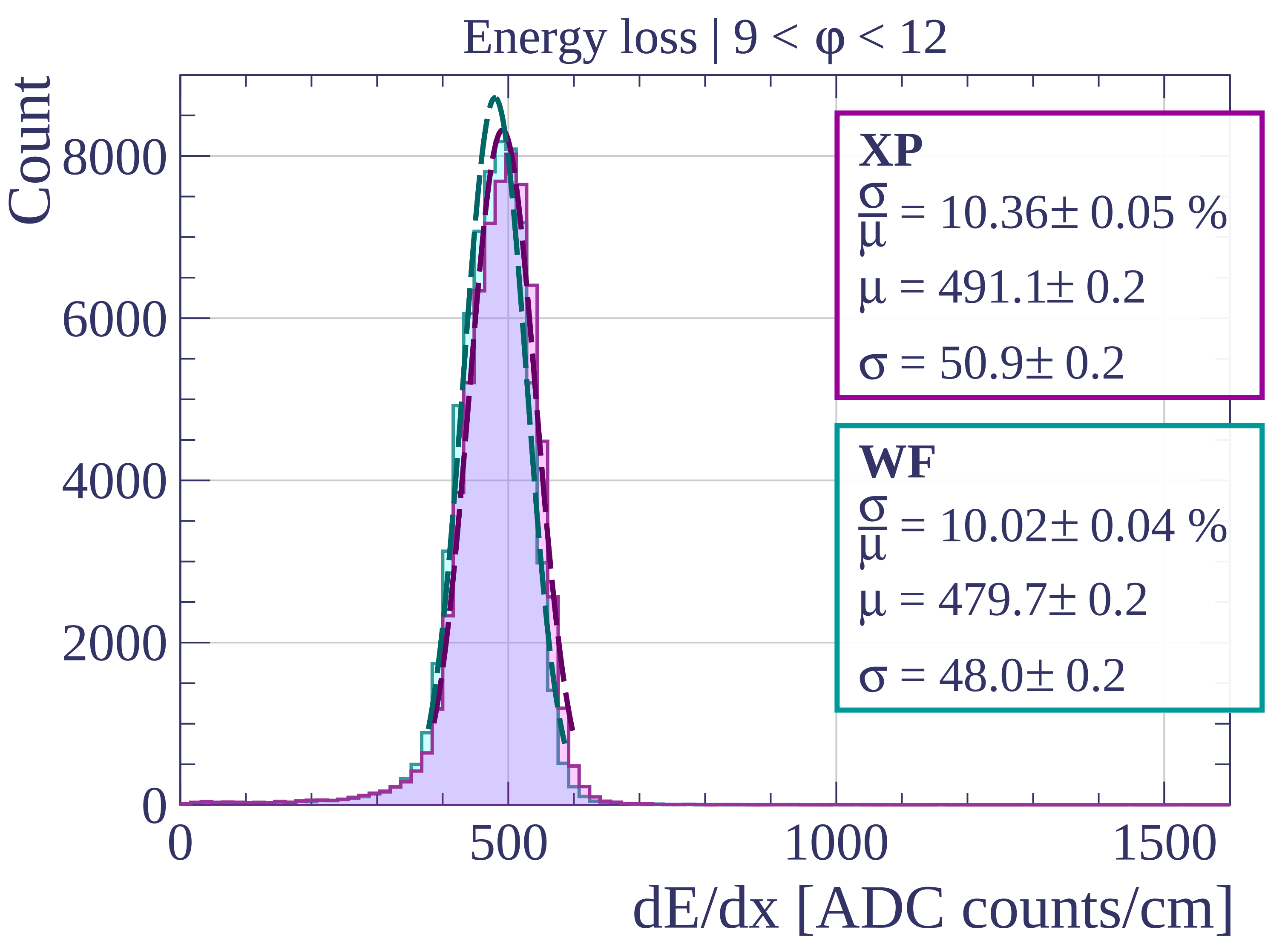
Energy loss | 0 < φ < 3
×10^3
Count
XP
σ/μ = 7.71± 0.03 %
μ = 513.7± 0.1
σ = 39.6± 0.1
WF
σ/μ = 7.70± 0.03 %
μ = 505.6± 0.1
σ = 38.9± 0.1
20
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]



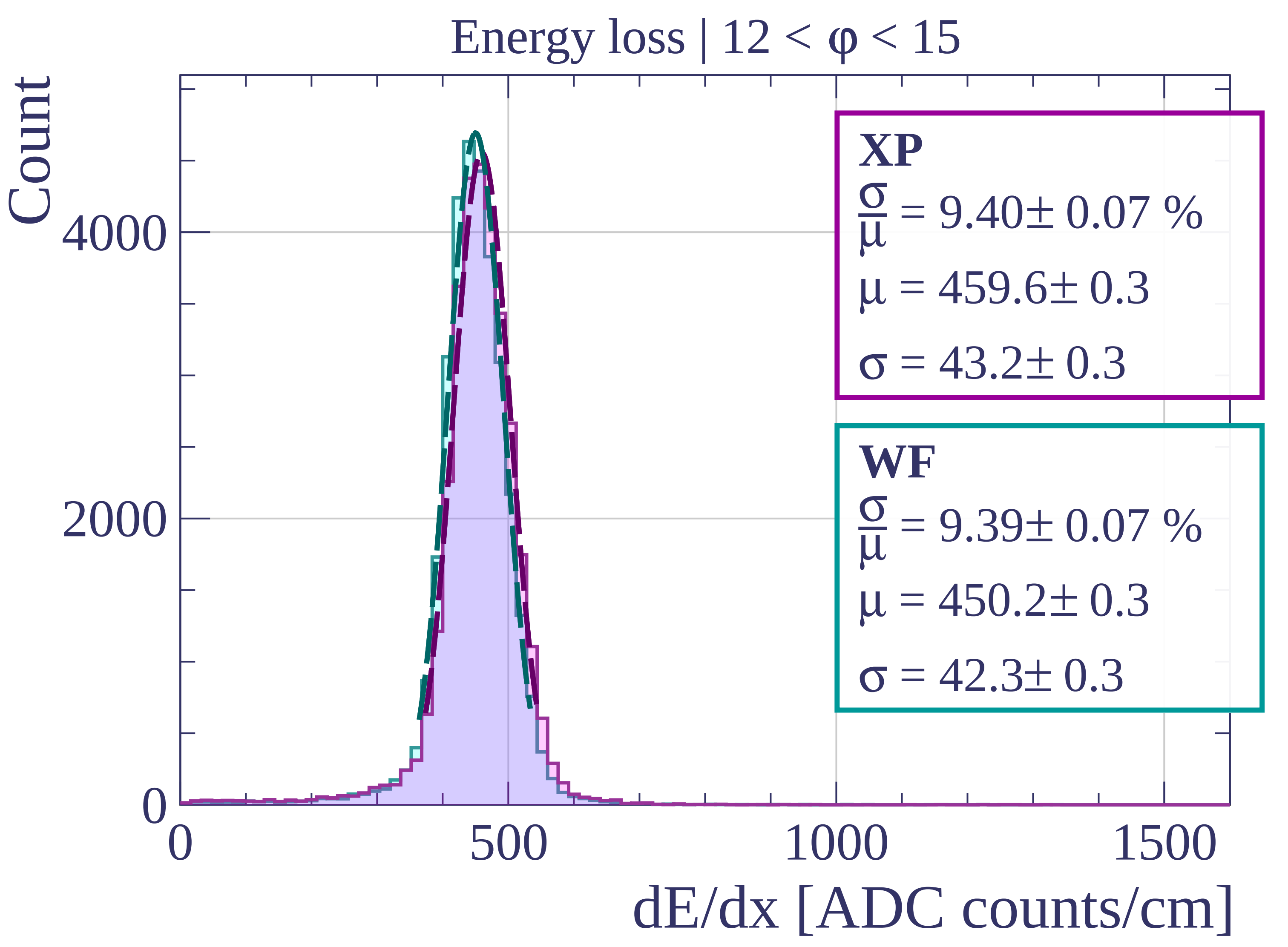
Energy loss | 3 < φ < 6
×10^3
Count
20
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 7.92± 0.03 %
μ = 514.2± 0.1
σ = 40.7± 0.1
WF
σ/μ = 7.74± 0.03 %
μ = 503.7± 0.1
σ = 39.0± 0.1



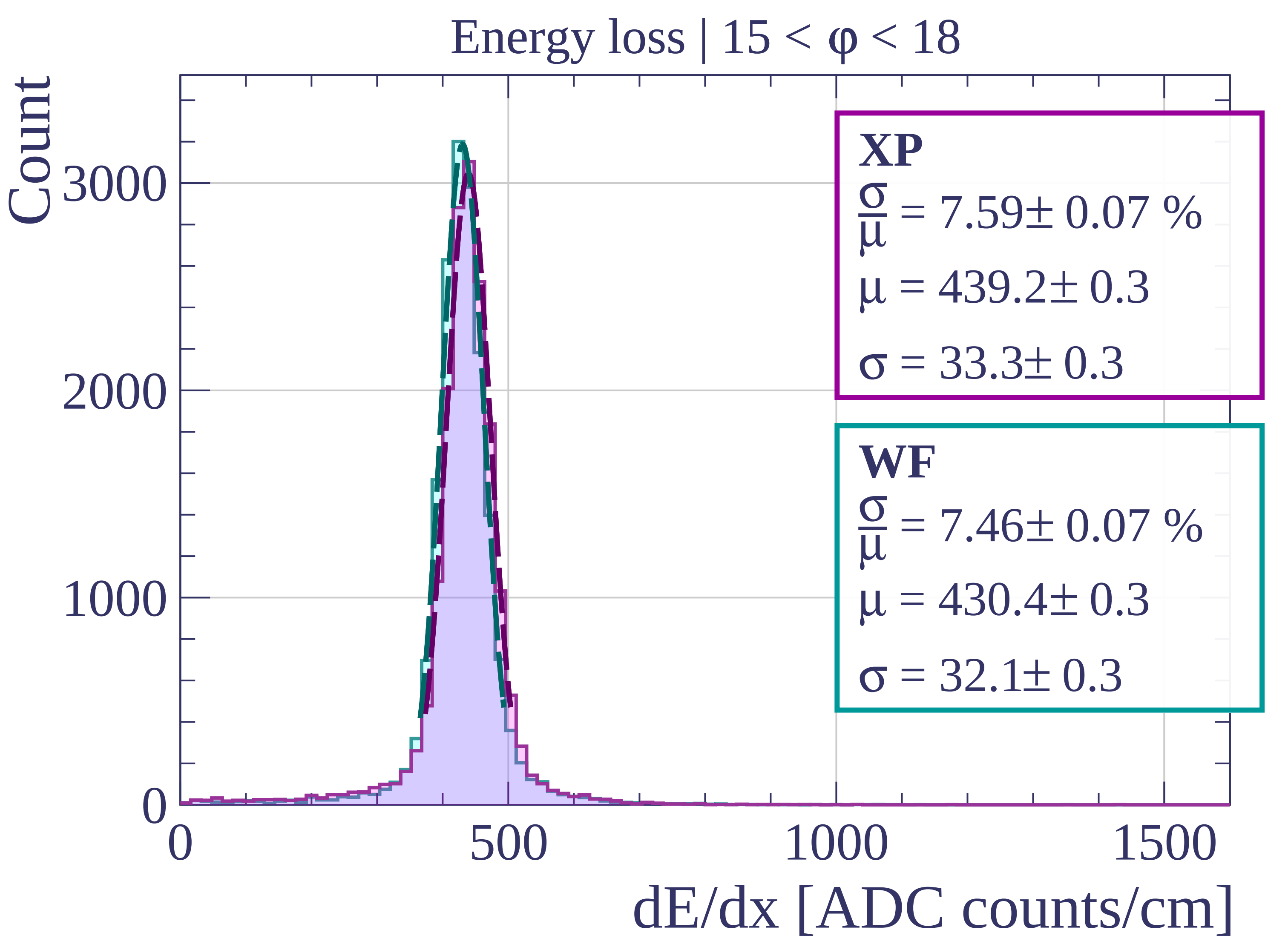
Energy loss | 6 < φ < 9
×10^3
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 9.26± 0.04 %
μ = 506.1± 0.2
σ = 46.8± 0.2
WF
σ/μ = 9.01± 0.03 %
μ = 493.6± 0.2
σ = 44.5± 0.1



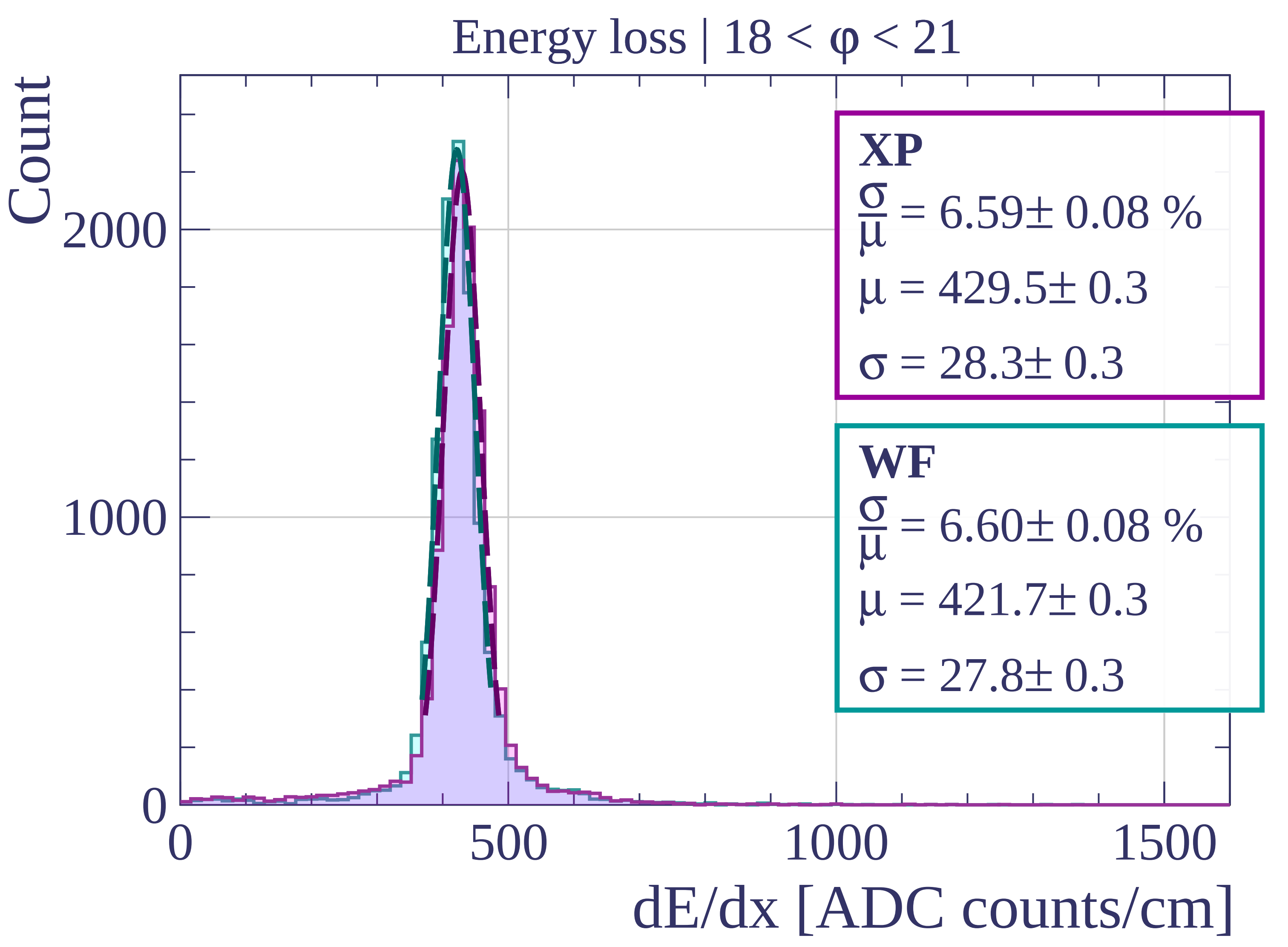
Energy loss | 9 < φ < 12
Count
8000
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 10.36± 0.05 %
μ = 491.1± 0.2
σ = 50.9± 0.2
WF
$\frac{\sigma}{\mu}$ = 10.02± 0.04 %
μ = 479.7± 0.2
σ = 48.0± 0.2



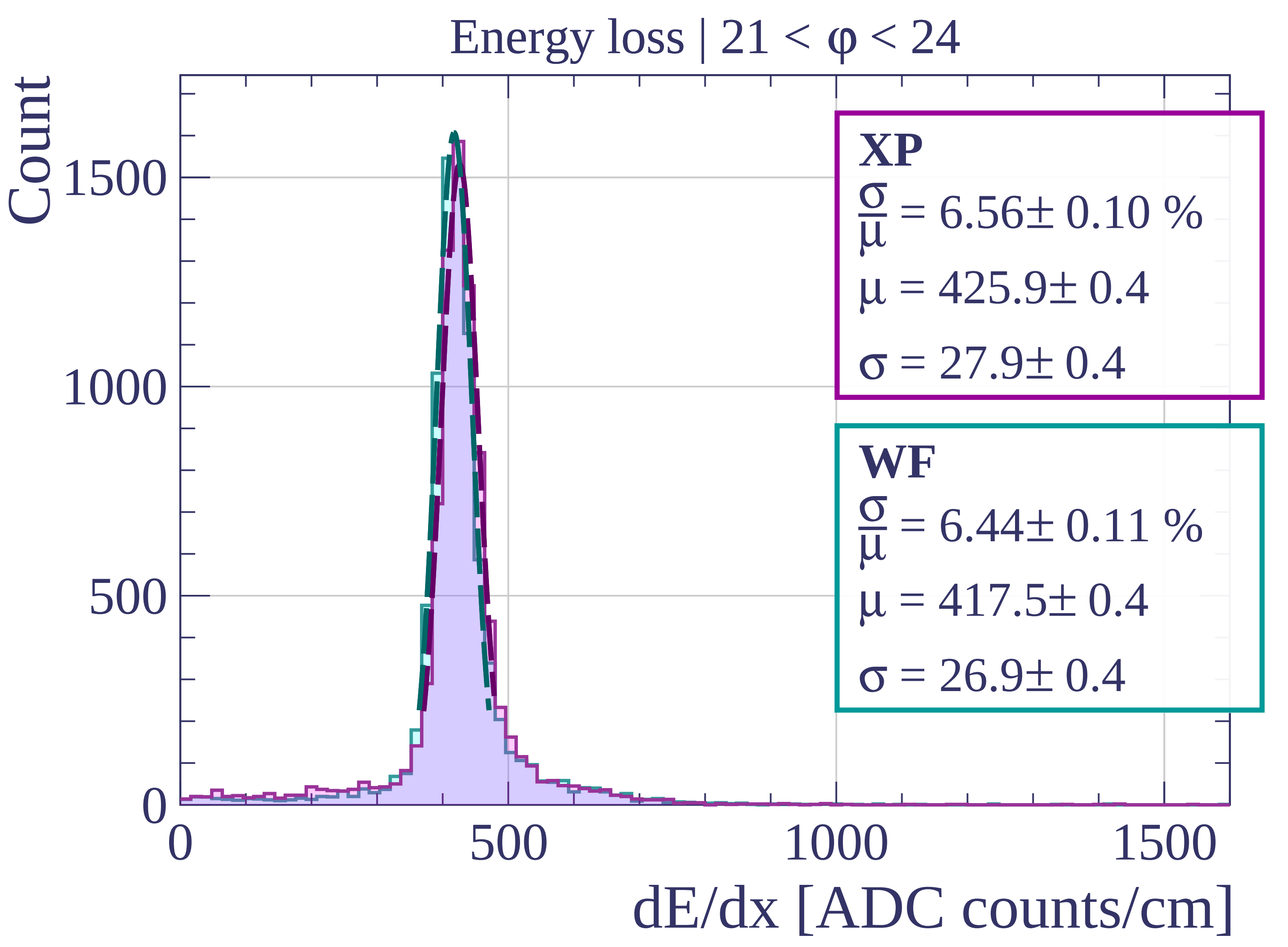
Energy loss | 12 < φ < 15
Count
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 9.40± 0.07 %
μ = 459.6± 0.3
σ = 43.2± 0.3
WF
σ/μ = 9.39± 0.07 %
μ = 450.2± 0.3
σ = 42.3± 0.3



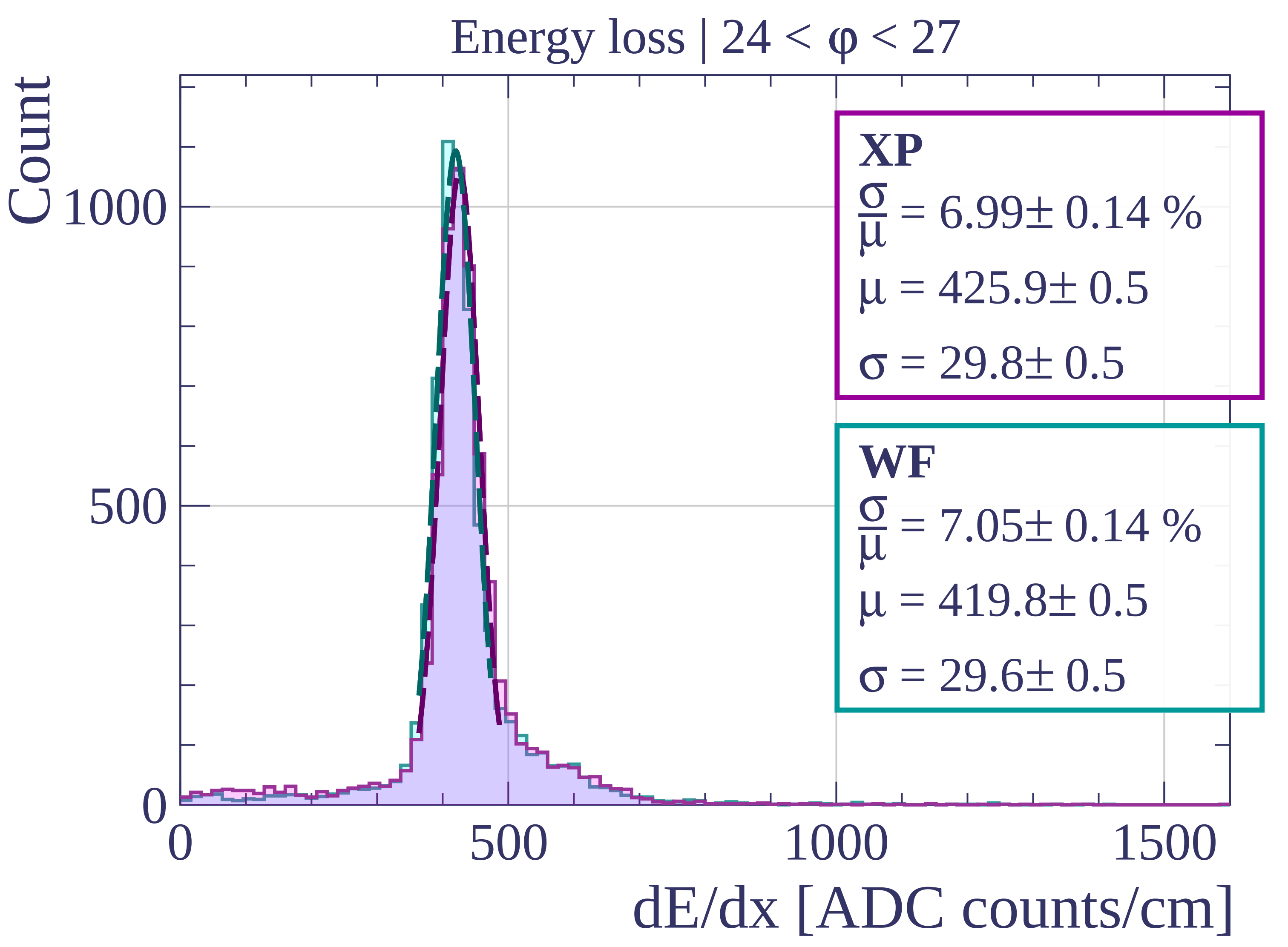
Energy loss | 15 < φ < 18
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 7.59± 0.07 %
μ = 439.2± 0.3
σ = 33.3± 0.3
WF
σ/μ = 7.46± 0.07 %
μ = 430.4± 0.3
σ = 32.1± 0.3



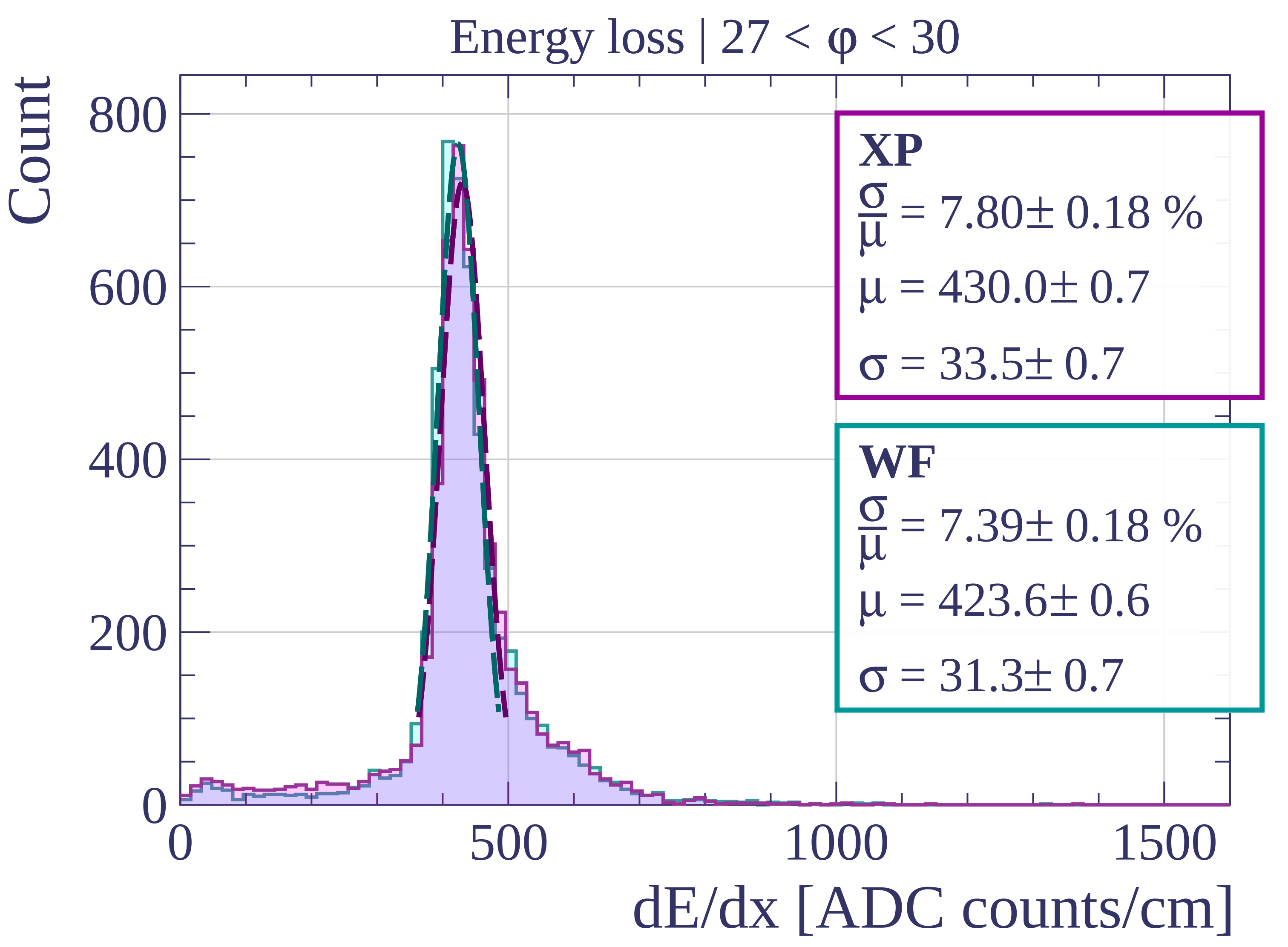
Energy loss | 18 < φ < 21
Count
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 6.59± 0.08 %
μ = 429.5± 0.3
σ = 28.3± 0.3
WF
σ/μ = 6.60± 0.08 %
μ = 421.7± 0.3
σ = 27.8± 0.3



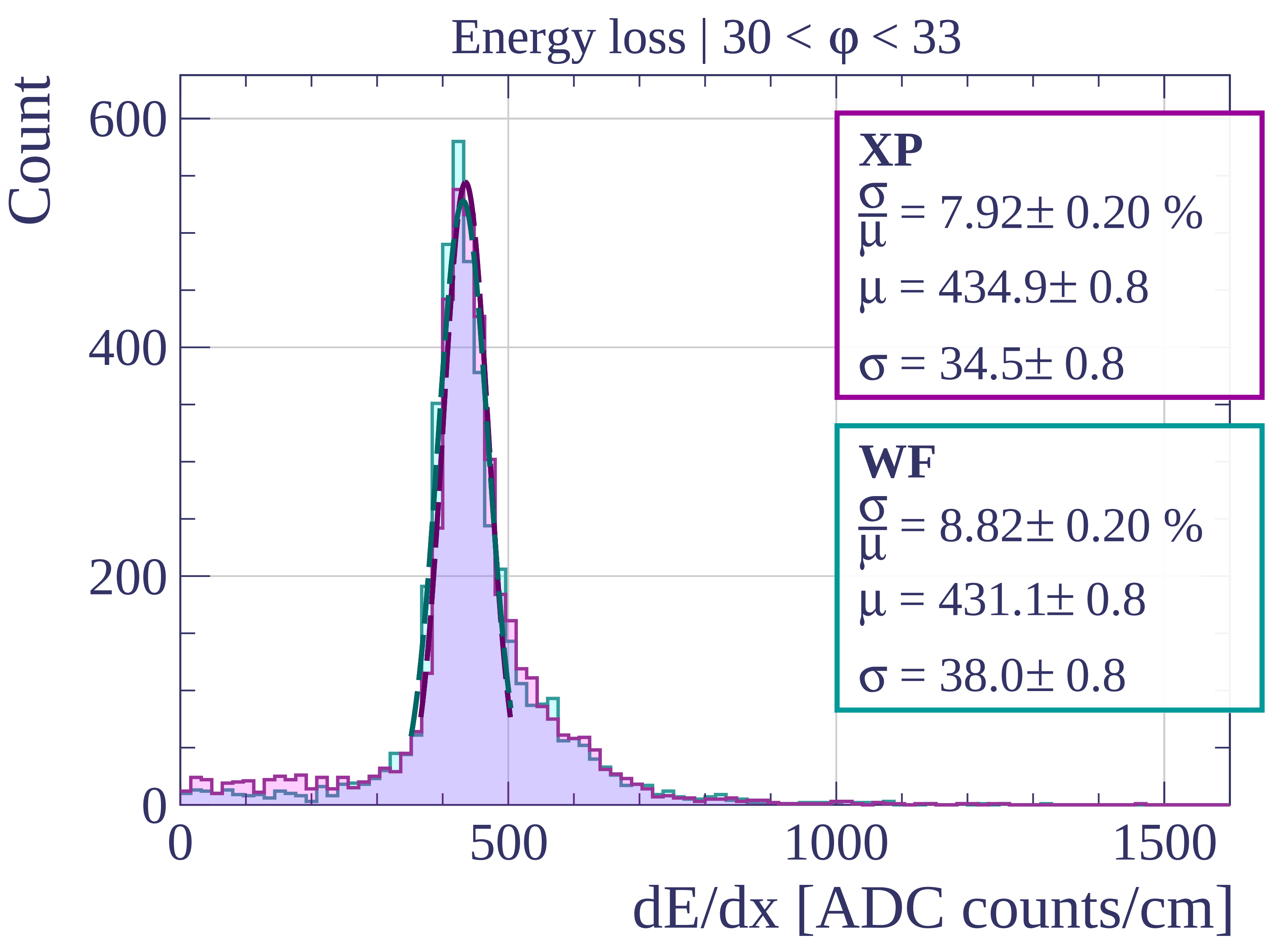
Energy loss | 21 < φ < 24
Count
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 6.56± 0.10 %
μ = 425.9± 0.4
σ = 27.9± 0.4
WF
σ/μ = 6.44± 0.11 %
μ = 417.5± 0.4
σ = 26.9± 0.4



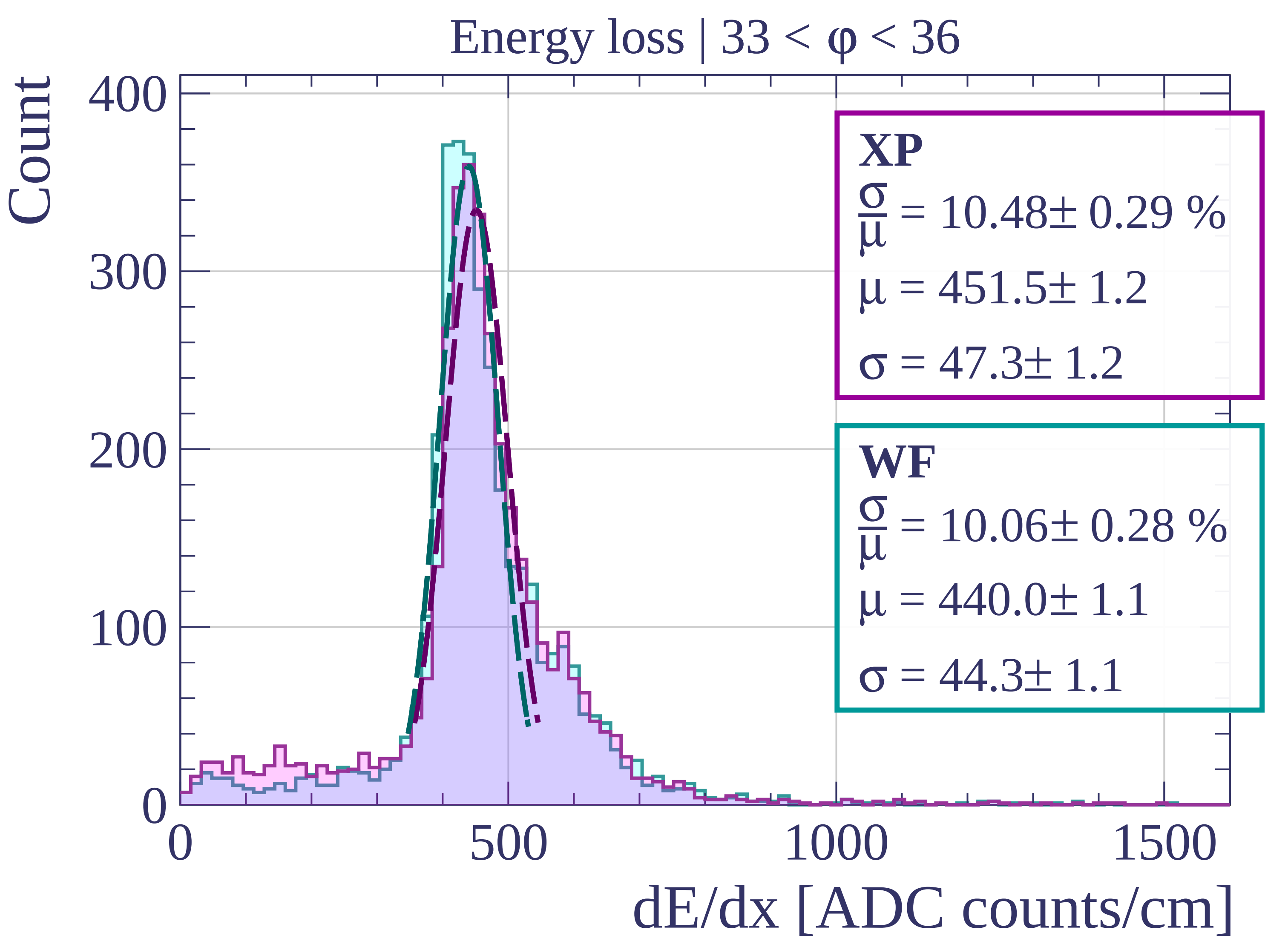
Energy loss | 24 < φ < 27
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 6.99± 0.14 %
μ = 425.9± 0.5
σ = 29.8± 0.5
WF
σ/μ = 7.05± 0.14 %
μ = 419.8± 0.5
σ = 29.6± 0.5



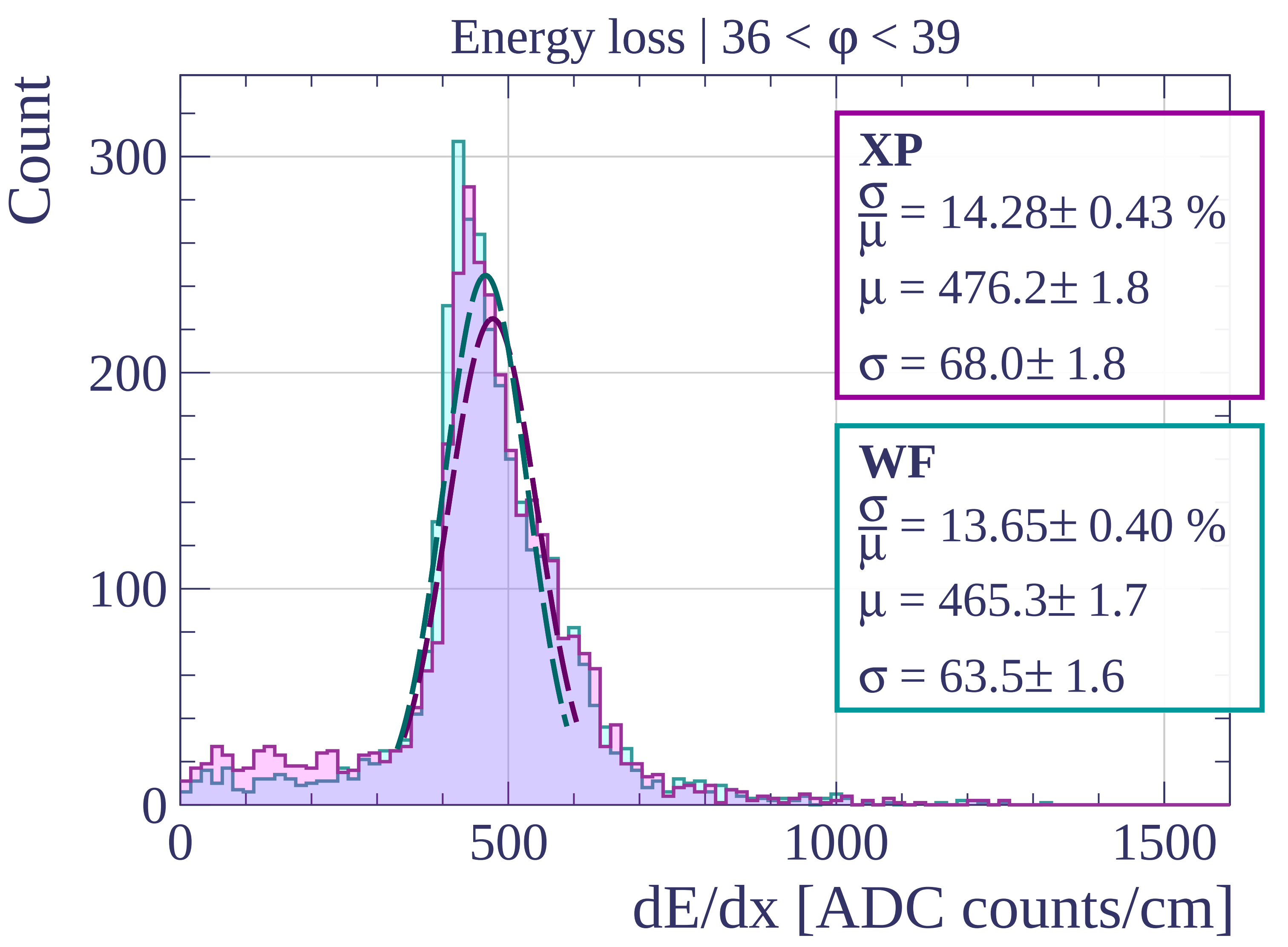
Energy loss | 27 < φ < 30
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 7.80± 0.18 %
μ = 430.0± 0.7
σ = 33.5± 0.7
WF
σ/μ = 7.39± 0.18 %
μ = 423.6± 0.6
σ = 31.3± 0.7



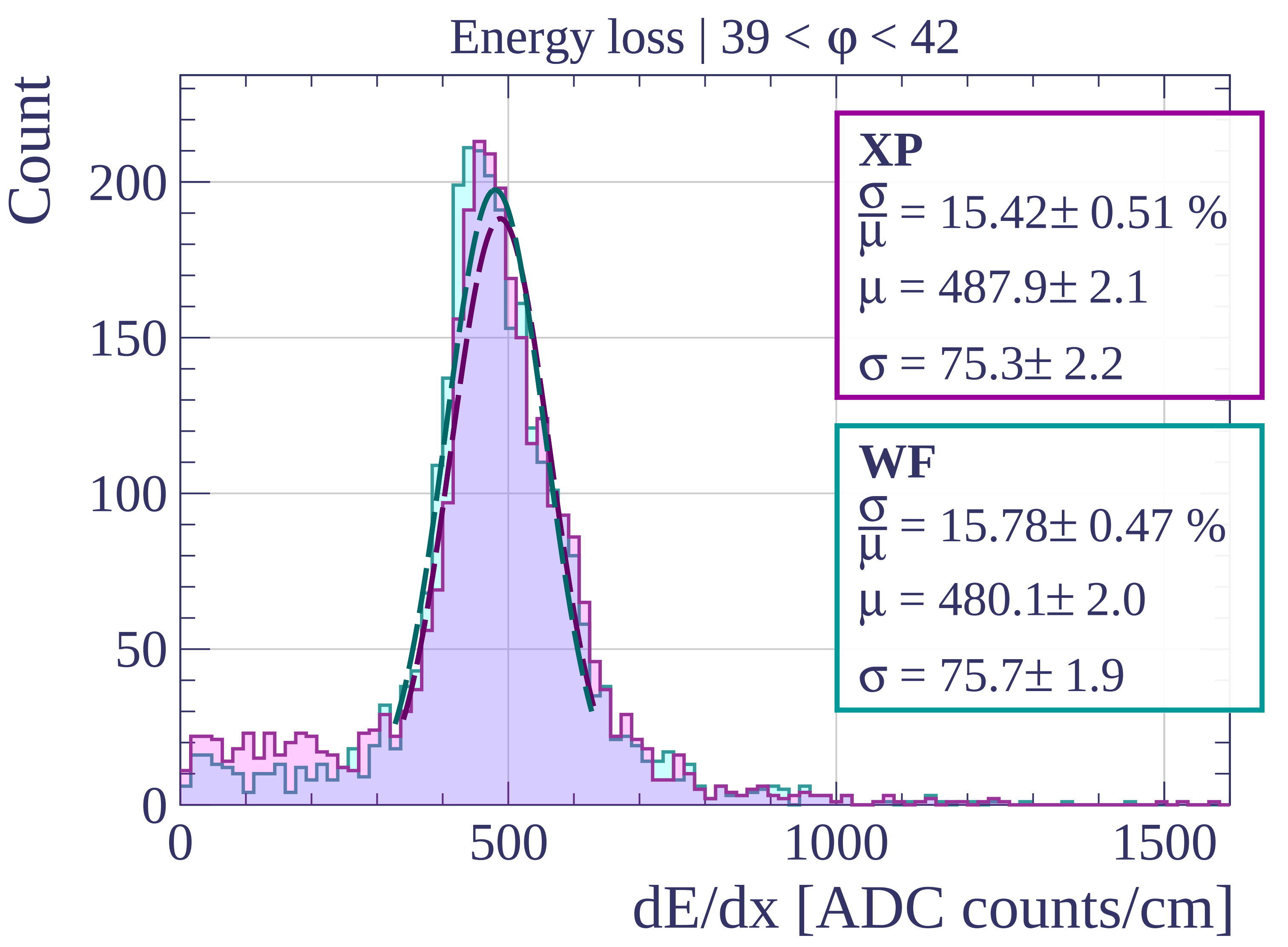
Energy loss | 30 < φ < 33
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 7.92± 0.20 %
μ = 434.9± 0.8
σ = 34.5± 0.8
WF
σ/μ = 8.82± 0.20 %
μ = 431.1± 0.8
σ = 38.0± 0.8



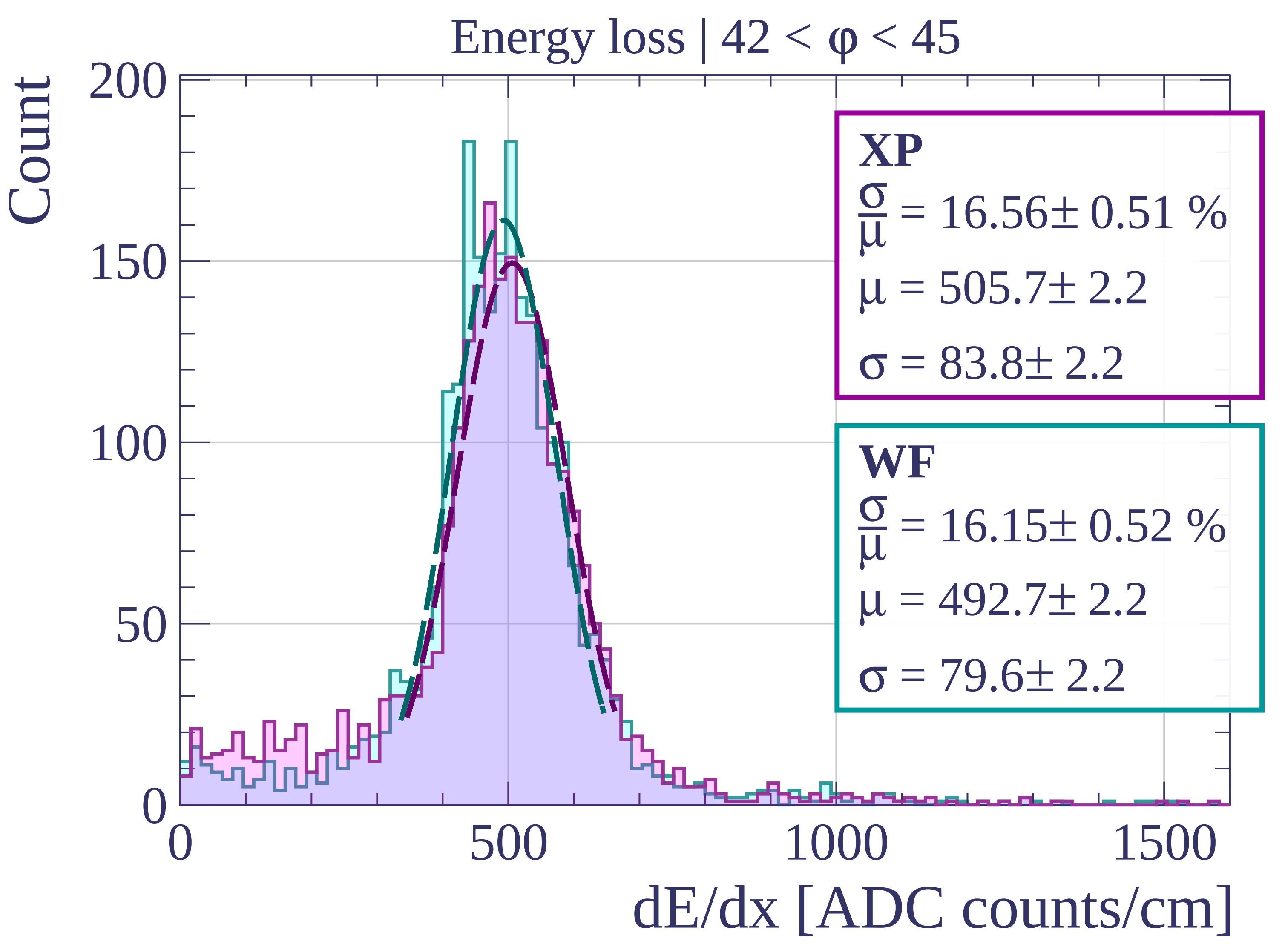
Energy loss | 33 < φ < 36
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 10.48± 0.29 %
μ = 451.5± 1.2
σ = 47.3± 1.2
WF
σ/μ = 10.06± 0.28 %
μ = 440.0± 1.1
σ = 44.3± 1.1



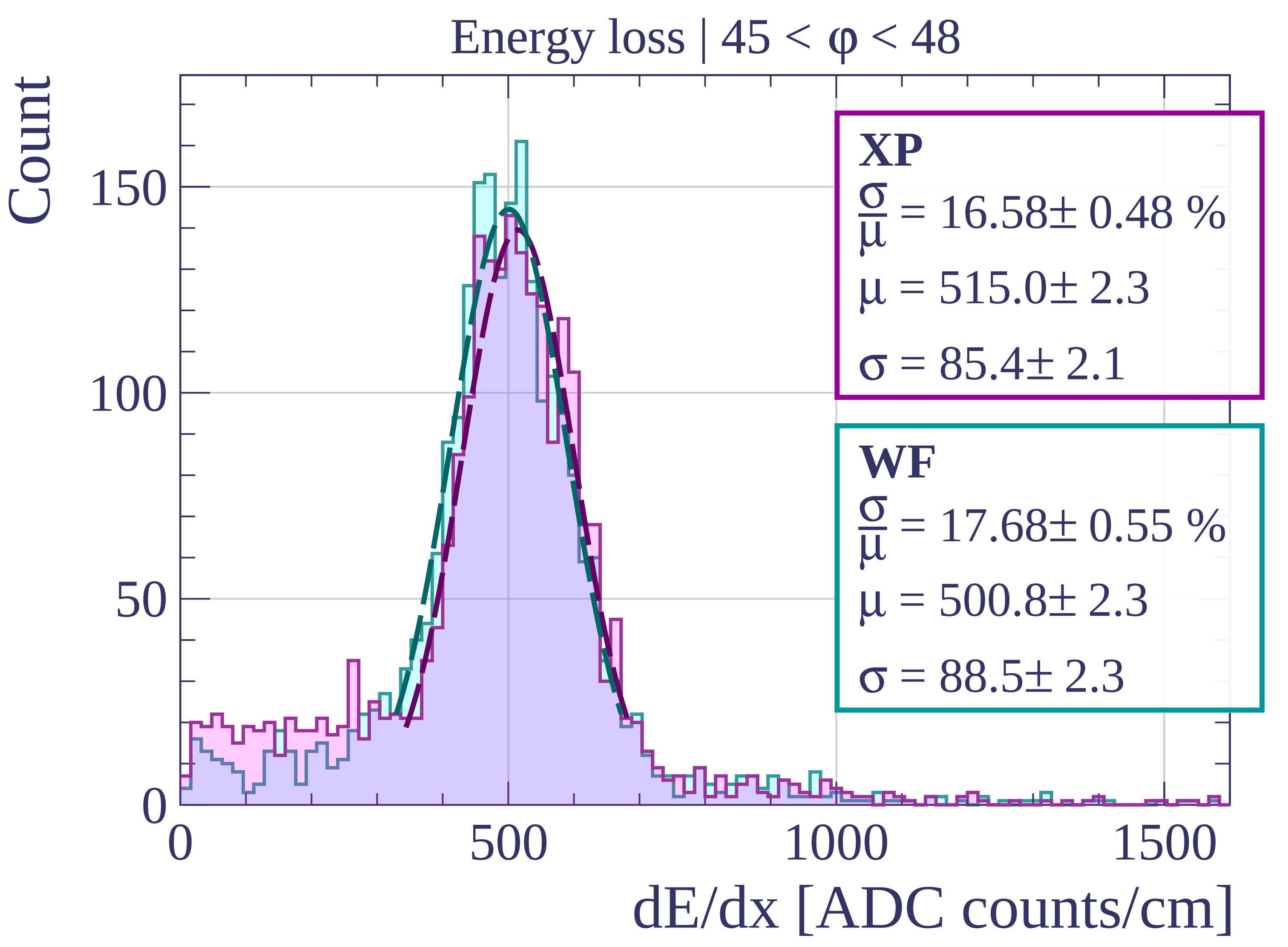
Energy loss | 36 < φ < 39
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 14.28± 0.43 %
μ = 476.2± 1.8
σ = 68.0± 1.8
WF
σ/μ = 13.65± 0.40 %
μ = 465.3± 1.7
σ = 63.5± 1.6

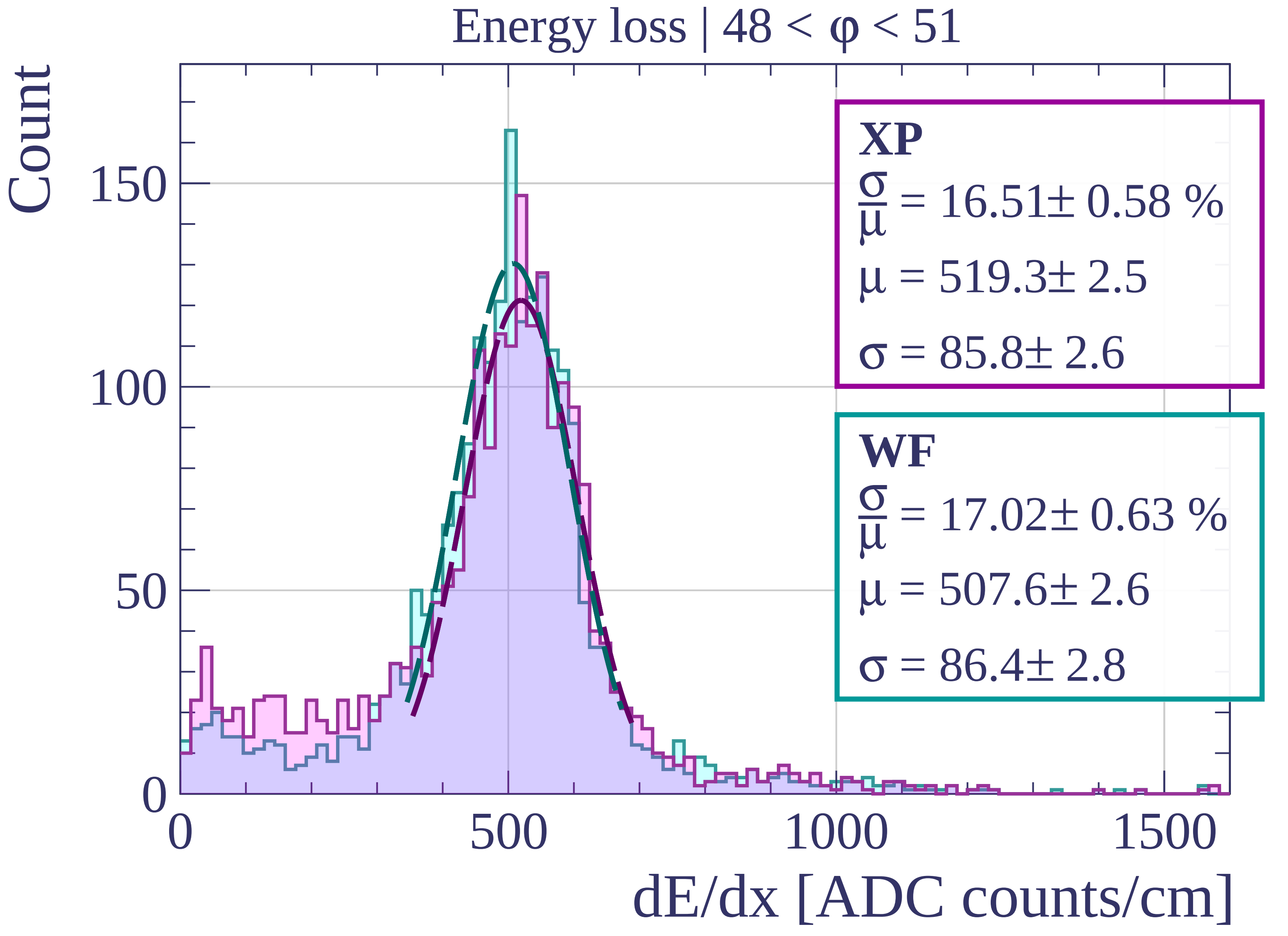


Energy loss | 39 < φ < 42
Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 15.42± 0.51 %
μ = 487.9± 2.1
σ = 75.3± 2.2
WF
σ/μ = 15.78± 0.47 %
μ = 480.1± 2.0
σ = 75.7± 1.9

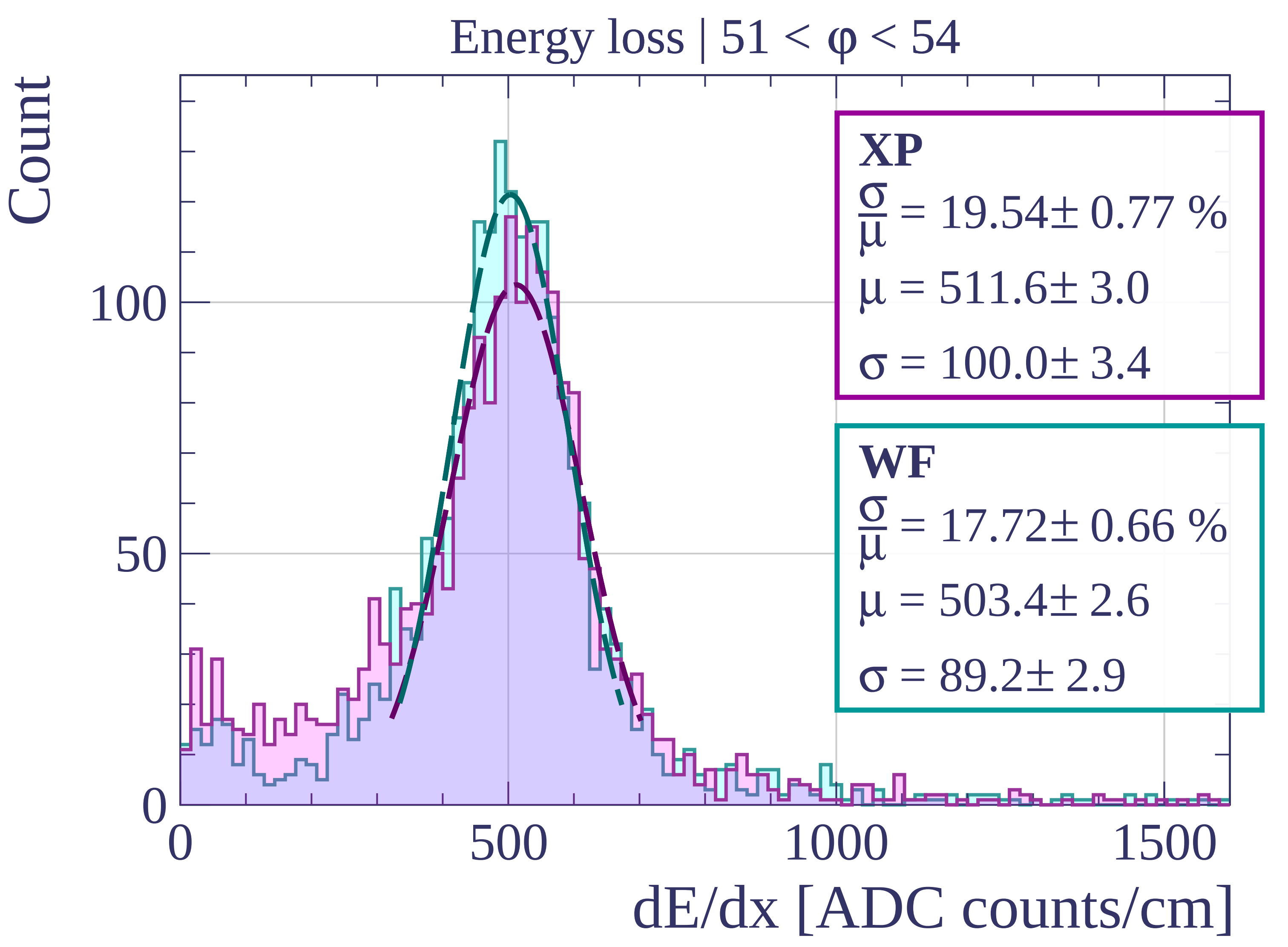


Energy loss | 42 < φ < 45
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
50
100
150
200
XP
σ/μ = 16.56± 0.51 %
μ = 505.7± 2.2
σ = 83.8± 2.2
WF
σ/μ = 16.15± 0.52 %
μ = 492.7± 2.2
σ = 79.6± 2.2

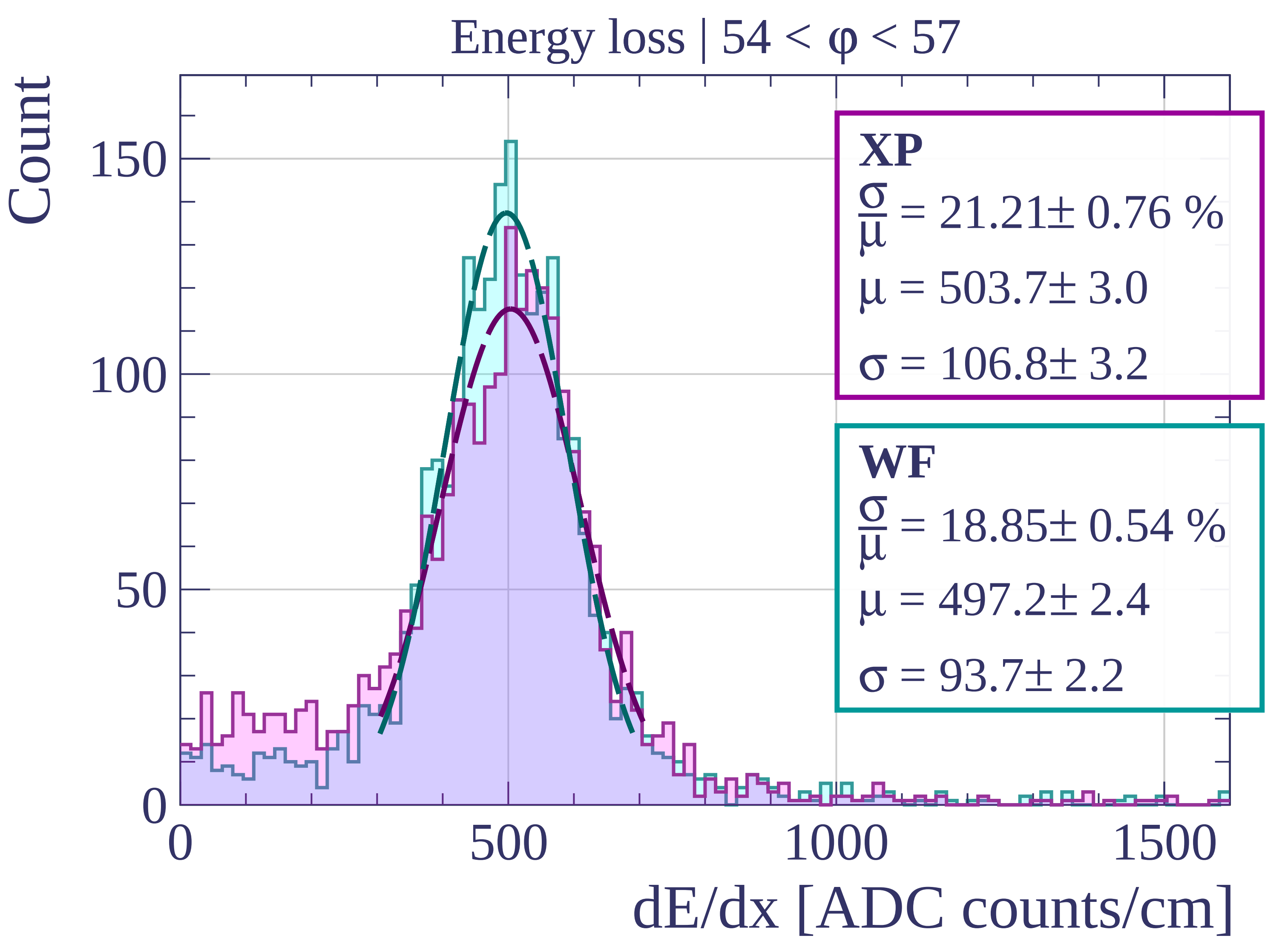

Energy loss | 45 < φ < 48
Count
dE/dx [ADC counts/cm]
0
50
100
150
0
500
1000
1500
XP
$\frac{\sigma}{\mu}$ = 16.58± 0.48 %
μ = 515.0± 2.3
σ = 85.4± 2.1
WF
$\frac{\sigma}{\mu}$ = 17.68± 0.55 %
μ = 500.8± 2.3
σ = 88.5± 2.3



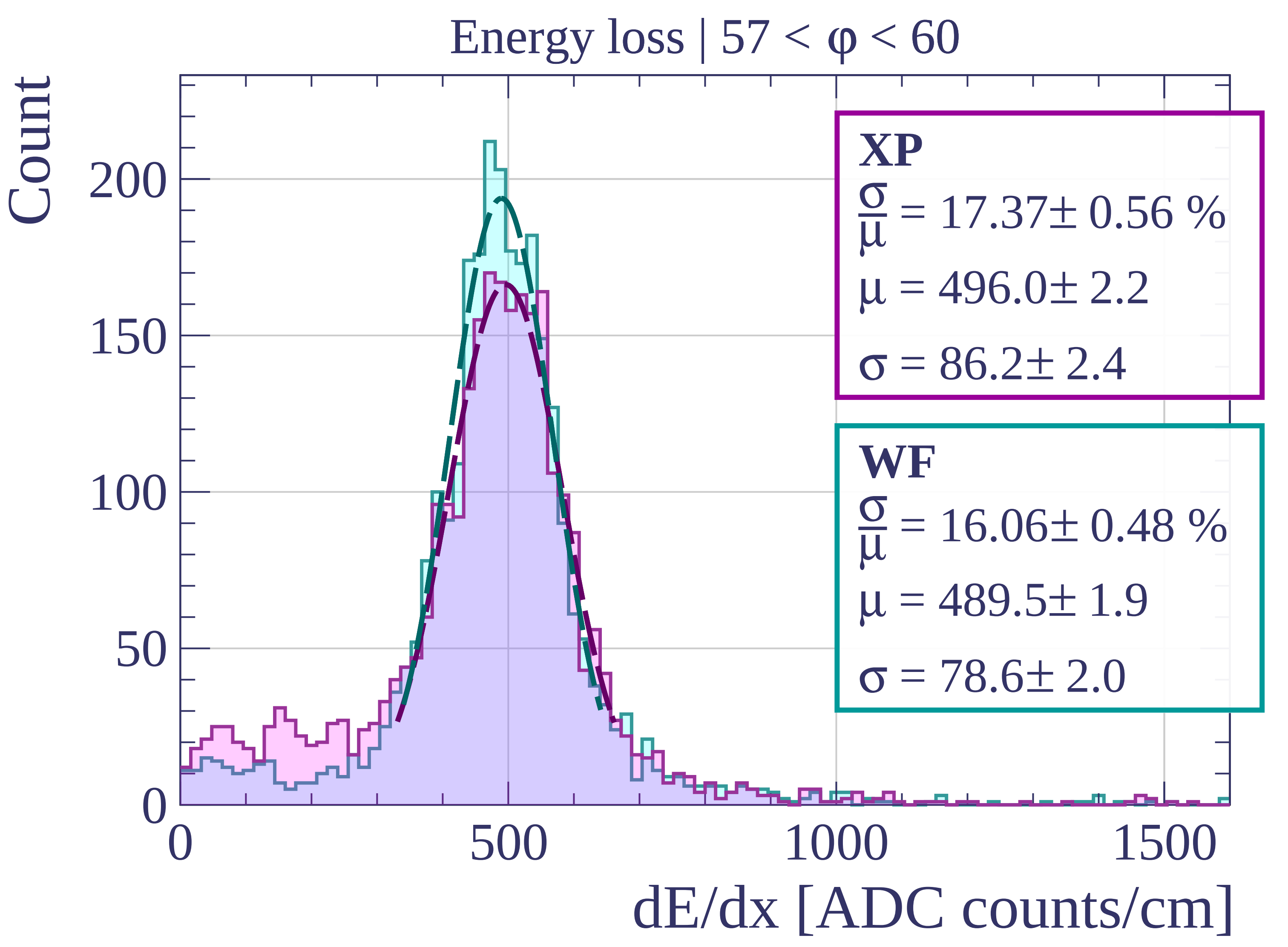
Energy loss | 48 < φ < 51
Count
dE/dx [ADC counts/cm]
XP
σ/μ = 16.51± 0.58 %
μ = 519.3± 2.5
σ = 85.8± 2.6
WF
σ/μ = 17.02± 0.63 %
μ = 507.6± 2.6
σ = 86.4± 2.8

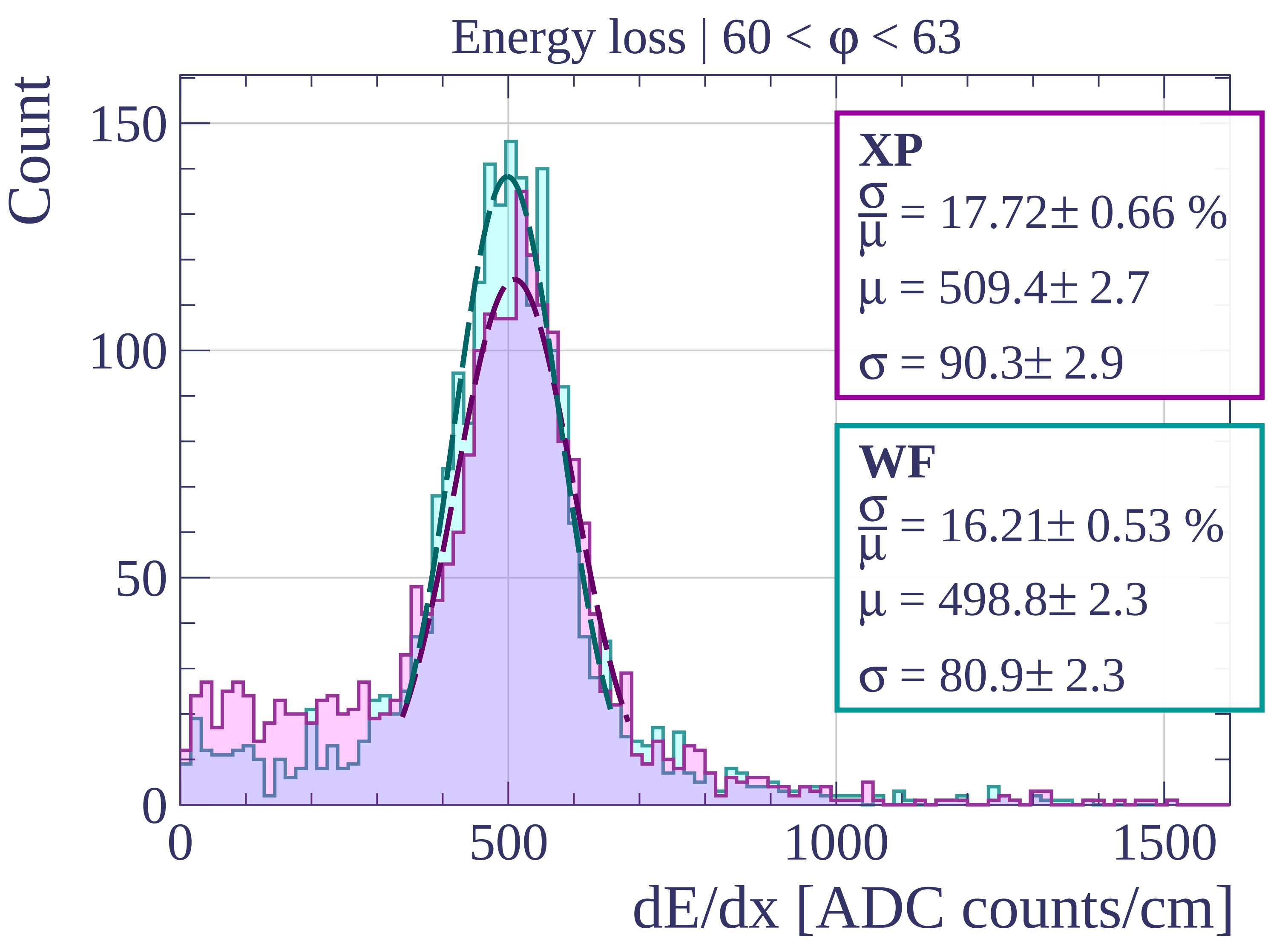


Energy loss | 51 < φ < 54
Count
100
50
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 19.54± 0.77 %
μ = 511.6± 3.0
σ = 100.0± 3.4
WF
σ/μ = 17.72± 0.66 %
μ = 503.4± 2.6
σ = 89.2± 2.9

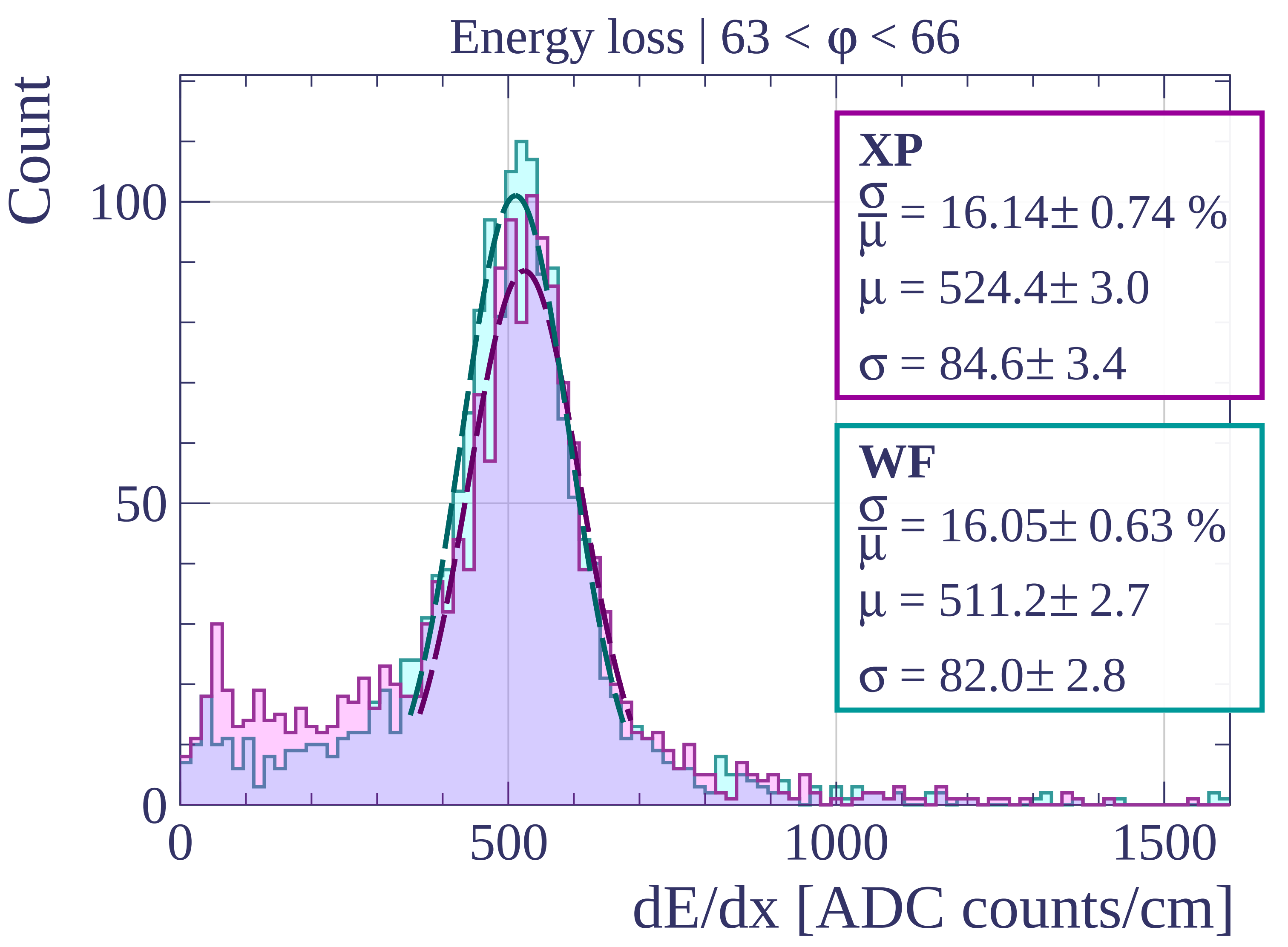


Energy loss | 54 < φ < 57
Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 21.21± 0.76 %
μ = 503.7± 3.0
σ = 106.8± 3.2
WF
σ/μ = 18.85± 0.54 %
μ = 497.2± 2.4
σ = 93.7± 2.2

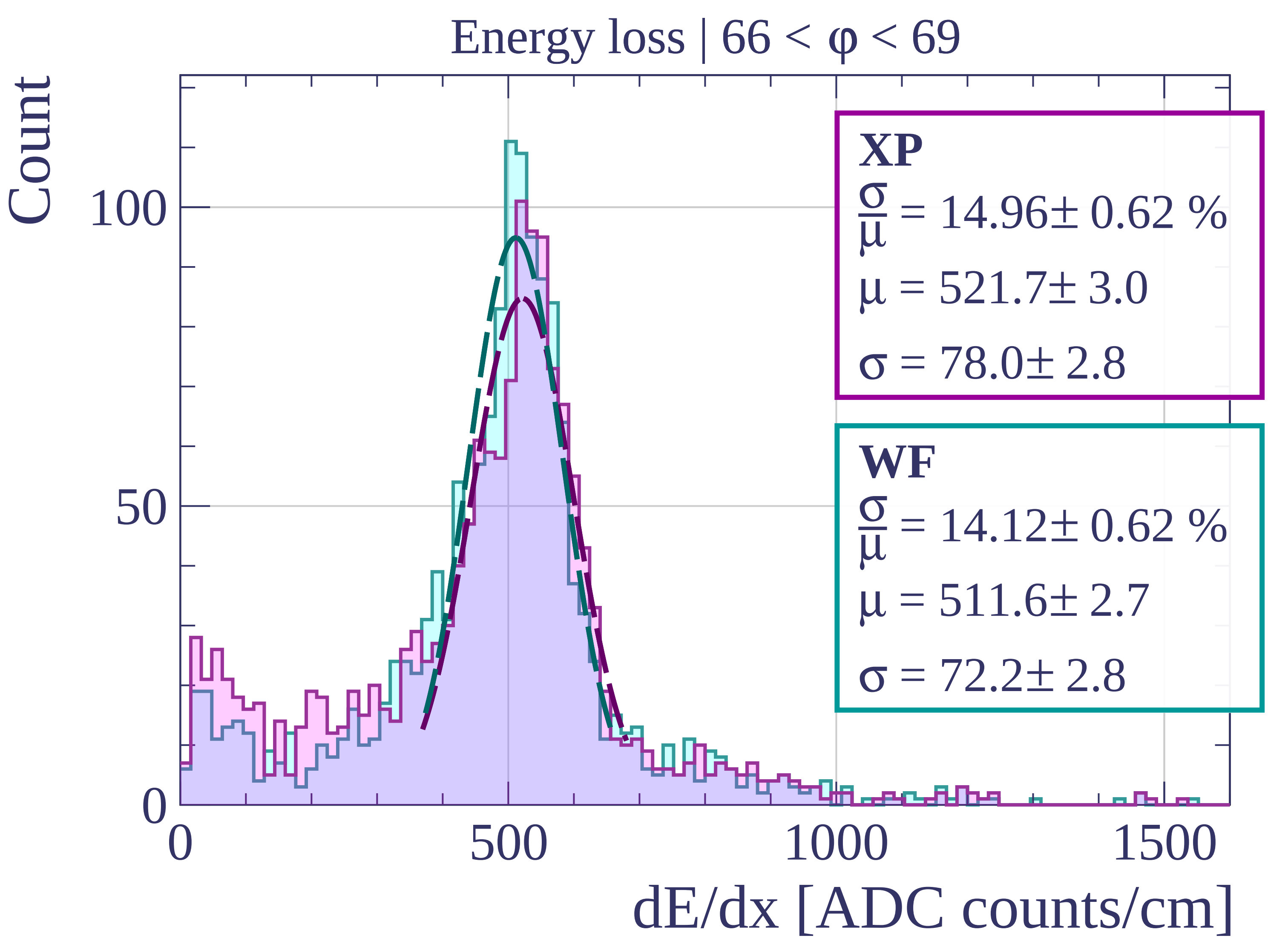

Energy loss | 57 < φ < 60
Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 17.37± 0.56 %
μ = 496.0± 2.2
σ = 86.2± 2.4
WF
$\frac{\sigma}{\mu}$ = 16.06± 0.48 %
μ = 489.5± 1.9
σ = 78.6± 2.0

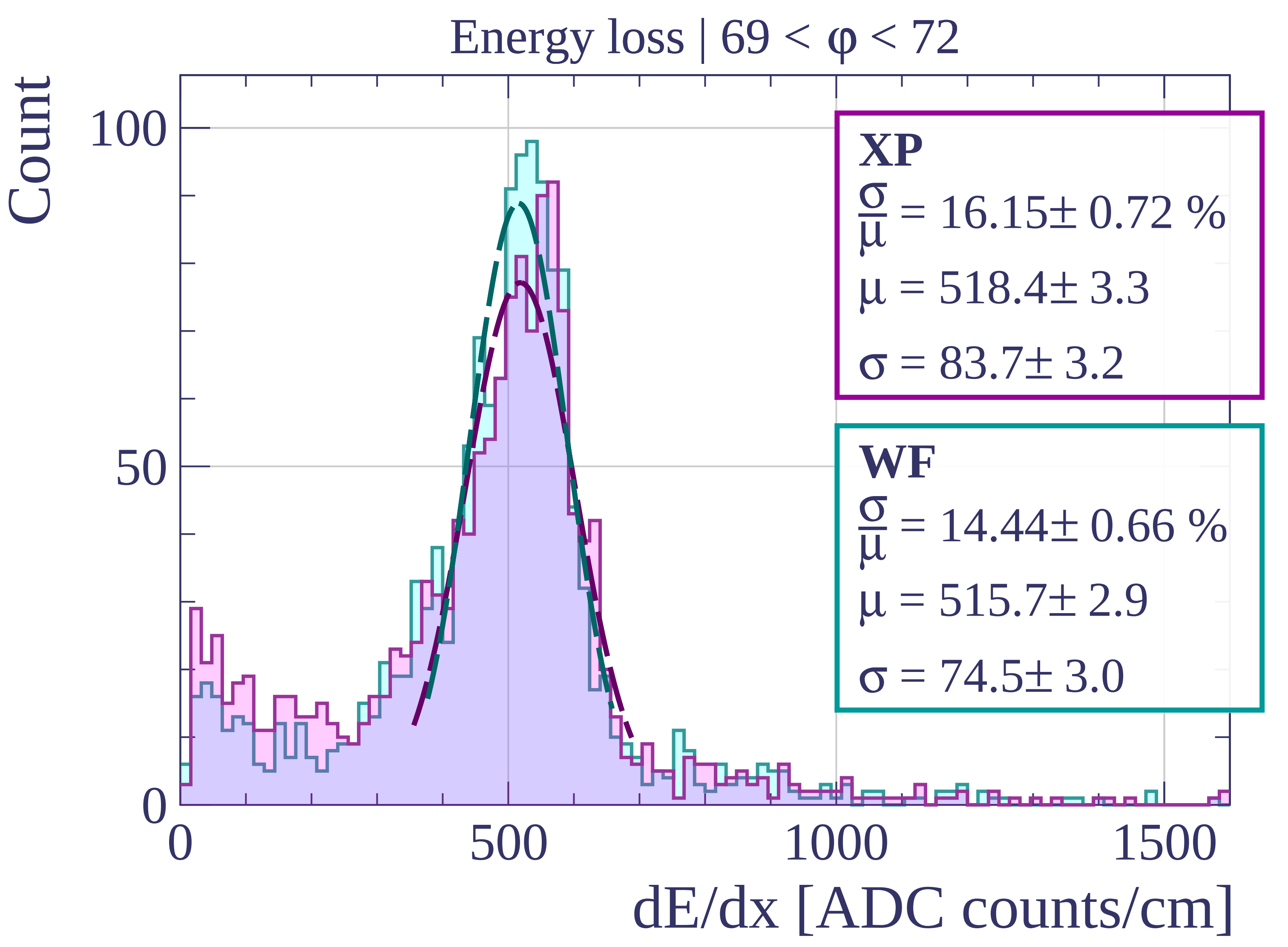


Energy loss | 60 < φ < 63
Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 17.72± 0.66 %
μ = 509.4± 2.7
σ = 90.3± 2.9
WF
σ/μ = 16.21± 0.53 %
μ = 498.8± 2.3
σ = 80.9± 2.3

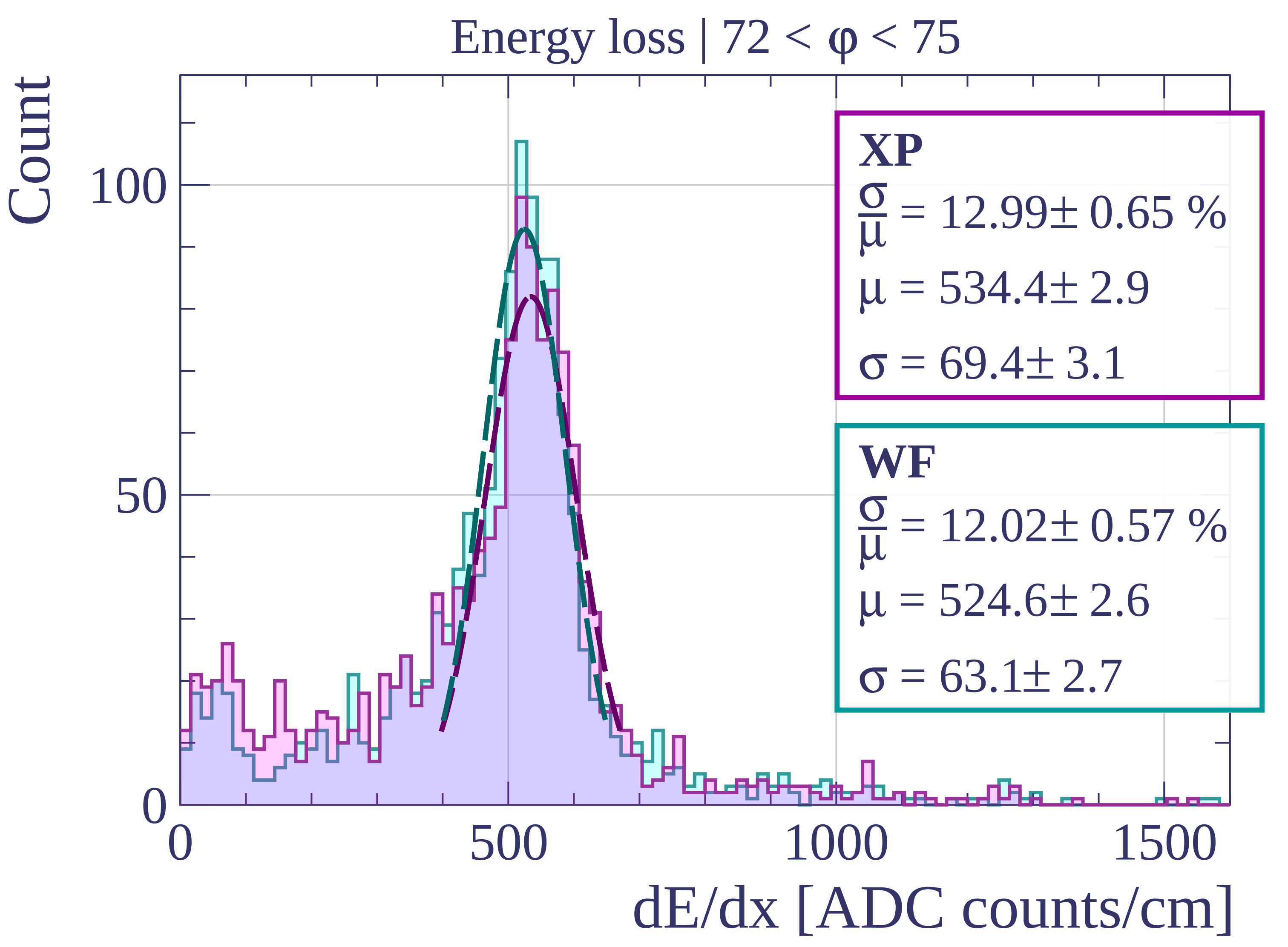


Energy loss | 63 < φ < 66
Count
dE/dx [ADC counts/cm]
0
50
100
0
500
1000
1500
XP
$\frac{\sigma}{\mu}$ = 16.14± 0.74 %
μ = 524.4± 3.0
σ = 84.6± 3.4
WF
$\frac{\sigma}{\mu}$ = 16.05± 0.63 %
μ = 511.2± 2.7
σ = 82.0± 2.8

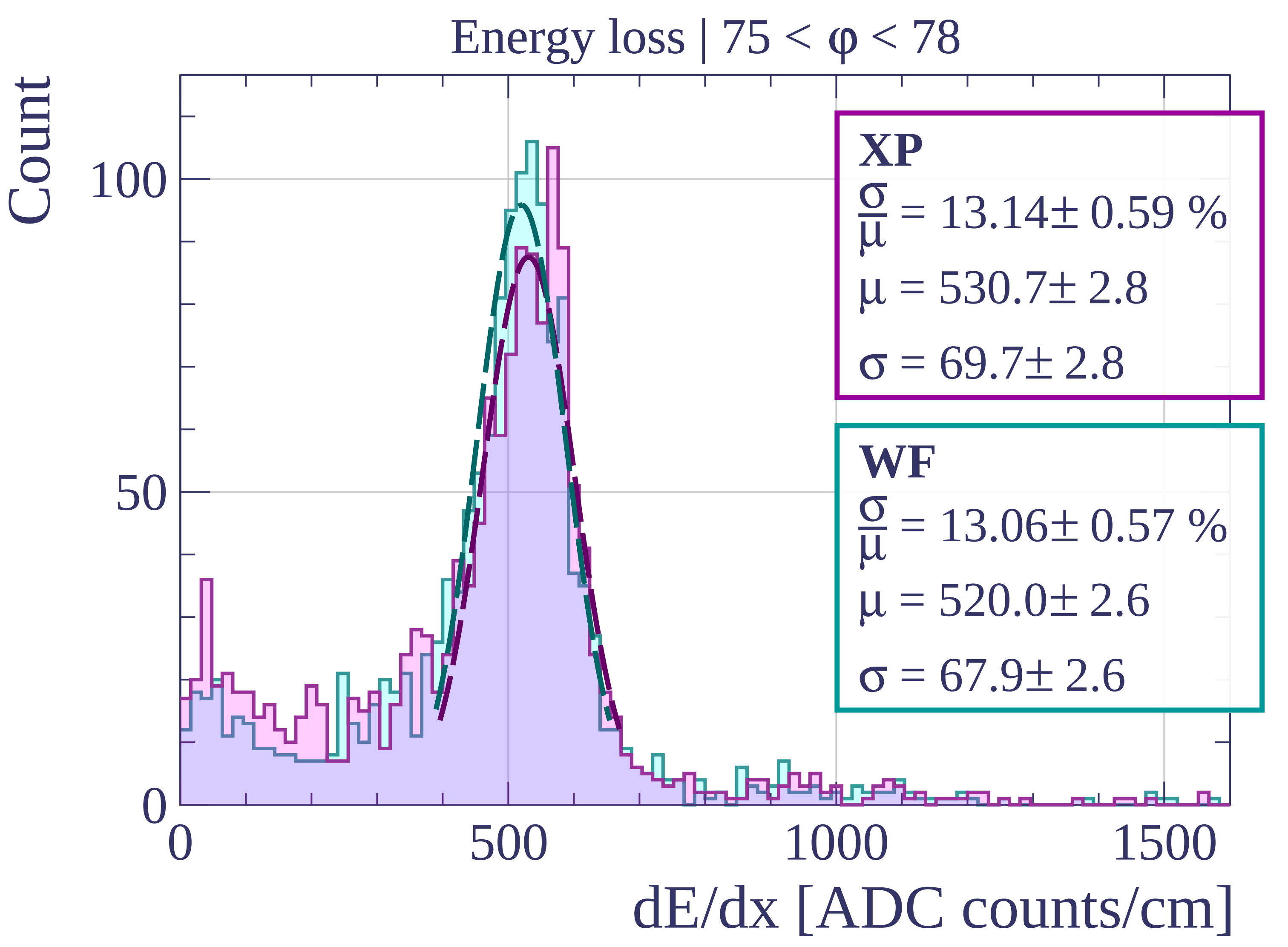

Energy loss | 66 < φ < 69
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 14.96± 0.62 %
μ = 521.7± 3.0
σ = 78.0± 2.8
WF
σ/μ = 14.12± 0.62 %
μ = 511.6± 2.7
σ = 72.2± 2.8



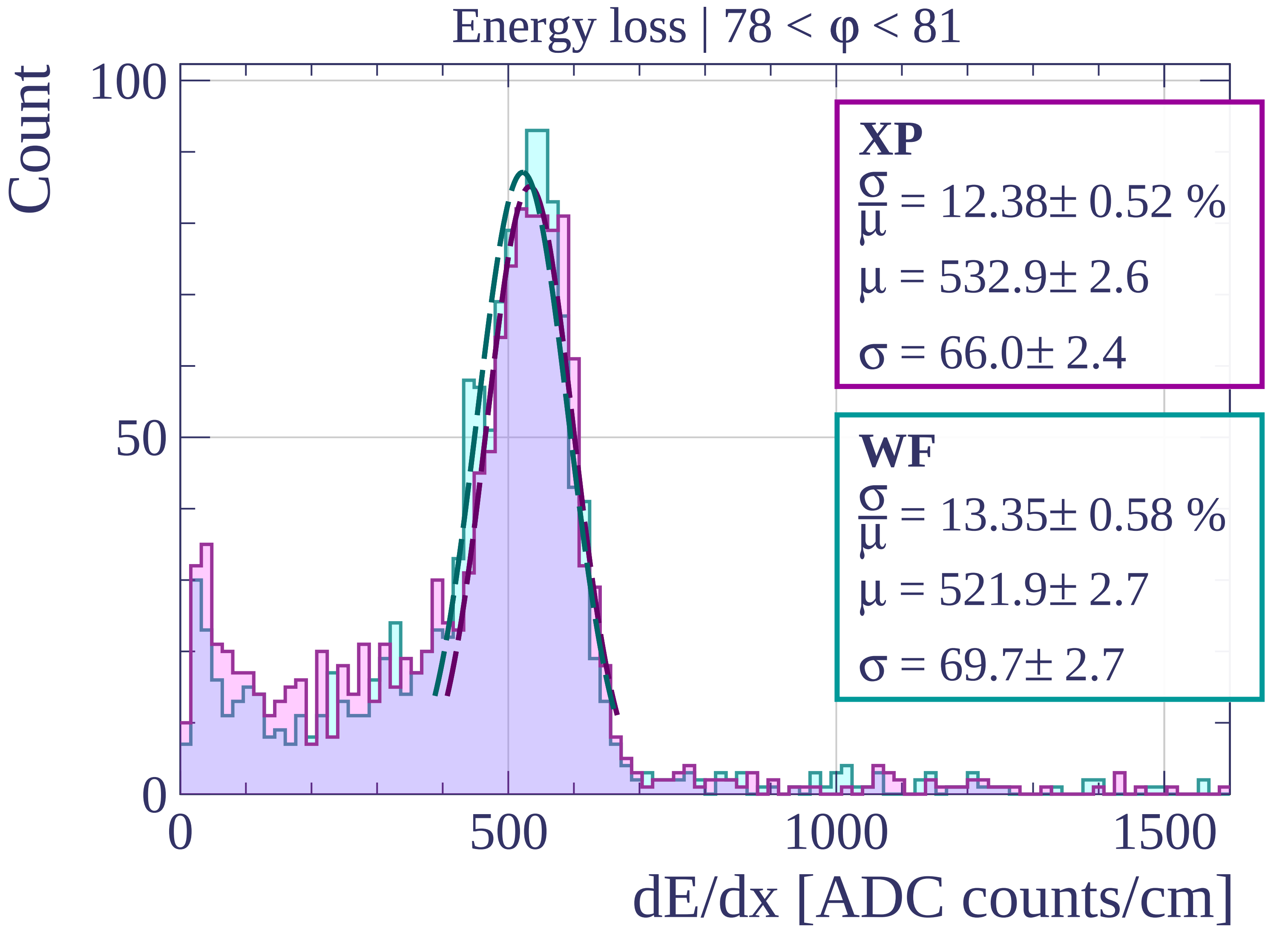
Energy loss | 69 < φ < 72
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 16.15± 0.72 %
μ = 518.4± 3.3
σ = 83.7± 3.2
WF
σ/μ = 14.44± 0.66 %
μ = 515.7± 2.9
σ = 74.5± 3.0



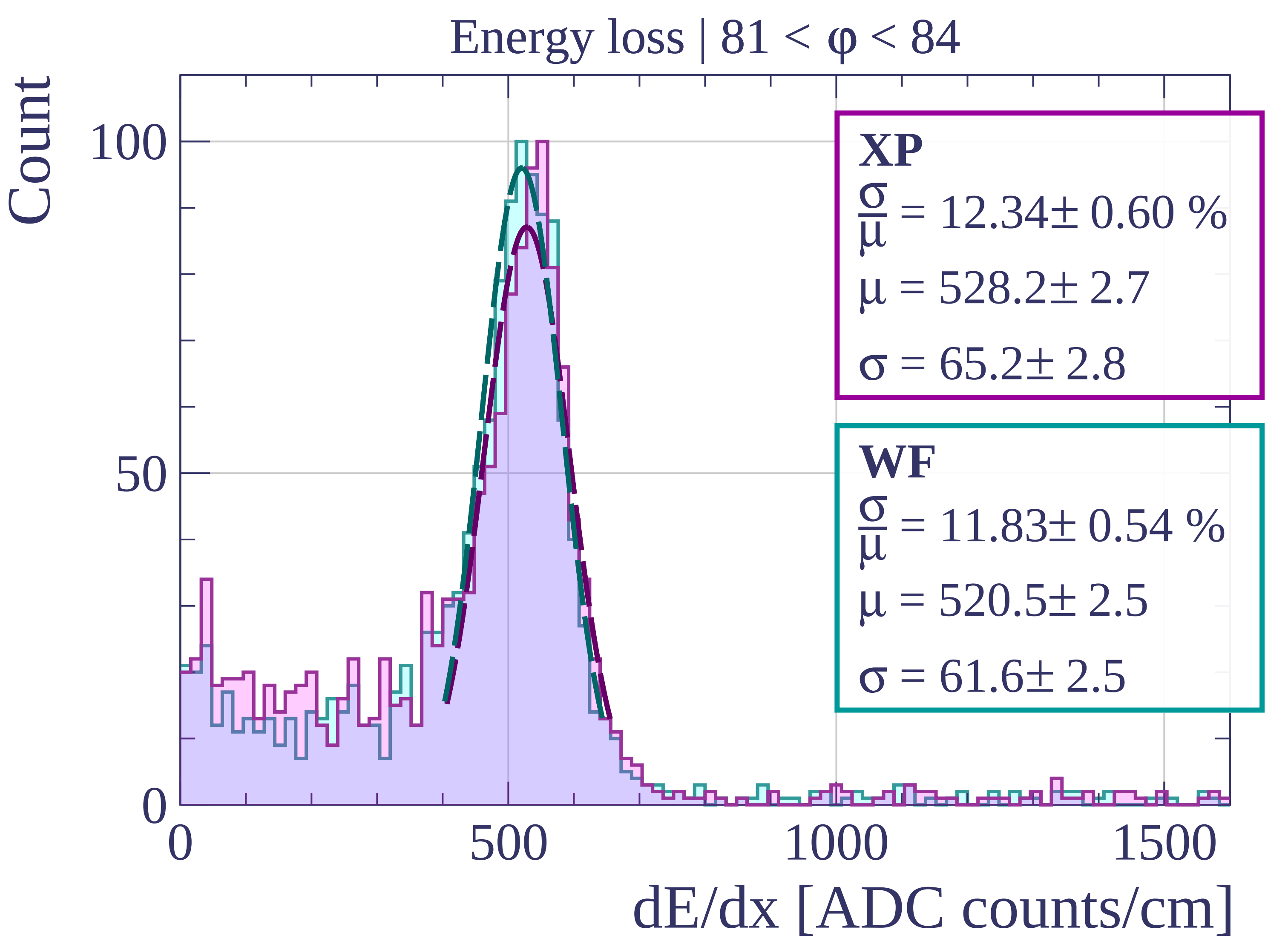
Energy loss | 72 < φ < 75
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 12.99± 0.65 %
μ = 534.4± 2.9
σ = 69.4± 3.1
WF
σ/μ = 12.02± 0.57 %
μ = 524.6± 2.6
σ = 63.1± 2.7



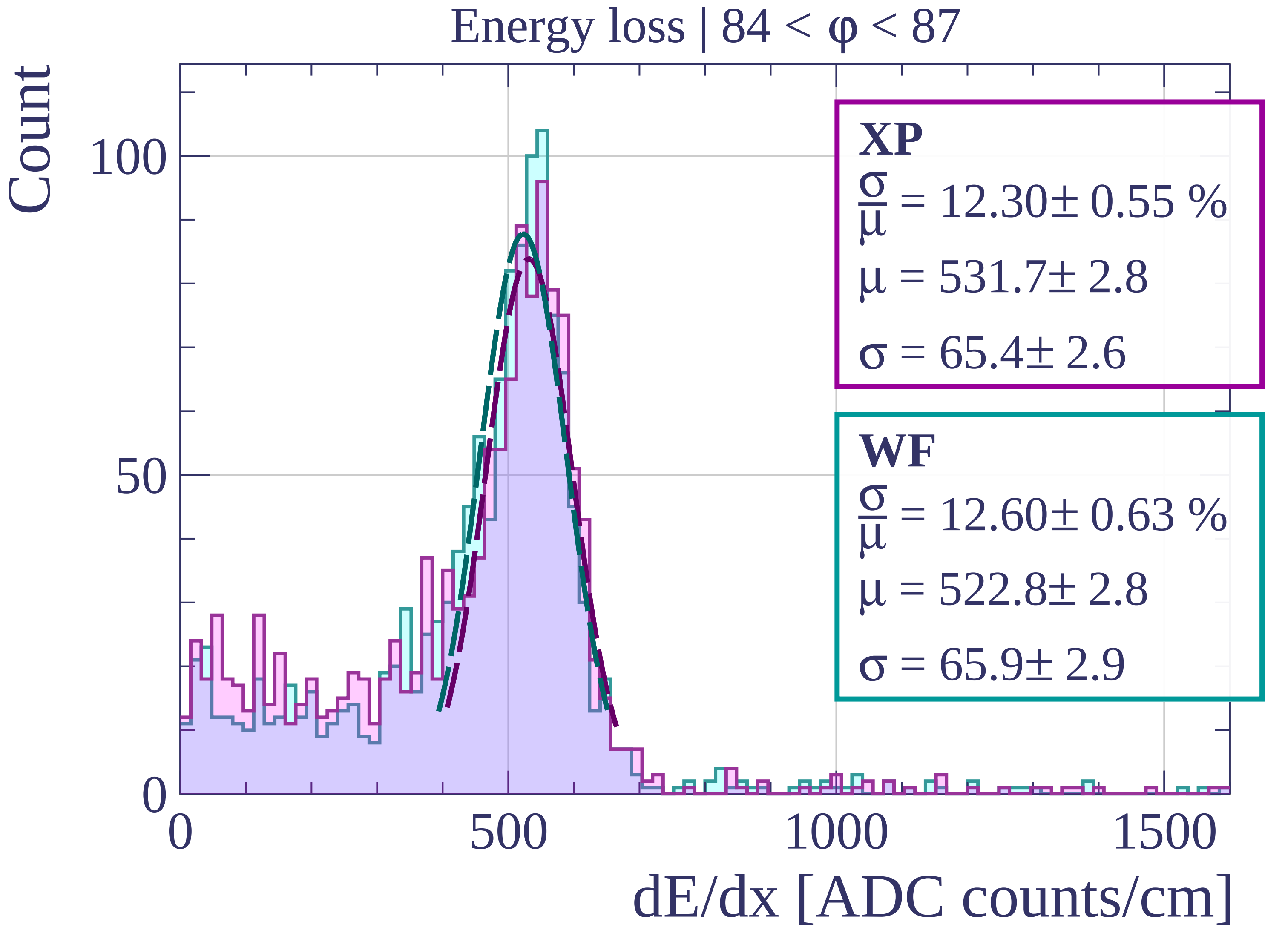
Energy loss | 75 < φ < 78
Count
dE/dx [ADC counts/cm]
0
50
100
0
500
1000
1500
XP
σ/μ = 13.14± 0.59 %
μ = 530.7± 2.8
σ = 69.7± 2.8
WF
σ/μ = 13.06± 0.57 %
μ = 520.0± 2.6
σ = 67.9± 2.6



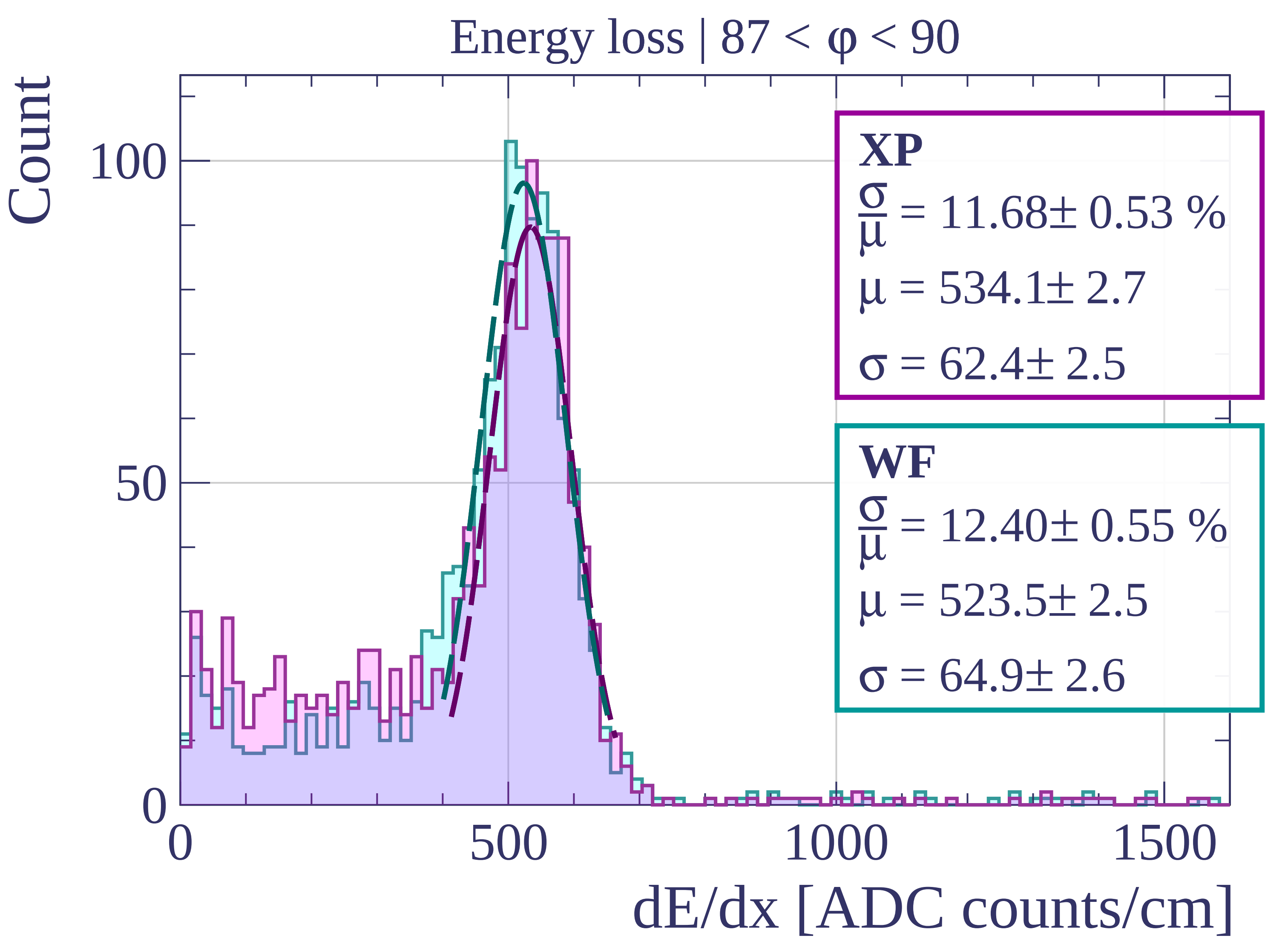
Energy loss | 78 < φ < 81
Count
100
50
0
XP
σ/μ = 12.38± 0.52 %
μ = 532.9± 2.6
σ = 66.0± 2.4
WF
σ/μ = 13.35± 0.58 %
μ = 521.9± 2.7
σ = 69.7± 2.7
0
500
1000
1500
dE/dx [ADC counts/cm]



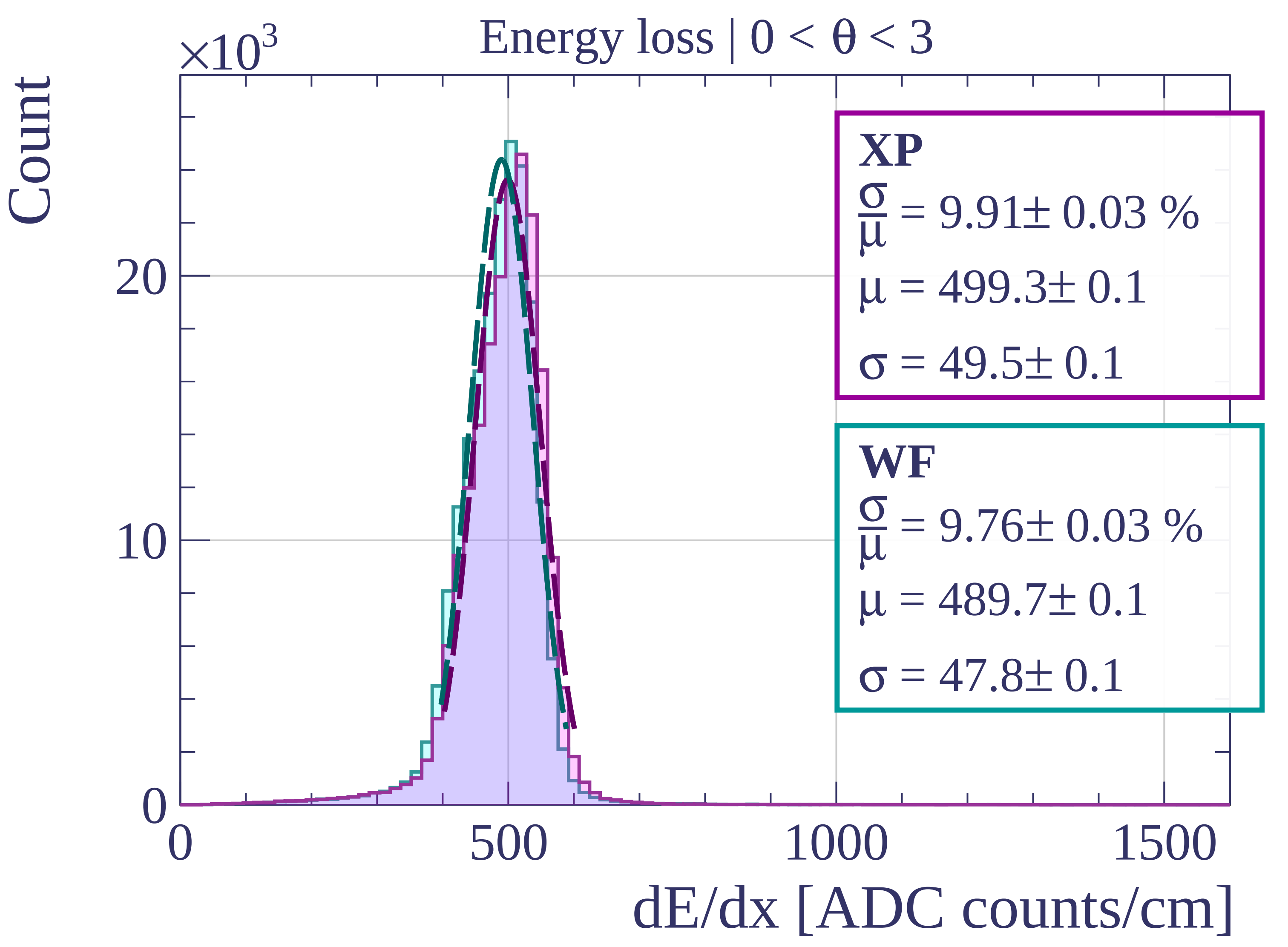
Energy loss | 81 < φ < 84
Count
dE/dx [ADC counts/cm]
0
50
100
0
500
1000
1500
XP
σ/μ = 12.34± 0.60 %
μ = 528.2± 2.7
σ = 65.2± 2.8
WF
σ/μ = 11.83± 0.54 %
μ = 520.5± 2.5
σ = 61.6± 2.5



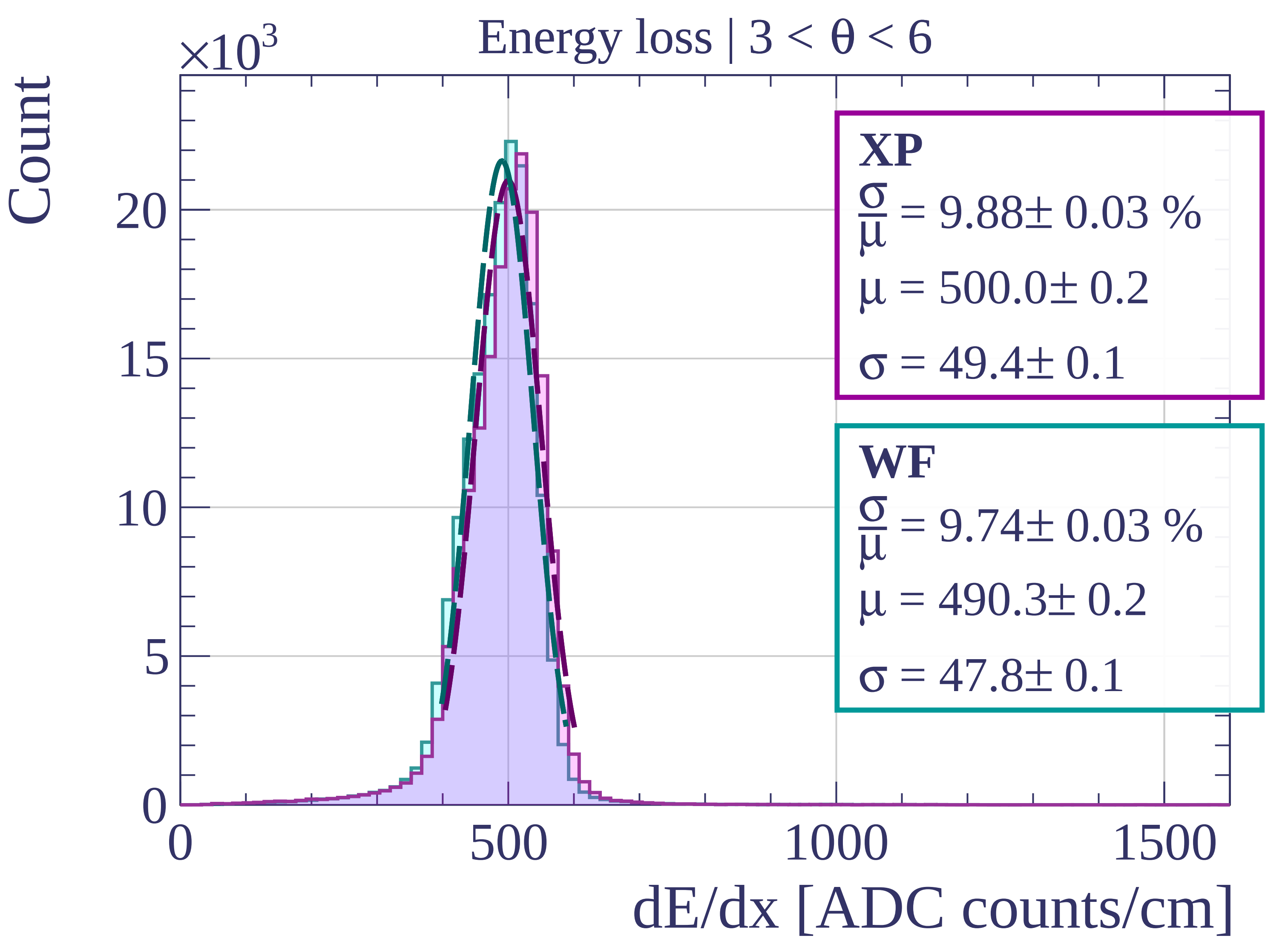
Energy loss | 84 < φ < 87
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 12.30± 0.55 %
μ = 531.7± 2.8
σ = 65.4± 2.6
WF
σ/μ = 12.60± 0.63 %
μ = 522.8± 2.8
σ = 65.9± 2.9



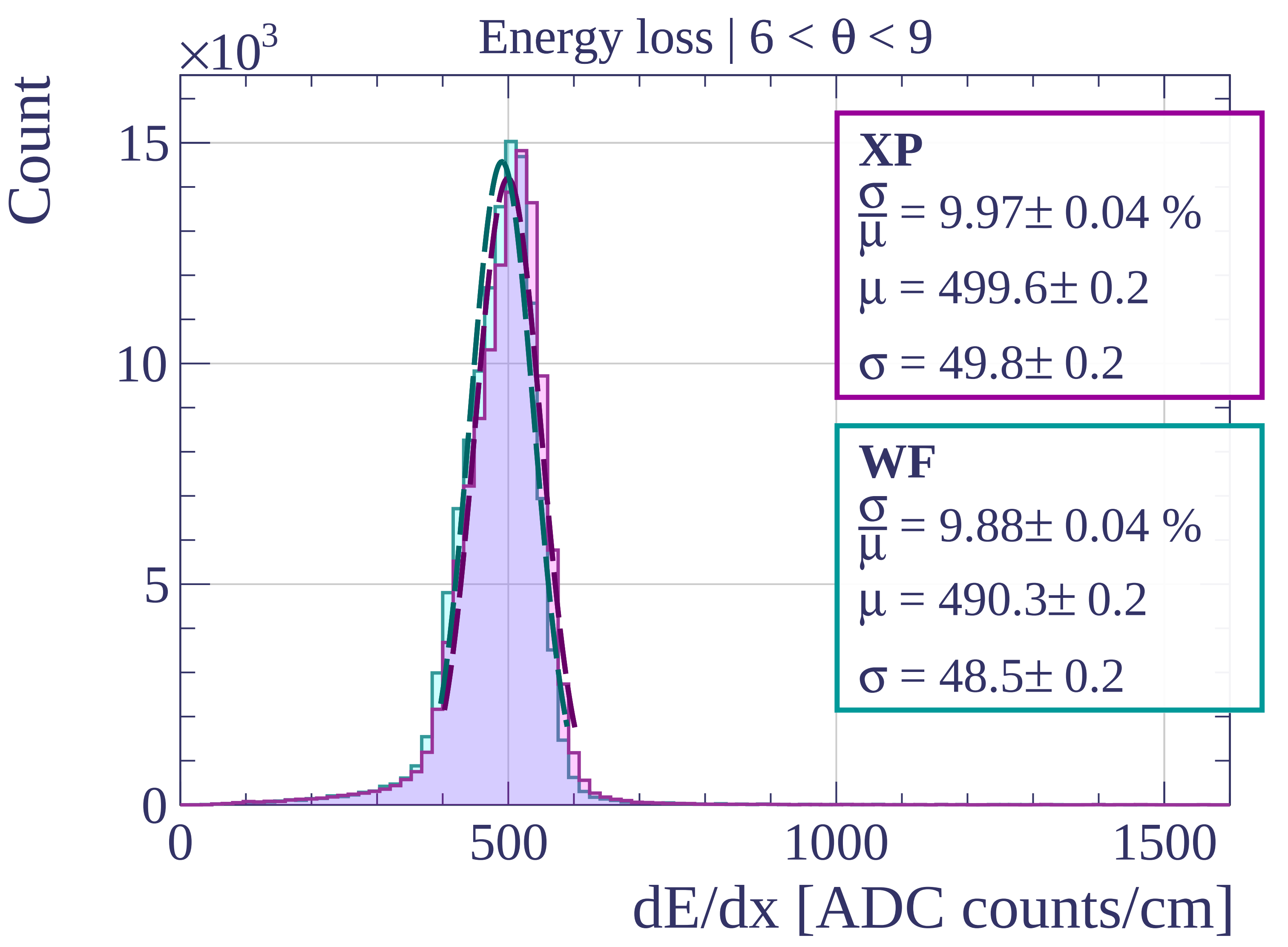
Energy loss | 87 < φ < 90
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 11.68± 0.53 %
μ = 534.1± 2.7
σ = 62.4± 2.5
WF
σ/μ = 12.40± 0.55 %
μ = 523.5± 2.5
σ = 64.9± 2.6



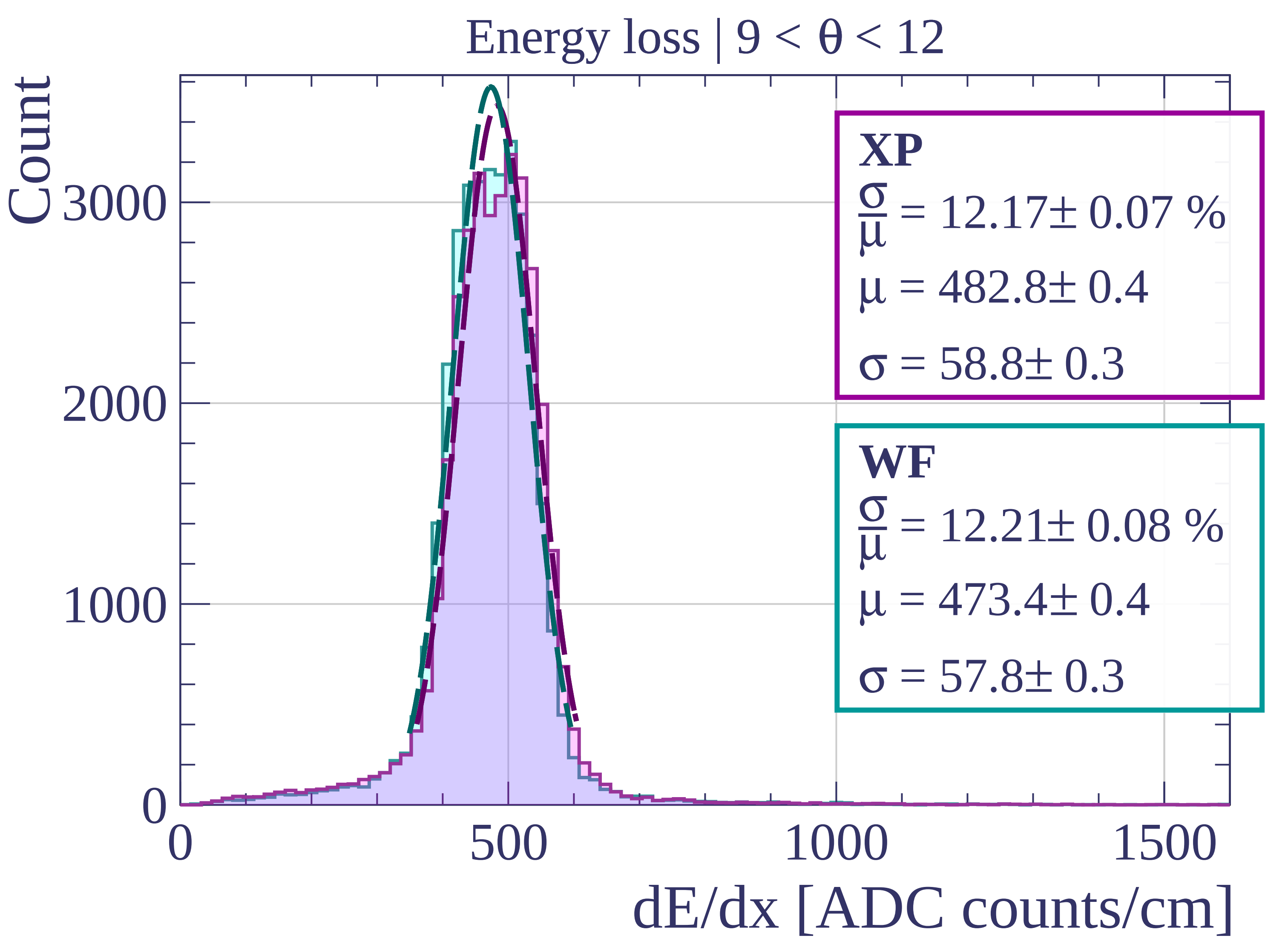
Energy loss | 0 < θ < 3
Count
×10^3
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 9.91± 0.03 %
μ = 499.3± 0.1
σ = 49.5± 0.1
WF
σ/μ = 9.76± 0.03 %
μ = 489.7± 0.1
σ = 47.8± 0.1



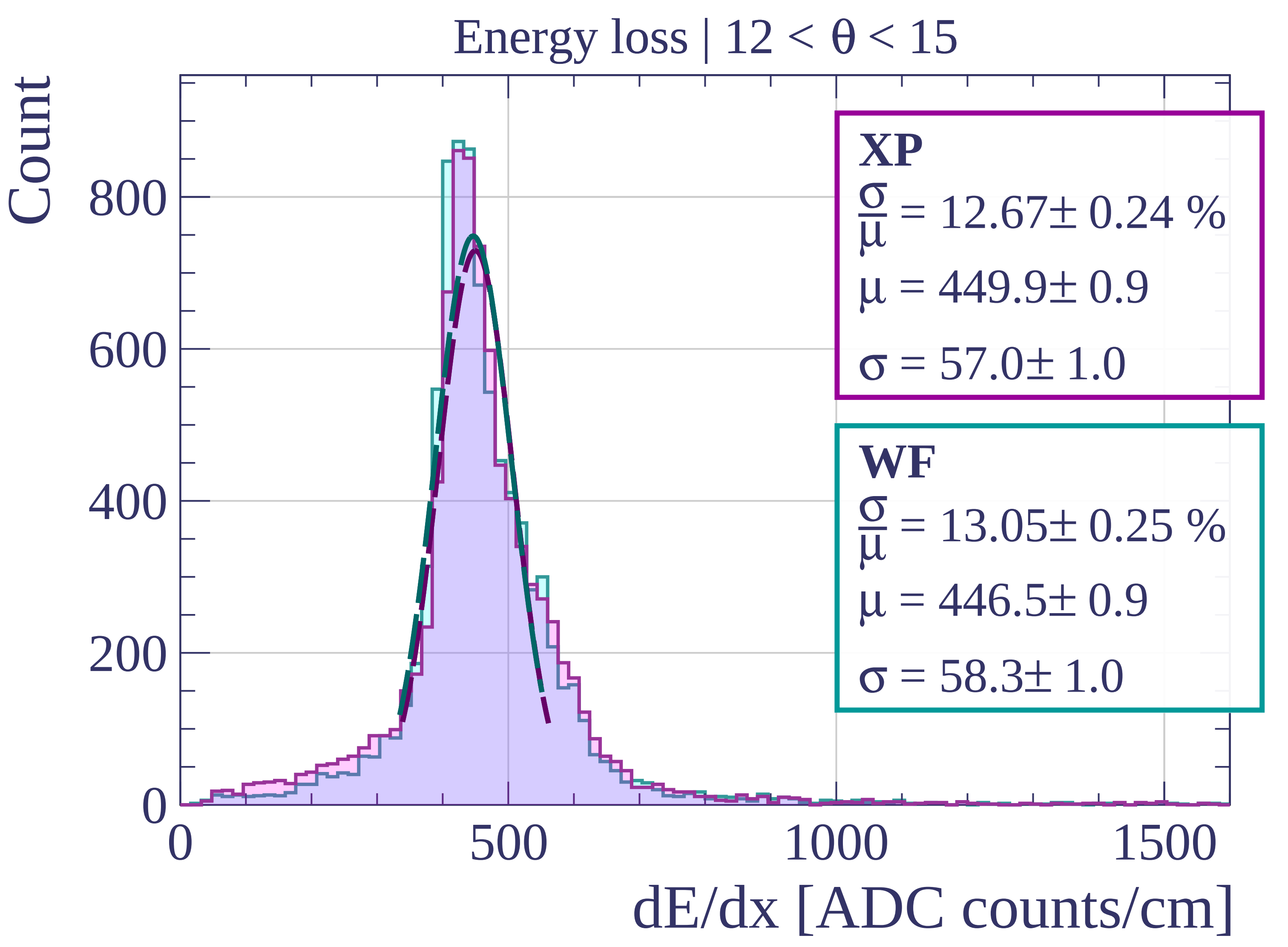
Energy loss | 3 < θ < 6
×10^3
Count
20
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 9.88± 0.03 %
μ = 500.0± 0.2
σ = 49.4± 0.1
WF
σ/μ = 9.74± 0.03 %
μ = 490.3± 0.2
σ = 47.8± 0.1

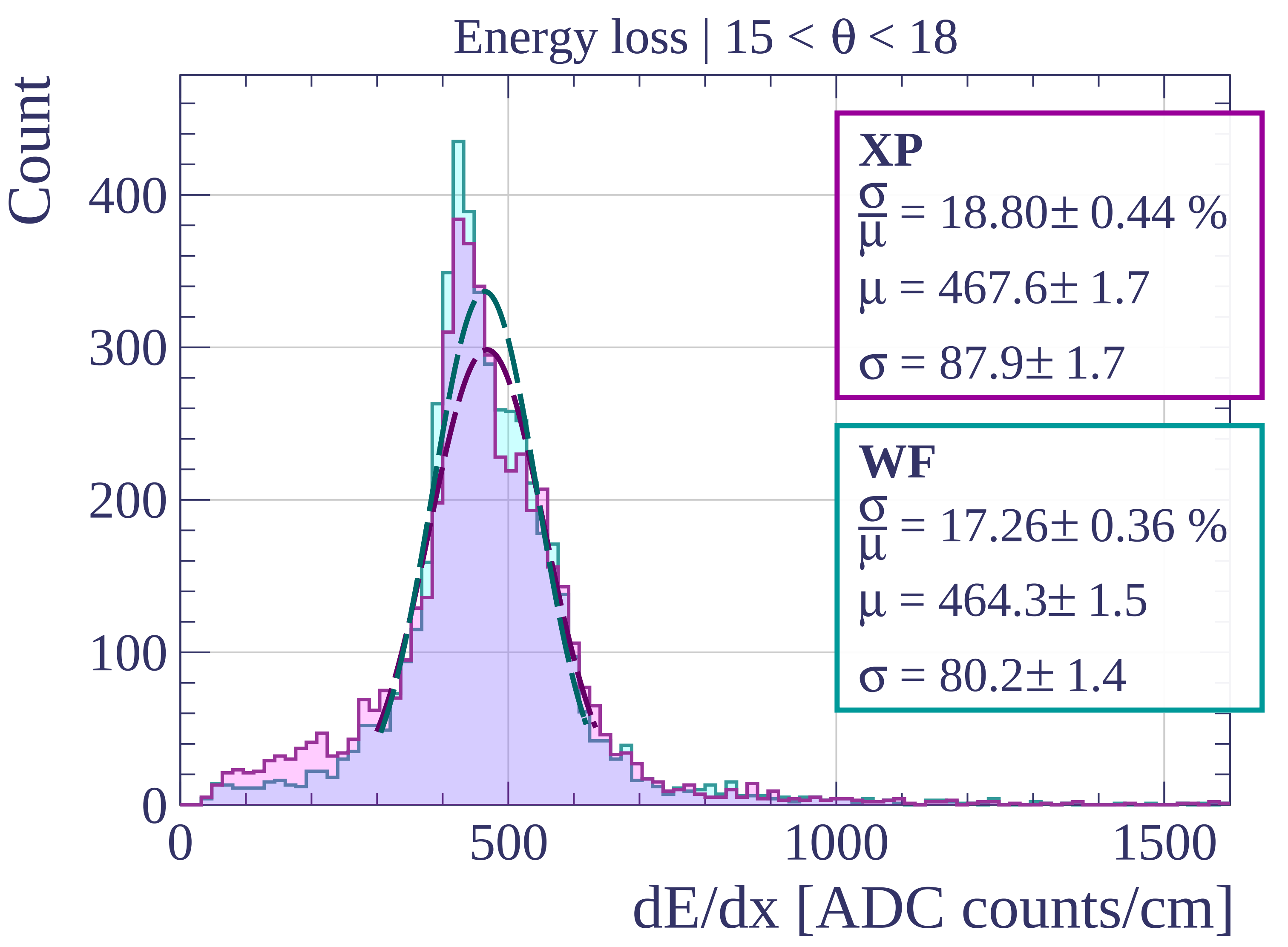


Energy loss | 6 < θ < 9
×10^3
Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 9.97± 0.04 %
μ = 499.6± 0.2
σ = 49.8± 0.2
WF
σ/μ = 9.88± 0.04 %
μ = 490.3± 0.2
σ = 48.5± 0.2

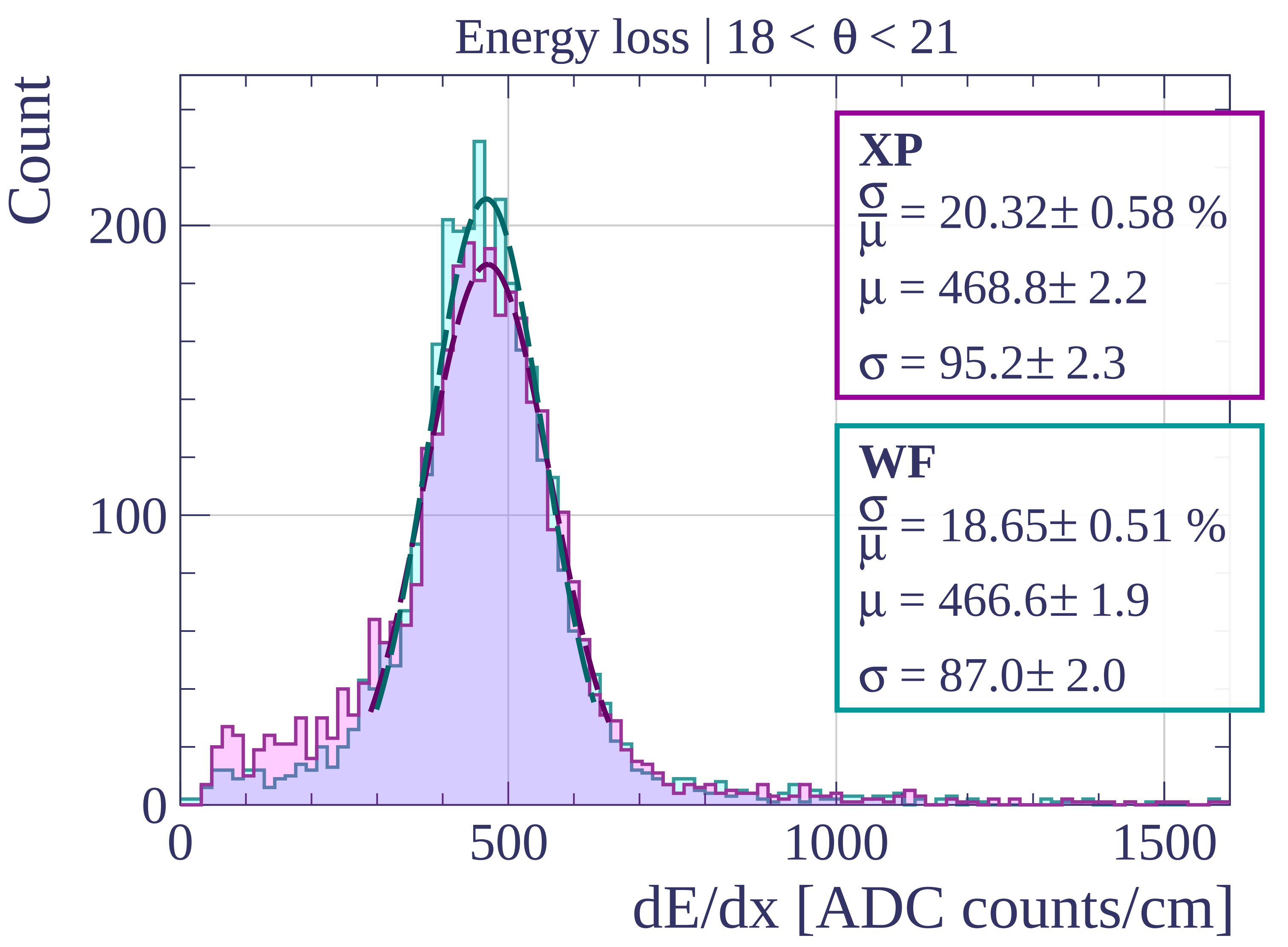


Energy loss | 9 < θ < 12
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 12.17± 0.07 %
μ = 482.8± 0.4
σ = 58.8± 0.3
WF
$\frac{\sigma}{\mu}$ = 12.21± 0.08 %
μ = 473.4± 0.4
σ = 57.8± 0.3

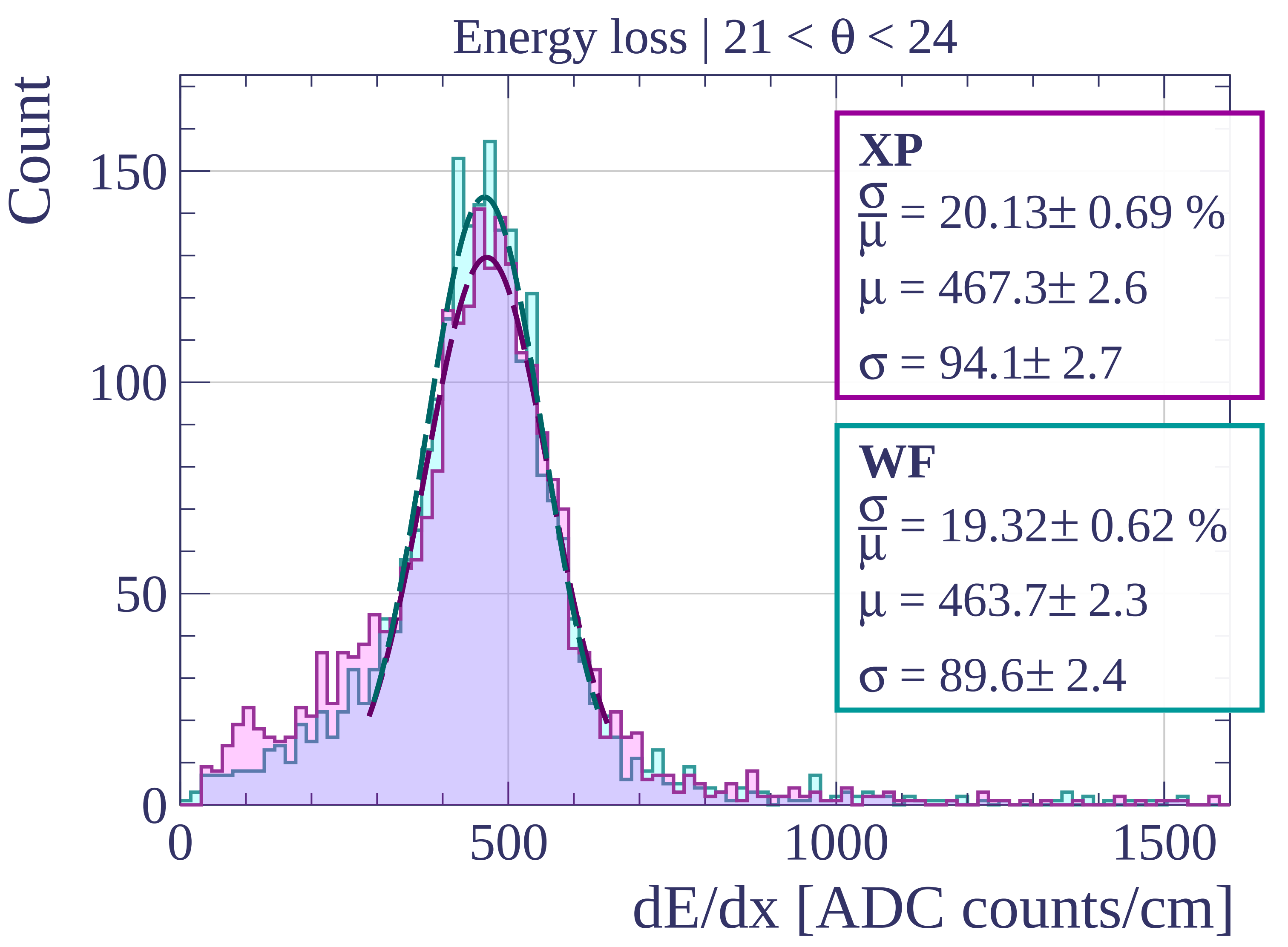

Energy loss | 12 < θ < 15
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 12.67± 0.24 %
μ = 449.9± 0.9
σ = 57.0± 1.0
WF
σ/μ = 13.05± 0.25 %
μ = 446.5± 0.9
σ = 58.3± 1.0

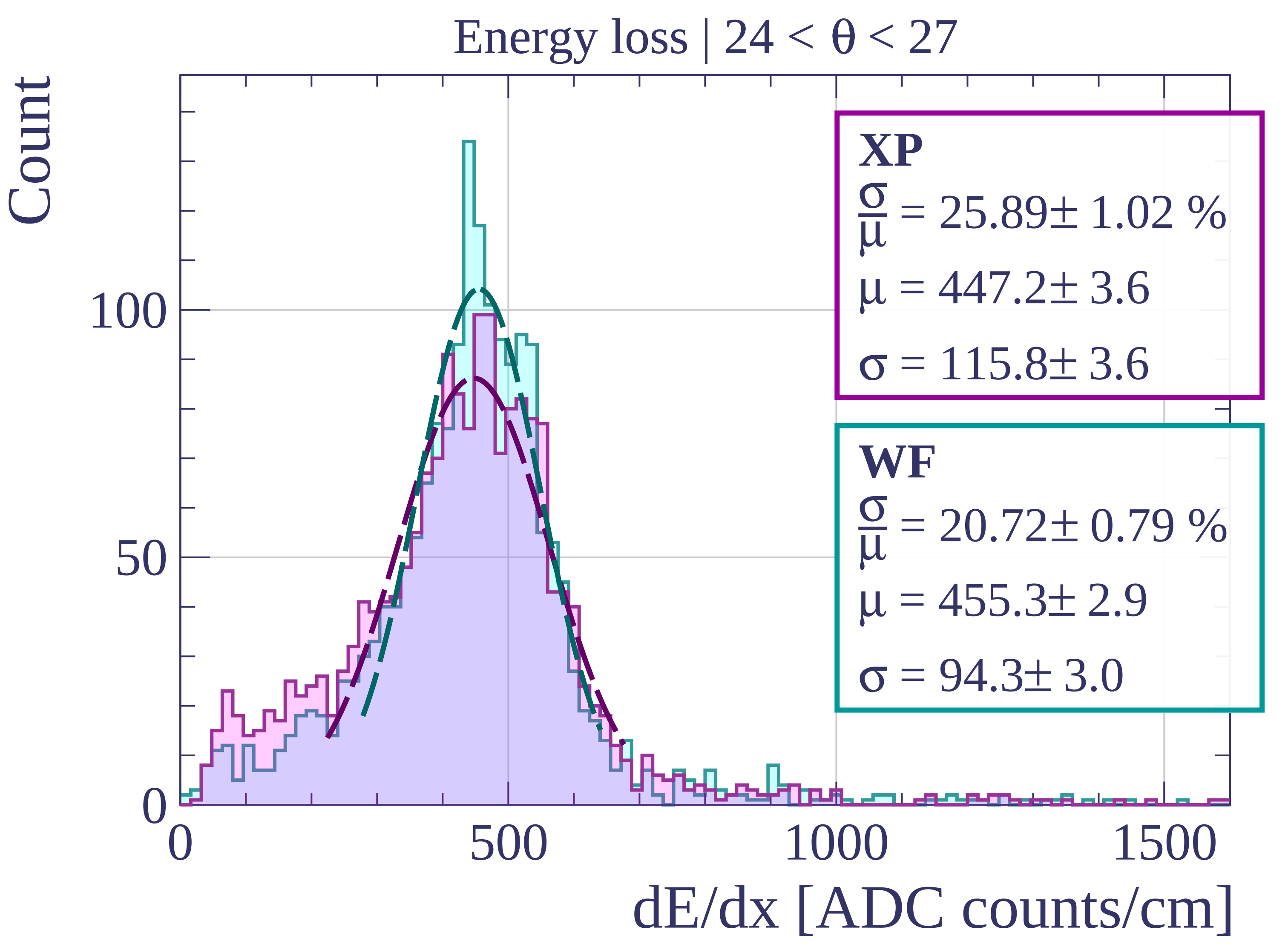


Energy loss | 15 < θ < 18
Count
0
100
200
300
400
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 18.80± 0.44 %
μ = 467.6± 1.7
σ = 87.9± 1.7
WF
σ/μ = 17.26± 0.36 %
μ = 464.3± 1.5
σ = 80.2± 1.4

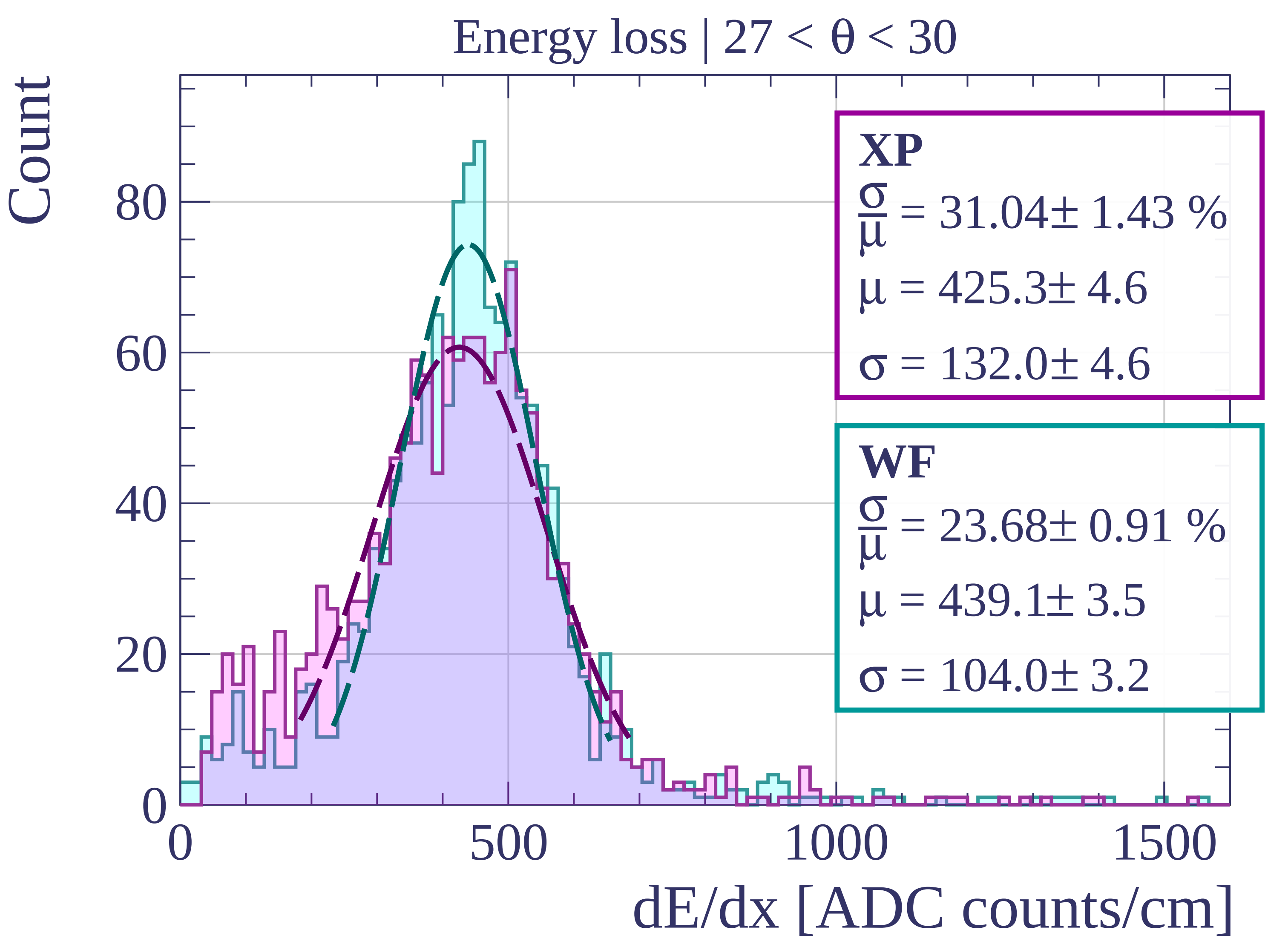


Energy loss | 18 < θ < 21
Count
0
100
200
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 20.32± 0.58 %
μ = 468.8± 2.2
σ = 95.2± 2.3
WF
σ/μ = 18.65± 0.51 %
μ = 466.6± 1.9
σ = 87.0± 2.0

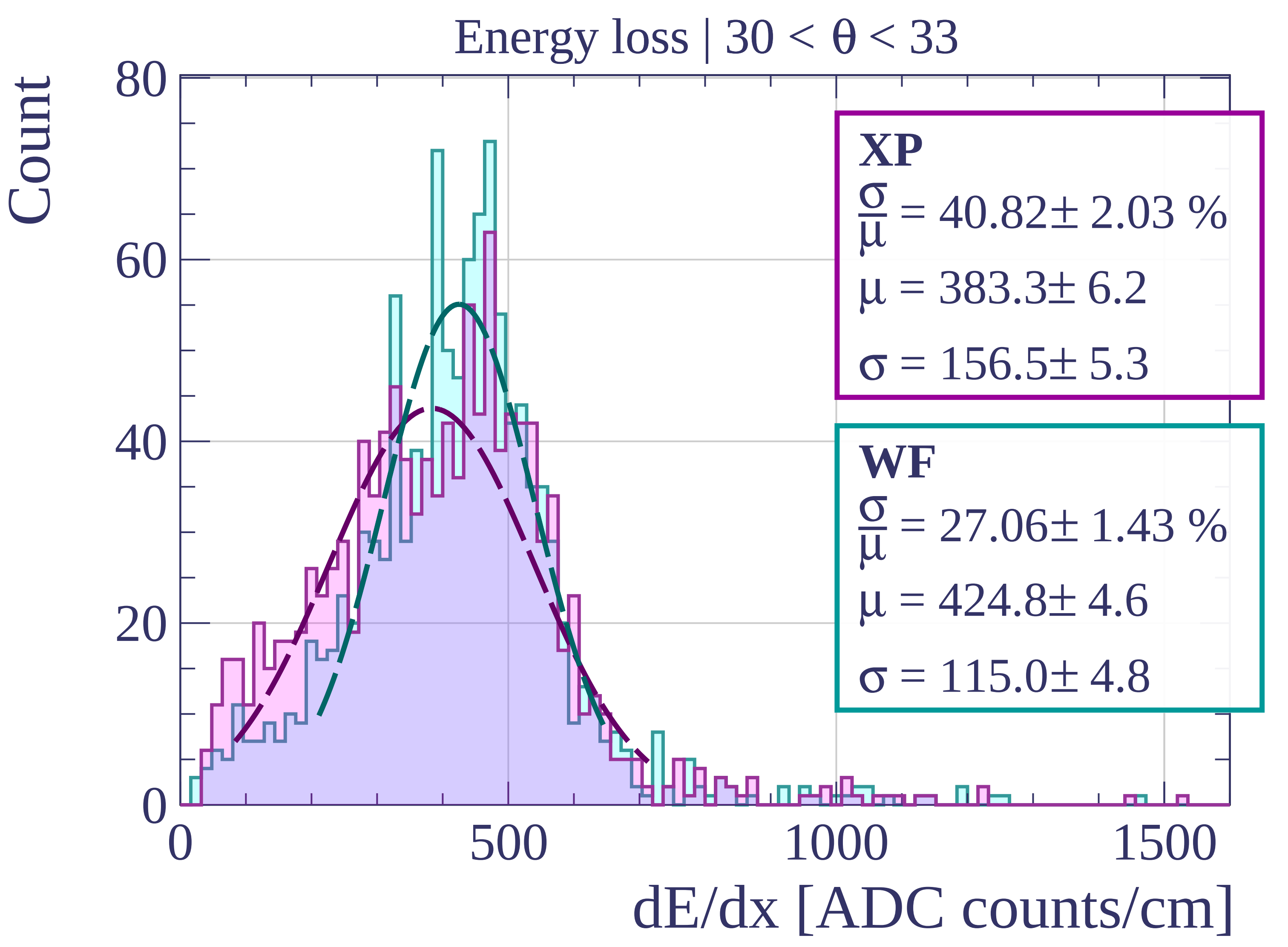

Energy loss | 21 < θ < 24
Count
dE/dx [ADC counts/cm]
0
50
100
150
0
500
1000
1500
XP
σ/μ = 20.13± 0.69 %
μ = 467.3± 2.6
σ = 94.1± 2.7
WF
σ/μ = 19.32± 0.62 %
μ = 463.7± 2.3
σ = 89.6± 2.4



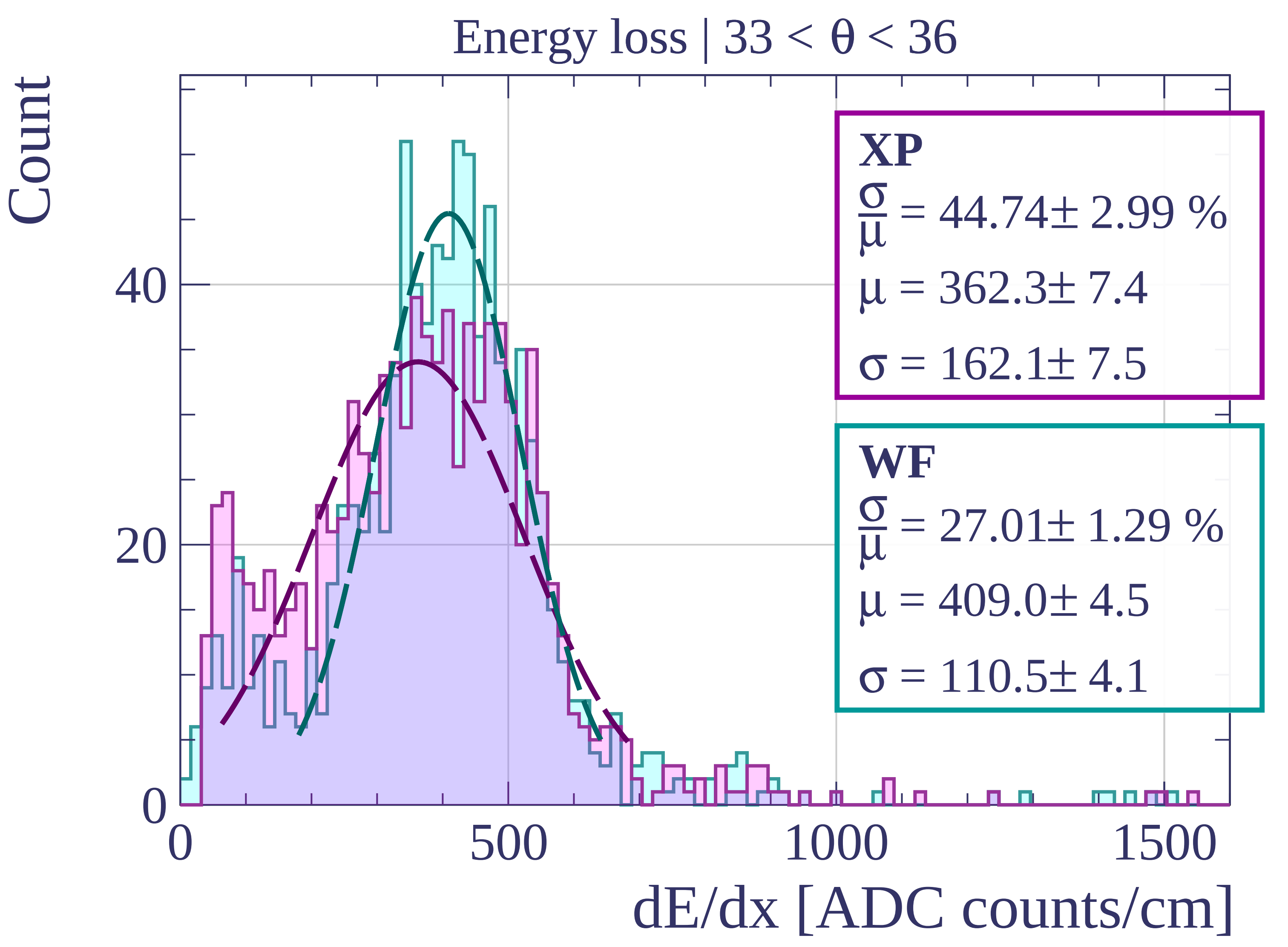
Energy loss | 24 < θ < 27
Count
dE/dx [ADC counts/cm]
0
50
100
0
500
1000
1500
XP
$\frac{\sigma}{\mu}$ = 25.89± 1.02 %
μ = 447.2± 3.6
σ = 115.8± 3.6
WF
$\frac{\sigma}{\mu}$ = 20.72± 0.79 %
μ = 455.3± 2.9
σ = 94.3± 3.0

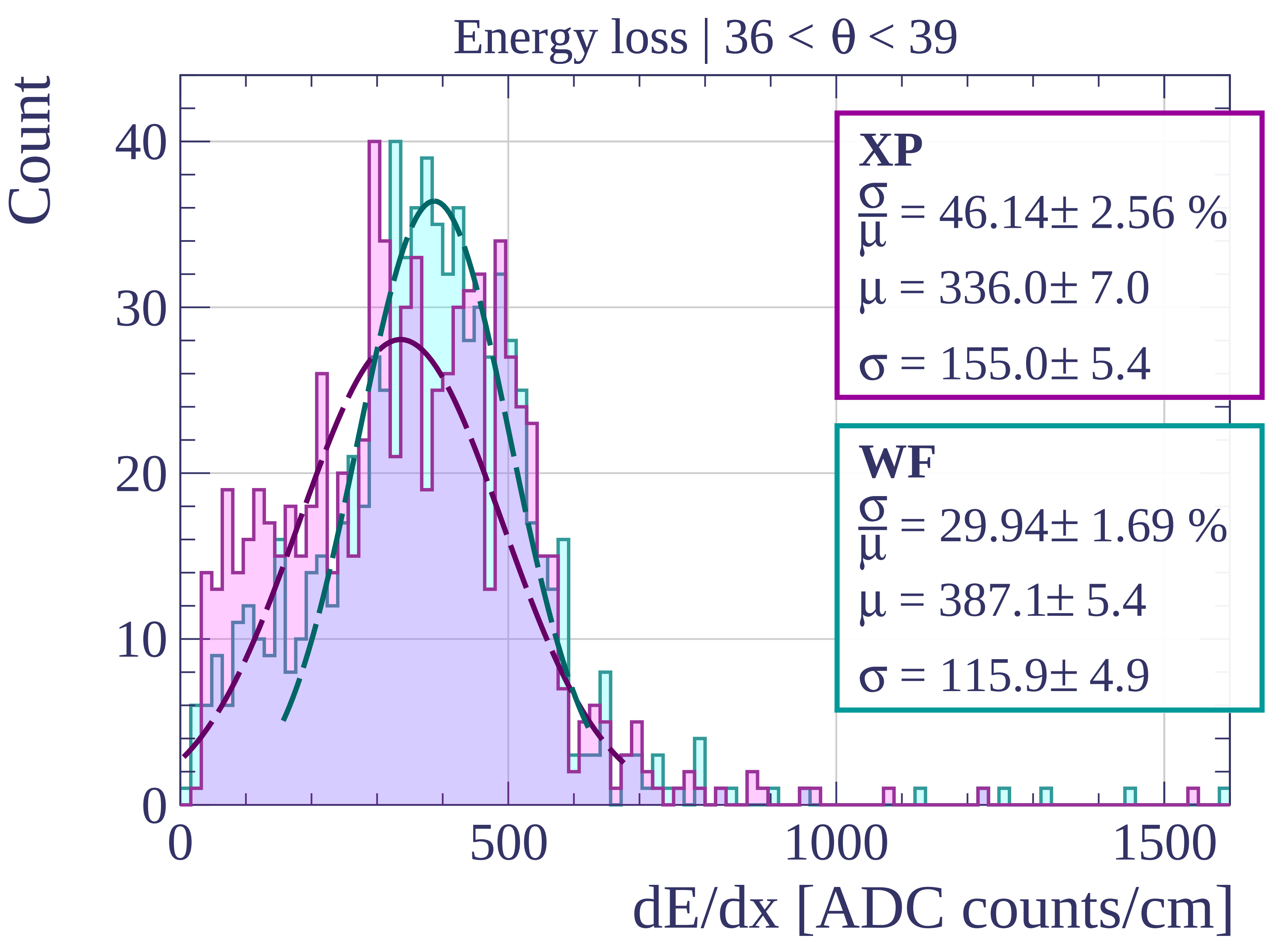


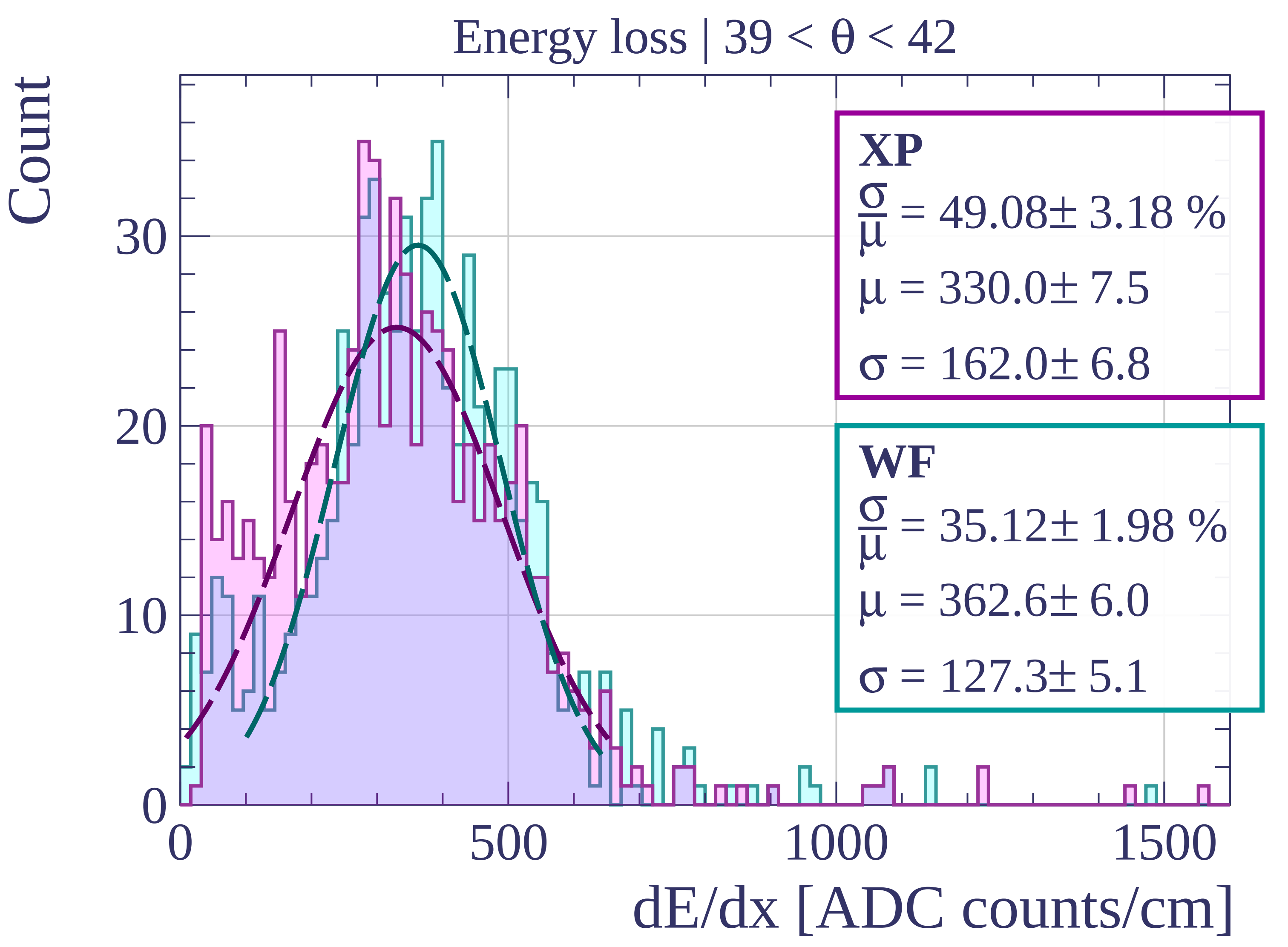
Energy loss | 27 < θ < 30
Count
dE/dx [ADC counts/cm]
0
20
40
60
80
0
500
1000
1500
XP
$\frac{\sigma}{\mu}$ = 31.04± 1.43 %
μ = 425.3± 4.6
σ = 132.0± 4.6
WF
$\frac{\sigma}{\mu}$ = 23.68± 0.91 %
μ = 439.1± 3.5
σ = 104.0± 3.2



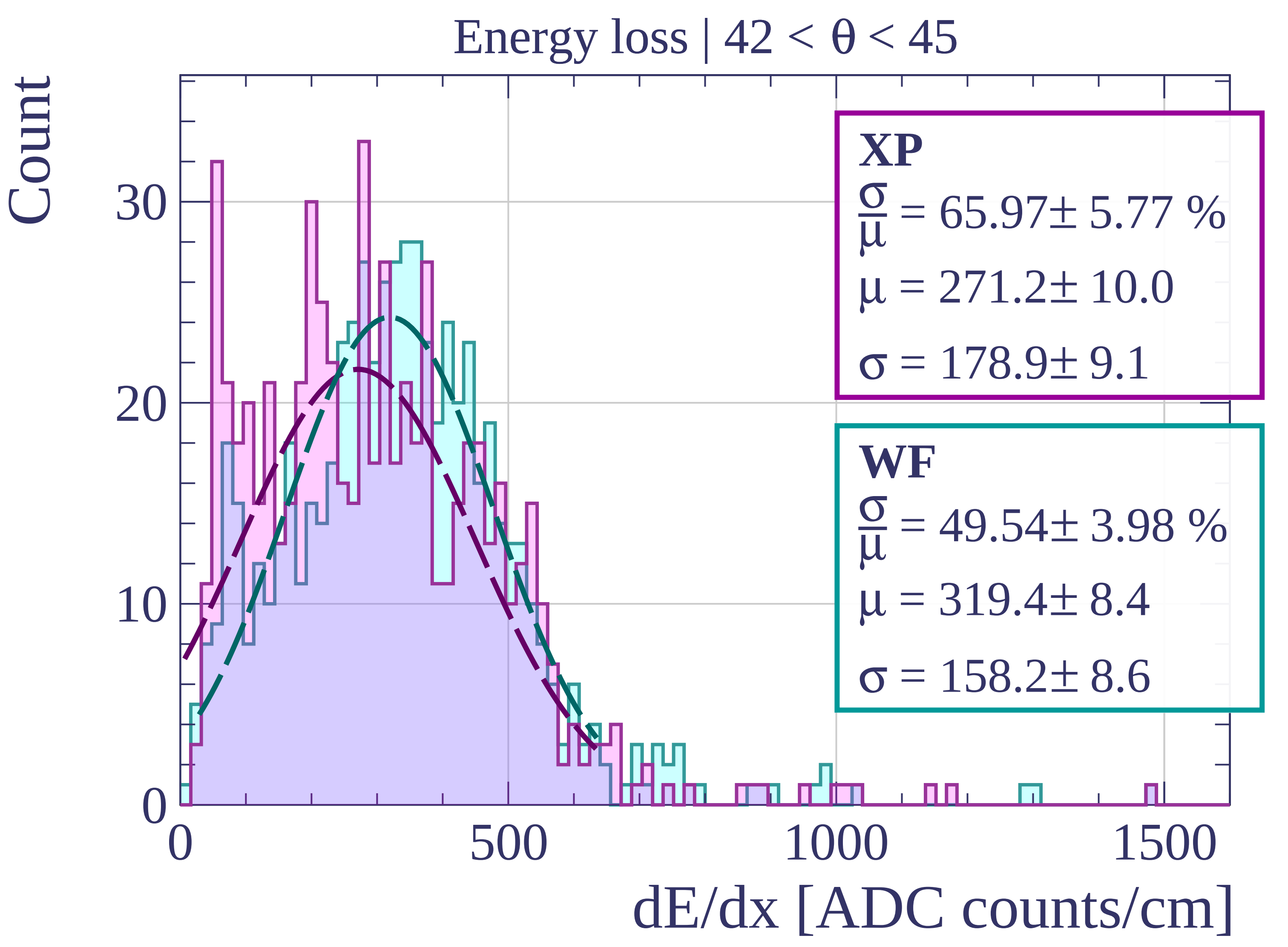
Energy loss | 30 < θ < 33
Count
dE/dx [ADC counts/cm]
0
20
40
60
80
0
500
1000
1500
XP
σ/μ = 40.82± 2.03 %
μ = 383.3± 6.2
σ = 156.5± 5.3
WF
σ/μ = 27.06± 1.43 %
μ = 424.8± 4.6
σ = 115.0± 4.8

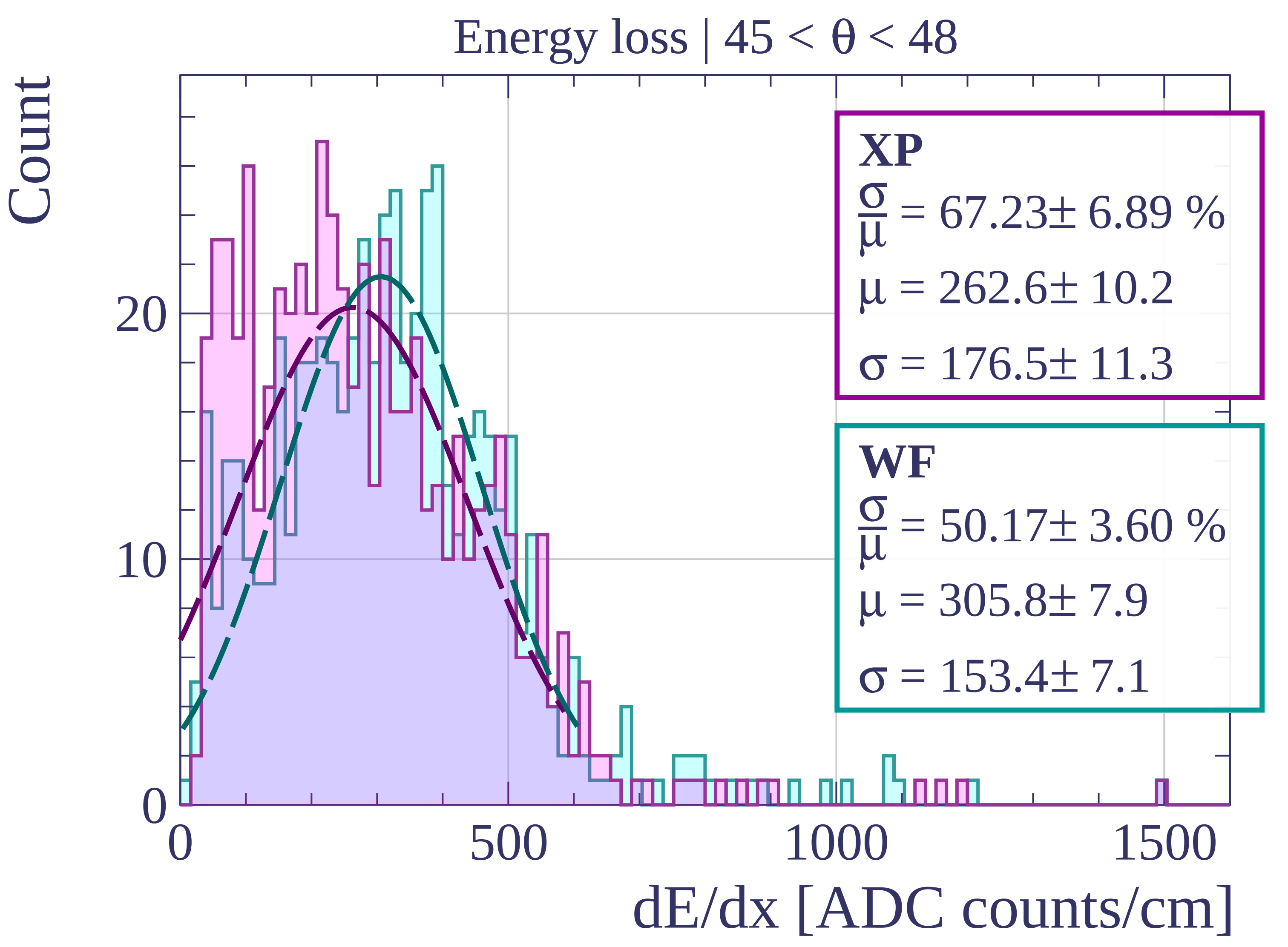

Energy loss | 33 < θ < 36
Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 44.74± 2.99 %
μ = 362.3± 7.4
σ = 162.1± 7.5
WF
σ/μ = 27.01± 1.29 %
μ = 409.0± 4.5
σ = 110.5± 4.1


Energy loss | 36 < θ < 39
Count
dE/dx [ADC counts/cm]
0
10
20
30
40
0
500
1000
1500
XP
σ/μ = 46.14± 2.56 %
μ = 336.0± 7.0
σ = 155.0± 5.4
WF
σ/μ = 29.94± 1.69 %
μ = 387.1± 5.4
σ = 115.9± 4.9

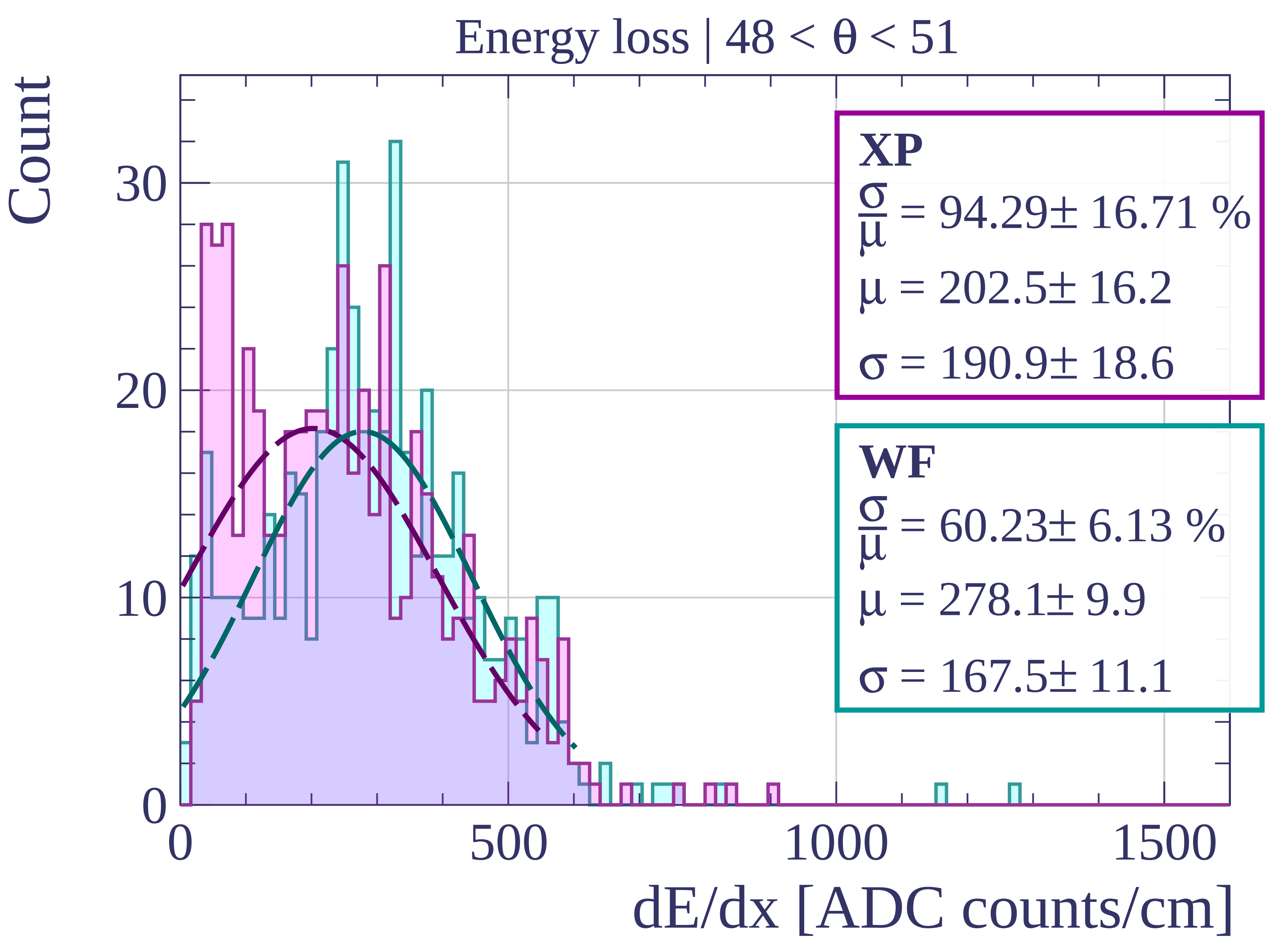


Energy loss | 39 < θ < 42
Count
dE/dx [ADC counts/cm]
0
10
20
30
0
500
1000
1500
XP
σ/μ = 49.08± 3.18 %
μ = 330.0± 7.5
σ = 162.0± 6.8
WF
σ/μ = 35.12± 1.98 %
μ = 362.6± 6.0
σ = 127.3± 5.1

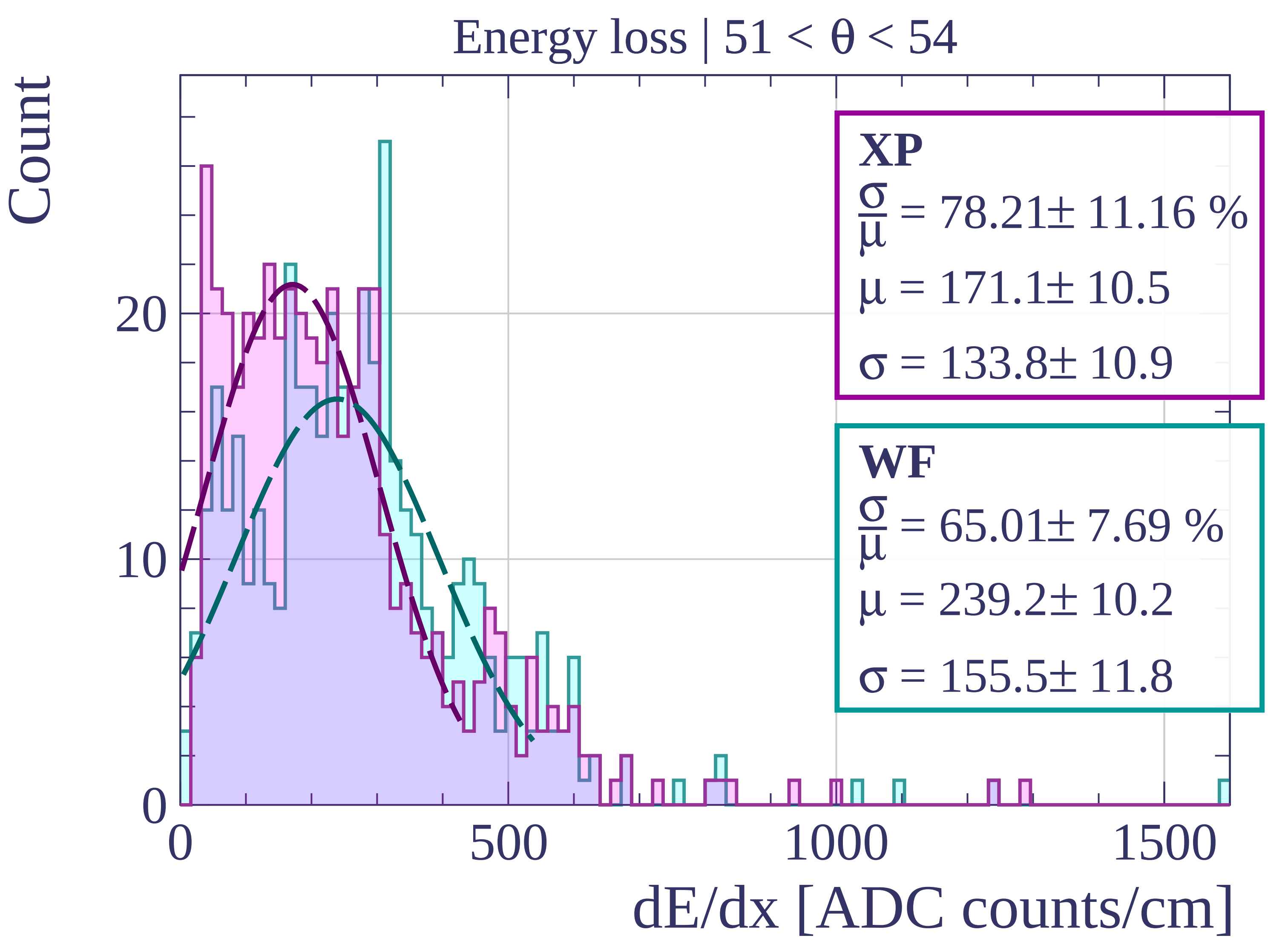

Energy loss | 42 < θ < 45
Count
0
10
20
30
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 65.97± 5.77 %
μ = 271.2± 10.0
σ = 178.9± 9.1
WF
σ/μ = 49.54± 3.98 %
μ = 319.4± 8.4
σ = 158.2± 8.6



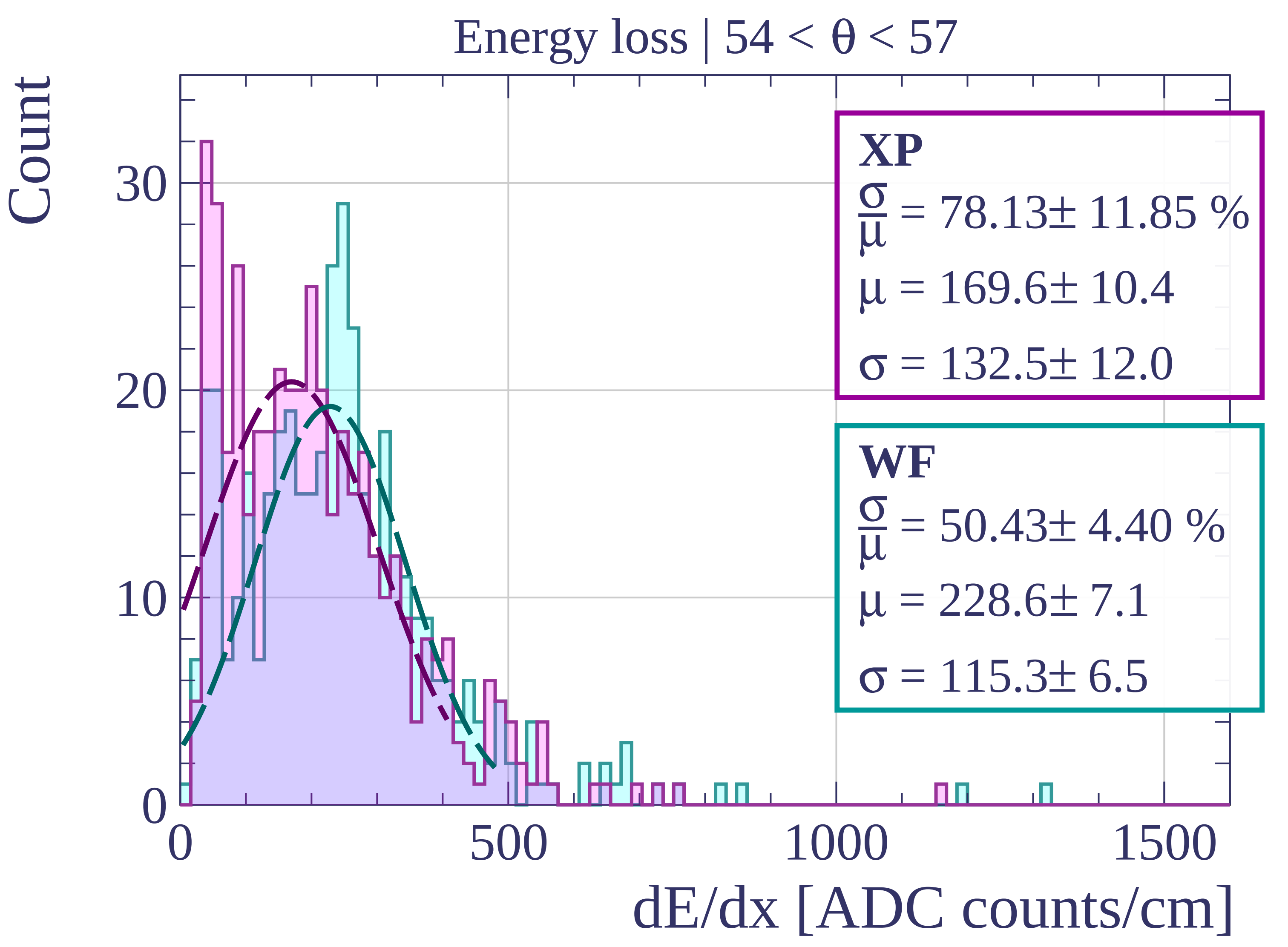
Energy loss | 45 < θ < 48
Count
dE/dx [ADC counts/cm]
0
10
20
0
500
1000
1500
XP
σ/μ = 67.23± 6.89 %
μ = 262.6± 10.2
σ = 176.5± 11.3
WF
σ/μ = 50.17± 3.60 %
μ = 305.8± 7.9
σ = 153.4± 7.1

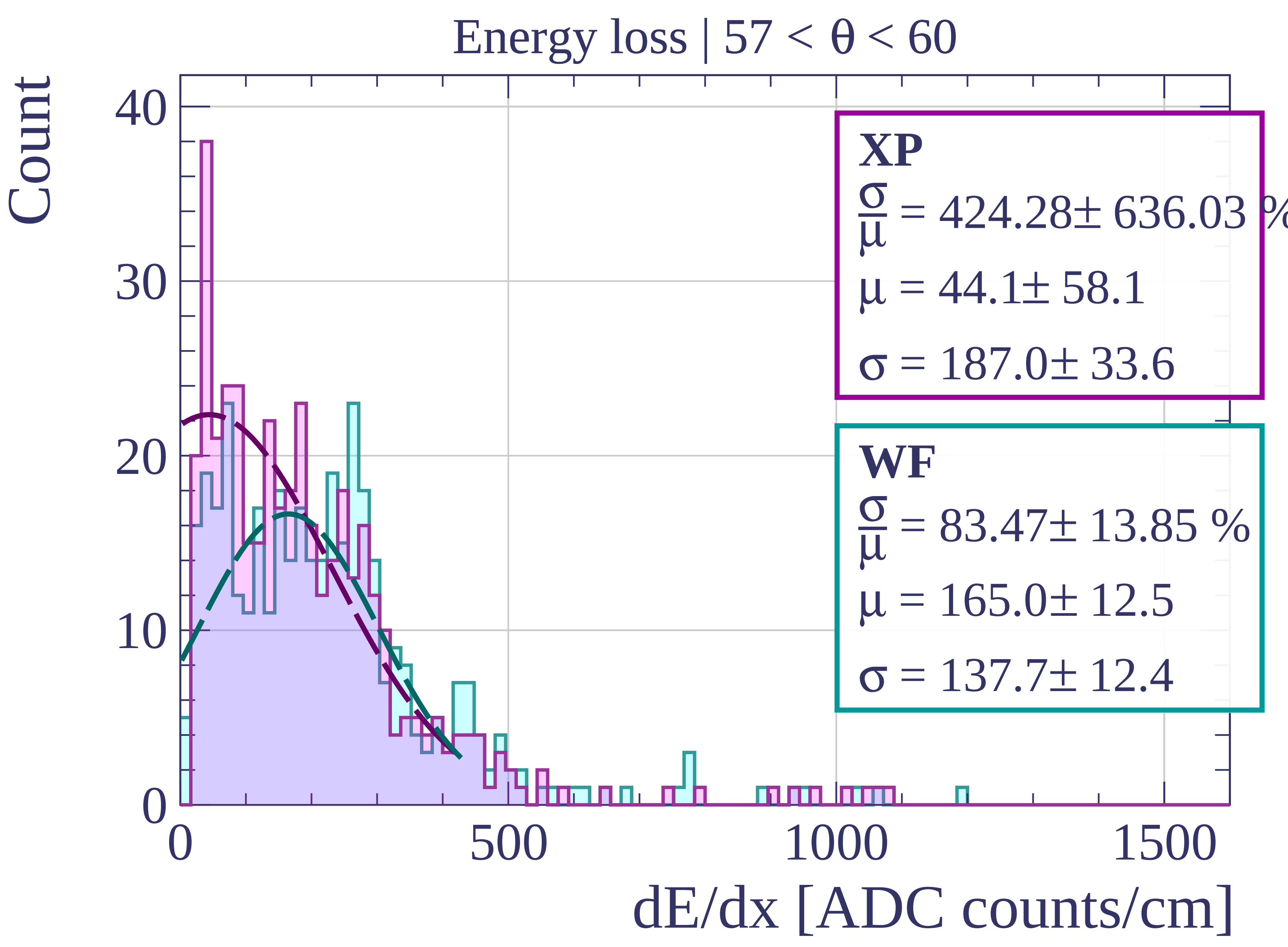


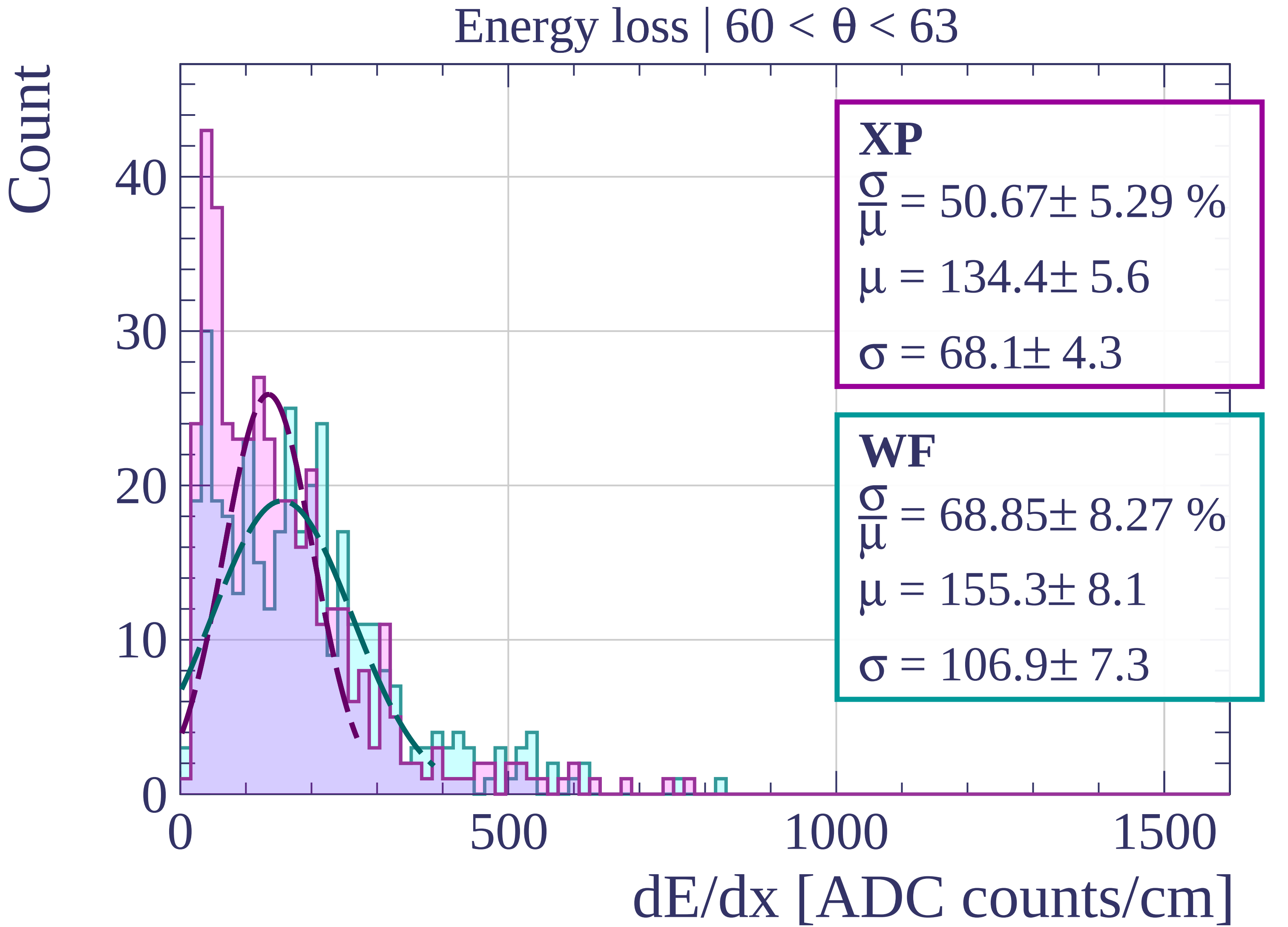
Energy loss | 48 < θ < 51
Count
dE/dx [ADC counts/cm]
0
10
20
30
0
500
1000
1500
XP
σ/μ = 94.29± 16.71 %
μ = 202.5± 16.2
σ = 190.9± 18.6
WF
σ/μ = 60.23± 6.13 %
μ = 278.1± 9.9
σ = 167.5± 11.1



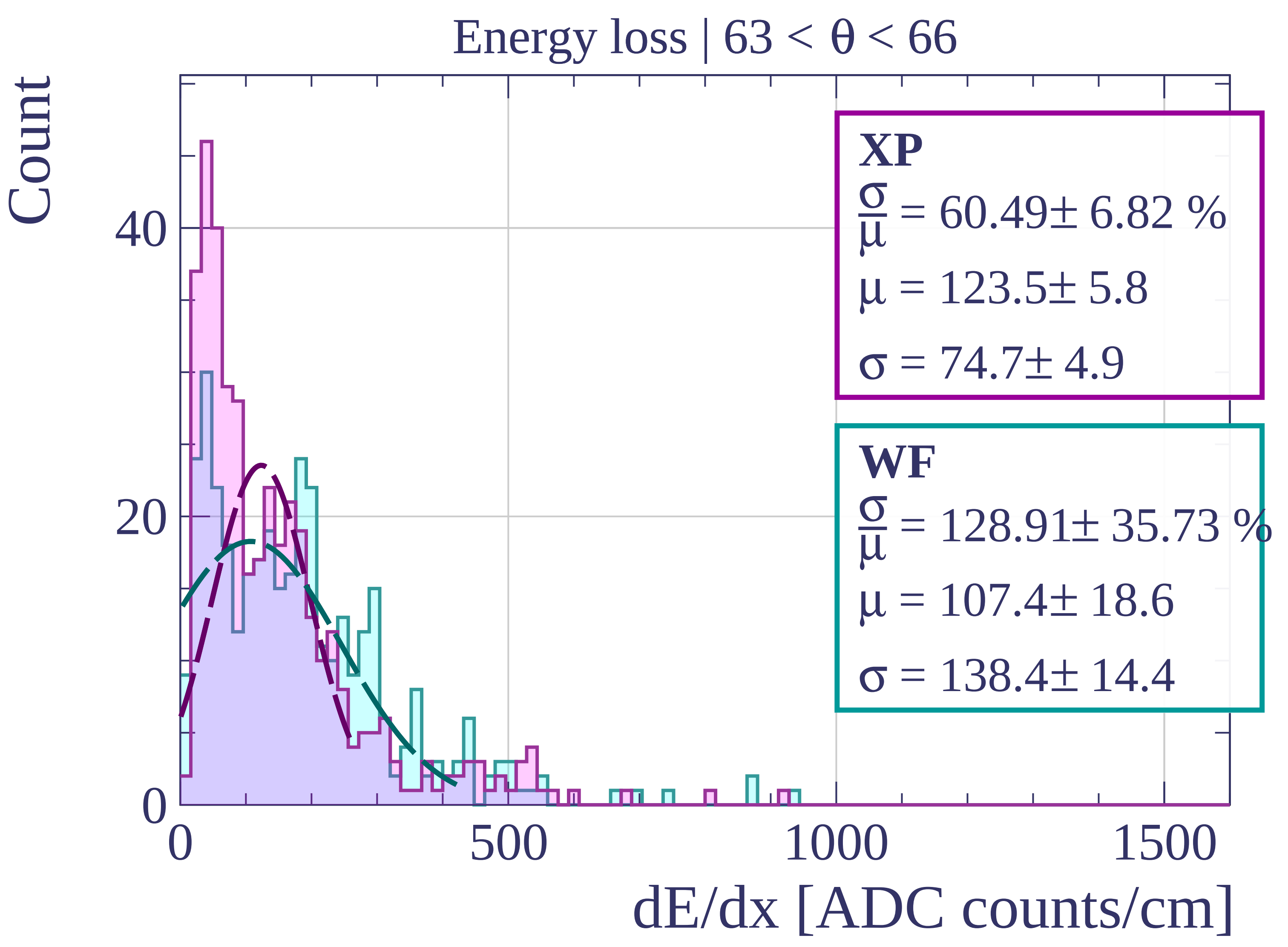
Energy loss | 51 < θ < 54
Count
dE/dx [ADC counts/cm]
0
10
20
0
500
1000
1500
XP
σ/μ = 78.21± 11.16 %
μ = 171.1± 10.5
σ = 133.8± 10.9
WF
σ/μ = 65.01± 7.69 %
μ = 239.2± 10.2
σ = 155.5± 11.8


Energy loss | 54 < θ < 57
Count
dE/dx [ADC counts/cm]
0
10
20
30
0
500
1000
1500
XP
σ/μ = 78.13± 11.85 %
μ = 169.6± 10.4
σ = 132.5± 12.0
WF
σ/μ = 50.43± 4.40 %
μ = 228.6± 7.1
σ = 115.3± 6.5

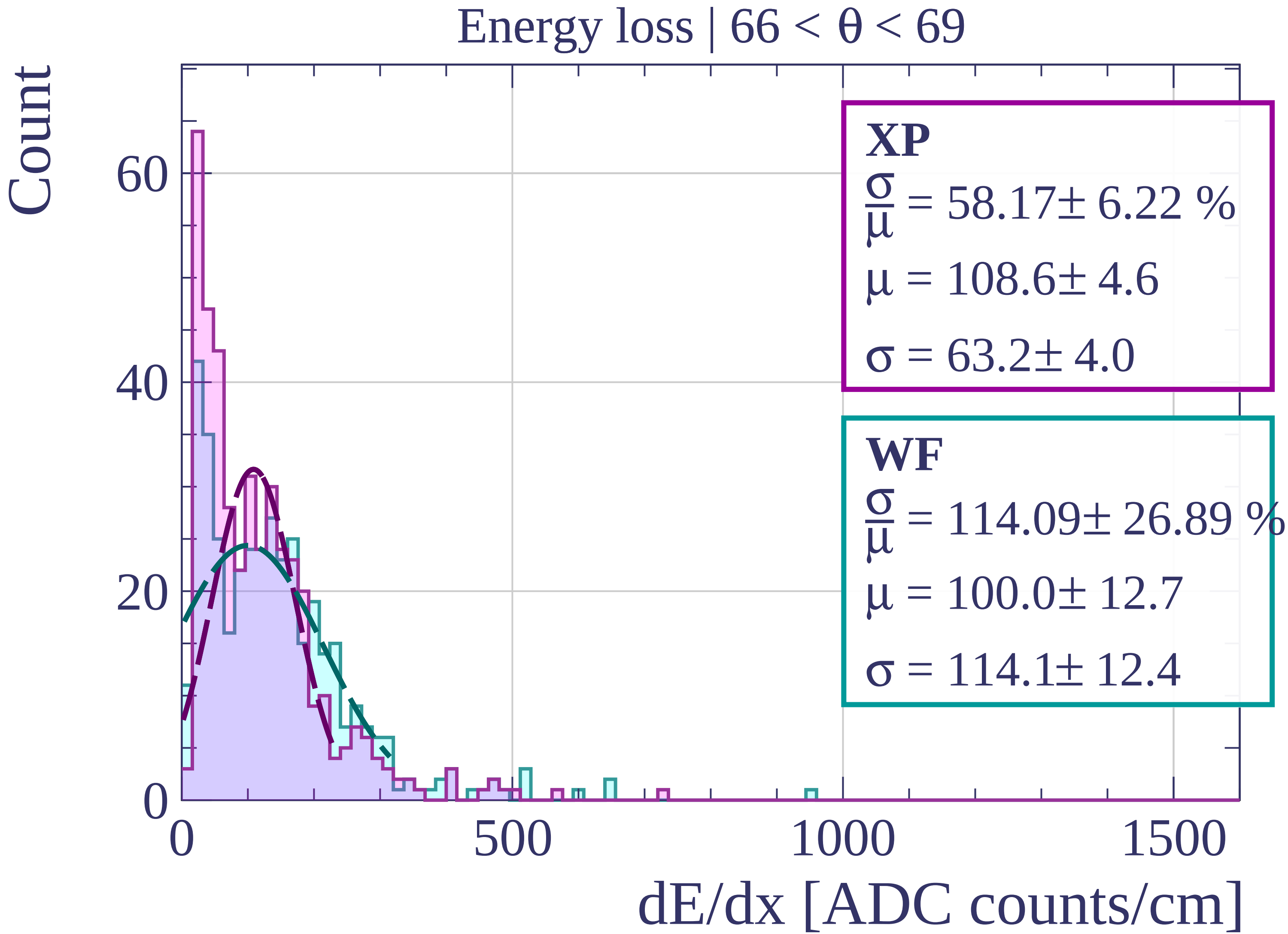

Energy loss | 57 < θ < 60
Count
40
30
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 424.28± 636.03 %
μ = 44.1± 58.1
σ = 187.0± 33.6
WF
σ/μ = 83.47± 13.85 %
μ = 165.0± 12.5
σ = 137.7± 12.4



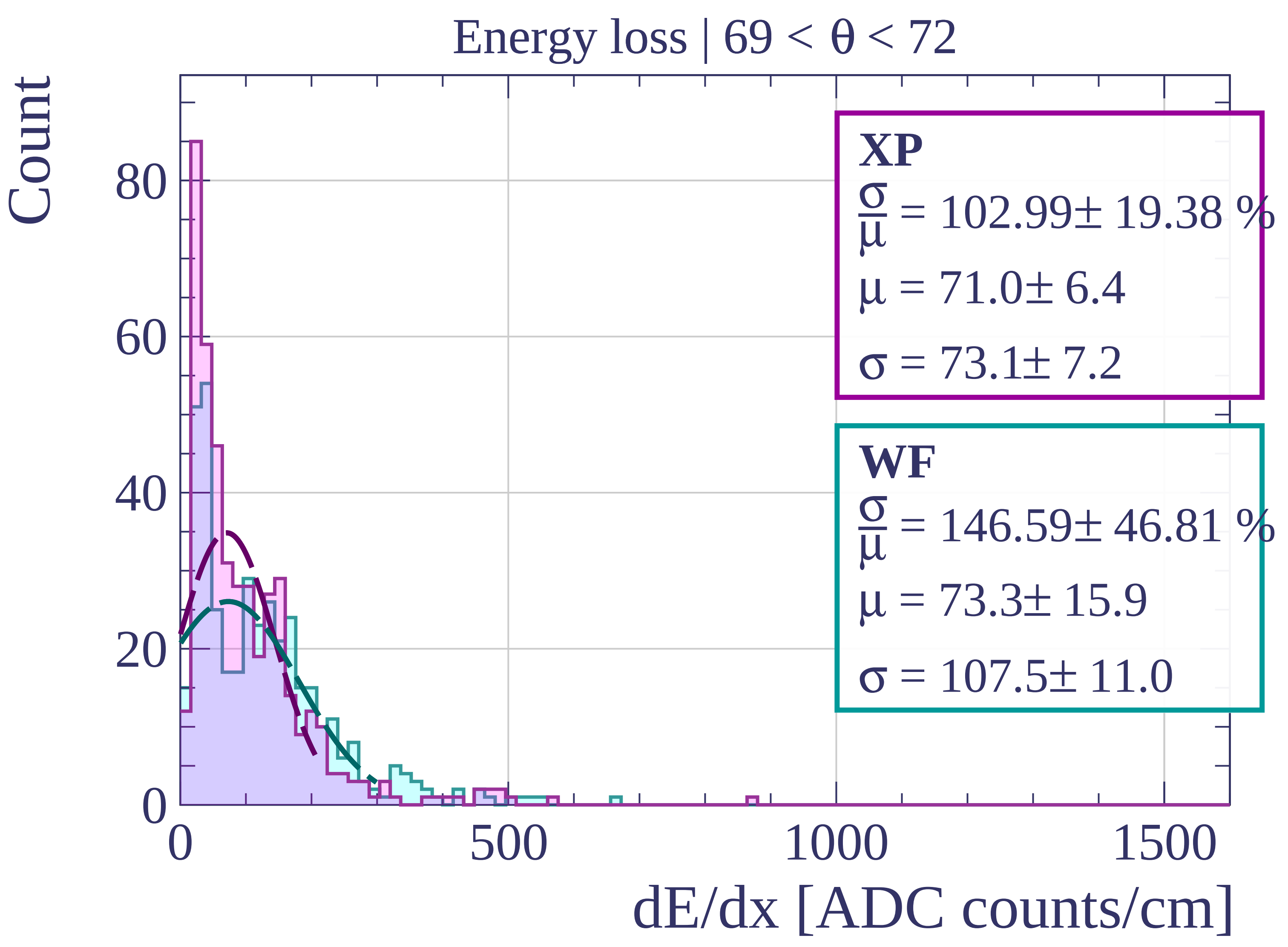
Energy loss | 60 < θ < 63
Count
0
10
20
30
40
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 50.67± 5.29 %
μ = 134.4± 5.6
σ = 68.1± 4.3
WF
σ/μ = 68.85± 8.27 %
μ = 155.3± 8.1
σ = 106.9± 7.3

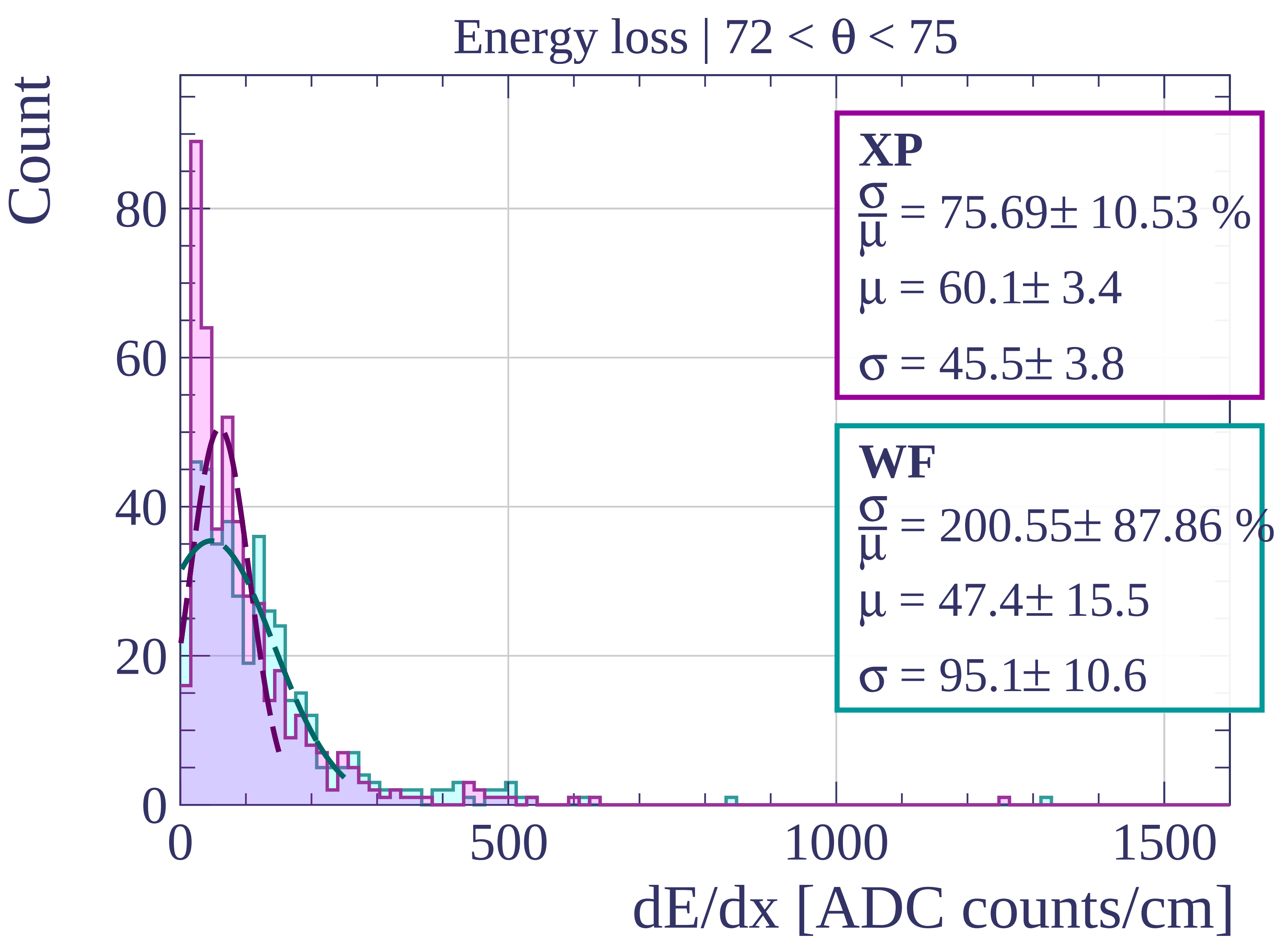


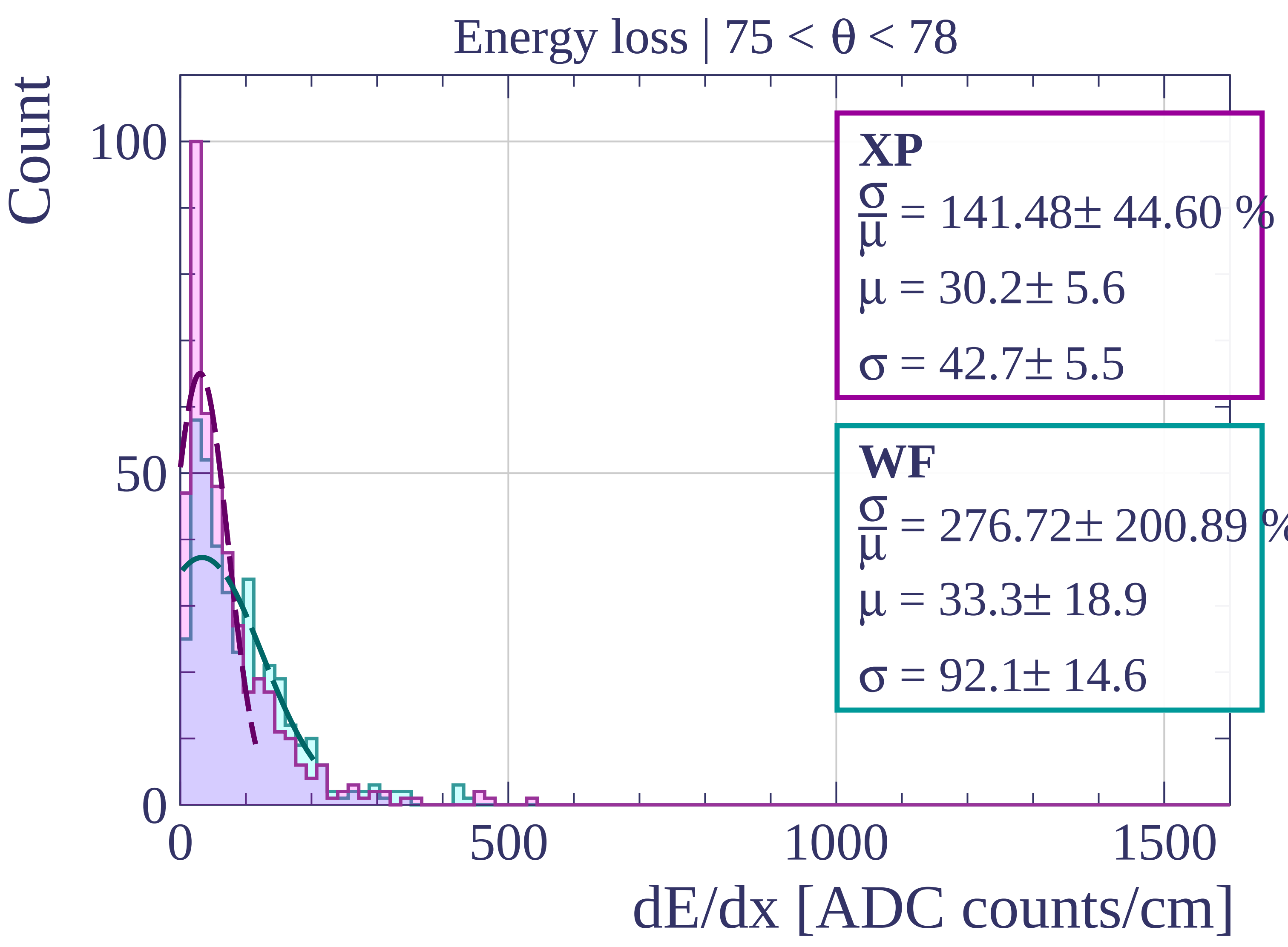
Energy loss | 63 < θ < 66
Count
0
20
40
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 60.49± 6.82 %
μ = 123.5± 5.8
σ = 74.7± 4.9
WF
σ/μ = 128.91± 35.73 %
μ = 107.4± 18.6
σ = 138.4± 14.4



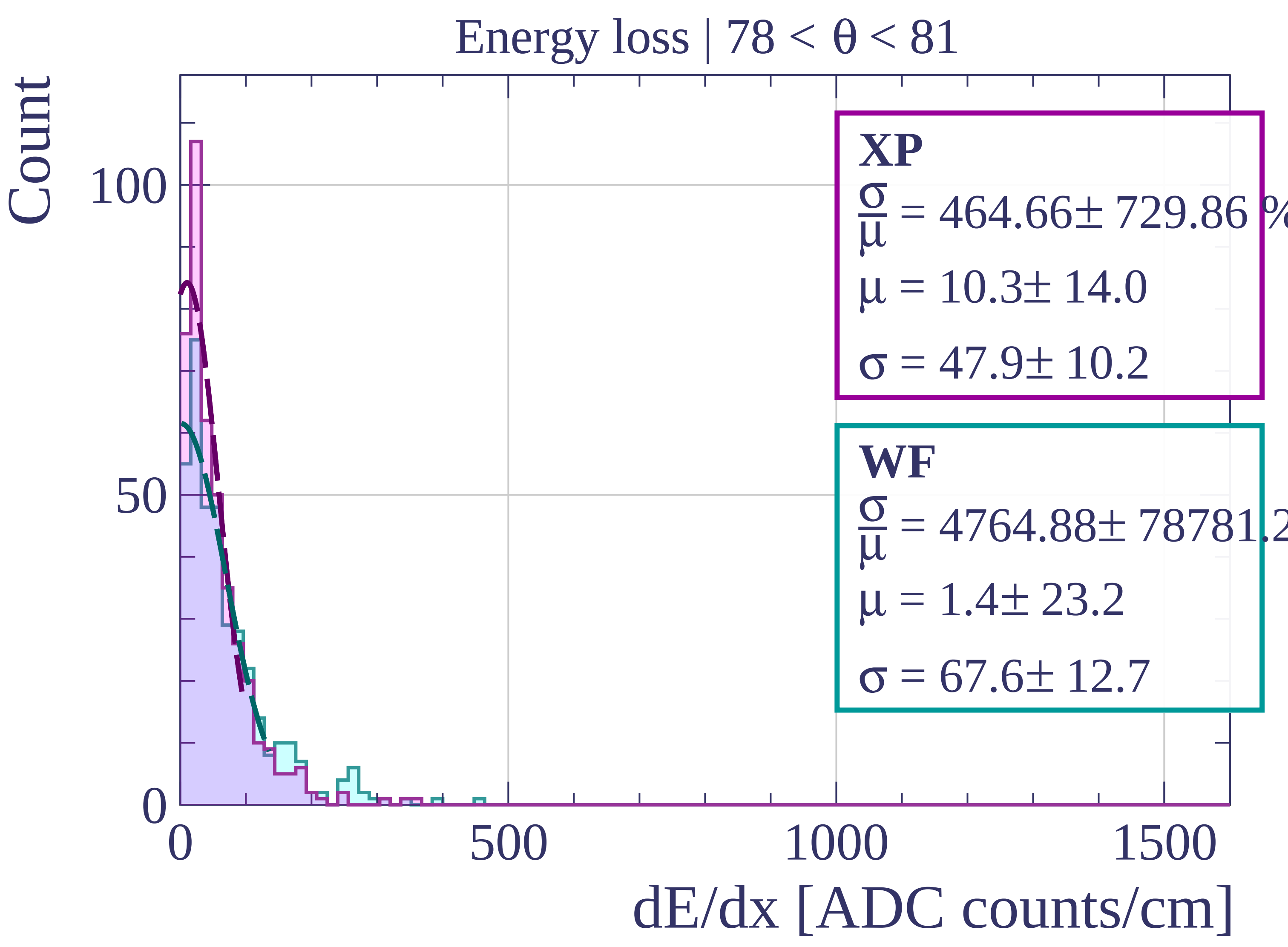
Energy loss | 66 < θ < 69
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 58.17± 6.22 %
μ = 108.6± 4.6
σ = 63.2± 4.0
WF
σ/μ = 114.09± 26.89 %
μ = 100.0± 12.7
σ = 114.1± 12.4


Energy loss | 69 < θ < 72
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 102.99± 19.38 %
μ = 71.0± 6.4
σ = 73.1± 7.2
WF
σ/μ = 146.59± 46.81 %
μ = 73.3± 15.9
σ = 107.5± 11.0

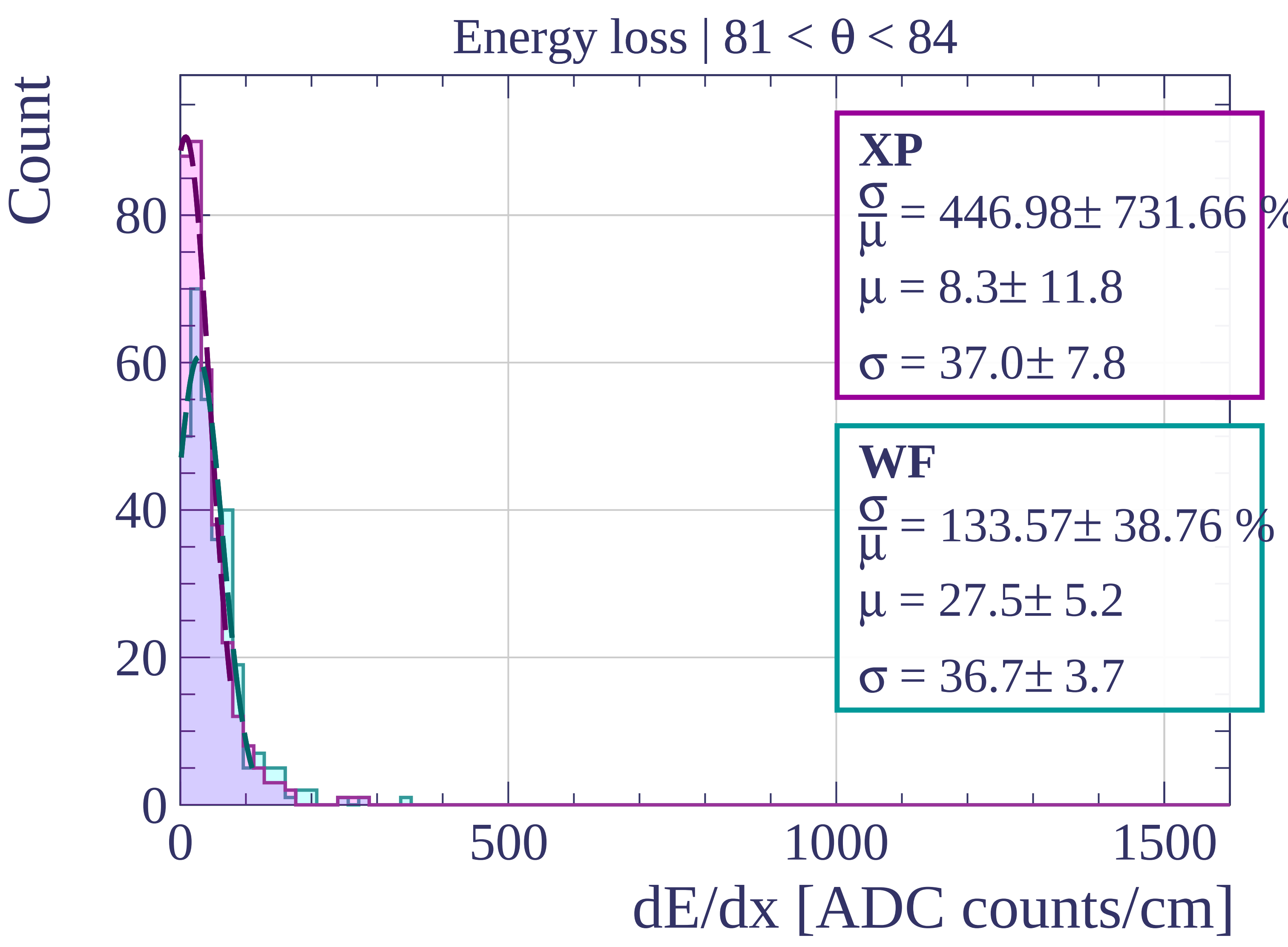

Energy loss | 72 < θ < 75
Count
dE/dx [ADC counts/cm]
XP
σ/μ = 75.69± 10.53 %
μ = 60.1± 3.4
σ = 45.5± 3.8
WF
σ/μ = 200.55± 87.86 %
μ = 47.4± 15.5
σ = 95.1± 10.6
0
20
40
60
80
0
500
1000
1500



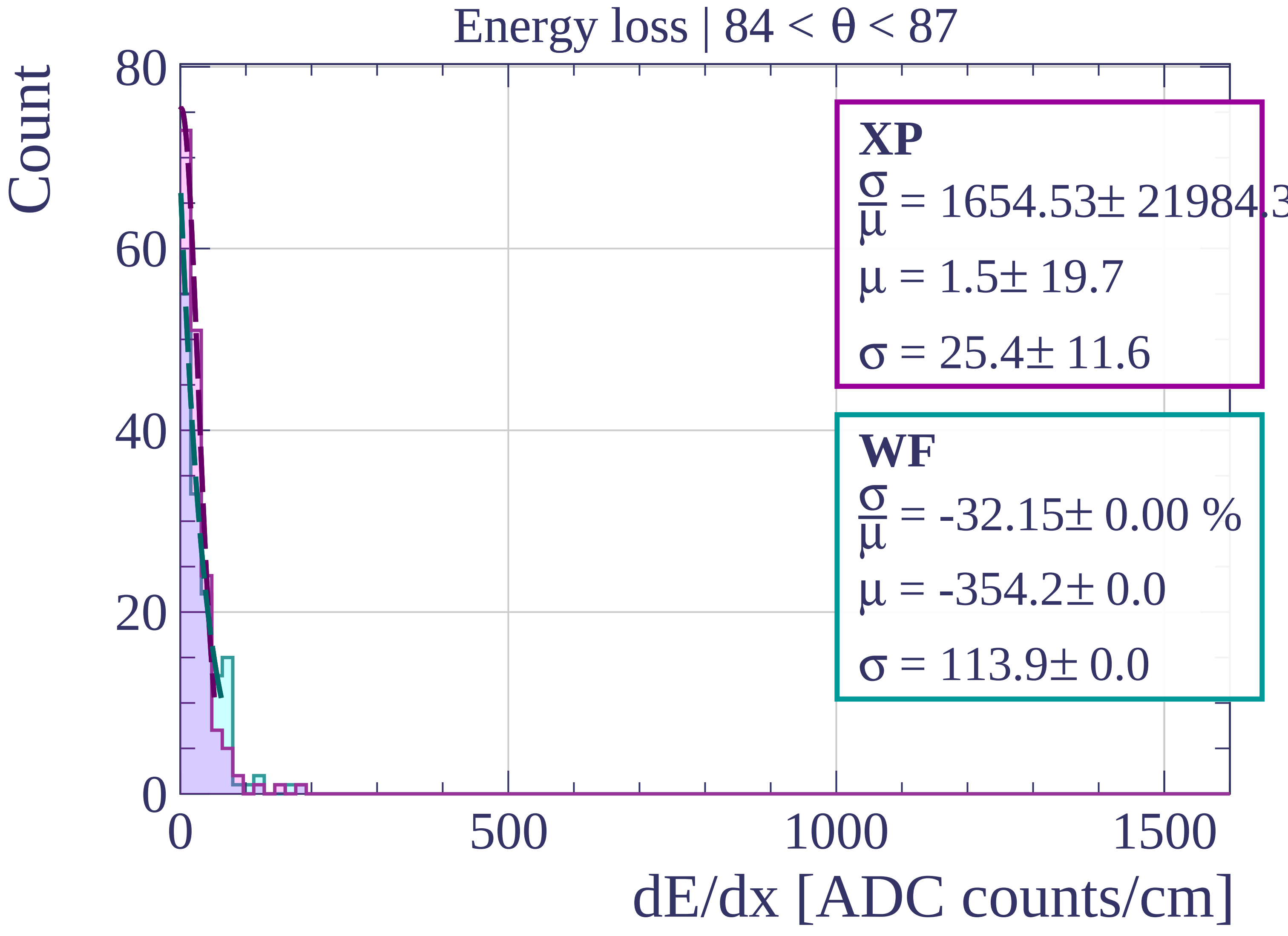
Energy loss | 75 < θ < 78
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 141.48± 44.60 %
μ = 30.2± 5.6
σ = 42.7± 5.5
WF
σ/μ = 276.72± 200.89 %
μ = 33.3± 18.9
σ = 92.1± 14.6



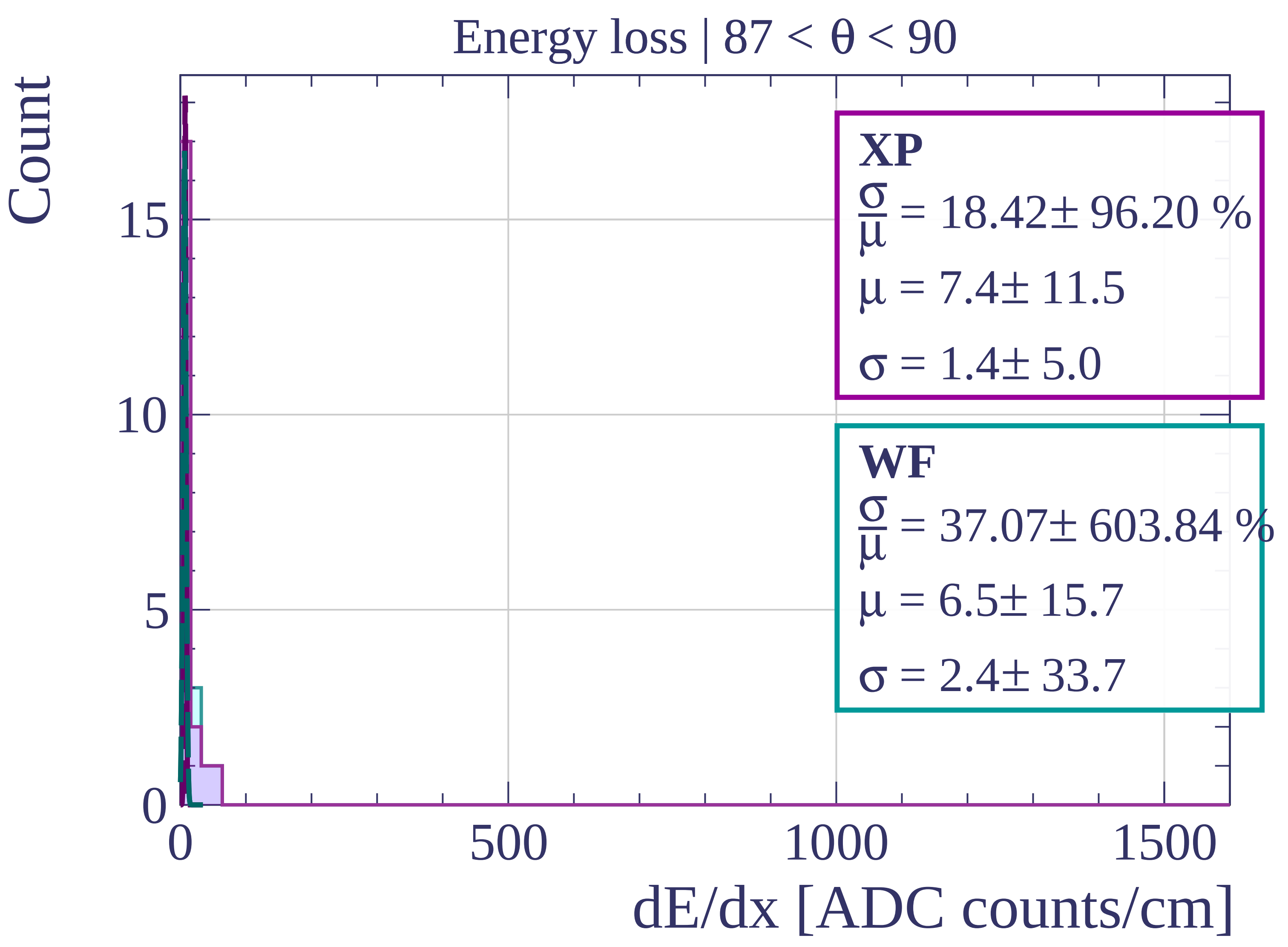
Energy loss | 78 < θ < 81
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 464.66± 729.86 %
μ = 10.3± 14.0
σ = 47.9± 10.2
WF
σ/μ = 4764.88± 78781.2
μ = 1.4± 23.2
σ = 67.6± 12.7



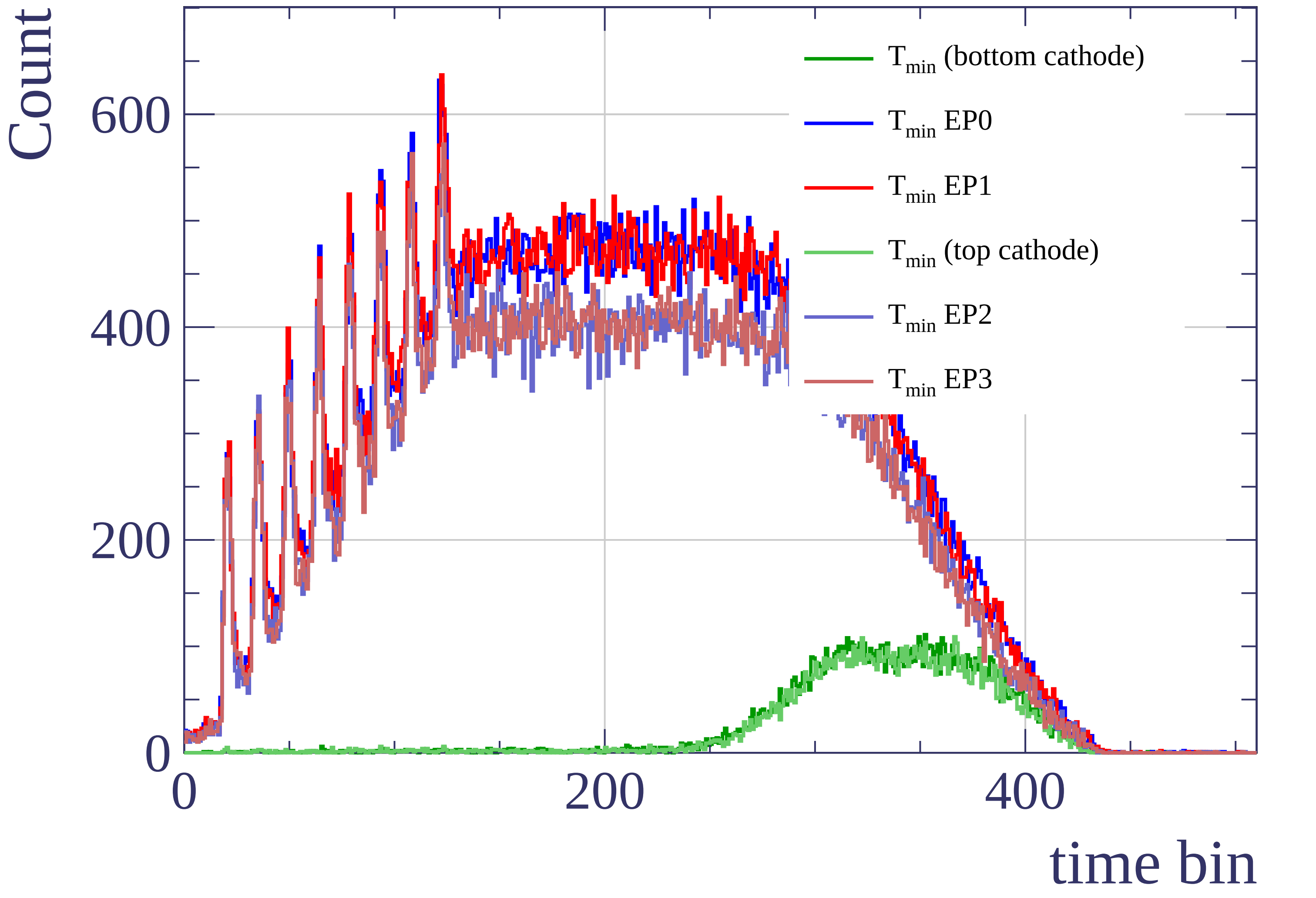
Energy loss | 81 < θ < 84
Count
0
20
40
60
80
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 446.98± 731.66 %
μ = 8.3± 11.8
σ = 37.0± 7.8
WF
σ/μ = 133.57± 38.76 %
μ = 27.5± 5.2
σ = 36.7± 3.7

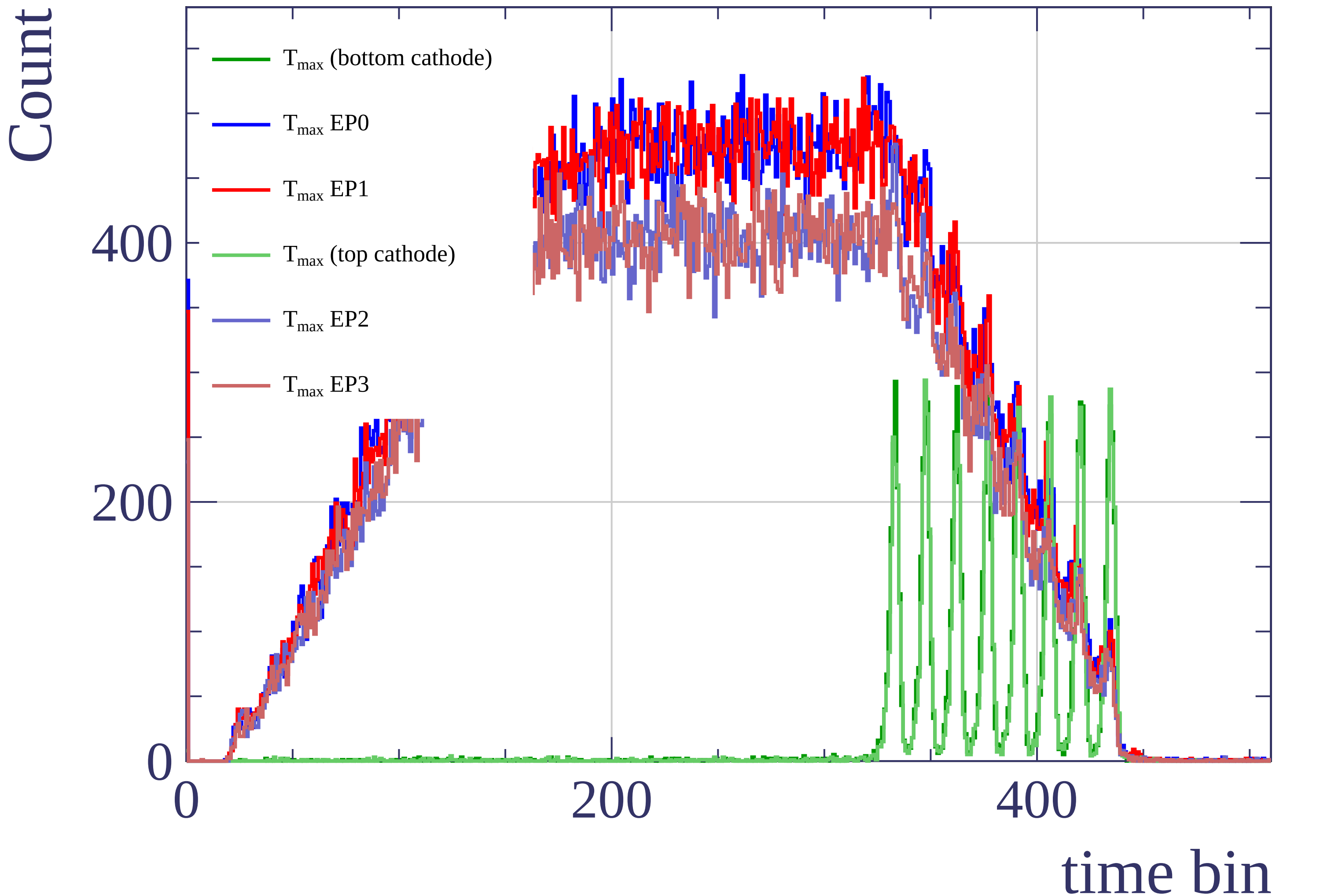


Energy loss | 84 < θ < 87
Count
dE/dx [ADC counts/cm]
0
20
40
60
80
0
500
1000
1500
XP
$\frac{\sigma}{\mu}$ = 1654.53± 21984.3
μ = 1.5± 19.7
σ = 25.4± 11.6
WF
$\frac{\sigma}{\mu}$ = -32.15± 0.00 %
μ = -354.2± 0.0
σ = 113.9± 0.0



Energy loss | 87 < θ < 90
Count
dE/dx [ADC counts/cm]
0
5
10
15
0
500
1000
1500
XP
$\frac{\sigma}{\mu}$ = 18.42± 96.20 %
μ = 7.4± 11.5
σ = 1.4± 5.0
WF
$\frac{\sigma}{\mu}$ = 37.07± 603.84 %
μ = 6.5± 15.7
σ = 2.4± 33.7


Count
600
400
200
0
0
200
400
time bin
$T_{min}$ (bottom cathode)
$T_{min}$ EP0
$T_{min}$ EP1
$T_{min}$ (top cathode)
$T_{min}$ EP2
$T_{min}$ EP3



Count
$T_{max}$ (bottom cathode)
$T_{max}$ EP0
$T_{max}$ EP1
$T_{max}$ (top cathode)
$T_{max}$ EP2
$T_{max}$ EP3
0
200
400
0
200
400
time bin